MASSIVE SU(2) GAUGE THEORY IN THREE DIMENSIONS

Abstract. We study a massive SU(2) gauge theory in three dimensions from the chamber lattice model with finitely many fundamental fermion flavours. At sufficiently weak gauge coupling, limits taken first in the volume and then along a spacing subsequence give centered correlations of the action density and scalar and pseudoscalar fermion bilinears. They are covariant under the full Euclidean group and reflection positive at every plane. They reconstruct a finite family of Hermitian local Wightman fields, each with nonzero two-point function. The proof combines the fermion block expansion with the pure gauge construction, controls the cumulative mass correction, and derives the insertion and shrinking-plate bounds needed for rotations and reconstruction.

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1. Introduction

We consider SU(2) gauge fields coupled to $N_f \geq 1$ massive four-component fermions in the fundamental representation. The lattice regularization uses the balanced chamber Dirac operator with Wilson parameter 1/2, defined in section 2. At spacing $a = L^{-K}$, the inverse gauge coupling and bare fermion mass are $\beta = \mathsf{b}/a$ and $m_0 = a\mathsf{m}$. The torus measure is

$$Z^{-1} e^{-\beta A(U)} |\det D_{U,+}|^{2N_f}\, dU, \qquad A(U) = \sum_p (1 - \tfrac{1}{2} \operatorname{tr} U_p).$$

Here $D_{U,+}$ is the $\tau_3 = 1$ internal block for one flavour of the massive chamber operator. Its determinant and that of the other block are complex conjugates.

Theorem 0. Fix the number of flavours and a compact positive mass interval. For sufficiently weak gauge coupling, the chamber model has subsequential continuum limits of the centered action-density and scalar and pseudoscalar bilinear correlations on separated supports, after the infinite-volume limit. Every such limit is Euclidean covariant and reflection positive at every plane. It reconstructs a finite family of Hermitian local Wightman fields, each with a nonzero two-point function. ■□

The precise statement, including the order of limits, field family and parameter assumptions, is Thm. A.

We retain the action density and a finite family of colour-singlet fermion bilinears. Let $\tau_0 = 1, \tau_1, \tau_2, \tau_3$ be the Pauli basis in the internal two-dimensional factor, and choose Hermitian flavour matrices t_j, $1 \le j \le N_f^2$, with $\operatorname{tr}(t_j t_l) = \delta_{jl}$. Define

$$(c_0, c_1, c_2, c_3) = (1, i, i, 1), \quad \Gamma_{a,j} = 1_{\mathrm{s}} \otimes c_a \tau_a \otimes 1_{\mathrm{c}} \otimes t_j, \quad \mathfrak{B} = \{\Gamma_{a,j}\}, \quad \mathcal{I} = \{0\} \cup \mathfrak{B}.$$

The corresponding observables, for real $f \in C_c^1(\mathbb{R}^3)$, are

$$\Phi_{0,K}(f) = a^{-1} \sum_p f(x(p))(1 - \tfrac{1}{2} \operatorname{tr} U_p), \qquad \Phi_{\Gamma,K}(f) = a \sum_x f(ax) \sum_{v : x_v = x} \bar{\psi}(v) \Gamma \psi(v). \tag{1}$$

In the bilinear sum $x \in \mathbb{Z}^3$ is a lattice site. Plaquettes are counted once in each positive coordinate orientation, and $x(p)$ is their physical center. The basis is reflection real: the Grassmann reflection sends each observable to its geometrically reflected test function without an index sign. An improper spatial isometry instead acts with signs $\pi_0 = 1$ and $\pi_{\Gamma_{a,j}} = (+1, -1, +1, -1)_a$.

At fixed K, take any weak limit ν_K of the gauge marginals as the physical torus sides $2L^m$ tend to infinity; spinors are antiperiodic in each coordinate. Local fermion expectations in this state use the Gaussian Berezin functional of D_U^{-1}. Let S^T_{n,K,ν_K} be the joint cumulants of (1). Only tuples with pairwise disjoint supports are considered. For a limiting family $(S_n)_{n \ge 2}$, its centered moments are

$$\sigma(\varnothing) = 1, \quad \sigma((\alpha, f)) = 0, \qquad \sigma(X) = \sum_{\substack{\mathcal{P} \text{ partition of } [n] \\ |B| \ge 2 \ (B \in \mathcal{P})}} \prod_{B \in \mathcal{P}} S_{|B|}(X_B). \tag{2}$$

Theorem A (The massive two-colour theory)**.** Fix $N_f \ge 1$ and $0 < m_- \le m_+ < 1/2$. Choose an odd integer $L \ge 13$ and the auxiliary constants in the order and numerical ranges of Prop. 10.20. Then there is $\mathsf{b}_0 < \infty$ with the following property. For every $\mathsf{b} \ge \mathsf{b}_0$, $\mathsf{m} \in [m_-, m_+]$, sequence $K_j \to \infty$ and choices of the states ν_{K_j} above, there is a common subsequence on which

$$S^T_{n,K_j,\nu_{K_j}}(X) \longrightarrow S_n(X) \qquad (n \ge 2)$$

for every separated tuple $X = ((\alpha_i, f_i))_{i=1}^n$. Every such limit has the following properties.

(i) It is symmetric and multilinear and defines distributions of locally finite order off the coincidence diagonals. For supports separated by $d > 0$, each in a ball of radius R,

$$|S_n(X)| \le A(1 + d^{-1})^3 B(R)^n (n!)^{\varkappa} \prod_i \|f_i\|_{\alpha_i},$$

where $A, B(R), \varkappa < \infty$ are independent of the positions of those balls and of the chosen subsequence, $\|f\|_0 = \max(\|f\|_\infty, 40M \operatorname{Lip} f)$ and $\|f\|_\Gamma = \|f\|_\infty$. The fixed constant M belongs to the construction.

(ii) Both S and σ are translation invariant and covariant under every $R \in \mathrm{O}(3)$:

$$\sigma(RX) = \pi(X)^{\delta(R)} \sigma(X), \qquad \overline{\sigma(X)} = \pi(X)\sigma(X) \quad \text{for real } X,$$

where $\pi(X) = \prod_i \pi_{\alpha_i}$ and $\delta(R) = 0$ for proper rotations and 1 otherwise; the same identities hold for S_n. The centered functional is reflection positive at every plane P:

$$\sum_{i,l} \bar{z}_i z_l \, \sigma(\theta_P X_i \cdot X_l) \ge 0$$

for every finite family of separated tuples on one open side and every choice of complex z_i.

(iii) There is a separable Hilbert space with a cyclic unit vector Ω and $1 + 4N_f^2$ Hermitian scalar operator-valued tempered distributions φ_α on $\mathbb{R}^{1+2}$, on a common invariant dense domain. They are covariant under a strongly continuous unitary representation of the proper orthochronous Poincaré group fixing Ω. Its translation spectrum lies in the closed forward

cone. Every pair of fields commutes at spacelike separation. Their Euclidean correlations agree with σ on separated supports. The two-point function of each φ_α is nonzero.

(iv) The flavour group $\mathrm{U}(N_f)^2$ acts unitarily, fixes Ω and commutes with the Poincaré representation. With $e_\pm = (1 \pm \tau_3)/2$ and $V = e_+ \otimes V_1 + e_- \otimes V_2$, its action is $\varphi_\Gamma \mapsto \varphi_{V\Gamma V^{-1}}$ and $\varphi_0 \mapsto \varphi_0$. Space reflection acts unitarily with the signs π_α and exchanges the flavour factors. Here φ_Γ is extended real linearly to the reflection-real span of $\mathfrak{B}$.

The assertions concern centered separated Euclidean correlations. They include neither convergence of lattice contact terms nor uniqueness of the vacuum, clustering or a mass gap. The constants are not uniform as $m_- \downarrow 0$. ■□

The proof has two model-dependent parts. First, the positive mass gives a bounded inverse at every lattice gauge field; relative determinants and periodic-state comparison then give reflection positivity at coordinate and diagonal mirrors. Second, the renormalization construction must control the determinant and the source energies uniformly as $a \downarrow 0$. The cumulative second-order mass identity and its cubic error keep the mass in its required window. Local source estimates and one common vacuum energy in the upper and lower partition bounds give compactness, insertion bounds and shrinking-plate bounds. These last two bounds permit the finite-family rotation argument and reconstruction. Nonvanishing follows from the short-distance two-point laws, not from reflection positivity alone.

We adapt the gauge construction and the general functional results of [3] to the chamber model defined in section 2. The fermion estimates and their use in the local and history expansions are proved below. Lem. 10.19 verifies the numerical and structural hypotheses of the gauge induction; Prop. 10.20 checks that the additional matter conditions preserve the required choice of constants. We use the observables in (1). The scalar field indexed by 0 is the action density. No gauge potential or coloured quark field is reconstructed here.

For the estimates below, the physical matrix source is $J(x) = \sum_{\Gamma\in\mathfrak{B}} f_\Gamma(x)\Gamma$ and

$$Z[w, J] = \Omega\left(\exp\{\Phi_{0,K}(w) + \sum_\Gamma \Phi_{\Gamma,K}(f_\Gamma)\}\right).$$

Its fine fermion matrix is $D_U - a_K\hat{J}$; the hat repeats the source at the chamber places of its site.

2. The chamber model and its massive lattice states

We use the chamber fermion with Wilson parameter $r = 1/2$. The gauge group is SU(2), and $N_f \geq 1$ is the number of four-component Dirac flavours in its fundamental representation. Throughout this section the spacing is one, $\beta \geq 0$, and the lattice mass m_0 is positive. The estimates are uniform in the gauge configuration and the volume; their constants may depend on m_0. Physical coordinates are recovered by $x \mapsto ax$, with $\beta = \mathsf{b}/a$ and $m_0 = a\mathsf{m}$.

2.1. Chambers, spinors and the probability measure.

Let W_3 be the group of signed permutation matrices. Its 48 elements label the chambers at a site of $\mathbb{Z}^3$. Put

$$A_0 = \{0 < z_3 < z_2 < z_1 < 1/2\}, \qquad (N_0, N_1, N_2, N_3) = (e_1, e_2 - e_1, e_3 - e_2, -e_3),$$

and let w_s be reflection in $N_s^\perp$. A chamber $v = (x, g)$ has alcove $x + gA_0$, site $x_v = x$, and neighbours

$$v \cdot s = (x + \delta_{s0} g e_1, g w_s), \qquad u_s(v) = gN_s/|N_s|, \qquad t_s = 3|N_s|. \tag{3}$$

Thus $v \mapsto v \cdot s$ is an involution and $u_s(v \cdot s) = -u_s(v)$. The four neighbours are distinct. Only $s = 0$ changes the site. Define

$$U(v, v \cdot s) = 1 \quad (s = 1, 2, 3), \qquad U(v, v \cdot 0) = U_{\langle x, x+ge_1\rangle}, \qquad U_{\langle y,x\rangle} = U^{-1}_{\langle x,y\rangle}.$$

The fibre is $\mathbb{C}^2_{\mathrm{s}} \otimes \mathbb{C}^2_{\tau} \otimes \mathbb{C}^2_{\mathrm{c}} \otimes \mathbb{C}^{N_f}$, of dimension $8N_f$. Spin and colour are separate factors. For the Pauli matrices σ_μ and τ_μ put

$$\gamma_\mu = \sigma_\mu \otimes \tau_3, \qquad \gamma(u) = \sum_\mu u_\mu \gamma_\mu, \qquad \gamma_4 = 1 \otimes \tau_1.$$

All these matrices act trivially on colour and flavour. The massive operator is

$$\begin{aligned} (D_U f)(v) &= M_0 f(v) - \sum_{s=0}^{3} h_s(v) U(v, v\cdot s) f(v\cdot s), \\ h_s(v) &= t_s(\tfrac{1}{2} - \gamma(u_s(v))), \qquad M_0 = m_0 + \tfrac{1}{2}\sum_s t_s. \end{aligned} \tag{4}$$

Since $\sum_s t_s u_s(v) = 0$, this is the chamber operator with bare mass m_0. On a finite domain we keep the same diagonal coefficient and retain only hops whose chambers and gauge link belong to the domain. A full domain contains every chamber at its sites and every link and plaquette contained in the region. On the cubic torus of even side $N \geq 4$, the gauge field is periodic and spinors are antiperiodic in each coordinate.

The variables $\psi_i(v)$ and $\bar\psi_i(v)$ are independent Grassmann generators. Normalize the Berezin integral to take value one on $\prod_{v,i} \psi_i(v)\bar\psi_i(v)$. With

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p, \qquad A(U) = \sum_p a_p(U), \qquad S_F = \bar\psi D_U \psi,$$

define

$$\Omega(F) = \int dU \int d\bar\psi\, d\psi\, e^{-\beta A - S_F} F, \qquad Z = \Omega(1).$$

The sums and integrals here are over the chosen finite domain or torus. Haar measures have total mass one.

Proposition 2.1 (The determinant and the massive inverse)**.** On every finite domain, on an antiperiodic torus, and on the infinite lattice, $\operatorname{Re} D_U \geq m_0$. The operator is invertible and $\|D_U^{-1}\| \leq m_0^{-1}$. In finite volume,

$$\det D_U = |\det D_{U,+}|^{2N_f} > 0,$$

where $D_{U,+}$ acts on the $\tau_3 = 1$ block for one flavour. Consequently $Z > 0$ and

$$\mu(dU) = Z^{-1} e^{-\beta A(U)} \det D_U \, dU$$

is a probability measure. ■

Proof. Let $(R_s f)(v) = U(v, v\cdot s) f(v\cdot s)$, with value zero at an inactive hop. Then $R_s^* = R_s$ and $\|R_s\| \leq 1$. The relation $u_s(v\cdot s) = -u_s(v)$ shows that $\gamma(u_s)R_s$ is skew-adjoint. Hence

$$\operatorname{Re} D_U = m_0 + \tfrac{1}{2}\sum_s t_s(1 - R_s) \geq m_0.$$

It follows that $\|D_U f\| \geq m_0 \|f\|$. The same estimate for D_U^* gives a closed, dense range, so D_U is onto even on the infinite lattice. Its inverse has the stated norm. The operator commutes with τ_3. Replacing all γ_μ by $-\gamma_\mu$ takes its adjoint, and therefore its two blocks are $D_{U,+}$ and $D_{U,+}^*$, each repeated N_f times. Their determinants are nonzero. The Gaussian Berezin formula gives $Z = \int e^{-\beta A} \det D_U \, dU > 0$. □

2.2. **Reflection positivity on finite domains.** A lattice mirror is one of the planes

$$x_\mu = c, \quad c \in \tfrac{1}{2}\mathbb{Z}, \qquad x_\mu \pm x_\nu = c, \quad c \in \mathbb{Z}, \quad \mu < \nu.$$

Fix a side H^+, reflection θ, and the unit normal n pointing into that side. No mirror meets the interior of an alcove. If the two alcoves of a hop lie on different sides, their common wall is the mirror, the second chamber is θv, and $u_s(v) = -n$ for the chamber on the positive side. A chamber has at most one such wall. These statements follow directly from the four inequalities defining A_0: the values of a coordinate in A_0 lie in $(0, 1/2)$, coordinate differences are nonzero and have modulus less than $1/2$, and sums of two coordinates lie in $(0, 1)$. The common face of adjacent alcoves then determines the crossed mirror.

Let $\mathfrak{A}_+(P)$ contain the generators of chambers on this side and coefficients depending on gauge links with both endpoints in H^+. Links in the plane are admitted. Define the antilinear, order-reversing involution

$$\Theta f = \overline{f \circ \theta}, \qquad \Theta\psi_i(v) = (\bar{\psi}(\theta v)\gamma(n))_i, \qquad \Theta\bar{\psi}_i(v) = (\gamma(n)\psi(\theta v))_i. \tag{5}$$

The order $F\Theta F$ and the normal into the side of F are part of this convention.

Theorem B (Finite-domain reflection positivity)**.** Let a finite domain be preserved by a lattice reflection. Then

$$\Omega(F\Theta F) \geq 0 \qquad (F \in \mathfrak{A}_+(P)).$$

The assertion holds for all nine families, either side, and bounded Borel coefficients, without a gauge-invariance assumption on F. ■□

Proof. The variables and the positive cone. If a gauge link crosses a coordinate mirror through links, write its variable as $V'^{-1}V$, with V, V' independent Haar variables. Integrating these two variables gives precisely Haar integration of the original link. Reflection exchanges V and V'. All other variables split into variables on the two sides and variables in the plane. Let J be Haar–Berezin integration in these variables. Let $\mathcal{K}_+$ be the cone of finite sums $G\Theta G$ with homogeneous G in the enlarged positive half algebra. It is closed under multiplication: for homogeneous G, H,

$$(G\Theta G)(H\Theta H) = (GH)\Theta(GH),$$

because $H\Theta H$ is even. If $K \geq 0$ is a finite matrix and b_i are odd positive-half elements, diagonalizing K gives $\sum_{i,j} b_i K_{ij} \Theta b_j \in \mathcal{K}_+$.

Let T_+ be the product $\prod \psi_i(v)\bar{\psi}_i(v)$ over the positive-side generators. Reflection maps it to the corresponding product on the negative side: the two determinants from $\gamma(n)$ cancel. If g is the coefficient of T_+ in G, then only the product of these two top coefficients contributes, and

$$J(G\Theta G) = \int dU_0 \left| \int dU_+ \, g \right|^2 \geq 0.$$

Thus J is nonnegative on $\mathcal{K}_+$.

The gauge action. Plaquettes on either side give $S_G^+ + \Theta S_G^+$ in $-\beta A$; plaquettes in the mirror are divided equally. For a cut plaquette, let A^p be its parallel transport along the positive half of its boundary. At a diagonal mirror this path joins the two opposite vertices in the mirror. At a link mirror include the two positive half-link variables. The reflected path is the other half of the boundary, with reverse orientation. Hence

$$\operatorname{tr} U_p = \sum_{a,b} A^p_{ab}\Theta(A^p_{ab}), \qquad -\beta A = S_G^+ + \Theta S_G^+ - \beta n_\times + Y, \quad Y = \frac{\beta}{2} \sum_{p \text{ cut}} \sum_{a,b} A^p_{ab}\Theta(A^p_{ab}) \in \mathcal{K}_+.$$

Here $n_\times$ counts cut plaquettes. No plaquette is cut by a coordinate mirror through sites. The stated decomposition exhausts the possibilities: along an edge the signed height changes by 0 or ± 1; a diagonal cut plaquette has heights $0, 1, 0, -1$, and a link cut plaquette has two heights $1/2$ and two $-1/2$.

The fermion action. Terms wholly on one side give $S_F^+ + \Theta S_F^+$ in $-S_F$. For a crossing pair let v be the chamber on the positive side and s its crossing wall. If the gauge link is bisected, put $\chi = V\psi(v)$ and $\bar\chi = \bar\psi(v)V^{-1}$; otherwise put $\chi = \psi(v)$ and $\bar\chi = \bar\psi(v)$. The two crossing hops are

$$\sum_{i,j} \bar\chi_i (t_s + \tfrac{1}{2} t_s \gamma(n))_{ij} \Theta \bar\chi_j, \qquad \sum_{i,j} \chi_j (t_s - \tfrac{1}{2} t_s \gamma(n))_{ij} \Theta \chi_i.$$

Indeed $h_s(v) = t_s(1/2 + \gamma(n))$ and $h_s(\theta v) = t_s(1/2 - \gamma(n))$; substituting (5) gives these formulas, with one transposition of odd generators in the second. Both coefficient matrices are positive, their eigenvalues being $t_s/2$ and $3t_s/2$. Their sum X_F over crossing pairs belongs to $\mathcal{K}_+$. The within-side identity follows also directly from $\gamma(n)\gamma(u)\gamma(n) = -\gamma(\theta^\circ u)$. Here θ° is the linear part of the affine reflection θ.

Put $S^+ = S_G^+ + S_F^+$. All exponents are even, so

$$e^{-\beta A - S_F} = e^{-\beta n_\times} e^{S^+} \Theta(e^{S^+}) e^{X_F} e^{Y}.$$

For homogeneous F, every term of $e^{S^+}\Theta(e^{S^+}) X_F^k Y^l F \Theta F$ belongs to $\mathcal{K}_+$. The X_F series is finite, the Y series converges uniformly in the gauge variables, and the coefficients $1/(k!l!)$ are nonnegative. Apply J term by term. For general $F = F_{\text{even}} + F_{\text{odd}}$ the mixed terms are odd and integrate to zero. This proves the assertion. □

The same proof gives positivity on the even cubic torus, with antiperiodic spinors, at its coordinate mirrors. The torus is cut by the chosen mirror and the parallel mirror at distance $N/2$. Use a positive slab for the half algebra. Across the first boundary the calculation above applies. At the far boundary, choose the representatives so that its reflected chamber is translated by one period in the normal direction. The outward normal there is the opposite normal, and the antiperiodic sign from that period reverses it once more. Thus the far crossing matrices are again $t_s \pm t_s\gamma(n)/2$, both positive; the gauge half-link factorisation and the top-monomial calculation are unchanged. This proves the antiperiodic coordinate-torus assertion, including mirrors through sites. A diagonal plane does not cut this cubic torus into two slabs.

2.3. **Decay, finite-rank perturbations and the specification.** Let $d(v, w)$ be distance in the chamber hopping graph and put

$$h_{\max} = \tfrac{3}{2}(6 + 6\sqrt{2}), \qquad \kappa_{\text{lat}} = \log\left(1 + \frac{m_0}{2h_{\max}}\right), \qquad S_{\kappa_{\text{lat}}} = 48\left(\frac{1 + e^{-\kappa_{\text{lat}}}}{1 - e^{-\kappa_{\text{lat}}}}\right)^3.$$

Since a hop changes its site by zero or one lattice step, $d(v, w) \geq |x_v - x_w|_1$ and $\sum_w e^{-\kappa_{\text{lat}} d(v,w)} \leq S_{\kappa_{\text{lat}}}$.

Lemma 2.2 (The propagator at a fixed configuration)**.** Writing $Q_U = D_U^{-1}$ on the infinite lattice, one has

$$\|Q_U(v, w)\| \leq \frac{2}{m_0} e^{-\kappa_{\text{lat}} d(v,w)}.$$

Every entry is continuous in U for the product topology and is a uniform limit of functions of finitely many links. If U has period $N\mathbb{Z}^3$, its antiperiodic torus inverse is

$$Q_U^N(v, w) = \sum_{n \in \mathbb{Z}^3} (-1)^{n_1+n_2+n_3} Q_U(v, w + Nn).$$

For fixed v, w, $Q_U^N(v, w) - Q_U(v, w) = O(e^{-\kappa_{\text{lat}} N})$ uniformly over periodic U. ■

Proof. For $f_R(v) = \min\{d(v, w), R\}$, conjugating D_U by $e^{\kappa_{\text{lat}} f_R}$ changes only the hopping terms. The absolute row and column sums of their change are at most $(e^{\kappa_{\text{lat}}} - 1)h_{\max} = m_0/2$. The conjugated operator therefore has real part at least $m_0/2$ and inverse norm at most $2/m_0$. Taking its (v, w) entry and then letting $R \to \infty$ proves the decay estimate. The same argument works on a torus.

The resolvent identity

$$Q_U - Q_{U'} = -Q_U(D_U - D_{U'})Q_{U'}$$

and the decay estimate give an absolutely summable series over the changed links. There are sixteen chamber hops over a link, each of norm at most $3(1+r)\|U_b - U_b'\|$, where $r = 1/2$. Consequently the effect of changing links outside an R-neighbourhood of v tends to zero uniformly as $R \to \infty$. On each fixed neighbourhood the finite sum is continuous in its link variables. This proves continuity and the local approximation. For periodic U the proposed image sum converges absolutely by the decay estimate. Applying D_U gives the antiperiodic delta function, so uniqueness of the torus inverse proves the formula. Its terms with $n \neq 0$ are bounded by a convergent geometric sum times $e^{-\kappa_{\mathrm{lat}} N}$, for fixed v, w. □

For configurations W, W' agreeing off a finite set of links define

$$\delta(W, W') = \det(D_W D_{W'}^{-1}) = \det(1 + (D_W - D_{W'})Q_{W'}).$$

This is a determinant of an identity plus a finite-rank operator. If B is a finite link set and $V \in \mathrm{SU}(2)^B$, let $V \vee U$ equal V on B and U elsewhere, and set $d_B(V \mid U) = \delta(V \vee U, U)$.

Lemma 2.3 (The relative determinant)**.** The relative determinant is positive and multiplicative; for each fixed finite B, the function $d_B(V \mid U)$ is jointly continuous in (V, U). The cocycle identity is

$$\delta(W, W')\delta(W', W'') = \delta(W, W'').$$

For $\lambda = 9/m_0$,

$$|\log d_B(V \mid U)| \leq 128 N_f \# B \log(1+\lambda).$$

Let R be a nonnegative integer. A link is within distance R of B if one of its endpoints has lattice ℓ^1 distance at most R from an endpoint of B. If $U' = U$ on all these links, then, uniformly in V,

$$|\log d_B(V \mid U') - \log d_B(V \mid U)| \leq 512 N_f \# B \lambda^2 S_{\kappa_{\mathrm{lat}}} e^{-\kappa_{\mathrm{lat}}(R+1)}. \tag{6}$$

The corresponding relative determinants on antiperiodic tori converge uniformly to d_B, over periodic configurations of the corresponding period, as that period tends to infinity. ■

Proof. The perturbation $\Delta = D_W - D_{W'}$ is supported on the sixteen chamber endpoints over each changed link, has rank at most $128 N_f \# B$, and norm at most 9. The last bound follows from the matching of chambers across the type-zero walls and $\|U_b - U_b'\| \leq 2$. Thus $|\det(1 + \Delta Q_{W'})| \leq (1+\lambda)^{128 N_f \# B}$. The inverse is obtained by exchanging W, W', which gives the lower bound. In the two τ_3 blocks, the finite-rank determinants are conjugates: $\det(1+AB) = \det(1+BA)$ changes the order in the adjoint block. Hence δ is a modulus squared, repeated N_f times, and is nonzero because $D_W D_{W'}^{-1}$ is invertible. Multiplicativity is the determinant product identity. After compressing to the finite set of perturbed endpoints, continuity follows from Lem. 2.2.

For the quantitative assertion, interpolate the link matrices on B linearly between U and $V \vee U$. These matrices are contractions, paired on reversed links by adjoint. Each R_s is still a self-adjoint contraction, so the real-part bound and the decay proof remain valid throughout the interpolation. With $U_s = (1-s)U + s(V \vee U)$ and $\Delta_B = D_{V \vee U} - D_U$,

$$\log d_B(V \mid U) = \int_0^1 \mathrm{Tr}(\Delta_B Q_{U_s})\, ds.$$

Use the same interpolation for U', and put $\Delta' = D_{U'} - D_U$. Subtracting gives

$$-\int_0^1 \mathrm{Tr}(\Delta_B Q_{U_s'} \Delta' Q_{U_s})\, ds.$$

Every nonzero term connects the endpoints over B to a changed link at distance at least $R+1$. The two kernels contribute $(2/m_0)^2$ and their exponential decays. Summing the remaining decay factor costs at most $S_{\kappa_{\mathrm{lat}}}$; the fibre trace and two hopping perturbations cost at most $32 N_f \lambda^2$ for each of the at most $16 \# B$ endpoints. This proves (6). For the torus comparison apply the same logarithmic

identity. The image formula compares the torus inverse to the inverse in the periodically changed configuration; the other translates of B are at distances tending to infinity, so the same resolvent estimate compares that inverse to Q_{U_s}. Both errors tend to zero uniformly. The determinant bounds just proved are uniform, so uniform convergence of logarithms also gives that of the ratios. □

Let A_B sum the plaquettes containing a link of B, and define

$$(\gamma_B F)(U) = \frac{\int F(V \vee U) e^{-\beta A_B(V \vee U)} d_B(V \mid U)\, dV}{\int e^{-\beta A_B(V \vee U)} d_B(V \mid U)\, dV}. \tag{7}$$

Lemma 2.4 (The specification)**.** These are probability kernels. They leave functions of the links off B unchanged, their values do not depend on $U|_B$, and $\gamma_{B'}\gamma_B = \gamma_{B'}$ when $B \subset B'$. They map continuous functions to continuous functions; on continuous local functions their values are uniform limits of continuous local functions. The set of laws ν satisfying $\nu\gamma_B = \nu$ for every finite B is convex and weakly compact. ■

Proof. The numerator has positive continuous weight and its denominator is positive. The positive relative weight

$$\mathrm{rw}(W, W') = e^{-\beta \sum_p (a_p(W) - a_p(W'))} \delta(W, W')$$

is a cocycle. If U is changed only on B, the resulting factor in $\mathrm{rw}(V \vee U, U)$ is independent of V and cancels in the quotient. This proves independence of $U|_B$; properness is immediate. For $B \subset B'$, insert the cocycle identity in the two numerators and denominators and apply Fubini over $\mathrm{SU}(2)^{B'} = \mathrm{SU}(2)^B \times \mathrm{SU}(2)^{B' \setminus B}$ to obtain consistency. Continuity follows by integration over a compact factor. For a local F, freeze all sufficiently distant links at one. The plaquette factor is then unchanged near B, and (6) makes the two normalized weights uniformly close. Indeed, if $|\log p - \log \widetilde{p}| \le \epsilon$, their normalized integrals against F differ by at most $\|F\|_\infty (e^{2\epsilon} - 1)$. This proves the local approximation. Finally the equations are linear in ν and weakly closed on continuous local tests, since $\gamma_B F - F$ is continuous. Such tests determine a measure; therefore the equations hold for all bounded Borel F as well. The space of probability measures on the compact metrizable configuration space is weakly compact. □

2.4. **Limits of tori and local fermion correlations.** For an infinite configuration define the Gaussian Berezin functional by Wick's rule with covariance Q_U:

$$\left\langle \psi_{p_1} \cdots \psi_{p_k} \bar{\psi}_{q_k} \cdots \bar{\psi}_{q_1} \right\rangle_U = \det[Q_U(p_i, q_j)]_{i,j \le k}.$$

It is zero for unequal numbers of barred and unbarred generators. For a local element F, with continuous link coefficients, put $h_F(U) = \langle F \rangle_U$ and $\omega_\nu(F) = \int h_F \, d\nu$. These formulas also define ω_ν for bounded Borel local coefficients. The functional is on a Grassmann algebra; only its restriction to gauge-field functions is a probability law.

Proposition 2.5 (Torus-limit states)**.** The periodically extended cubic torus marginals have weak limit points as their even side tends to infinity. Every such ν solves (7) and is invariant under gauge transformations and under $\mathbb{Z}^3 \rtimes W_3$. Along each sequence defining ν,

$$\langle F \rangle_N \longrightarrow \omega_\nu(F)$$

for every local Grassmann element with bounded Borel local coefficients. For a monomial Ψ of degree $2k$,

$$|\omega_\nu(f\Psi)| \le \|f\|_\infty m_0^{-k}.$$

At every coordinate lattice mirror, with either side,

$$\omega_\nu(F \Theta F) \ge 0 \qquad (F \in \mathfrak{A}_+(P)). \tag{8}$$

In particular the set of solutions of the specification is nonempty. ■

Proof. The configuration space is a countable product of compact copies of SU(2) and is compact metrizable. Its laws are weakly compact. Gram's inequality and $\|Q_U\| \le m_0^{-1}$ give $|\langle\Psi\rangle_U| \le m_0^{-k}$, since each column of the Wick matrix is a subvector of $Q_U e_q$. The same argument holds on the torus. By Lem. 2.2, h_F is continuous for continuous local coefficients and

$$\sup_{U\ N\text{-periodic}} |\langle F\rangle_U^N - h_F(U)| \longrightarrow 0.$$

Thus weak convergence of the gauge marginals proves convergence of these local expectations.

For a finite B and continuous local scalar f, finite-volume conditional integration gives exactly $\mu_N(\gamma_B^N f) = \mu_N(f)$. The affected plaquettes agree with those defining A_B once N exceeds the diameter of the data. By Lem. 2.3, $\gamma_B^N f - \gamma_B f$ tends to zero uniformly on the periodic configurations. The continuity of $\gamma_B f$ allows passage to the weak limit, proving $\nu\gamma_B f = \nu f$. The equality then holds on all bounded Borel tests because both sides define laws agreeing on the continuous local functions.

We record why Borel coefficients also converge. For a finite link set B, let $k_B(V,\xi)$ be the normalized positive weight of (7), and let k_B^N be its torus analogue. The latter converges uniformly to the former, and k_B is continuous in V and the infinite gauge configuration ξ on their compact product space. The B-marginal of μ_N has Haar density

$$p_N(V) = \int k_B^N(V,\xi)\,\mu_N(d\xi).$$

Weak convergence and uniform continuity in V imply uniform convergence to $p(V) = \int k_B(V,\xi)\,\nu(d\xi)$: use a finite net in the compact V space to make the weak convergence uniform. Since ν solves the specification, p is its B-marginal density. Hence the marginals converge in total variation. This proves convergence on bounded Borel local scalar functions. For a local Grassmann element $f\Psi$, h_Ψ is uniformly approximable by continuous functions of finitely many links. Apply the scalar assertion to f times each approximation and then remove the approximation. The torus covariance error is uniform, so it gives the claimed Grassmann convergence.

For a signed permutation R, choose $s_R \in \mathrm{SU}(2)$ with $s_R\sigma(u)s_R^{-1} = (\det R)\sigma(Ru)$, and put $S_R = s_R \otimes \tau_2^{\delta(R)}$, where $\delta(R) = 0$ for proper R and 1 otherwise. Then $S_R\gamma(u)S_R^{-1} = \gamma(Ru)$. Together with the action $R(x,g) = (Rx, Rg)$ on chambers, this intertwines D_U and D_{RU}. Translations act by reindexing. Thus the cubic torus marginals are invariant under translations and the signed coordinate permutations. These maps act continuously on the infinite configuration space, so the same holds in the limit. For a gauge transformation and a continuous local test, choose a periodic gauge transformation agreeing with it at every site read by that test. Finite-volume gauge invariance then passes to the weak limit. The lifts of lattice isometries on spinors intertwine the corresponding Dirac operators, so Wick's rule gives the same covariance of ω_ν; a lift's sign cancels on even elements, and odd elements have zero expectation.

Finally, for a fixed coordinate mirror and local F in its half algebra, all sufficiently large cubic tori contain F in the appropriate open slab. Their antiperiodic reflection forms are nonnegative by the torus argument following Thm. B. The just-proved convergence, applied to $F\Theta F$, gives (8). □

3. Reflection positivity of the infinite-volume lattice states

We keep $r = 1/2$, $m_0 > 0$ and now assume $\beta > 0$. The states obtained from cubic tori in the preceding section are reflection positive at the coordinate mirrors. A diagonal plane does not split these tori. We first prove an assertion about all periodic solutions of the specification; this permits comparison with states obtained from tori adapted to a diagonal normal.

Write $\mathcal{G}$ for the solutions of (7), and $\mathcal{G}^{\mathrm{per}}$ for those invariant under $N_0\mathbb{Z}^3$ for some integer $N_0 \ge 1$. For the proof we use the reflection

$$\vartheta_P = (-1)^{\deg}\Theta_P,$$

where Θ_P is formed with the normal into the upper side. Thus $\vartheta_P\psi = -\bar{\psi}\gamma(n)$ and $\vartheta_P\bar{\psi} = -\gamma(n)\psi$, at reflected positions. For an upper-side element F,

$$\omega_\nu(\vartheta_P(F)F) = \omega_\nu(F\Theta_P F). \tag{9}$$

For homogeneous F this follows by interchanging the two factors; for a general F, the mixed-parity terms vanish. The convention with ϑ_P is convenient for the crossing Fock space.

Proposition 3.1 (All lattice mirrors)**.** Every $\nu \in \mathcal{G}^{\mathrm{per}}$ is invariant under $\mathbb{Z}^3 \rtimes W_3$. For every lattice mirror P, either side, and every local half-algebra element F with bounded Borel link coefficients, and every local element G with bounded Borel coefficients,

$$\omega_\nu(\Theta_P G) = \overline{\omega_\nu(G)}, \qquad \omega_\nu(F\Theta_P F) \geq 0.$$

The reflection form is Hermitian and vanishes between elements of different Grassmann parity. In particular these conclusions hold for every weak limit of the cubic torus marginals. ■

The proof uses two different comparisons. Insertions of unitary operators at a bond mirror force a reflection identity on its two neighbouring layers. The field and link equations then propagate that identity to the whole lattice. This makes the periodic-state class compact. A second comparison, on tori adapted to a prescribed mirror, then allows a separation argument within that class.

3.1. **The local equations and a surface comparison.** We identify the chamber (k, g) with the point $x = k + gv_0$, where $v_0 = (v_1, v_2, v_3) = (3/8, 1/4, 1/8)$ lies in A_0, and write $\pi(x) = k$. The four neighbours $x^{(i)}$ are numbered in the reverse order to (3): type i is wall $3 - i$. Their weights are

$$(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6), \qquad w(x, x') = \frac{w_i}{2}(r - \gamma(\widehat{x' - x})).$$

There are eight type-three edges over each gauge link. Only these edges carry a gauge transporter. In this notation $D_U = M_0 - \sum_i w(x, x^{(i)})U(\pi(x), \pi(x^{(i)}))$. For an oriented chamber edge put

$$\mathrm{hop}_V(x, x') = \bar{\psi}(x)[w(x, x') \otimes V(\pi(x), \pi(x')) \otimes 1_{N_f}]\psi(x'),$$

where $V(\pi(x), \pi(x')) = 1$ when the two chambers have the same site. No result depends on the auxiliary choice of v_0.

Every ω_ν, $\nu \in \mathcal{G}$, obeys the field equations

$$\omega_\nu((D_U\psi)_p G) = \omega_\nu(\partial_{\bar{\psi}_p} G), \qquad \omega_\nu((\bar{\psi}D_U)_p G) = -\omega_\nu(\partial_{\psi_p} G). \tag{10}$$

Here and below derivatives are left derivatives. For a finite gauge link set B, put $H_B = -\beta A_B$ and

$$R_B(V; U) = e^{H_B(V \vee U)} \exp\left(\sum_{\text{oriented edges over } B} \mathrm{hop}_V(x, x')\right), \quad \mathrm{Res}_B\, G = \frac{\int R_B(V; U)G(V \vee U)\, dV}{\int R_B(V; U)\, dV}.$$

The denominator is an even Grassmann element with strictly positive body. The link equations are

$$\omega_\nu(G) = \omega_\nu(\mathrm{Res}_B\, G). \tag{11}$$

For clarity, these equations follow from the specification, not from an assumed fermionic probability law. Wick's determinant recursion gives (10). If a finite chamber set Y contains the endpoints over B and the fields of G, let $C = P_Y Q_U P_Y$, $C' = P_Y Q_{V \vee U} P_Y$ and $\Delta = P_Y(D_{V \vee U} - D_U)P_Y$. The resolvent identity gives $C'^{-1} = C^{-1} + \Delta$; these compressions are invertible since

$$\mathrm{Re}\langle f, Cf\rangle = \mathrm{Re}\langle D_U Q_U f, Q_U f\rangle \geq m_0\|Q_U f\|^2 \geq m_0\|f\|^2/\|D_U\|^2.$$

The finite Gaussian formula consequently gives, for a polynomial ξ with coefficients independent of the gauge field, the bridge identity

$$d_B(V \mid U)\langle\xi\rangle_{V \vee U} = \left\langle \xi \exp\left(\sum \mathrm{hop}_V - \sum \mathrm{hop}_U\right)\right\rangle_U. \tag{12}$$

Apply this coefficientwise to $G(V \vee U)$ and to $\operatorname{Res}_B G$ under the numerator of γ_B. The two numerators agree after multiplication by the common normalization; $\operatorname{Res}_B G$ is independent of $U|_B$. Integration against $\nu\gamma_B = \nu$ proves (11).

A torus is specified by a rank-three period lattice Γ and a sign character $\chi : \Gamma \to \{\pm 1\}$. Its systole is $\min_{g\neq 0} |g|_\infty$. The decay and finite-rank determinant arguments of the preceding section apply with arbitrary such periods and signs: the image sum uses $\chi(g)$, and the number of periods of size at most js is at most $(2j+3)^3$ when the systole is s. In particular local covariances and determinant ratios converge uniformly as $s \to \infty$.

Lemma 3.2 (Surface comparison)**.** Let $\mathcal{D}$ be a set of site representatives for a torus, $B = \{\langle y, y+e_j\rangle : y \in \mathcal{D},\ j = 1,2,3\}$, and let $s(\mathcal{D})$ count links with exactly one endpoint in $\mathcal{D}$. Set

$$q_0 = 1 + \frac{9}{2m_0}, \qquad K(\mathcal{D}) = (4\beta + 192 N_f \log q_0) s(\mathcal{D}).$$

For nonnegative Borel f depending on B,

$$e^{-2K(\mathcal{D})} \gamma_B f(U) \le \mu_{\mathbb{T}}(f^{\mathrm{per}}) \le e^{2K(\mathcal{D})} \gamma_B f(U).$$

Suppose $\Gamma \subset \Gamma'$, $\nu \in \mathcal{G}$ is Γ'-invariant, and A is real continuous and local. For $\mathbb{A} = \sum_{v \in \Gamma'/\Gamma} \tau_v A$, let M be the number of terms and M_{in} the number with a representative supported on B. Then

$$\log \mu_{\mathbb{T}}(e^{t\mathbb{A}}) \ge tM\nu(A) - 2K(\mathcal{D}) - 2|t|\|A\|_\infty (M - M_{\mathrm{in}}). \tag{13}$$

■

Proof. Let $Y = \pi^{-1}(\mathcal{D})$. Cutting the type-three edges at its boundary gives the common interior operator $D_Y(V)$. The torus differs from it by a perturbation of rank at most $64 N_f s(\mathcal{D})$. The infinite operator differs from $D_Y(V) \oplus P_{Y^c} D_U P_{Y^c}$ by a perturbation of rank at most $128 N_f s(\mathcal{D})$. Both perturbation norms are at most $w_3(1+r)/2 = 9/2$, and the operators before and after cutting have real part at least m_0. Therefore

$$(1 + \|R\|/m_0)^{-k} \le |\det((A+R)A^{-1})| \le (1 + \|R\|/m_0)^k$$

when $\operatorname{rank} R \le k$: use the finite-rank determinant bound in one direction and $1 - R(A+R)^{-1}$ in the other. Factor the infinite relative determinant through the cut operators. Its interior factor is $\det D_Y(V)/\det D_Y(U|_B)$; the second denominator and exterior factors do not depend on V. The torus determinant has the same interior factor, leaving the two rank bounds above. At most $2s(\mathcal{D})$ plaquettes differ, and $0 \le a_p \le 2$. Thus the unnormalized torus and conditional weights differ by a positive factor independent of V times a factor in $[e^{-K}, e^K]$. Normalization proves the first assertion.

Let $\Phi = t \sum_{\mathrm{in}} \tau_v A$. The first assertion and Jensen's inequality give $\log \mu_{\mathbb{T}}(e^\Phi) + 2K \ge \gamma_B \Phi(U)$. Integrate against ν, use its specification and Γ' invariance, and bound the omitted terms by $|t|\|A\|_\infty(M - M_{\mathrm{in}})$ on each side. This gives (13). □

We will also use a slowly varying scalar weight on tori. If $W_j > 0$ is continuous and

$$\sup_{U=U' \text{ off } B+\Gamma_j} |\log W_j(U) - \log W_j(U')| \longrightarrow 0 \tag{14}$$

for every finite B, then every weak limit of $W_j \mu_{\mathbb{T}_j} / \mu_{\mathbb{T}_j}(W_j)$, with systole tending to infinity, solves the same specification. Indeed its finite-volume conditional kernel differs uniformly from γ_B by the ratio estimate following Lem. 2.4, the error being the sum of (14) and the torus determinant error. Conditional integration is exact before passage to the limit. The image formula also proves convergence of local continuous Grassmann expectations to ω_ν along these weighted laws.

3.2. **Operators inserted at a bond mirror.** Fix a coordinate direction, write its coordinate as j, and put $c = t+1/2$. Write $u_j(b) = U(b^{(j)})$ for horizontal gauge links and $w_j(z) = U((z,j),(z,j+1))$ for vertical ones. For a finite horizontal link set S choose a finite nonempty set Λ containing its endpoints. Let $X^-(z)$ be the eight chamber sites over (z,t) at height $t+v_1$, and $X^+(z) = \theta X^-(z)$. These are exactly the endpoints of the edges crossing the mirror. Put $X^\pm(\Lambda) = \bigcup_{z\in\Lambda} X^\pm(z)$. Two-layer elements have continuous coefficients in $(u_t|_S, w_t|_\Lambda, u_{t+1}|_S)$ and generators at $X^-(\Lambda)\cup X^+(\Lambda)$.

Let $\mathcal{F}_\Lambda$ be the Grassmann algebra on the upper generators, made a Hilbert space by declaring its ordered monomials orthonormal. An orthonormal change of the one-generator basis preserves this inner product on every exterior power, by the determinant formula for its Gram matrix. The endomorphism O induced on this algebra by

$$\bar\psi e \longmapsto \bar\psi\, 3(1+r\gamma(n))e, \qquad e^*\psi \longmapsto (3(1-r\gamma(n))e)^*\psi$$

is positive and invertible: its one-generator eigenvalues are $3(1\pm r)>0$. For $\Upsilon \in \operatorname{End}\mathcal{F}_\Lambda$ define

$$\operatorname{Ins}[\Upsilon] = \sum_{a,a'} \langle a, \Upsilon a'\rangle \vartheta(a') a, \qquad \operatorname{Cx}_\Lambda = \operatorname{Ins}[\mathsf{O}]. \tag{15}$$

Here a, a' range over an orthonormal monomial basis. This map is independent of the basis and is a linear bijection onto the Grassmann algebra of the two sides. It satisfies

$$\vartheta \operatorname{Ins}[\Upsilon] = \operatorname{Ins}[\Upsilon^*], \qquad \operatorname{Cx}_\Lambda = \exp\left(\sum_{x\in X^+(\Lambda)} \operatorname{hop}_1(x,\theta x) + \operatorname{hop}_1(\theta x, x)\right).$$

The subscript 1 means that the crossing transport has been set to one. To check the second identity, diagonalize $\gamma(n)$. The crossing terms become $3(1+\epsilon r)\vartheta(\bar\psi e)\bar\psi e$ and $3(1-\epsilon r)\vartheta(e^*\psi)e^*\psi$ for $\gamma(n)e = \epsilon e$. They are commuting square-zero elements. Expanding their exponential gives the exterior powers of the same one-generator matrix. On disjoint upper sets, the product of even insertions is the insertion of their tensor product: the two signs from interchanging odd factors cancel. These facts also show $\operatorname{Ins}[\Upsilon]$ is even precisely when Υ preserves parity.

For the gauge factor let $k_\beta(g) = e^{-\beta(1-\operatorname{tr} g/2)}$. Write χ_l for the spin-l character of $\mathrm{SU}(2)$, $l \in \frac12\mathbb{N}_0$, and I_n for the modified Bessel function. Its convolution eigenvalues on spin-l matrix coefficients are

$$\lambda_l = \frac{2}{\beta} e^{-\beta} I_{2l+1}(\beta) > 0, \qquad \lambda_l \le \frac{\beta^{2l}}{(2l+1)!}. \tag{16}$$

The Weyl integration formula gives this expression; positivity also follows from the power series of I_{2l+1}. Consequently K_S, convolution with $k_S(u,u') = \prod_{b\in S} k_\beta(u(b)u'(b)^{-1})$, maps the trigonometric polynomials bijectively onto themselves. The series $q_\beta = \sum_l (2l+1)\lambda_l^{1/2}\chi_l$ converges uniformly. If $q_u = \prod_b q_\beta(u(b)\ \cdot\ ^{-1})$, then

$$\langle q_u, q_{u'}\rangle = k_S(u,u'), \qquad \|q_u\| = 1. \tag{17}$$

On $\mathcal{H}_{S,\Lambda} = L^2(\mathrm{SU}(2)^S)\otimes\mathcal{F}_\Lambda$ let $R_g = L_g \otimes \mathsf{s}_g^{-1}$, where $(L_g f)(u) = f(g^{-1}.u)$ and s_g is the diagonal gauge action on upper generators. Let $\mathcal{W}$ be the algebra of bounded operators commuting with every R_g. For $X\in\mathcal{W}$ define

$$\langle a, \kappa_X(u',u)a'\rangle = \langle q_{u'}\otimes a, X(q_u\otimes a')\rangle, \quad \widehat{G}_X(u,u') = \frac{\operatorname{Ins}[\mathsf{O}^{1/2}\kappa_X(u',u)\mathsf{O}^{1/2}]}{k_S(u,u')\operatorname{Cx}_\Lambda}.$$

The lift of a reduced element is

$$(\operatorname{Lf}\widehat{Y})(U) = \mathsf{s}^+_{w_t}\widehat{Y}(u_t, w_t.u_{t+1}), \qquad G_X = \operatorname{Lf}\widehat{G}_X. \tag{18}$$

The superscript + means gauge substitution on the upper generators only. The crossing factor has body one and is invertible. For a Grassmann polynomial, $|\cdot|_1$ denotes the sum of the absolute values of its coefficients in the ordered monomial basis.

Lemma 3.3 (Insertions and their range)**.** The element G_X is continuous, local, even and gauge invariant; $G_1 = 1$, $\vartheta G_X = G_{X^*}$, and $\sup_U |G_X(U)|_1 \le C_{S,\Lambda}\|X\|$. Every gauge-invariant two-layer element is a limit in coefficient supremum norm of elements G_X. There is an antiunitary involution J on $\mathcal{H}_{S,\Lambda}$ such that $\mathsf{J}G_X = G_{JXJ}$, where J is the antilinear algebra automorphism induced on generators by $C_+ = (\sigma_2\otimes\tau_1)\otimes\sigma_2$ and on coefficients by complex conjugation. For every configuration, $\langle \mathsf{J}F\rangle_U = \overline{\langle F\rangle_U}$. ■

Proof. The identities follow from (17), the insertion identities, and $\kappa_X(u,u')^* = \kappa_{X^*}(u',u)$. Gauge covariance follows from $L_g q_u = q_{g.u}$, by moving the gauge action through X. The element is even since R_{-1} acts as parity. For its norm, $k_S \ge e^{-2\beta \# S}$, $\|\kappa_X\| \le \|X\|$, and all other maps act on a fixed finite Grassmann algebra; multiplication by the inverse crossing factor and the lift therefore cost a constant depending only on the finite pair. Each upper generator is sent by the lift to two colour components, so this cost is uniform in the gauge field.

A gauge-invariant two-layer element has the form $\mathrm{Lf}\,\widehat{Y}$ with reduced $\widehat{Y}$ invariant under diagonal gauge substitutions. Indeed set its middle gauge variables w_t to one by an upper-layer gauge transformation; the remaining simultaneous transformations on the two layers give this invariance, and reversing the substitution gives (18). Expand $k_S\widehat{Y}$ in Grassmann monomials and approximate each coefficient uniformly by trigonometric polynomials in (u,u'). Gauge averaging preserves these polynomial spaces and the approximation. To represent an approximant exactly, write $m\mathrm{Cx}_\Lambda = \mathrm{Ins}[\Upsilon'_m]$ and $\Upsilon_m = \mathsf{O}^{-1/2}\Upsilon'_m\mathsf{O}^{-1/2}$. If e_α is a normalized matrix-coefficient basis,

$$\langle q_{u'}, (\lambda_\alpha\lambda_{\alpha'})^{-1/2}|e_\alpha\rangle\langle e_{\alpha'}|q_u\rangle = e_\alpha(u')\overline{e_{\alpha'}(u)}.$$

Thus finite sums of these rank-one operators tensored with Υ_m represent every approximant. Average the resulting operator over R_g to put it in $\mathcal{W}$. This step uses both $r<1$, for $\mathsf{O}^{-1/2}$, and $\beta>0$, for $\lambda_\alpha > 0$.

Finally $C_+\overline{D_U} = D_U C_+$ and $C_+^2 = 1$. The automorphism J fixes the Gaussian action and the top monomial, hence has the asserted expectation identity. It commutes with gauge substitutions and with ϑ. On the upper Fock space it is antiunitary; tensor it with complex conjugation on L^2, where q_u is real. The same insertion calculation gives $\mathsf{J}G_X = G_{JXJ}$. □

For a torus $\mathbb{T}$, write $Z_\mathbb{T}[F] = \Omega_\mathbb{T}(F)$ for its unnormalized Haar–Berezin functional, and let $C(N)$ denote the antiperiodic cubic torus of side N.

Lemma 3.4 (The operator in the mirror)**.** On an antiperiodic cubic torus, let finitely many disjoint pairs (S_a,Λ_a) lie in one bond mirror. For even, gauge-invariant slab elements F, F' there are vectors $\Phi_F, \Phi_{F'}$, independent of the X_a, such that

$$Z_\mathbb{T}\left[\vartheta(F')\prod_a G_{X_a}F\right] = \langle \Phi_F, (\bigotimes_a X_a\otimes 1)\Phi_{F'}\rangle. \tag{19}$$

Consequently its modulus is at most $\prod_a \|X_a\|\, Z_\mathbb{T}[\vartheta(F)F]^{1/2} Z_\mathbb{T}[\vartheta(F')F']^{1/2}$. ■

Proof. Gauge-fix the vertical crossing links at the two boundary mirrors to one. Gauge invariance of the integrand and Haar invariance make this change exact: fix each crossing variable V, perform the gauge transformation V^{-1} at its lower endpoint, and integrate the remaining lower variables; their Haar measure is unchanged. The transformation is one on the upper slab.

Let U_+, U'_+ be the two slab configurations, with the latter reflected into the upper slab. The Wilson weight is

$$W_+(U_+)W_+(U'_+)k_{S_0}(u^{\mathrm{dn}}, u'^{\mathrm{dn}})k_{S_0}(u^{\mathrm{up}}, u'^{\mathrm{up}}),$$

where S_0 is the full finite horizontal section. The fermion weight is $e^{-S_+}\vartheta(e^{-S_+})\,\mathrm{Ins}[\mathsf{O}_Y]$, with Y the upper crossing generators at both boundaries. At the far boundary the antiperiodic sign changes the apparent negative crossing coefficient to a positive one, so $\mathsf{O}_Y \ge 0$. Insertion G_{X_a} cancels its factor $k_{S_a}\mathrm{Cx}_{\Lambda_a}$ and replaces it by the kernel of X_a between q_u and the corresponding Fock modes.

For $\xi = Fe^{-S_+}$ define $v_\xi \in \mathrm{Gr}(Y)$ by $\langle v_\xi, a\rangle = \int_{\mathrm{upper}} \xi a$. The top-monomial normalization and evenness give exactly

$$\int \vartheta(\xi')\xi \,\mathrm{Ins}[\Upsilon] = \langle v_\xi, \Upsilon v_{\xi'}\rangle.$$

There is no sign: products of even insertions on disjoint sets are tensor products, as checked after (15). Factor $L^2(\mathrm{SU}(2)^{S_0})$ over the S_a and its complement, and factor the Fock space over the disjoint mode sets. In that order set

$$\Phi_F = \int dU_+ \, W_+(U_+)\, q_{u^{\mathrm{dn}}} \otimes \mathsf{O}_Y^{1/2} v_{Fe^{-S_+}} \otimes q_{u^{\mathrm{up}}}.$$

In these finite-dimensional Fock factors and compact gauge integrals, Fubini proves (19). Setting all $X_a = 1$ identifies $\|\Phi_F\|^2$ with the reflected form, and Cauchy–Schwarz gives the bound. □

Proposition 3.5 (A periodic array of contraction insertions)**.** Let $\|X\| \le 1$, and suppose a cube of side $2r_0$ contains the support of G_X. If $N' > 2r_0$ is even, i is odd, and $iN' \ge 4$, then on the cubic torus of side iN',

$$\left| Z_{\mathbb{T}} \left[\prod_{v \in N'\mathbb{Z}^3/(iN')\mathbb{Z}^3} \tau_v G_X \right] \right| \le Z_{\mathbb{T}}[1].$$

■

Proof. Group the insertions by their i normal layers, and replace the operator on a layer by a label in $\{1, X, X^*\}$. Let $F(\sigma)$ be the resulting partition function. The supports of distinct insertions are disjoint. At layer k use Lem. 3.4 with the two boundaries $c + kN'$ and $c + kN' + iN'/2$. Since i is odd, the second boundary lies between insertion layers. Reflection and Cauchy–Schwarz give

$$|F(\sigma)|^2 \le F(R_k^+\sigma)F(R_k^-\sigma), \qquad F(R_k^\pm\sigma) \ge 0,$$

where $R_k^\pm$ replaces label k by 1, retains one half of the other labels, and reflects their adjoints into the other half. On the finite set of label arrays choose a maximizer of $|F|$. Both reflected arrays also maximize it. A string of l successive labels 1 can be enlarged, by reflection at its next boundary, to a string of length $\min(2l+1, i)$. Starting with the one 1 created by $R_k^\pm$ gives a constant 1 maximizer. Its value is $Z_{\mathbb{T}}[1]$. □

3.3. **The lower bound for the insertion array.** The following estimate concerns a normalized Gaussian Berezin functional in a fixed gauge configuration, on any torus. It uses the mass bound alone.

Lemma 3.6 (Separated local elements)**.** Fix a cube size. There are $\epsilon_0 > 0$ and $C < \infty$, depending on that size, m_0 and N_f but not on the torus, the gauge field or the number R of cubes, with the following property. Let the cubes and all their period translates be disjoint, and let V_j be even elements supported in them with $|V_j|_1 \le 1$. Then

$$F(z) = \left\langle e^{z\sum_{j=1}^R V_j} \right\rangle_U^{\mathbb{T}}$$

has no zero for $|z| < \epsilon_0$, and its branch $L(0) = 0$ of the logarithm satisfies

$$\left| L(z) - z\sum_j \langle V_j\rangle_U^{\mathbb{T}} \right| \le C|z|^2 R. \tag{20}$$

■

Proof. Write the covariance as $C(p,q) = \langle f_p, g_q\rangle$ with $\|f_p\| = 1$, $\|g_q\| \le m_0^{-1}$. Its block row and column sums are at most $2S_{\kappa_{\mathrm{lat}}}/m_0$. For a covariance K and left Grassmann derivatives, put $\Delta_K = \sum K(p,q)\partial_{\bar\psi_q}\partial_{\psi_p}$. Wick's rule is $\langle a\rangle_K = [e^{\Delta_K} a]_0$. In particular differentiation in the

covariance inserts $\Delta_{\dot K}$, and block diagonal covariances factorize expectations of even elements in distinct blocks.

We use the forest interpolation identity [1, Thm. III.1] for a polynomial H in edge variables on n vertices:

$$H(1) = \sum_{\mathcal{F}} \int_{[0,1]^{\mathcal{F}}} dt \left(\prod_{e \in \mathcal{F}} \partial_{s_e} H \right) (s^{\mathcal{F}}(t)). \tag{21}$$

The off-diagonal value $s_{ij}^{\mathcal{F}}$ is the minimum of the parameters on the forest path from i to j, and zero if there is no path; the diagonal values are one. This matrix is positive semidefinite: integrate over $u \in [0,1]$ the block matrices of ones corresponding to components of forest edges with $t_e \geq u$. Hence $s_{ij} = \langle e_i, e_j \rangle$ for unit vectors e_i. The interpolated covariance is represented by $e_i \otimes f_p, e_j \otimes g_q$, so Gram's inequality has the same constant at every interpolation point.

Set $W_j = e^{zV_j} - 1$, so $|W_j|_1 \leq 2|z|$ for $|z| \leq 1$. Apply (21) to $\langle \prod_{j \in J} W_j \rangle_{C_s}$ for every subset J. Factorization over forest components gives a hard-core polymer sum with nonempty vertex sets P and activities

$$\zeta(P) = \sum_{T \text{ tree on } P} \int dt \left\langle \prod_{e \in T} \Delta_e \prod_{j \in P} W_j \right\rangle_{C_{s^T(t)}}.$$

Let N_b bound the number of generators in one cube and set $K_G = \max(1, m_0^{-1})^{N_b/2}$. The sums of the cross-covariance absolute values over a second cube are bounded by a constant g_b, uniformly in the first cube: one may take $g_b = 256 N_f^2 n_b S_{\kappa_{\mathrm{lat}}}/m_0$ if n_b bounds its chamber sites. Expanding the derivatives and applying Gram's inequality bounds a tree by the product of its edge sums times $(2|z|K_G)^{\#P}$. Each differentiation removes specified generators, and there is no additional factorial at a vertex. Root at a specified vertex; there are at most 4^{n-1} rooted plane-tree shapes with n vertices, and summing the cube labels along each tree costs at most g_b^{n-1}. Thus

$$\sum_{P \ni j,\ \#P = n} |\zeta(P)| \leq (2|z|K_G)^n (4g_b)^{n-1}. \tag{22}$$

Choose ϵ_0 so small that $\sum_{P \ni j} |\zeta(P)| 2^{\#P-1} \leq 1/2$. For a finite vertex set I, let $Z(I)$ be its polymer sum. Deleting j gives

$$\frac{Z(I)}{Z(I \setminus \{j\})} = 1 + y_j, \qquad y_j = \sum_{P \subset I,\, j \in P} \zeta(P) \frac{Z(I \setminus P)}{Z(I \setminus \{j\})}.$$

Induction on $\#I$ bounds the quotient in each summand by $2^{\#P-1}$, proves $|y_j| \leq 1/2$, and proves nonvanishing. The terms $\#P \geq 2$ in (22) give $|y_j - \zeta(\{j\})| \leq C_1 |z|^2$. Also $|\zeta(\{j\}) - z\langle V_j \rangle| \leq C_2 |z|^2$. Order the vertices, multiply these successive ratios, and use $|\log(1+y) - y| \leq |y|^2$ for $|y| \leq 1/2$. Summation proves (20). □

Proposition 3.7 (The lower tangent for insertions)**.** Let $\nu \in \mathcal{G}^{\mathrm{per}}$ have period number N_0, and let $D \in \mathcal{W}$ satisfy $D^* = -D$ and $JDJ = D$. For an even multiple N' of N_0 large enough to separate the supports, let $P_i(\epsilon) = \prod_{v \in N'\mathbb{Z}^3/(iN')\mathbb{Z}^3} \tau_v G_{e^{\epsilon D}}$. For all sufficiently small real ϵ, its normalized fermion expectation is positive at every gauge configuration, and

$$\liminf_{i \to \infty} i^{-3} \log \frac{Z_{C(iN')}[P_i(\epsilon)]}{Z_{C(iN')}[1]} \geq \epsilon \omega_\nu(G_D) - C_D \epsilon^2.$$

The constant and the permitted interval for ϵ are independent of i, N', ν and the configuration. ■

Proof. Linearity and the norm bound of Lem. 3.3 show

$$G_{e^{\epsilon D}} = 1 + \epsilon G_D + O_{|\cdot|_1}(\epsilon^2).$$

It is fixed by J, since $Je^{\epsilon D}J = e^{\epsilon D}$. For small ϵ its finite Grassmann logarithm $V_\epsilon = \log G_{e^{\epsilon D}}$ exists, is even and J-fixed, and satisfies

$$|V_\epsilon|_1 \le c_2|\epsilon|, \qquad |V_\epsilon - \epsilon G_D|_1 \le c_3\epsilon^2.$$

Apply Lem. 3.6 to its disjoint translates, divided by $c_2\epsilon$, with parameter $z = c_2\epsilon$. The expectation is real by the J identity, has no zero on the small real interval, and is one at zero. It is therefore positive, and its real logarithm gives, uniformly in U,

$$\langle P_i(\epsilon)\rangle_U^{C(iN')} \ge \exp\left(\epsilon \sum_v \langle \tau_v G_D\rangle_U^{C(iN')} - C_D\epsilon^2 i^3\right).$$

The function h_{G_D} is real continuous and uniformly approximable by real local functions. The image formula replaces each torus expectation by its infinite-volume value with an error $O(e^{-\kappa_{\mathrm{lat}} iN'})$ uniform in its position. Approximate h_{G_D} by a real local A within δ and integrate the preceding inequality against the torus gauge law. Apply (13) to the array of A on the cube $\{0,\dots,iN'-1\}^3$. Its surface is $6(iN')^2$; the number of array terms meeting a fixed-width boundary layer is $O((iN')^2)$. Divide by i^3 and let $i \to \infty$ at fixed N', then $\delta \to 0$. The surface, boundary and image errors vanish, and $\nu(h_{G_D}) = \omega_\nu(G_D)$. □

Theorem C (The two-layer principle)**.** For every periodic solution and every continuous two-layer element Y at a bond mirror,

$$\omega_\nu(\vartheta Y) = \overline{\omega_\nu(Y)}.$$

■□

Proof. For D as in Prop. 3.7, $e^{\epsilon D}$ is unitary. Apply the upper bound Prop. 3.5 for odd i, and the lower tangent along the same indices. Their quotient is positive, so $0 \ge \epsilon\omega_\nu(G_D) - C_D\epsilon^2$ for both signs of small real ϵ. Thus $\omega_\nu(G_D) = 0$. Let $\phi(X) = \omega_\nu(G_X)$. It is linear and $\phi(JXJ) = \overline{\phi(X)}$. For $JXJ = X$, its value is real, and $(X - X^*)/2$ is an allowed D, giving $\phi(X^*) = \phi(X)$. Decompose a general X into its two J-real parts to conclude $\phi(X^*) = \overline{\phi(X)}$.

For a gauge-invariant Y, the range assertion of Lem. 3.3 now proves the result by uniform coefficient approximation. The expectation is continuous in this norm by the mass bound on its finitely many generators. For general Y, average over gauge transformations on the finite sites of the two layers which it reads. Gibbs states are gauge invariant: resampling the links incident to one site, Haar substitution and the relative-determinant cocycle give this assertion first for one-site transformations, then for arbitrary finite sets. The averaged element is gauge invariant and remains two-layer; the expectation and its reflected expectation are unchanged. This proves the identity for Y. □

3.4. **Two layers determine the chain.** Fix the same coordinate direction and $c = t_0 + 1/2$. For each horizontal site let $\widehat{w}_{t_0} = 1$ and $\widehat{w}_{j+1}(z) = \widehat{w}_j(z)w_j(z)$. Temporal gauge replaces all vertical gauge links by one and horizontal layers by

$$\widetilde{u}_j(b) = \widehat{w}_j(b_-)u_j(b)\widehat{w}_j(b_+)^{-1}.$$

Each such coordinate and each $\widehat{w}_j$ reads finitely many links. Pull a local element of the new variables back by this substitution and the corresponding gauge substitution on its generators; denote the map by tg^*. It is a unital homomorphism and sends a continuous local element to a continuous local element.

The fermion sites above a gauge plane have six thin layers, ordered by their normal displacements $-v_1, -v_2, -v_3, v_3, v_2, v_1$. Write them $X_{j,k}$, $1 \le k \le 6$, or $\mathbb{L}_{6j+k}$. Every site has one neighbour in each adjacent thin layer and two in its own thin layer. The upward edge types are $2,1,0,1,2,3$ as k runs from 1 to 6, and the downward types are $3,2,1,0,1,2$. In temporal gauge these upward and downward transports are one. At layers $k = 1, 6$ every transport is one; at each other layer the only nontrivial transport reads one horizontal coordinate of plane j. These statements follow by applying

the four wall reflections to the signed coordinates of gv_0. In particular the endpoints of the fermion edges over a horizontal link in plane j belong to its layers $2, 3, 4, 5$.

The chain algebra consists of continuous functions of finitely many $\widetilde{u}_j(b)$ with local Grassmann polynomials. In it the field equations are (10) with the temporal-gauge operator $\widetilde{D}$. The link equation at horizontal S in plane j resamples $\widetilde{u}_j|_S$ with the even Grassmann kernel

$$\widetilde{k}_{S,j}(v) = k_S(\widetilde{u}_{j-1}|_S, v)e^{-\beta V_S(v\vee\widetilde{u}_j)}k_S(v, \widetilde{u}_{j+1}|_S)e^{-S_F^{S,j}(v)}. \tag{23}$$

Here V_S sums the horizontal plaquettes containing S, and $S_F^{S,j}$ is minus the sum of the oriented fermion hops over those links. Denote its normalized integration by $\widetilde{\gamma}_{S,j}$. Its denominator is invertible, having body at least $e^{-8\beta\#S}$.

The functional $\widehat{\omega} = \omega_\nu \circ \mathrm{tg}^*$ obeys these chain equations and is bounded on each fixed finite Grassmann sector. Indeed temporal gauge intertwines D_U and $\widetilde{D}$ by the colour matrices $\widehat{w}_j$. The chain rule for left derivatives gives the same intertwining for the field equations. Resampling a horizontal link does not change $\widehat{w}_j$; Haar substitution $v(b) = \widehat{w}_j(b_-)V(b^{(j)})\widehat{w}_j(b_+)^{-1}$ therefore intertwines its link equation with (23). The sector bounds follow from the fixed-degree Wick bound and bounded colour substitutions.

Proposition 3.8 (Uniqueness from two layers)**.** Two locally bounded functionals on the chain algebra satisfying the field and link equations agree everywhere if they agree on gauge planes t_0, t_0+1 and thin layers $X_{t_0,6} \cup X_{t_0+1,1}$. ∎

Proof. Subtract the functionals and call the difference δ. It obeys the homogeneous chain equations. There are two propagation steps.

Adding a fermion site. Suppose the null algebra includes a neighbour x' of a new site x, the three other neighbours of x', and the coordinates read by their transports. The hopping matrix at the missing edge is invertible,

$$w(x', x)^{-1} = \frac{2}{w_i(r^2-1)}(r + \gamma(\widehat{x - x'})).$$

Solve the field equation at x' for $\psi(x)$, multiplying by the inverse hopping and the inverse transporter; do the same on the right for $\bar{\psi}(x)$. After multiplication by a monomial and application of δ, the field equations replace $\widetilde{D}\psi$ and $\bar{\psi}\widetilde{D}$ by left derivatives. Each term has one fewer generator at x. Induction on that number proves nullity after adjoining x.

Adding a gauge plane. Let $\epsilon = \pm 1$. Suppose the null algebra contains $\widetilde{u}_j|_S$, $\widetilde{u}_{j-\epsilon}|_S$, all horizontal plaquette coordinates in plane j needed by V_S, and the endpoints of the fermion hops over S in that plane; the new coordinates are $\widetilde{u}_{j+\epsilon}|_S$. Factor

$$\widetilde{k}_{S,j}(v) = k_S(v, \widetilde{u}_{j+\epsilon}|_S)\Phi_\epsilon(v),$$

where Φ_ϵ is even, invertible and does not read the new plane. For χ in the old null algebra and free of $\widetilde{u}_j|_S$, resampling $G = \varphi(\widetilde{u}_j|_S)\chi\Phi_\epsilon^{-1}$ gives

$$\widetilde{\gamma}_{S,j}G = Z_{S,j}^{-1}(K_S\varphi)(\widetilde{u}_{j+\epsilon}|_S)\chi.$$

Its value under δ is zero by the link equation. Since K_S maps trigonometric polynomials onto themselves, Peter–Weyl density [2, Thm. 37.5(ii)] and local boundedness imply $\delta(Z_{S,j}^{-1}H) = 0$ for every H in the new coordinates and the old coordinates other than $\widetilde{u}_j|_S$. For an arbitrary element of the enlarged algebra, its resampling is of this form. Its value under δ is therefore zero.

To check exhaustion, write $[p, q; a, b]$ for gauge planes $p, \dots, q$ and thin layers $\mathbb{L}_a, \dots, \mathbb{L}_b$. A null interval extends by a thin layer at its upper end if that end has $k = 1$ or 6, or if its gauge plane lies in $[p, q]$; the same holds at its lower end. This is precisely the first step and the thin-layer geometry

above. It extends by a gauge plane above q once its thin layers $2,\dots,5$ are included, and similarly below p, by the second step. Starting from $[t_0, t_0+1; 6t_0+6, 6t_0+7]$, obtain

$$[t_0, t_0+1; 6(t_0-1)+5, 6(t_0+2)+2].$$

From $[t_0-s, t_0+1+s; 6(t_0-s-1)+5, 6(t_0+s+2)+2]$ add the two outside gauge planes, then six thin layers on each side, to obtain the interval for $s+1$. These intervals exhaust every local element. □

Proposition 3.9 (Bond-reflection invariance)**.** Every periodic Gibbs law is invariant under every coordinate bond reflection. Moreover

$$\mathcal{G}^{\mathrm{per}} = \{\nu \in \mathcal{G} : \nu \text{ is } 2\mathbb{Z}^3\text{-invariant}\},$$

so this set is nonempty, convex and weakly compact. ■

Proof. Let $\nu_1 = \nu$ and $\nu_2 = \theta_*\nu$ for a bond mirror. The reflected law is again periodic Gibbs, by reflection covariance of the specification. Pull both functionals back by the same temporal gauge. On the two-layer algebra in Prop. 3.8, its pullback is an element Y of the physical two-layer algebra. Reflection covariance of the Gaussian functional and Thm. C give

$$(\omega_{\nu_2} \circ \mathrm{tg}^*)(G) = \overline{\omega_\nu(\vartheta Y)} = \omega_\nu(Y) = (\omega_{\nu_1} \circ \mathrm{tg}^*)(G).$$

Both pulled-back functionals obey the chain equations, so Prop. 3.8 identifies them everywhere. For a gauge-invariant local scalar F, evaluate it on temporal-gauge configurations to get f with $\mathrm{tg}^* f = F$; hence $\nu_1(F) = \nu_2(F)$. Both laws are gauge invariant, so averaging any local scalar over its finite gauge variables identifies them on all local continuous functions. Thus $\nu = \theta_*\nu$. The product of reflections in consecutive bond mirrors is a translation by two. Every periodic solution is therefore $2\mathbb{Z}^3$-invariant; the converse is immediate. The right side is a weakly closed convex subset of the compact Gibbs class of Lem. 2.4, and contains a cubic-torus limit. □

3.5. **Adapted tori and comparison states.** Fix a mirror $P = \{n \cdot x = c\}$, with $n \in \{e_\mu, e_\mu \pm e_\nu\}$, and put $q = |n|^2$. Take orthogonal integer vectors $f_0 = n, f_1, f_2$: for a coordinate normal use the remaining axes; for $n = e_\mu \mp e_\nu$ use $f_1 = e_\mu \pm e_\nu$ and the remaining axis as f_2. For fixed N' and integer $N \geq 2$, put

$$a = 2NN', \quad h = qN', \quad \Gamma = \bigoplus_{j=0}^{2} a\mathbb{Z} f_j, \quad \Gamma' = \bigoplus_{j=0}^{2} N'\mathbb{Z} f_j, \quad \chi(af_j) = -1.$$

The torus has systole a and normal height period qa; the parallel mirrors $c_j = c + jh$ split it into $2N$ slabs. Write $g_1 = 1$, $g_{j+1} = \theta_{n,c_j} g_j$. Then g_j takes the first slab to the jth and

$$\theta_{n,c_m} g_j = g_{2m+1-j}, \quad g_{j+2N} = \tau_{af_0} g_j.$$

For odd j, $g_j = \tau_{(j-1)N'f_0}$; for even j this translation is $g_j \mathrm{fl}$, where $\mathrm{fl} = \theta_{n,c+h/2}$ reflects the first slab.

The finite-domain positivity proof in the preceding section applies to these tori and slabs. Represent the generators in one half-period slab; at the far boundary, the normal reverses and the character $\chi(af_0) = -1$ reverses the crossing sign. Both boundaries therefore have positive crossing matrices. The gauge plaquette and Haar factorizations are the same as in Thm. B. In the present order convention this gives $Z_{\mathbb{T}}[\vartheta(F)F] \geq 0$ for every local slab element F, including odd ones.

Write $\Gamma_\parallel = \Gamma \cap n^\perp$ and $\Gamma'_\parallel = \Gamma' \cap n^\perp$. For a real local A supported in the first slab define

$$\mathbb{A} = \sum_{v \in \Gamma'/\Gamma} \tau_v A, \qquad \mathbb{A}^{\mathrm{f}} = \sum_{j=1}^{2N} \sum_{v \in \Gamma'_\parallel/\Gamma_\parallel} g_j \tau_v A, \qquad M = (2N)^3.$$

The folded law $\mu_N^{\mathrm{f}} = e^{t\mathbb{A}^{\mathrm{f}}}\mu_{\mathbb{T}}/\mu_{\mathbb{T}}(e^{t\mathbb{A}^{\mathrm{f}}})$ satisfies

$$\mu_{\mathbb{T}}(e^{t\mathbb{A}}) \le \mu_{\mathbb{T}}(e^{t\mathbb{A}^{\mathrm{f}}}), \qquad \log \mu_{\mathbb{T}}(e^{t\mathbb{A}^{\mathrm{f}}}) \le tM\mu_N^{\mathrm{f}}(A). \tag{24}$$

Here are the finite-array details behind the first inequality. For slab functions f_j, write

$$\mathrm{cy}(f_1, \dots, f_{2N}) = \mu_{\mathbb{T}}(\prod_j g_j f_j).$$

Cauchy–Schwarz at each wall bounds its squared modulus by the two arrays obtained by copying either half into the other. Their values are nonnegative. Normalize each f by $\|f\| = \mathrm{cy}(f, \dots, f)^{1/(2N)}$ when this is positive. Among the finite arrays with entries taken from a fixed list, maximize the modulus after this normalization. Both half-reflections of a maximizer remain maximizers. Choose one with a longest constant cyclic string. Reflecting at its end doubles that string if its length is at most N, and makes it constant otherwise. Thus the maximum is a constant array and equals one. It follows that

$$|\mathrm{cy}(f_1, \dots, f_{2N})| \le \prod_j \|f_j\|.$$

For the present application take $f = \exp(t\sum_v \tau_v A)$ and $f' = \mathrm{fl} f$; their norms are strictly positive and equal, since translation by $N' f_0$ carries one full fold to the other. The unfurled array alternates f, f', and the folded array is constant. This proves the first inequality in (24).

The folded law is invariant under all walls c_j and under $\Gamma^{\mathrm{f}} = \Gamma'_{\parallel} \oplus 2N'\mathbb{Z} f_0$: the displayed identities for g_j permute its terms. Each term therefore has expectation $\mu_N^{\mathrm{f}}(A)$. Finally Jensen's inequality applied to $\mu_N^{\mathrm{f}}(e^{-t\mathbb{A}^{\mathrm{f}}}) = \mu_{\mathbb{T}}(e^{t\mathbb{A}^{\mathrm{f}}})^{-1}$ gives the second inequality. The folded law is reflection positive, with fermions, at P: $e^{t\mathbb{A}^{\mathrm{f}}} = \Phi\vartheta\Phi$, where Φ is the exponential of the terms in one half-period slab. It is a scalar, so the tilted form is the original positive form applied to ΦF.

Lemma 3.10 (Comparison states)**.** For $\nu \in \mathcal{G}^{\mathrm{per}}$ and real continuous local A supported in the upper half of P, there is $\nu_0 \in \mathcal{G}^{\mathrm{per}}$ such that

$$\nu(A) \le \nu_0(A), \qquad \omega_{\nu_0}(\vartheta(F)F) \ge 0$$

for every continuous local upper-half element F. ■

Proof. Choose N' a multiple of a period number of ν, large enough that the support of A lies in the first slab above P. In the preceding construction put $t = a^{-1/2}$ and choose site representatives

$$\mathcal{D} = \{y \in \mathbb{Z}^3 : 0 \le f_j \cdot y < a|f_j|^2,\ j = 0, 1, 2\}.$$

There are qa^3 sites. Each fixed value of $f_j \cdot y$ has at most qa^2 representatives; therefore $s(\mathcal{D}) \le 36qa^2$. If the support of A fits in a cube of side r_A, the translates not wholly inside this domain number at most $9r_A qa^2$. Combine (13) and (24):

$$tM\nu(A) - 2K(\mathcal{D}) - 2t\|A\|_\infty(M - M_{\mathrm{in}}) \le tM\mu_N^{\mathrm{f}}(A).$$

Here $M = (a/N')^3$. Dividing by tM makes the errors $O(a^{-1/2}) + O(a^{-1})$. Let ν_0 be a weak limit of the folded laws. A link belongs to at most $2\#B_A$ folded support sets, since each odd or even family is an array of distinct translations. Hence a change on a fixed finite B changes $t\mathbb{A}^{\mathrm{f}}$ by at most $4t\|A\|_\infty \#B_A \#B \to 0$. The slowly varying weight argument after (14) proves $\nu_0 \in \mathcal{G}$ and convergence of its local continuous Grassmann expectations. The group Γ^{f} is independent of N and contains $4N'\mathbb{Z}^3$, so ν_0 is periodic. Continuity of A gives $\nu(A) \le \nu_0(A)$. The finite folded forms are positive, and their local continuous Grassmann convergence proves the second assertion. □

3.6. **Separation and bounded Borel coefficients.**

Proof of Prop. 3.1. Fix P and let $\mathcal{R}$ be the periodic Gibbs laws whose forms $\omega_\nu(\vartheta(F)F)$ are nonnegative for every continuous upper-half element. It is a nonempty compact convex set: each condition is affine and weakly closed because the fixed-configuration Wick expectation is continuous; nonemptiness follows from Lem. 3.10; compactness follows from Prop. 3.9.

If $\nu \in \mathcal{G}^{\mathrm{per}} \setminus \mathcal{R}$, there is a real continuous scalar f and $\delta > 0$ with $\nu(f) \geq \nu'(f) + 3\delta$ for all $\nu' \in \mathcal{R}$. Indeed finitely many continuous integrals separate the point from the compact convex set: otherwise the finite-intersection property would put the point in it. In that finite-dimensional space the nearest-point separating functional gives f. Approximate f uniformly by a real local A_0 within δ, and translate it by $2kn$ into the upper half of P. All periodic Gibbs laws are $2\mathbb{Z}^3$-invariant by Prop. 3.9, so the translated A still satisfies $\nu(A) \geq \nu'(A) + \delta$ on $\mathcal{R}$. But Lem. 3.10 supplies $\nu_0 \in \mathcal{R}$ with $\nu(A) \leq \nu_0(A)$, a contradiction. Thus every periodic solution is positive at P.

Scalar reflection positivity now gives reflection invariance. The form $s(F', F) = \nu(\overline{F' \circ \theta} F)$ on continuous upper-half scalar functions is Hermitian, so $\nu(F) = \nu(F \circ \theta)$ there. Translate an arbitrary local scalar function far enough into that half by $2kn$, and use $\theta\tau_{2kn} = \tau_{-2kn}\theta$ and translation invariance to get the same equality for every local function. The reflections in all nine mirror families generate $\mathbb{Z}^3 \rtimes W_3$, proving the full lattice invariance. Reflection covariance of the Dirac inverse and Wick's rule give $\omega_\nu(\vartheta G) = \overline{\omega_\nu(G)}$. The same identity holds with Θ_P, since the two differ only on odd elements, whose values vanish.

For bounded Borel coefficients, write $F = \sum_j f_j \xi_j$ with finitely many half-side links and monomials. Approximate f_j by continuous $f_{j,k}$ in L^2 of their finite-link marginal of ν. Reflection invariance gives the same convergence for $f_{j,k} \circ \theta$. Each Wick kernel $h_{\vartheta(\xi_i)\xi_j}$ is bounded. Cauchy–Schwarz therefore shows convergence of every term of the reflected form. This extends positivity and Hermitian symmetry. Terms between different parities are odd and vanish. On the lower half, apply the upper form to ϑF; interchanging homogeneous odd factors produces its parity sign. Changing the normal in Θ_P produces precisely the same sign. Thus (9) gives $\omega_\nu(F\Theta_P F) \geq 0$ with the normal into either chosen side. Cubic-torus limits are periodic Gibbs laws by Prop. 2.5, so the last assertion follows. □

4. The pure gauge input and local expansions

We use the finite gauge construction of [3] with the following conventions and hypotheses. Fix its odd blocking integer $L \geq 13$, an integer $r_c \geq 2$, an integer $\mathfrak{p}_0 \geq 6r_c + 1$, and a power $M = L^{m'}$. Fix also the positive fermion block penalty b_F, subject to the bounds used below. These choices precede the cutoff and volume. Here $K \geq 1$ and the fine spacing is $a_K = L^{-K}$, the physical torus has side $\ell = 2L^m$, and level j has spacing $\lambda_j = L^{j-K}$ and $n_j = \ell^3 L^{3(K-j)}$ sites. The calibration and running coefficients are

$$\mathsf{t}_j = L^{K-j}\mathsf{b}, \qquad x_0 = \beta = L^K \mathsf{b}, \qquad x_{j+1} = L^{-1}x_j - b_{j+1}, \quad |b_{j+1}| \leq B_\sharp.$$

The last bound implies $|x_j - \mathsf{t}_j| \leq \bar{B}$ for a fixed $\bar{B}$. Require $x_j \geq \gamma^{-2}$ and $MR(x_K) \leq L^m$, where

$$R(x) = \min\{L^q : q \in \mathbb{Z}_{\geq 0},\ L^q \geq (\log x)^{r_c}\}, \quad \epsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}, \quad \delta(x) = A_1 x^{-1/2}(\log x)^{\mathfrak{p}_0}.$$

For the continuum construction fix $0 < g \leq \gamma$ and take b in the closure of $(g^{-2} + \bar{B}, L(g^{-2} + B_\sharp) - \bar{B}]$. Then $g^{-2} \leq x_K \leq L(g^{-2} + B_\sharp)$. Write $X_g = L(g^{-2} + B_\sharp)$. We take the allowed physical sides to be the cofinal set $\mathfrak{L}(g) = \{2L^m : m \geq m_g\}$, where

$$m_g = 1 + m' + \max\left\{\left\lceil \log_L R(X_g + \bar{B}) \right\rceil, \lceil \log_L C_{\mathrm{s}} \rceil\right\}.$$

Here $C_{\mathrm{s}} \geq 2$ is the fixed chart-room constant of [3, Cor. 4.167] and its preceding torus-room calculation. This is the room for the larger local charts as well as the finite induction. Omitting

finitely many sides does not alter the class of infinite-volume torus limits. The fermion mass is $m_0 = L^{-K}\mathsf{m}$, with $\mathsf{m} \in [m_-, m_+]$ and $0 < m_- \le m_+ < 1/2$.

We use the threshold and growth functions

$$\mathfrak{b}(t) = t^{1/2}\delta(t), \qquad \phi(t) = t^{-1/2}(1+\log t)^{\mathfrak{p}_0}, \qquad \widehat{\phi}(t) = (1+\log t)^{\mathfrak{p}_0},$$

$$\phi_\sharp(t) = t^{-1/2}(1+\log t)^{3\mathfrak{p}_0}, \qquad \psi_\sharp(t) = t^{-1/2}(1+\log t)^{9\mathfrak{p}_0}, \qquad \mathfrak{m}_j = L^{j-K}\mathsf{m}.$$

The blocks of one step are the cubes of L^3 sites centered at the next lattice. The gauge map $\mathcal{M}$ is the global real block average of [3, eq. 3.xvii] and [3, eq. 3.xviii]. On its small-curvature domain it is the symmetrized contour mean: for a coarse bond c its fine contours start at c_-, follow the straight fine path from x to $x + L\hat{\mu}$, and end at c_+, with $x \in B(c_-)$. The symmetrized path transports give matrices W_x near one, and

$$\mathcal{M}(U)(c) = \exp\left[L^{-3}\sum_{x\in B(c_-)} \log W_x\right] U[c].$$

The continuation is used only on the logarithm branch specified there. Outside the small-curvature domain use the Borel real extension of [3, Prop. 3.4] and its preceding definition. This fixes the global map as well as its analytic branch; it does not choose a branch separately in each history.

At each level let $\mathcal{P}_j$ be the partition into cubes of side M in level-j units. A polymer $X \in \mathcal{D}_j$ is a nonempty wall-connected union of these cubes. Its size $d_j(X)$ is the least number of edges in a tree formed from edges of the cubes of X and meeting every closed cube of X. A single corner is allowed as a tree. We use the entropy and joining bounds, in particular

$$\sum_{X \ni \square} e^{-c_E d_j(X)} \le C_E, \qquad d_j(X_1 \cup X_2) \le d_j(X_1) + d_j(X_2) + c_0$$

for intersecting polymers, with the fixed geometric constant c_0. The closed enlargement $X^\sharp$ has the reach fixed by that construction. Parameters on cubes outside this enlargement localize a kernel; a finite difference in a cube set $\mathcal{K}$ is assigned to $X^\sharp \cup \bigcup \mathcal{K}$.

In level-j units, a regular pair $(\mathbf{U}, \mathbf{J})$ is represented by a triple $(U, A', \mathbf{J})$, with U real and $\mathbf{U} = e^{iL^{-j}A'}U$ complex. It belongs to the analytic core $\mathcal{U}_j^\circ(X; b_0, b_1)$ of [3, Def. 4.26]: its curvature and current have the indicated small bounds, its local real gauge potentials have the stated bounds, and every required reduced block tower exists on the prescribed logarithm branch and satisfies the critical-background bounds at every earlier level. The core is not defined by curvature alone. Two representatives of the same pair have the same $(\mathbf{U}, \mathbf{J})$; their real reference configurations may differ. At radius $a \le \alpha_0$ use $(b_0, b_1) = (a, (\alpha_1/\alpha_0)a)$ at all levels. In a restriction from level j' to level $j \le j'$, put $\lambda = L^{-(j'-j)}$. The forms A' and $\mathbf{J}$ scale by λ and λ^3, respectively, while $\mathbf{U}$ is unchanged, as in [3, eq. 4.cxxiv].

For a regular coarse field V, $U_j(V)$ denotes the admitted small-curvature critical gauge orbit of Wilson's action on the fibre $\mathcal{M}^j U = V$. At a representative of this orbit the shift system $\mathcal{A}(B; P)$ is the holomorphic coordinate change for the gauge fluctuation: the fine field read by a step is $U' = U \exp \mathcal{A}(B; P)$, with the associated site gauge factor. Each term of a kernel expansion is assigned to the polymer hull containing its path and parameter cubes. The shift is bounded by $C_{\mathcal{A}}\delta(\mathfrak{t}_j)$ on the fluctuation polydisc and has the exponential tails across the layers. We retain the site gauge factor in the fermion averages; gauge covariance carries it to their generators.

The following results of [3] are the external pure gauge input. The verification in Prop. 10.20, using Lem. 10.19, supplies the hypotheses of its finite-renormalization theorem [3, Thm. C] on the sequences used here. It gives the critical backgrounds, shift systems, histories, coordinate maps, positive gauge-fixing weights, retained forms and pivot determinants; its local terms are holomorphic on the cores, local and gauge covariant, with exponential polymer bounds. Its history decomposition is linear in a bounded Borel density, coefficientwise for a finite-dimensional algebra-valued density.

We use [3, Prop. 4.54] and [3, Def. 4.53]: holomorphic local Gaussian kernels and interaction, the real-reference bounds, the gauge covariance and independence of the triple representing a given pair $(\mathbf{U}, \mathbf{J})$, the polymer bounds and its stated numerical inequalities. The output has rate $\kappa_+ = (1 + 4\beta_s)\kappa$. The quadratic term caused by changing the gauge Gaussian coefficient, denoted by q in the expansion, is kept as a separate scalar family. We also use [3, Prop. 4.52]: at fixed radius the change of an old array gains the size of the shift, and at radius a its bound is

$$\frac{6}{5} K_\Lambda \frac{C_{\mathcal{A}} \delta(\mathfrak{t}_k)}{a} \sum_{i \le k} L^{-(k-i)} \epsilon_i \, e^{-(1-2\delta_w)\kappa d_k(Y)}.$$

Linearity permits explicit normalizer families and small fluctuation families to be localized separately. These assertions include their original existence, domain and numerical hypotheses; the constants are chosen in their admitted order. The matter-specific uses of them are verified below.

4.1. **The centered conditional integral.** On a history, some gauge variables remain outside the integration region. The conditional Gaussian therefore has a nonzero mean. The expansion must be estimated about that mean: a bound obtained by taking the absolute value of an uncentered density would introduce an extensive error depending on the external field.

We recall the conditional-integral estimate of [3, Prop. 5.12], [3, Lem. 5.13], and [3, Prop. 5.14]. Let $(C, \Gamma, \mathbf{m})$ be a holomorphic Gaussian system satisfying its mean, error, geometry and tested-bond conditions. Suppose $F(B; n)$ has the required holomorphy in B on the prescribed tube and is holomorphic in the center n on its prescribed domain for each real B. If

$$|F(B; n)| \le F_0 \exp\{\rho_+ \|B - n\|^2\}, \qquad \rho_+ = (LM)^{-3},$$

then the centered conditional integral satisfies

$$|J^{\mathbf{m}}[F(\cdot; \mathbf{m})]| \le F_0 e^{c_P^+ N_\Lambda(X)} e^{-\gamma_P \# e^+}, \qquad c_P^+ = c_P + C_{\mathrm{mn}}^3/\lambda_r. \tag{25}$$

Here $N_\Lambda(X)$ counts the integration cubes and e^+ is the failed-bond set of the conditional integral. A finite difference in a set $\mathcal{K}$ of parameter cubes gains $e^{-(\bar\kappa - 1)\#\mathcal{K}}$. The estimate uses the actual complex Gaussian system at the point of evaluation. In the contour proof its centered argument is $\beta' - i\eta$, with $\|\eta\|^2 \le 16 d_*^{-2} \epsilon_m^2 \|\beta'\|^2$. Thus the bound contains no factor depending on the size of the conditional mean.

Lemma 4.1 (Algebra-valued conditional integration)**.** The estimate (25), including its finite-difference version, holds for a holomorphic integrand with values in a finite-dimensional normed algebra, with its norm in place of the absolute value. A subsequent Gaussian Berezin integration preserves the pointwise norm estimate whenever it is a contraction for the fibre norm on the whole parameter domain. The joined finite-difference and component statements additionally require the holomorphy, locality and component restrictions specified in Lem. 9.1. ■

Proof. Apply the scalar conditional-integral theorem to $\lambda(F)$ for each norm-one linear functional λ. Its growth bound is the same as that of F, and the contour integral commutes with λ. Taking the supremum gives the norm bound, with no factor for the dimension of the algebra. The same argument applies before taking finite differences. The pointwise Berezin extension follows from its uniform contraction bound; the parameter-dependent joined expansion is treated in Lem. 9.1. In the even Grassmann algebra, products commute and the fibre norm is submultiplicative; hence the scalar Mayer majorants also bound the corresponding products. □

This functional estimate is only one input to a history step. The gauge domain, the fermion covariance, the origin assigned to each quadratic term, and the identities of the actual density must also be checked. We make those checks in Lems. 9.1, 9.4 and 9.9 and Prop. 9.11.

5. Fermion block integration

We now express a fermion block integral as a Schur complement and a determinant. Coercivity gives the inverse needed in this identity; the fibre norm controls its Grassmann coefficients, and the normalized block average fixes the Gaussian block kernel.

Throughout this section the fermion is the balanced chamber operator with $r = \frac{1}{2}$, and $N_f \geq 1$ is fixed. Its fibre dimension is $\nu = 8N_f$. The maps $\iota\varphi(x,g) = 48^{-1/2}\varphi(x)$, $\Pi_0 = \iota\iota^*$ and $\Pi_1 = 1 - \Pi_0$ distinguish the chamber mean from the other components. In this section $D[U]$ is massless: it is the foundational operator $D_U - m_0$. Write $\Delta[U] = \operatorname{Re} D[U]$ and $\theta = b_F > 0$. For an oriented site bond $b = \langle x, y\rangle$ set $\nabla_U\varphi(b) = U_{\langle x,y\rangle}\varphi(y) - \varphi(x)$; its squared norm is summed over both orientations of each bond. Restricted gradient norms use both orientations of bonds with both endpoints in the indicated set. The lattice operator satisfies

$$\Delta \geq 0, \quad D^*D \leq c_{\mathrm{disc}}\Delta, \quad c_{\mathrm{disc}} = 9(6+6\sqrt{2}), \qquad \langle f, \Delta f\rangle \geq \frac{\lambda_\rho}{2}\|\Pi_1 f\|^2 + c_\Delta\|\nabla_U\iota^* f\|^2, \quad c_\Delta = \frac{\lambda_\rho}{96}.$$

Here $\lambda_\rho > 0$ is the first nonzero eigenvalue of the within-site chamber Laplacian. Every constant below is independent of the cutoff, volume and history unless a dependence is displayed.

Here are proofs of these elementary bounds. Write $D = \sum_s h_s(1 - R_s)$, where R_s is a self-adjoint involution, $h_s = t_s(r - \gamma_s)$ and $\Delta = \sum_s \rho_s(1 - R_s)$, $\rho_s = rt_s > 0$. Thus $\|(1-R_s)f\|^2 = 2\langle f, (1-R_s)f\rangle$. Weighted Cauchy–Schwarz gives

$$\|Df\|^2 \leq 2\Big(\sum_s \frac{t_s^2(1+r)^2}{\rho_s}\Big)\langle f, \Delta f\rangle = c_{\mathrm{disc}}\langle f, \Delta f\rangle.$$

For the balanced chamber weights $\sum_s t_s = 6 + 6\sqrt{2}$ and $r = 1/2$, giving the displayed constant. In particular $\|D\| \leq c_{\mathrm{disc}}$ and $0 \leq \Delta \leq c_{\mathrm{disc}}$. The three within-site reflections generate the chamber group. Their graph Laplacian has constants as its kernel, and its least positive eigenvalue is λ_ρ; hence $\sum_{s\geq 1}\rho_s(1 - R_s) \geq \lambda_\rho\Pi_1$. For $f = \iota\varphi + f'$, $f' = \Pi_1 f$, the remaining hop gives

$$\langle f, (1-R_0)f\rangle \geq \tfrac{1}{4}\|(1-R_0)\iota\varphi\|^2 - \tfrac{1}{2}\|(1-R_0)f'\|^2 \geq \tfrac{1}{24}\|\nabla_U\varphi\|^2 - 2\|f'\|^2.$$

Each of the six signed coordinate directions occurs eight times among the 48 chambers. With both bond orientations in the norm, this gives $\|(1-R_0)\iota\varphi\|^2 = \frac{1}{6}\|\nabla_U\varphi\|^2$ and hence the coefficient $1/24$. This hop form is also nonnegative. Take its fraction $\tau = \lambda_\rho/(4\rho_0) \leq 1$ and add the within-site bound to obtain the stated coercivity. Finally, for every real $m \geq 0$,

$$(D+m)^*(D+m) \leq (c_{\mathrm{disc}} + m)(\Delta + m),$$

because the difference after using $D^*D \leq c_{\mathrm{disc}}\Delta$ is $m(c_{\mathrm{disc}} - \Delta) \geq 0$. The massive operator therefore belongs to the disc of diameter $c_{\mathrm{disc}} + m$ and has positive definite real part when $m > 0$.

5.1. **The fibre norm and Gaussian integration.** Let P be a finite set of places, with independent barred and unbarred copies of the fibre at each place. For positive weights $h(s)$, equal on the two copies, define

$$\|F\|_h = \inf\left\{\sum_k |a_k| \prod_l h(s_{kl}) : F = \sum_k a_k e_{k1}\cdots e_{kn_k}, \quad e_{kl} \text{ a unit vector in a fibre at } s_{kl}\right\}.$$

Lemma 5.1 (The fibre norm)**.** The even Grassmann algebra with this norm is a commutative unital Banach algebra. The norm is unchanged by fibrewise unitary substitutions. If $T_{s's}$ are the blocks of a linear substitution and $\sum_{s'} h'(s')\|T_{s's}\| \leq h(s)$, then $\|F \circ T\|_{h'} \leq \|F\|_h$. For Berezin integration over a subset P_0,

$$\|\mathrm{Ber}_{P_0} F\|_h \leq \prod_{s\in P_0} h(s)^{-2\nu}\|F\|_h.$$

If μ_C is Gaussian Grassmann integration with covariance C on P_0 and $h \geq \|C\|^{1/2}$ there, then $\|\mu_C F\|_h \leq \|F\|_h$. More generally,

$$\left\| \int F(\Psi + \Psi')\, d\mu_C(\Psi') \right\|_h \leq \|F\|_{h+\|C\|^{1/2}\mathbf{1}_{P_0}}.$$

■

Proof. Concatenating representations proves submultiplicativity. If $|F|_h$ denotes the coefficient norm in an orthonormal basis, then $\|F\|_h \leq |F|_h \leq \|F\|_{\sqrt{\nu}h}$: the second inequality follows by expanding unit vectors, whose coefficient sums are at most $\sqrt{\nu}$. In particular the infimum defines a norm. The even algebra is commutative and finite dimensional, hence complete. A fibrewise unitary carries the defining unit vectors to unit vectors; applying the inverse gives equality of the norms. For a unit vector e at s, write $Te = \sum_{s'} T_{s's} e$ and apply the assumed weighted row bound. Multiplication and the defining infimum give the substitution estimate.

Berezin integration of a product of unit vectors vanishes unless there are 2ν factors at each integrated place. Its top coefficient is a determinant of unit columns, with modulus at most one by Hadamard's inequality. This removes exactly the weights displayed in the estimate. Gaussian integration contracts m unbarred and m barred factors to $\det[\langle a_i, Cb_j\rangle]$. Its modulus is at most $\|C\|^m$, by the exterior-product form of Gram's inequality; unbalanced products have zero integral. The integrated weights pay this factor. Finally, expand each factor of $F(\Psi + \Psi')$ into its two terms. Summing the bounds for the subsets taken at Ψ' replaces each weight $h(s)$ by $h(s)+\|C\|^{1/2}\mathbf{1}_{P_0}(s)$. □

We normalize the Berezin integral so that $\operatorname{Ber} e^{-\bar{\Psi} A \Psi} = \det A$. At invertible A, Gaussian moments are the minors of A^{-1}. Integration in disjoint sets of barred–unbarred pairs commutes; each such pair is an even operation. Thus all the following identities hold coefficientwise also for measures with Grassmann values. In finite dimension this last assertion follows by applying scalar integration to each coefficient.

5.2. **Averages defined at every gauge field.** For a block $B(y)$ and its contour families $\mathcal{G}(y,x)$ put

$$S(V;y,x) = \frac{1}{\#\mathcal{G}(y,x)} \sum_{\Gamma \in \mathcal{G}(y,x)} V(\Gamma),$$

$$n(V;y) = L^{-3} \sum_{x \in B(y)} \det S(V;y,x), \qquad q(V;y,x) = n(V;y)^{-1/2} S(V;y,x).$$

$$(Q(V)\varphi)(y) = L^{-3/2} \sum_{x \in B(y)} q(V;y,x)\varphi(x).$$

The contours run through the coordinate directions in all orders; on the coordinate lines through the block centre there is one contour.

Lemma 5.2 (The normalized block average)**.** At every real gauge field,

$$(3L-2)L^{-3} \leq n(V;y) \leq 1, \qquad Q(V)Q(V)^* = 1.$$

The average is block local, Borel, gauge covariant and covariant under the lattice isometries preserving the block partition. It preserves the spinor boundary signs. For $R_k = Q(\mathcal{M}^{k-1}U)\cdots Q(U)\iota^*$ one has $R_k R_k^* = 1$ and $R_k \Pi_1 = 0$. ■

Proof. A real linear combination of SU(2) matrices is a real quaternion. Hence $SS^* = (\det S)1$ and $0 \leq \det S \leq 1$ for the average of unit quaternions. At the $3L-2$ sites on the three coordinate lines, S is a single transporter and its determinant is one. This proves both bounds on n. Since the blocks are disjoint,

$$(QQ^*)(y,y') = \delta_{yy'} L^{-3} n(V;y)^{-1} \sum_x \det S(V;y,x) = \delta_{yy'}.$$

Path transporters are gauge covariant, and an isometry permutes the contours. The formulas are local and Borel; translating a whole block by a period gives the same formula with the prescribed spinor sign. The assertions for R_k follow by composition and $\iota^*\iota = 1$, $\iota^*\Pi_1 = 0$. □

Lemma 5.3 (The first fluctuation at an arbitrary field)**.** On a block $B(y)$,

$$\|\varphi\|_B^2 \le 2|Q(V)\varphi(y)|^2 + 12L^3(L-1)\sum_{b\subset B}|\nabla_V\varphi(b)|^2.$$

Here the explicit sums over $b \subset B$ count each bond once. Consequently the first fluctuation operator, on any union of blocks and with zero continuation off that union, is $\Gamma_0 = (D[V] + m_0)_{AA} + \theta q_1^* q_1$, where A is the set of chambers over the union and $q_1 = L^{-1/2}Q(V)\iota^*$. It satisfies

$$\operatorname{Re}\Gamma_0 \ge m_0 + c_{\mathrm{fl}}, \qquad c_{\mathrm{fl}} = \min\left\{\frac{\theta}{2L}, \frac{c_\Delta}{6L^3(L-1)}\right\} > 0.$$

These assertions hold for every real SU(2) configuration. Let $c_{\mathrm{hop}} = 9/2$ be the bound on the site-changing hopping coefficient, and put $\delta_0 = c_{\mathrm{fl}}/[4e(c_{\mathrm{hop}} + 6\theta)]$. For every set Y of chambers, the compression $\Gamma_{0,Y}$ obeys

$$|\Gamma_{0,Y}^{-1}(v,v')| \le \frac{2}{c_{\mathrm{fl}}} e^{-\delta_0|x_v - x_{v'}|_1}.$$

The same estimate holds on the mirror tori used below. ■

Proof. Let $\delta^2 = \sum_{b\subset B}|\nabla_V\varphi(b)|^2$ and $c_L^2 = 3(L-1)/2$. Telescoping along a contour gives $|V(\Gamma)\varphi(x) - \varphi(y)| \le c_L\delta$. Average this identity over the contours. Since $n \le 1$,

$$|Q\varphi(y)| \ge L^{3/2}(|\varphi(y)| - c_L\delta), \qquad |\varphi(x)| \le |\varphi(y)| + c_L\delta.$$

Hence $\|\varphi\|_B^2 \le L^3(L^{-3/2}|Q\varphi(y)| + 2c_L\delta)^2$, which is the first inequality. Write $f = \iota\varphi + f'$. The chamber coercivity has coefficient $\lambda_\rho/2$ on $\|f'\|^2$ and $2c_\Delta$ on the internal, once-counted bond differences of φ. Since $q_1 f = L^{-1/2}Q\varphi$, summing the block inequality gives

$$\|f\|^2 \le c_{\mathrm{fl}}^{-1}(\langle f, \Delta f\rangle + \theta\|q_1 f\|^2).$$

Adding $m_0\|f\|^2$ proves the fluctuation floor.

For a real weight η changing by at most δ_0 across a bond, twisting D changes its norm by at most $c_{\mathrm{hop}}(e^{\delta_0} - 1)$. Write $q_1^* q_1 = L^{-1}\iota P\iota^*$, where P is the sum of the block projections. The variation of η from a block centre is at most $3L\delta_0/2 \le 1$, so

$$\|e^{\eta}Pe^{-\eta} - P\| \le e^{3L\delta_0} - 1 \le 6eL\delta_0.$$

The two changes together cost at most $c_{\mathrm{fl}}/2$. Compression commutes with multiplication by e^{η}; hence the twisted compressed operator has real part at least $m_0 + c_{\mathrm{fl}}/2$ and inverse norm at most $2/c_{\mathrm{fl}}$. Take $\eta(x) = \delta_0\,\mathrm{dist}_1(x, x_{v'})$ to obtain the kernel bound. The argument is local to blocks and walls, so it also applies on every mirror torus. □

Let $q_1 = L^{-1/2}Q(U)\iota^*$ and $q_{j+1} = L^{-1/2}Q(\mathcal{M}^j U)$ for $j \ge 1$. They satisfy $q_{j+1}q_{j+1}^* = L^{-1}$. The normalized kernel of one step is

$$\mathrm{Ker}_j = \theta^{-\nu n_{j+1}}\exp\{-\theta\langle\bar\Psi_{j+1} - \bar\Psi_j q_{j+1}^*, \Psi_{j+1} - q_{j+1}\Psi_j\rangle\}.$$

Its integral in the new variables is one: translate these variables by $q_{j+1}\Psi_j$ and use the Gaussian determinant. Gauge covariance is immediate from that of q_{j+1} and the contragredient barred action. Thus the transformation $\mathbf{T}_F\nu = \mathcal{M}_*(\mathrm{Ber}_j(\mathrm{Ker}_j\nu))$ commutes with full Berezin integration. This exact identity is made before the small- and large-field tests.

Lemma 5.4 (Composition of Gaussian block kernels)**.** Two normalized block kernels with parameters (b, Q) and (b', Q'), where $Q'Q'^* = c'1$, compose to the normalized kernel with average $Q'Q$ and parameter b'' given by

$$(b'')^{-1} = (b')^{-1} + c'b^{-1}.$$

In particular the parameter after k steps is

$$\theta_k = \frac{b_F(1 - L^{-1})}{1 - L^{-k}}, \qquad \mathfrak{a}_k = L^{-k}\theta_k.$$

■

Proof. Translate the intermediate variables by $Q\Psi$. For $w = \Psi'' - Q'Q\Psi$ their Gaussian integration leaves the quadratic form

$$b'\bar{w}\{1 - b'Q'(b + b'Q'^*Q')^{-1}Q'^*\}w = \frac{bb'}{b + b'c'}\bar{w}w.$$

The identity $Q'(b + b'Q'^*Q')^{-1}Q'^* = c'/(b + b'c')$ also gives the determinant $b^{\nu(n'-n'')}(b + b'c')^{\nu n''}$, which converts the two normalizations into $(b'')^{-\nu n''}$. Iterating with $c' = L^{-1}$ gives the displayed geometric sum. □

Set

$$c_b = \frac{b_F(1 - L^{-1})}{b_F(1 - L^{-1}) + 1/2}, \qquad c_{\mathrm{tw}} = \max\{c_{\mathrm{disc}} + 1, b_F\}.$$

The latter is a uniform tower bound on the mass range used here. For a site matrix C write

$$\mathsf{S}_1(C\psi) = \sum_v \bar{\psi}(v)C(x_v)\psi(v), \qquad \mathsf{S}_1(\psi) = \mathsf{S}_1(1\psi).$$

5.3. **The Schur complement and its determinant.** For $c > 0$ let $\mathsf{C}(c)$ denote the operator disc $\{X : X^*X \le c\,\mathrm{Re}\,X\}$. At a step split the old variables into stored variables S and integrated variables A. If their quadratic form is $\mathbb{D}_j$, let

$$\Gamma_j = (\mathbb{D}_j)_{AA} + \theta q_{j+1}^* q_{j+1}.$$

Write $q = q_{j+1}$ and $\Psi^\bullet = (\Psi_S, \Psi')$. The two background substitutions on the integrated variables are

$$\Psi_A^{\mathrm{bg}} = \Gamma_j^{-1}(\theta q^*\Psi' - (\mathbb{D}_j)_{AS}\Psi_S), \qquad \bar{\Psi}_A^{\mathrm{bg}} = (\theta\bar{\Psi}'q - \bar{\Psi}_S(\mathbb{D}_j)_{SA})\Gamma_j^{-1}.$$

The stored variables are unchanged. The second substitution is the row twin of the first. On complex gauge fields the symbol q^* in these formulas is continued by its holomorphic row extension q^+, which agrees with the adjoint at real fields. With $W = \Psi_A - \Psi_A^{\mathrm{bg}}$ and $\bar{W} = \bar{\Psi}_A - \bar{\Psi}_A^{\mathrm{bg}}$, completing the square gives

$$\bar{\Psi}\mathbb{D}_j\Psi + \theta(\bar{\Psi}' - \bar{\Psi}_A q^*)(\Psi' - q\Psi_A) = \bar{\Psi}^\bullet\mathbb{D}_{j+1}\Psi^\bullet + \bar{W}\Gamma_j W.$$

The row twin above is not the adjoint of the unbarred background map.

Lemma 5.5 (Exact factorization)**.** The tower is formed at every real gauge configuration when $m_0 > 0$. Writing $D_0 = D[U] + m_0$, one has

$$\det D_0 = \left(\prod_{i<j} \theta^{-\dim H_{i+1}} \det\Gamma_i\right) \det\mathbb{D}_j.$$

If $X^*X \le c\,\mathrm{Re}\,X$ and $\mathrm{Re}\,X > 0$, then $\mathrm{Re}(X^{-1}) \ge c^{-1}$. For $D_0 = D[U] + m_0$, it holds with $c = c_{\mathrm{disc}} + m_0$. Writing $c_* := \max\{c, \theta\}$, the tower forms obey $\|\mathbb{D}_j\| \le c_*$, whereas $\|\Gamma_j\| \le c_* + \theta$. More precisely, on the accumulated stored variables and the current variables, write S_i for the one-particle space first stored at level i and H_j for the current level-j space. Then

$$\mathrm{Re}\,\mathbb{D}_j^{-1} \ge c^{-1}1_{S_0} \oplus \theta^{-1}1_{S_1\oplus\cdots\oplus S_{j-1}\oplus H_j}.$$

■

Proof. The matrix before eliminating Ψ_A is $\mathsf{T}^*(\mathbb{D}_j \oplus \theta)\mathsf{T}$ with $\mathsf{T}(\Psi, \Psi') = (\Psi, \Psi' - q\Psi_A)$ and $\det \mathsf{T} = 1$. Its Schur complement is $\mathbb{D}_{j+1}$; therefore

$$\det \mathbb{D}_j = \theta^{-\dim H_{j+1}} \det \Gamma_j \det \mathbb{D}_{j+1}, \qquad \mathbb{D}_{j+1}^{-1} = \mathsf{q}\mathbb{D}_j^{-1}\mathsf{q}^* + (0 \oplus \theta^{-1}),$$

with q the identity on all previously stored variables and q on the integrated variables. The inverse inequality is the identity

$$\operatorname{Re}(X^{-1}) - c^{-1} = X^{-*}(\operatorname{Re} X - c^{-1}X^*X)X^{-1}.$$

Suppose the displayed block floor holds at level j. On every stored block q is the identity, so its floor is preserved. On the new variables the transported positive term is supplemented by θ^{-1} times the identity. This proves the floor at level $j+1$ even though $qq^* = L^{-1}$; no isometry of q is being assumed. The uniform lower bound is consequently c_*^{-1}. The inverse identity makes this equivalent to $X^*X \le c_* \operatorname{Re} X$, or $\|X - c_*/2\| \le c_*/2$, for $X = \mathbb{D}_j$. It gives $\|\mathbb{D}_j\| \le c_*$. A compression preserves this disc because $\|PXPf\|^2 \le \|XPf\|^2$. The penalty θq^*q is nonnegative with norm $\theta/L \le \theta$; its disc diameter is at most θ. Discs add, so $\Gamma_j \in \mathsf{C}(c_* + \theta)$. Moreover the compressed real part of $\mathbb{D}_j$ is positive definite, so Γ_j is invertible. The Schur complement has positive real part: evaluate the original positive-real-part form on its background vector. This supplies the next invertible form and closes the induction. The determinant identity then iterates as claimed. □

On the history without large-field regions the closed form is

$$\mathcal{K}_k = D[U] + m_0 + \mathfrak{a}_k R_k^* R_k, \qquad \mathbb{D}_k = L^k \mathsf{D}^{(k)}, \qquad \mathsf{D}^{(k)} = \mathfrak{a}_k - \mathfrak{a}_k^2 R_k \mathcal{K}_k^{-1} R_k^*.$$

6. Fluctuation operators in a regular gauge background

The block integral requires estimates on its covariance that remain uniform as the lattice spacing decreases. The small mass does not provide such estimates. The block penalties control the light modes, while the chamber gap controls the heavy modes. We first combine these two estimates, then localize the inverse. The localization will also let us integrate over a polymer using only its configuration.

For a history with regions $\Omega_1 \supset \cdots \supset \Omega_k$, put $X_n^i = \Omega_n \setminus \Omega_{n+1}$ for $n < i$ and $X_i^i = \Omega_i$. These sets partition Ω_1. On X_n^i let $\mathsf{L}_i = L^n$, and let it be one outside Ω_1. The separation and regularity conditions used below are

$$\operatorname{dist}_\infty(\Omega_{n+1}, \Omega_n^c) > RL^n, \qquad |\partial U(p) - 1| \le \epsilon L^{-2n} \quad \text{if a corner of } p \text{ lies in } X_n^{j+1}.$$

There is no regularity requirement outside Ω_1. Here $\partial U(p) = U_p$ is the oriented plaquette product. Define

$$\mathsf{s}_i(x) = \min_y \{\mathsf{L}_i(y) + |x-y|_1/R\}, \qquad \mathcal{K}_i = (D[U] + m_0)_{\Omega_1\Omega_1} + \mathsf{W}_i, \qquad \mathsf{W}_i = \sum_{n=1}^{i} \sum_{B \subset X_n^i} \theta_n L^{-n} \iota P_B \iota^*.$$

Here $P_B = Q_B^* Q_B$, and Q_B is the row of the iterated site average on the block B of side L^n. Equivalently set $T_n^i = 1_{X_n^i} q_n \cdots q_1$; then $\mathsf{W}_i = \sum_n (T_n^i)^+ \theta_n T_n^i$, where $+$ denotes the row twin on the complex class and the adjoint at real fields. At real fields its norm squared on a block is L^{-n}. At the top integrated places of step j write $T_A = 1_{A_j} q_j \cdots q_1$ and $c_j = \theta_j + \theta q_{j+1}^+ q_{j+1}$. The block inverse identity gives

$$\Gamma_j^{-1} = c_j^{-1} + c_j^{-1}\theta_j T_A \mathcal{K}_{j+1}^{-1} T_A^+ \theta_j c_j^{-1}.$$

One way to verify it is to eliminate the bare field first or the last coarse field first in the same three-block Gaussian matrix. The bare Schur form after the second elimination is $\mathcal{K}_j - T_A^+ \theta_j c_j^{-1} \theta_j T_A$, which is $\mathcal{K}_{j+1}$. The corresponding block of its inverse is the displayed formula. At $j = 0$, $\Gamma_0^{-1} = \mathcal{K}_1^{-1}$. Put $\mathcal{E}_U(g) = \|\Pi_1 g\|^2 + \|\nabla_U \iota^* g\|^2$. We use $c_G = c_\Delta$ in the bound $\langle f, \Delta f\rangle \ge c_G \mathcal{E}_U(f)$. The separation

gives $1 \le \mathsf{s}_i \le \mathsf{L}_i \le L\mathsf{s}_i$ and $|\mathsf{s}_i(x) - \mathsf{s}_i(x')| \le R^{-1}$ on a bond. The distance $d_{h,i}$ is the shortest-path distance whose bond $\langle x, x'\rangle$ has length $1/\max\{\mathsf{s}_i(x), \mathsf{s}_i(x')\}$. A weight has rate δ if its difference across every bond is at most δ times this length.

Lemma 6.1 (Regular averages)**.** Assume that the admitted pure-gauge block-average estimate applies in a block B of side L^n, so that

$$|\partial(\mathcal{M}^i U)(p) - 1| \le 2\epsilon L^{2(i-n)} \quad (0 \le i < n), \qquad \epsilon \le \min\{\varepsilon_\flat, (400L^2)^{-1}\}.$$

Then

$$\|(1 - P_B)\varphi\|_B^2 \le c_P L^{2n} \|\nabla_U \varphi\|_B^2 + \epsilon_P \|\varphi\|_B^2, \quad c_P = \frac{e^{1/(L-1)} L^2}{2(L^2-1)}, \quad \epsilon_P = 50(64L^2)^2 L^{-2}\epsilon^2.$$

■

Proof. Write $V_i = \mathcal{M}^i U$, $\bar\omega_i = 2\epsilon L^{2(i-n)}$ and $r_i = e^{9L^2\bar\omega_i/8} - 1$. Rearranging the steps of two contours bounds the difference of their transporters by r_i. For unit quaternions,

$$\left|\frac{1}{m}\sum_a g_a\right|^2 = 1 - \frac{1}{2m^2}\sum_{a,b} |g_a - g_b|^2.$$

Consequently $q(V_i; y, x) = w(y,x)p(y,x)$, with $p(y,x) \in \mathrm{SU}(2)$, $|w-1| \le r_i^2$ and $L^{-3}\sum_x w(y,x)^2 = 1$. The same contour rearrangement, and the formula for the admitted gauge block average on the double block of a bond, give

$$|p(y,c_-)V_i(c) - p(y,c_+)| \le c_T\bar\omega_i,$$
$$|V_{i+1}(c)q(V_i; c_+, z') - q(V_i; c_-, z)V_i([z,z'])| \le c_T\bar\omega_i, \qquad c_T = 64L^2.$$

For the second inequality the two paths have rearranged step sequences of length at most $(5L-3)/2$; their difference is at most $e^{25L^2\bar\omega_i/8} - 1$, and $\bar\omega_i \le (200L^2)^{-1}$ bounds the resulting exponentials linearly. These comparisons use only the block or double block in question, including bonds crossing the faces of B.

Let $Q_i = Q(V_{i-1})\cdots Q(V_0)$ and $\mathsf{P}_i = Q_i^* Q_i$. These are nested orthogonal projections, so $1 - P_B = \sum_{i<n}(\mathsf{P}_i - \mathsf{P}_{i+1})$ orthogonally. For $u = Q_i\varphi$ on one block put $g(z) = p(y,z)u(z)$ and $\tilde w = L^{-3/2}w(y,\cdot)$. Since $\|\tilde w\| = 1$,

$$\|g\|^2 - |\langle \tilde w, g\rangle|^2 \le 2\|g - \bar g\|^2 + 2r_i^4\|g\|^2.$$

The cube Poincaré inequality and the first transport comparison therefore yield

$$\|(\mathsf{P}_i - \mathsf{P}_{i+1})\varphi\|^2 \le \tfrac{1}{2}L^2\|\nabla_{V_i} Q_i\varphi\|^2 + (3L^2c_T^2\bar\omega_i^2 + 2r_i^4)\|\varphi\|^2.$$

The second transport comparison, Cauchy–Schwarz along the straight paths of length L, and the fact that each fine bond belongs to exactly L of these paths give, for every $\eta_i > 0$,

$$\|\nabla_{V_{i+1}} Q(V_i)u\|^2 \le (1+\eta_i)L^2\|\nabla_{V_i} u\|^2 + 6(1+\eta_i^{-1})c_T^2\bar\omega_i^2\|u\|^2.$$

Take $\eta_i = L^{-(n-i)}$. The product $\prod_{i<n}(1+\eta_i)$ is at most $e^{1/(L-1)}$, and iteration gives

$$\|\nabla_{V_i} Q_i\varphi\|^2 \le e^{1/(L-1)}L^{2i}\|\nabla_U\varphi\|^2 + 30c_T^2\epsilon^2 L^{3(i-n)-3}\|\varphi\|^2.$$

Sum over $i < n$. The gradient sum is at most $L^{2n}/(L^2-1)$ and the remaining geometric sums are bounded by ϵ_P. This proves the assertion. □

Lemma 6.2 (Localized ellipticity)**.** There is $c_e > 0$, depending only on the chamber operator, such that $\|D[1]g\|^2 \ge c_e\mathcal{E}_1(g)$ for finitely supported g. Let $\mathsf{s} \ge 1$ have bond differences at most R^{-1}, with $R \ge 6R'$ and $R' \ge 8$. Suppose the cube of side $2R'\mathsf{s}(x)$ around every x in the support of g has plaquette bound $\epsilon'\mathsf{s}(x)^{-2}$. Then

$$\|D[U]g\|^2 \ge \frac{c_e}{16}\mathcal{E}_U(g) - C_{\mathrm{pu}}(1+c_e)c_{\mathrm{hop}}^2(R'^{-2} + R'^2\epsilon'^2)\|\mathsf{s}^{-1}g\|^2.$$

■

Proof. The free chamber symbol has heavy block of real part at least λ_ρ, off-diagonal light–heavy blocks of size $O(|p|)$, and light Schur complement $i\gamma(p)+O(|p|^2)$. The two block equations therefore bound $|p|\,|\iota^*\hat g|+|\Pi_1\hat g|$ by a constant times $|\hat D(p)\hat g|$ near zero. Away from zero the coercive real part bounds the inverse uniformly. Fourier transformation gives the stated free inequality.

Choose the partition of unity on the covering lattice with

$$\rho(c)=R'\mathsf{s}(c)/8,\quad \beta_c(x)=\beta((x-c)/\rho(c)),\quad Z=\sum_c\beta_c^2,\quad \chi_c=\beta_c Z^{-1/2},$$

where β is one on the half unit cube, vanishes outside the unit cube and has Lipschitz constant two. Slow variation of s gives

$$\sum_c\chi_c^2=1,\qquad \sum_c|\chi_c(x')-\chi_c(x)|^2\le C(R'\mathsf{s}(x))^{-2}.$$

Indeed, the centres with nonzero $\beta_c(x)$ lie within $2R'\mathsf{s}(x)/15$ of x, while those within $R'\mathsf{s}(x)/17$ give $\beta_c(x)=1$; these two counts and the difference formula for $\beta_cZ^{-1/2}$ give the bound. The family is covariant under the periods, so it can be summed over centres modulo the periods even if a localizing cube exceeds the torus side.

Only wall-zero hops change the site, and hence $\|[D,\chi]f\|\le c_{\mathrm{hop}}\||\nabla\chi|^\sharp f\|$. Thus

$$\|Dg\|^2\ge\tfrac12\sum_c\|D\chi_cg\|^2-Cc_{\mathrm{hop}}^2R'^{-2}\|\mathsf{s}^{-1}g\|^2.$$

On the support cube of χ_c, axial gauge represents each link as a product of at most $2R'\mathsf{s}(c)$ plaquette boundaries, so $|U^u-1|\le 4R'\epsilon'\mathsf{s}(c)^{-1}$. Apply the free inequality to $u\chi_cg$, continued by zero, and compare $D[U^u]$ with $D[1]$. Comparing the corresponding gradient energies and summing the partition gives

$$\sum_c\mathcal{E}_U(\chi_cg)\ge\tfrac12\mathcal{E}_U(g)-CR'^{-2}\|\mathsf{s}^{-1}g\|^2.$$

The two comparisons each use $\|a+b\|^2\ge\frac12\|a\|^2-\|b\|^2$. Together with the first localization inequality they give the factor $c_e/16$ and the displayed error. □

Proposition 6.3 (Weighted energy and fluctuation accretivity)**.** There are $R_0,C_S<\infty$ and $\epsilon_*>0$, depending only on L,b_F and the fixed chamber operator, such that if $R\ge R_0$ and $0<\epsilon\le\epsilon_*$ in the stated separation and regularity conditions, then for $1\le i\le j+1$,

$$\|\mathsf{s}_i^{-1/2}f\|\le C_S\|\mathsf{s}_i^{1/2}\mathcal{K}_if\|,\qquad \|\mathsf{s}_i^{-1/2}\mathcal{K}_i^{-1}\mathsf{s}_i^{-1/2}\|\le C_S.$$

The same holds for $\mathcal{K}_i^*$. Consequently $\operatorname{Re}\Gamma_j\ge 2/C_D$ for a uniform C_D. Every compression of Γ_j has an inverse with exponentially decaying kernel in the units of level j. ■

Proof. Fix i, suppress its subscript and continue f by zero outside Ω_1. Write $N=\|\mathsf{s}^{-1/2}f\|$, $E=\|\mathsf{s}^{1/2}\mathcal{K}f\|$ and $g=\mathsf{s}^{1/2}f$. The real part gives

$$\langle f,\Delta f\rangle+\langle f,\mathsf{W}f\rangle+m_0\|f\|^2\le NE.$$

The penalty blocks have $\|\mathsf{s}^{1/2}\mathsf{W}f\|^2\le\theta\langle f,\mathsf{W}f\rangle$. The commutator with $\mathsf{s}^{1/2}$ costs at most $2c_{\mathrm{hop}}R^{-1}N$. Across the boundary of Ω_1 the scale is at most $1+R^{-1}$ and the real part pays for the crossing differences, without a regularity condition on the exterior bonds. These facts give

$$\|Dg\|^2\le 3E^2+(3\theta+C_{\mathrm{bh}})NE+12c_{\mathrm{hop}}^2R^{-2}N^2,\qquad C_{\mathrm{bh}}=12(1+r)^2/r=54.$$

The constants are independent of m_0: accretivity permits dropping the positive mass in this inequality.

Let $d_1=2R'L$ and choose $\zeta=0$ within d_1 of Ω_1^c, $\zeta=1$ beyond $2d_1$ and $|\nabla\zeta|\le d_1^{-1}$. Separation puts this boundary zone in X_1, where $\mathsf{L}=L$. In the zone the unweighted real part already controls

the gradient. On the support of ζ, the regularity condition allows Lem. 6.2 with $\epsilon' = L^2\epsilon_*$. It follows that the weighted gradient $Y = \sum_b \mathsf{s}(b_-)|\nabla_U \iota^* f(b)|^2$, with both orientations included in this sum, satisfies

$$Y \le \frac{64}{c_e}\|Dg\|^2 + C_0(R^{-2} + R'^{-2} + R'^2 L^4 \epsilon_*^2)N^2 + 2Lc_G^{-1}NE.$$

Here $c_G > 0$ is the coercivity constant in the real part, and C_0 depends on that constant, c_e and c_{hop}. Apply Lem. 6.1 block by block and use the penalty on the averaged component and the heavy-component gap. This gives

$$N^2 \le (2/\lambda_\rho + 2L/\theta)NE + L^2 c_P Y + L\epsilon_P N^2.$$

Combining the last three inequalities yields $(1-\vartheta)N^2 \le AE^2 + BNE$. Choose R' first so its R'^{-2} error is at most $1/8$; then impose the separation floor on R_0 so its R^{-2} error is at most $1/8$; finally decrease ϵ_* so its $R'^2L^4\epsilon_*^2$ and $L\epsilon_P$ errors are each at most $1/8$. Thus $\vartheta \le 1/2$ and $N \le (2B + \sqrt{2A})E$. The adjoint replaces γ by $-\gamma$ and has the same proof.

The closed Schur formula is

$$\Gamma_j^{-1} = c_j^{-1} + c_j^{-1}\theta_j T_A \mathcal{K}_{j+1}^{-1} T_A^* \theta_j c_j^{-1}, \qquad c_j = \theta_j + \theta q_{j+1}^* q_{j+1}.$$

It gives $\|\Gamma_j^{-1}\| \le 2/\theta + LC_S$. At $j = 0$ the all-field block Poincaré inequality gives $\operatorname{Re}\Gamma_0 \ge m_0 + c_{\mathfrak{B}}$. The disc inequality then gives

$$\operatorname{Re}\Gamma_j \ge \{(c_{\mathrm{tw}} + \theta)\|\Gamma_j^{-1}\|^2\}^{-1}.$$

Choose C_D to make this at least $2/C_D$.

For the decay, conjugate the fine operator by e^η, with $|\eta(x) - \eta(x')| \le \delta/\max\{\mathsf{s}(x), \mathsf{s}(x')\}$. The weighted norm of its change is at most $3e(c_{\mathrm{hop}} + \theta)L\delta$. The Neumann series therefore preserves the weighted inverse bound for a fixed small $\delta > 0$. Lift a Lipschitz weight on level j to the fine lattice by $\tilde\eta(x) = \min_y\{\eta(y) + \alpha L^{-j}|x - y|_1\}$. Across a block of level j it differs from its centre value by at most $3\alpha/2$; the Schur formula consequently gives $\|e^\eta \Gamma_j e^{-\eta} - \Gamma_j\| \le C_\Gamma \alpha$. Choose $C_\Gamma \alpha \le 1/C_D$. Compression commutes with the weights and preserves the real-part floor. The inverse of every weighted compression has norm at most C_D; taking $\eta(y) = \alpha|y - y'|_1$ proves its kernel bound. □

6.1. **Complex fields, sources, and local kernels.** For the complex class at depth j, give sites outside Ω_1 depth zero and sites in X_n^{j+1} depth n. For a bare bond b, let $n(b)$ be the smaller of its endpoint depths. We set $\mathsf{U}(b) = e^{iX(b)}U(b)$ on positive bonds, $\mathsf{U}(\bar b) = \mathsf{U}(b)^{-1}$, with $X(b) \in \mathfrak{sl}(2,\mathbb{C})$, and require $|X(b)| \le \alpha L^{-n(b)}$.

Lemma 6.4 (Complete fermion averages at complex fields)**.** There are $a_Q, \sigma_Q > 0$ and $C_Q < \infty$, depending only on L, with the following property. Let $1 \le n \le K$ and let $B = B^n(y)$ be a block of side L^n, let U be real on its inner bonds, and suppose its plaquettes satisfy $|U_p - 1| \le aL^{-2n}$, $a \le a_Q$. Let $\mathsf{U} = e^{iX}U$ there, with $L^n \sup|X| \le \sigma \le \sigma_Q$. Put $V_i = \mathcal{M}^i U$, $V_i^{\mathrm{c}} = \mathcal{M}^i \mathsf{U}$ for $0 \le i < n$. The normalized one-step kernels and their row twins are holomorphic on this common domain, with

$$q_i^{\mathrm{c}} = n(V_i^{\mathrm{c}})^{-1/2} S(V_i^{\mathrm{c}}), \qquad q_i^{+} = n(V_i^{\mathrm{c}})^{-1/2}[(\operatorname{tr} S(V_i^{\mathrm{c}}))1 - S(V_i^{\mathrm{c}})].$$

Their complete products $q_n^{\mathrm{c}}(y,x)$, along the unique nested chain from x to y, and the reversed products $(q_n^{\mathrm{c}})^+(x,y)$ satisfy

$$\sup_{x\in B}|q_n^{\mathrm{c}}(y,x) - q_n(U;y,x)| + \sup_{x\in B}|(q_n^{\mathrm{c}})^+(x,y) - q_n(U;y,x)^*| \le C_Q\sigma.$$

For the rows normalized by $L^{-3n/2}$, the same comparison holds in operator norm, and $Q_n^{\mathrm{c}}(Q_n^{\mathrm{c}})^+ = 1$. Consequently the chamber average $R_n^{\mathrm{c}} = Q_n^{\mathrm{c}}\iota^*$ has $R_n^{\mathrm{c}}(R_n^{\mathrm{c}})^+ = 1$, with $(R_n^{\mathrm{c}})^+ = \iota(Q_n^{\mathrm{c}})^+$. The kernels read only the bonds of B, preserve the prescribed boundary signs, and have the gauge and lattice covariance of the real averages.

For a regular history with $\epsilon \le a_Q$, $\alpha \le \sigma_Q$ and $C_Q\alpha \le 1/32$, the transported comparison holds as well. If $m \le n$, $c = \langle y, y' \rangle$ is a level-m bond with both endpoints in Ω_n, $x \in B^m(y)$ and $x' = x + L^m\hat{c}$, then both the real and complex transported comparisons hold:

$$V_m(c)q_m(y', x') = q_m(y, x)U([x, x']) + E, \qquad |E| \le c_T\epsilon L^{2(m-n)},$$

$$V_m^{\mathrm{c}}(c)q_m^{\mathrm{c}}(y', x') = q_m^{\mathrm{c}}(y, x)\mathsf{U}([x, x']) + E^{\mathrm{c}}, \qquad |E^{\mathrm{c}}| \le c_T\epsilon L^{2(m-n)} + C_Q\alpha L^{m-n+1}.$$

This includes bonds crossing the faces of the level-n blocks. At a bond from Ω_n to its adjacent lower layer, with $m \le n-1$, the same statement holds with $n-1$ in place of n. ∎

Proof. We first estimate the iterated gauge fields and then the products defining the fermion averages. By [3, eq. 4.cdxxx], on the double block, after reducing fixed radii, the iterates are holomorphic and

$$V_i^{\mathrm{c}} = e^{iX_i}V_i, \quad \xi_i := \sup|X_i| \le C_{\mathrm{av}}\sigma L^{i-n}, \qquad |(V_i)_p - 1| \le \omega_i := 2aL^{2(i-n)}.$$

Write $c_{\mathrm{ia}}, c_\sharp$ for its curvature and perturbation radii, and choose

$$a_Q = \min\{\varepsilon_\flat, c_{\mathrm{ia}}/2, (400L^2)^{-1}\}.$$

To use that double-block statement on B, reflect its data into the neighbouring block, put $U = 1$ and $X = 0$ on the crossing bonds, and observe that every crossing plaquette is one. The average in B is independent of this extension by [3, Lem. 3.41].

The contour rearrangement estimate gives $r_i = e^{9L^2\omega_i/8} - 1$. The real quaternion identity used in Lem. 6.1 gives

$$n(V_i) \ge 1 - r_i^2/2 \ge 1/2, \qquad |q(V_i)| \le 1 + r_i^2.$$

A one-step contour has at most $3(L-1)/2$ bonds. Telescoping its transports gives $|S(V_i^{\mathrm{c}}) - S(V_i)| \le CL\xi_i$. The determinant formula for two-by-two matrices consequently gives $|n(V_i^{\mathrm{c}}) - n(V_i)| \le C_0L\xi_i$, where $C_0 \ge 1$ also bounds the square-root and kernel changes on $\{z : |z - r| \le 1/4,\ 1/2 \le r \le 1\}$. This constant depends only on L. Take C_Q larger than $32C_0LC_{\mathrm{av}}$ and than the fixed constants in the transport comparison below, and define

$$\sigma_Q = \min\{c_\sharp/2, (60LC_{\mathrm{av}})^{-1}, (4C_0LC_{\mathrm{av}})^{-1}, (32C_Q)^{-1}\}.$$

Thus $3L\xi_i \le 1/20$ and the determinant change is at most $1/4$. All these values lie in the right half-plane, so use its single square-root branch, positive at real fields. The displayed formulas for q_i^{c} and q_i^+ are holomorphic and differ from their real values by at most $CL\xi_i$.

For every two-by-two matrix S, $S[(\operatorname{tr} S)1 - S] = (\det S)1$. Thus

$$L^{-3}\sum_{x \in B(z)} q_i^{\mathrm{c}}(z, x)q_i^+(x, z) = 1.$$

Iterating this identity gives the complete normalization. Moreover

$$\sum_{i<n} r_i^2 \le 13a^2, \qquad \sum_{i<n} L\xi_i \le C\sigma,$$

with constants independent of n. The products of all factor norms are bounded by $\exp\sum_{i<n}(r_i^2 + CL\xi_i) \le 2$. Telescoping the complete product now proves the asserted $C_Q\sigma$ comparison; the reversed row product has the same bound. The normalized operator comparison follows by Cauchy–Schwarz over the L^{3n} sites. Locality, covariance and boundary signs pass through these same formulas.

For the history assertion, inner bare bonds of Ω_n have $n(b) \ge n$. Apply the block result at level m with $a = \epsilon L^{2(m-n)}$ and $\sigma = \alpha L^{m-n}$. Here is the iteration of the real one-step comparison in the proof of Lem. 6.1. Writing E_m for its complete defect, product substitution gives

$$|E_m| \le (1 + r_{m-1}^2)L\sup|E_{m-1}| + 2c_T\omega_{m-1}.$$

The real product bounds and $\sum_{i<m} L^{m-1-i}\omega_i \le 2\epsilon L^{2(m-n)}/[L(L-1)]$ give $|E_m| \le c_T\epsilon L^{2(m-n)}$ with the same c_T, since the real product is at most $21/20$ and $L \ge 13$. This iteration uses all the bonds of

the double block, including those crossing faces. Write q_m for the complete real kernels. Subtracting its transport identity from the complex one gives

$$E^{\mathrm{c}} - E = (V_m^{\mathrm{c}}(c) - V_m(c))q_m^{\mathrm{c}}(y', x') + V_m(c)(q_m^{\mathrm{c}}(y', x') - q_m(y', x')) \\ - (q_m^{\mathrm{c}}(y, x) - q_m(y, x))\mathsf{U}([x, x']) - q_m(y, x)(\mathsf{U}([x, x']) - U([x, x'])).$$

The pure double-block bound also estimates $V_m^{\mathrm{c}}(c)$ when c crosses a block face or $m = n$. The straight path has L^m bonds, so its complex transport changes by at most $C\alpha L^{m-n}$. Each of the four terms above is therefore bounded by $C_Q\alpha L^{m-n}$, with fixed product bounds. The sum is at most $5C_Q\alpha L^{m-n}$, and $L \geq 13$ gives the stated weaker bound with L^{m-n+1}. For a bond into the lower layer, both blocks and their double block lie in Ω_{n-1}; the same argument uses that region. No product of unrestricted one-step Lipschitz constants is used. □

Reduce ϵ_*, if necessary, so that $\epsilon_* \leq a_Q/2$.

Lemma 6.5 (Stability)**.** The preceding inverse estimates persist jointly on a sufficiently small complex gauge class and at every complex matrix source $m_0 - a_K\hat{J}$, where J acts on spin, the internal doublet and flavour and is the identity on colour, satisfying

$$\|J\|_{[j]} := \sup_x \|L^{j-K}J(x)\| \leq \varrho_\sharp.$$

The radii $\alpha_1^* > 0$ and $\varrho_\sharp > 0$ depend on the earlier operator constants, not on m_0, the volume, the cutoff or N_f. In particular no condition on the real part of the source mass is imposed. Every fluctuation compression is invertible and holomorphic there, with the same exponential kernel estimate after enlarging its constant by a fixed factor. ■

Proof. For $\widetilde{\mathcal{K}} = \mathcal{K} + E$, the weighted inverse identity and a Neumann series apply if

$$\|\mathsf{s}^{1/2}E\mathsf{s}^{1/2}\| \leq (2C_S)^{-1}.$$

They give $\|\mathsf{s}^{-1/2}\widetilde{\mathcal{K}}^{-1}\mathsf{s}^{-1/2}\| \leq 2C_S$. A twist of rate δ changes the fine operator by at most $3e(c_{\mathrm{hop}} + \theta)L\delta$ in this norm. A complex gauge change costs at most $8(1 + \theta)(c_{\mathrm{hop}} + C_Q)\alpha$; the comparison of Lem. 6.4 is used for the block penalty. The source costs at most $L\|J\|_{[j]}$. Thus one can first choose

$$\delta_* \leq [24eC_S(c_{\mathrm{hop}} + \theta)L]^{-1}, \qquad \bar{\alpha} \leq \min\{\sigma_Q, (32C_Q)^{-1}, [64C_S(1 + \theta)(c_{\mathrm{hop}} + C_Q)]^{-1}\},$$

and then lower the source radius. These choices apply on the local mirror tori as well as on the original history.

Put $C_{\mathrm{inv}} = \max\{2/\theta + 2LC_S, 2/c_{\mathrm{fl}}\}$. The Schur formula bounds the fluctuation inverse by C_{inv}. Choose C_D to dominate $2(c_{\mathrm{tw}} + \theta)C_{\mathrm{inv}}^2$ and $4/c_{\mathrm{fl}}$, choose the fluctuation twist rate $0 < \alpha_D \leq \delta_*/4$ after C_D, and increase R_0 until

$$\alpha_D R_0 \geq \max\{2\log L, 32\}, \qquad R_0 \geq 1 + 16/\delta_*.$$

Increasing R_0 does not increase C_S. The source radius may be taken below

$$\min\{(8L^2C_S)^{-1}, (4C_\Gamma'' C_D)^{-1}, c_{\mathrm{fl}}/4, [1 + 9C_S^2(c_{\mathrm{hop}} + \theta)^2]^{-1}\}, \qquad C_\Gamma'' = 16\theta^2 C_S^2.$$

Finally choose $\alpha_1^* \leq \bar{\alpha}$ so that the change of the fluctuation form has norm at most $(4C_D)^{-1}$. The real-part floor of the real form and these perturbation bounds give the inverse bounds for every compression. Inserting exponential weights gives their kernel bounds. In finite volume all entries are holomorphic matrix functions; the locally uniform bounds give the same assertion for the infinite-volume kernels. □

We record the free fact needed both in localization and in the mass calculation. For the balanced chamber operator, with $t = 3$ and $\rho_0 = 3/2$,

$$\iota^*\widehat{D}(p)\iota = i\sum_\mu \gamma_\mu \sin p_\mu + \tfrac{1}{2}\sum_\mu (1 - \cos p_\mu), \qquad \Pi_1 \operatorname{Re}\widehat{D}(p)\Pi_1 \geq \lambda_\rho \Pi_1.$$

The light–heavy couplings are $O(|p|)$. Indeed the balanced within-site hops cancel on a chamber-independent field, leaving only $(\rho_0 - 3\gamma(e))\nabla_e\varphi$; the same argument for D^* bounds both couplings. The heavy block has inverse norm at most λ_ρ^{-1}. Its Schur complement on the light space is therefore

$$m + i\gamma \cdot p + O(|p|^2),$$

uniformly for $m \geq 0$ in a bounded interval. Near zero its inverse differs from $(m + i\gamma \cdot p)^{-1}$ by at most $C|p|^2/(m^2 + |p|^2)$. To see this, write the complement as $m + i\gamma \cdot p + E$, $\|E\| \leq C|p|^2$, and apply the resolvent identity when $C|p|^2 \leq \frac{1}{2}(m^2 + |p|^2)^{1/2}$. Away from zero, the real part has a positive gap: equality in $\sum_s \rho_s(1 - \widehat{R}_s(p)) \geq 0$ forces the vector to be constant in the chamber variable and $e^{ip\cdot e} = 1$ for all coordinate directions. Compactness supplies the remaining bound. This proves the free ellipticity used above and excludes additional zero modes. Differentiating the Schur formula gives, for each multi-index ζ,

$$\|\partial_p^\zeta(\widehat{D}(p) + m)^{-1}\| \leq C_\zeta(m + |p|)^{-1-|\zeta|}.$$

The singular light contribution to the heavy output has one additional factor $|p|$, since the coupling is a difference. The heavy-heavy inverse also has a regular, uniformly bounded part.

Lemma 6.6 (Local regularity)**.** There are $\epsilon_{\mathrm{lr}}, C_{\mathrm{lr}} > 0$ with the following property. For a site x_0 put $Q_r = \{x : |x - x_0|_\infty \leq r\}$. Let $l \geq 4$, let U be real on the bonds meeting Q_{2l}, and suppose the curvature at every plaquette meeting Q_{2l} is at most $\epsilon_{\mathrm{lr}} l^{-2}$. Let F, s be defined on the places over Q_{2l} and its neighbours, with $(D[U] + \mu)F = s$ over Q_{2l} and $0 \leq \mu \leq 1$. Write $\|F\|_2 = \|F\|_{\ell^2(Q_{2l})}$ and $\|s\|_\infty = \sup_{Q_{2l}} |s|$. At the centre,

$$|F| \leq C_{\mathrm{lr}}(l^{-3/2}\|F\|_2 + l\|s\|_\infty),$$

$$|\nabla_U \iota^* F| + |\Pi_1 F| \leq C_{\mathrm{lr}}\{l^{-5/2}\|F\|_2 + (1 + \log l)\|s\|_\infty\}.$$

Put $\varphi = \iota^* F$. For $x \in Q_{l/4}$, $x' = x + \rho e$, where e is a coordinate unit vector and $1 \leq \rho \leq l/4$, the straight-path transport satisfies

$$|U([x, x'])\varphi(x') - \varphi(x)| \leq C_{\mathrm{lr}}\rho \left\{ l^{-5/2}\|F\|_2 + (1 + \log(l/\rho))\|s\|_\infty + \epsilon_{\mathrm{lr}} l^{-1} \sup_{Q_{3l/2}} |F| \right\}.$$

■

Proof. The free symbol estimates just proved, a dyadic partition around $p = 0$, and integration by parts on each dyadic annulus give

$$|G_\mu(x)| \leq C(1 + |x|)^{-2} e^{-\mu|x|/C}, \qquad |\nabla G_\mu(x)| + |\Pi_1 G_\mu(x)| \leq C(1 + |x|)^{-3} e^{-\mu|x|/C}.$$

For completeness, on an annulus $|p| \asymp 2^{-n}$ the symbol has size $O(2^n)$ and its qth derivative size $O(2^{(q+1)n})$. Its Fourier coefficient is bounded by $C_q 2^{-2n}(1 + 2^{-n}|x|)^{-q}$. Summing with $q > 3$ gives the first bound. A difference or a heavy projection removes 2^n, giving the second. A contour shift of size $c\mu$ gives the exponential factor. Subtracting the kernels at distance ρ and splitting at $|x| = \rho$ gives the asserted logarithmic modulus.

Choose a rooted axial gauge on Q_{2l}; summing plaquette curvatures over its coordinate surfaces gives $|U^u(b) - 1| \leq C\epsilon_{\mathrm{lr}} l^{-1}$, and at the root the sharper bound proportional to its distance from the root. For a cutoff χ, equal to one on Q_l and with $|\nabla\chi| \leq C/l$, extend by zero and write

$$\chi u F = G_\mu\{\chi u s + [D[1], \chi]uF + \chi(D[1] - D[U^u])uF\}.$$

The source term uses respectively $\sum_{|x|\leq Cl}(1 + |x|)^{-2} \leq Cl$ and $\sum_{|x|\leq Cl}(1 + |x|)^{-3} \leq C(1 + \log l)$. The commutator is on the cutoff shell; Cauchy–Schwarz with the free kernel gives $Cl^{-3/2}\|F\|_2$, or $Cl^{-5/2}\|F\|_2$ after a difference or heavy projection. The field term is absorbed in the weighted local supremum $\sup_x \operatorname{dist}(x, \partial Q_{2l})^{3/2}|F(x)|$ by taking ϵ_{lr} small. For the differentiated estimate use the

axial gauge rooted at the point being estimated; the additional distance in $U^u - 1$ gives precisely the same absorption. The kernel difference calculation gives the modulus of continuity. □

Proposition 6.7 (Local walks and background maps)**.** Choose a power $M_w \geq L$ of L after the operator and local-regularity constants, and assume $R \geq LM_w$. There are local walks $\mathcal{W}_\omega$, $\omega = (\square_0, \dots, \square_n)$, such that

$$\mathcal{K}_i^{-1} = \sum_\omega \mathcal{W}_\omega, \quad \|\mathsf{s}^{-1/2}\mathcal{W}_\omega \mathsf{s}^{-1/2}\| \leq 2LC_S q_w^n, \quad q_w = C_q/M_w,$$

and their kernels and gauge/source dependence are contained in $X_\omega = \bigcup_l W_{\square_l}$. With $|g|_{(a)} = \sup_x \mathsf{s}(x)^a |g(x)|$,

$$|\mathcal{W}_\omega g|_{(1)} \leq C_{\sup} q_{\sup}^n |1_{W_{\square_n}} g|_{(2)}, \qquad q_{\sup} = C_q^{\sup}/M_w.$$

The constants are chosen so that $C_{\mathrm{nb}} \max\{q_w, q_{\sup}\} \leq e^{-3}$. The same bounds hold for the row twins and jointly on the complex gauge class and the source ball. Every covariance obtained by switching off walk cubes has a uniform Gram constant Γ_F on the closed decoupling polydisc, factorizes over the remaining components, and agrees with the original covariance when all parameters are one. For the Gaussian integrations we take the mean of the right and left localized inverses. This covariance, denoted by S_j^{sym} below, also preserves conjugation and charge conjugation at every value of the decoupling parameters.

For the background map to level n in a history ending at k, write $\mathcal{H}_{(n)} = \mathcal{H}_{(n)}^{\mathrm{loc}} + \sum_\omega \mathcal{H}_{(n);\omega}$. Its local part reads one block and has norm at most $2L^{-2(n_P - n)}$ on a column of level n_P. The walk part obeys

$$\sum_P |\mathcal{H}_{(n);\omega}(x; P)| \leq C_H \frac{L^n}{\mathsf{s}_k(x)} q_{\sup}^{|\omega|}.$$

For neighbouring level-n sites x, x', let $U_n(x, x')$ be their level-n link transporter. Then

$$\sum_P |\mathcal{H}_{(n);\omega}(x; P) - U_n(x, x')\mathcal{H}_{(n);\omega}(x'; P)| \leq C_H \Big(\frac{L^n}{\mathsf{s}_k(x)}\Big)^2 \Big(1 + \log_+ \frac{\mathsf{s}_k(x)}{L^n}\Big) q_{\sup}^{|\omega|}.$$

At level zero the heavy part has this latter gain. In particular, for every $0 < \alpha < 1$ the second bound is at most $C(\alpha)(L^n/\mathsf{s}_k(x))^{1+\alpha} q_{\sup}^{|\omega|}$. On the history without large fields $\mathsf{s}_k = L^k$. All these estimates are uniform in k, K, the volume and N_f. ■

Proof. At the first step $\mathcal{K}_1 = \Gamma_0$ and $1 \leq \mathsf{s}_1 \leq L$. The preceding all-field bound also admits a matrix source of norm at most $\varrho_\sharp \leq c_{\mathrm{fl}}/4$: the twisted real part loses at most $c_{\mathrm{fl}}/4$, so the kernel prefactor becomes $4/c_{\mathrm{fl}}$. For a weight of rate at most $\min\{\delta_*, \delta_0\}/8$, the twisted kernel is bounded by $(4/c_{\mathrm{fl}}) e^{-7\delta_0 |x_v - x_{v'}|_1/8}$. Summing over the $\nu_0 = 48$ chambers at each site gives

$$|e^\eta \mathcal{K}_1^{-1} e^{-\eta} g|_{(1)} \leq \frac{4L\nu_0}{c_{\mathrm{fl}}} \left(\coth \frac{7\delta_0}{16}\right)^3 |g|_{(2)}.$$

Include this number in $C_{\sup}$ before choosing the walk constants. It applies to the full and mirror inverses at every real configuration. The regular complex domains are treated by the perturbation bounds below.

Partition the regular layers into cubes $\square$ of side $M_w L^{n(\square)}$ and choose $\sum h_\square^2 = 1$, with $h_\square$ supported in the concentric cube $W_\square$ of triple side and $|\nabla h_\square| \leq CL/(M_w L^{n(\square)})$. Separation allows the history and the configuration on $W_\square$ to be reflected to a mirror torus. The reflected layers retain separation and regularity and their smoothed scales differ by at most L; hence its local inverse $\mathrm{Sl}_\square$ has the weighted bound $2LC_S$. The right parametrix $\sum h_\square \mathrm{Sl}_\square h_\square$ has error made of $[\mathcal{K}_i, h_\square]\mathrm{Sl}_\square h_\square$. The finite-range hops and the block penalties give

$$\|\mathsf{s}^{1/2}[\mathcal{K}_i, h_\square]\mathsf{s}^{1/2}\| \leq C/M_w.$$

Indeed the penalty kernel has row sum at most two and its commutator costs its row sum times $\operatorname{osc}_B h_\square \le 3L^{n(B)}|\nabla h_\square|$. The error norm is below one after M_w is chosen. Its Neumann series gives exactly the stated walks and the factor q_w per step. Only consecutive overlapping cubes occur, with at most C_{nb} continuations.

Here is the supremum estimate needed for the legs. Split g into the blocks B of the layers. The twisted energy estimate gives

$$\|\mathsf{s}^{-1/2}e^{\delta_* d(\cdot,y_B)}\mathcal{K}_i^{-1}1_B g\| \le CL^2|g|_{(2)}.$$

At $l=\varepsilon_0\mathsf{s}(x)\ge 4$, keep the real operator $D[U]+m_0$ in the local regularity estimate and write

$$(D[U]+m_0)F = 1_B g - \mathsf{W}_i F + YF, \qquad Y = a_K\hat{J} - (D[\mathsf{U}]-D[U]).$$

On this cube the source and complex-hop bounds give $\|Y\| \le (L\varrho_\sharp + 3c_{\mathrm{hop}}\alpha_1^*)/\mathsf{s}(x)$. In the weighted supremum $\sup_x \mathsf{s}(x)e^{\delta_* d(x,y_B)/2}|F(x)|$, the penalty, mass and complex-field terms have coefficients at most $1/18$, $5/32$ and $1/96$, respectively; their sum is at most $1/4$. Choose ε_0 first and then lower ϵ_* and α_1^* to secure these inequalities. Where $l<4$, use the weighted energy bound directly. It follows that

$$|\mathcal{K}_i^{-1}1_B g(x)| \le C\mathsf{s}(x)^{-1}e^{-\delta_* d(x,y_B)/2}|g|_{(2)}.$$

The number of layer places within distance t is bounded by $C(1+te^{t/R})^3$. The floor $R\ge 1+16/\delta_*$ and the distance R between successive complete layers make its sum against the displayed exponential finite, uniformly in the history. This proves the supremum bound for the full inverse and for the local mirror inverse, independently of M_w. Moreover

$$\|[\mathcal{K}_i, h_\square]\|_{(2)\leftarrow(1)} \le C(c_{\mathrm{hop}}+\theta)L^2/M_w.$$

Multiplying these bounds along a walk gives $q_{\sup}$ per step. For $\omega=(\square_0,\dots,\square_m)$, write $\mathcal{W}_\omega g = h_{\square_0}F$, where $F=(\mathcal{K}^{\mathrm{mir}}_{\square_0})^{-1}g_1$. Here $g_1 = h_{\square_0}g$ if $m=0$, and $g_1 = -h_{\square_0}[\mathcal{K}_i, h_{\square_1}]F_1$ otherwise, where $\mathcal{W}_{(\square_1,\dots,\square_m)}g = h_{\square_1}F_1$. The preceding bounds give $|g_1|_{(2)}+|F|_{(1)} \le Cq_{\sup}^{|\omega|}G$, with $G=|1_{W_{\square_m}}g|_{(2)}$. Apply the real local estimate to F in the first cube. Put $s_*=\mathsf{s}_i(x)$ and $l=\varepsilon_0 s_*$ for a fixed small $\varepsilon_0>0$. The energy and supremum bounds just proved give

$$\|F\|_{\ell^2(Q_{2l})} \le Cq_{\sup}^{|\omega|}s_*^{1/2}G, \qquad \sup_{Q_{3l/2}}|F| \le Cq_{\sup}^{|\omega|}s_*^{-1}G.$$

In the real local equation the forcing is $g_1 - \mathsf{W}_iF + YF$; its supremum on Q_{2l} is at most $Cq_{\sup}^{|\omega|}s_*^{-2}G$. Thus all three terms in the modulus of Lem. 6.6, including the curvature term, are bounded by $Cq_{\sup}^{|\omega|}\rho s_*^{-2}(1+\log(s_*/\rho))G$. Multiplication by $h=h_{\square_0}$ preserves this bound: the extra term $(h(x')-h(x))U([x,x'])F(x')$ has norm at most $C\rho s_*^{-2}|F|_{(1)}$, by the gradient bound for h. For a level-n difference take $\rho=L^n$. The normalized average contributes L^n, giving the stated squared scale ratio and logarithm. If $4L^n > \varepsilon_0 s_*$, or $l<4$, use the size bound instead: $t=L^n/s_*$ is then bounded below by a fixed positive constant, and $t\le Ct^2$. The complete-average comparison pays the transport defect, including bonds crossing layer faces. The heavy-component estimate in Lem. 6.6, with the same forcing and energy bounds, gives

$$|\Pi_1\mathcal{W}_\omega g(x)| \le Cq_{\sup}^{|\omega|}\mathsf{s}_i(x)^{-2}(1+\log\mathsf{s}_i(x))G.$$

For $l<4$ this follows from the size bound, since $\mathsf{s}_i(x)$ is then bounded. All these estimates retain the same $q_{\sup}^{|\omega|}$; no general complex-source extension of the real lemma is used.

For the background maps take $i=k$. The background equations have the form

$$\psi = \mathcal{K}_k^{-1}S\Psi, \quad S\Psi = \sum_m (T_m^k)^+\theta_m\Psi_m - D_{A_0S_0}\Psi_{S_0}, \qquad \|S\|_{(2)\leftarrow(0)} \le C.$$

At level $n \geq 1$, eliminating the block equations gives

$$\mathcal{H}_{(n);\omega} = \left(q_n \cdots q_1 - \theta_n^{-1} \sum_{m>n} (q_m \cdots q_{n+1})^+ 1_{X_m^k} \theta_m T_m^k \right) \mathcal{W}_\omega S.$$

The leftover term is $\mathcal{H}^{\mathrm{loc}}_{(n)}$. In the per-chamber supremum norm, ι^* has norm $\sqrt{48}$: it is $48^{-1/2}$ times the sum over the chambers at a site. Thus each average is at most $2\sqrt{48}L^n$ times the supremum; $L^{m-1} \leq \mathsf{s} \leq L^m$ on its layer gives the size bound. Write $\psi_\omega = \mathcal{W}_\omega S\Psi$ and $\varphi_\omega = \iota^*\psi_\omega$. For neighbouring blocks the real complete-average defect is at most $c_T\epsilon_* L^{2(n-m)}$. On the complex class, Lem. 6.4 adds $C_Q\alpha_1^* L^{n-m+1}$. Multiplying by the normalized average factor L^n and the size bound $|\varphi_\omega| \leq 2\sqrt{48}|\psi_\omega|_{(1)}/\mathsf{s}_k(x)$ bounds this contribution by

$$C(c_T\epsilon_* L^2 + C_Q\alpha_1^* L^2) \left(\frac{L^n}{\mathsf{s}_k(x)} \right)^2 |\psi_\omega|_{(1)}.$$

For $\rho \leq \varepsilon_0 \mathsf{s}_k(x)/4$, the complex and real transports along a straight path of length ρ differ by at most $C\alpha_1^* L\rho/\mathsf{s}_k(x)$. At $\rho = L^n$ this difference, multiplied by the same size bound and average factor, has the same squared gain. Changing the level-n link transporter costs at most $CC_Q\alpha_1^* L\, L^n/\mathsf{s}_k(x)$ and hence also has this gain after multiplication by the background size. Together with the logarithmic difference bound in the first cube, these estimates give the stated bound, with the same walk factor, including across faces between layers. The local term has size $O(L^{-2(m-n)})$, so its possible jump has the same gain. At level zero apply the heavy-component bound above to $g = S\Psi$; the bound on S gives the asserted gain.

At zero source all kernels are trivial in flavour. The row sum of the norms of their matrix entries is bounded by $\nu' = 8$ times the norm on vector-valued bounded functions: expand a row vector in an orthonormal basis and choose each column vector with its maximizing phase. At a source use instead

$$\mathcal{K}(J)^{-1} = [1 - \mathcal{K}(0)^{-1} a_K \hat{J}]^{-1} \mathcal{K}(0)^{-1}$$

in the row norm $\|A\|_{\mathrm{row}} = \sup_x \mathsf{s}(x) \sum_y \|A(x,y)\| \mathsf{s}(y)^{-2}$. Since $\|a_K \hat{J}\|_{(2)\leftarrow(1)} \leq L\|J\|_{[j]}$, lower the source radius so the Neumann ratio is at most $1/2$. For a difference or heavy component of a background row, put the zero-source inverse in the first factor of the resolvent identity. Its gain is retained; only the remaining factors cost two. Thus the column sums have no extra factor N_f.

To file walks into polymers of side $M \geq 3M_w$, let c_0 satisfy $d(X \cup Y) \leq d(X) + d(Y) + c_0$ for meeting polymers and $d(Y) \leq c_0$ in a cube of side $2M$. Divide a walk into segments of at most $\lfloor M/(3M_w) \rfloor$ cubes. Each segment lies in a cube of side M. Consequently

$$d(Y_\omega) \leq c_0 + 6c_0(|\omega| + 1)M_w/M.$$

Put $q = \max\{q_w, q_{\mathrm{sup}}\}$. With $c_f M \geq C_{\mathrm{dc}}(\kappa + 2)$ and $c_f = (3LM_w)^{-1}$, choose C_{dc} so that $6c_0(\kappa+2)M_w/M \leq 1/8$. For each fixed first cube $\square$, counting continuations gives

$$\sum_{\substack{\omega:\square_0=\square \\ Y_\omega = Y}} q^{|\omega|} \leq 22 e^{(\kappa+2)c_0} e^{-(\kappa+2)d(Y)},$$

with the same bound for a fixed last cube. For a fixed place x of a polymer X, sum instead over walks whose first cube is one of the at most C_{nb} cubes for which $h_{\square_0}$ does not vanish on the layer block containing x. Then

$$\sum_\omega e^{|\omega| + \kappa[d(X\cup Y_\omega) - d(X)]_+} q^{|\omega|} \leq A_\kappa := 2C_{\mathrm{nb}} e^{(\kappa+2)c_0}.$$

The same bound holds with the admissible last cubes of x. The offset c_0 is essential for walks of no step.

For decoupling, let $Z_\circ$ be a union of fine cubes, enlarge it by D_{ex} to $Z_\circ^\sharp$, and put $\bar s(\Delta)=1$ on its big cubes and $\bar s(\Delta)=s(\Delta)$ elsewhere. Here $|s(\Delta)|\le e^{\bar\kappa}$. Weaken a walk by $\bar s(\omega)=\prod_{\Delta\cap X_\omega\neq\varnothing}\bar s(\Delta)$, each cube counted once. Define

$$\Gamma_j^{-1}(Z_\circ,s)=\chi_{Z_\circ}\left[c_j^{-1}+c_j^{-1}\theta_jT_A\Big(\sum_\omega \bar s(\omega)\mathcal{W}_\omega\Big)T_A^+\theta_jc_j^{-1}\right]\chi_{Z_\circ};$$

at $j=0$ use the restricted weakened fine inverse directly. A chain of $n+1$ cubes meets at most $8+CnM_w/M$ big cubes. If it starts and ends in $Z_\circ$ and leaves $Z_\circ^\sharp$, then $n+2\ge 2D_{\mathrm{ex}}/(3LM_w)$, in level-$j$ units. Indeed a walk cube has side at most $3LM_w$ in those units, and the walk goes outside the enlargement and returns. Therefore

$$|\bar s(\omega)|(C_{\mathrm{nb}}q_w)^n\le e^{8\bar\kappa}(e^{C\bar\kappa M_w/M}e^{-3})^n\le e^{8\bar\kappa}e^{-2n}.$$

The floor $c_f\min\{M,D_{\mathrm{ex}}\}\ge C_{\mathrm{dc}}\bar\kappa$, with C_{dc} sufficiently large, ensures both inequalities. In particular $D_{\mathrm{ex}}\ge 15LM_w\bar\kappa$ gives $n\ge 10\bar\kappa-2\ge 8\bar\kappa$, so the last bound is at most e^{-n} for walks which leave the enlargement. For the other walks every weakening factor is one, and the same bound follows from $C_{\mathrm{nb}}q_w\le e^{-3}$. The series converges uniformly on the larger polydisc $|s(\Delta)|\le e^{\bar\kappa+1}$. On the complex class, the block-average bounds and $C_Q\alpha_1^*\le 1/32$ give

$$\|c_j^{-1}\theta_jT_A\mathsf{s}^{1/2}\|^2,\ \|\mathsf{s}^{1/2}T_A^+\theta_jc_j^{-1}\|^2\le\tfrac54L,\qquad \|c_j^{-1}\|\le\tfrac65\theta^{-1}.$$

Schur's test over the first and last cubes bounds the weighted norm of the walk sum by

$$2LC_S\sum_{n\ge0}e^{-n}\le 16LC_S/5.$$

At $j=0$ its plain norm is at most $16/(5c_{\mathrm{fl}})$. Thus

$$\|\Gamma_j^{-1}(Z_\circ,s)\|\le\Gamma_F^2,\qquad \Gamma_F^2=2/\theta+16C_{\mathrm{nb}}L^2C_S+10C_{\mathrm{nb}}/c_{\mathrm{fl}}.$$

A walk contributing to the entry at (y,y') satisfies $3M_w(n+1)+2\ge|y-y'|$ in level-$(j+1)$ units. The endpoint offsets are absorbed by the geometric sum:

$$\sum_{\substack{n\ge0\\3M_w(n+1)+2\ge|y-y'|}}e^{-n}\le\frac{e^{1+2/(3M_w)}}{1-e^{-1}}e^{-Lc_f|y-y'|}\le 5e^{-c_f|y-y'|}.$$

The entry prefactor is at most $\frac52C_{\mathrm{nb}}L^2C_S$ for $j\ge1$ and $2C_{\mathrm{nb}}/c_{\mathrm{fl}}$ for $j=0$. The block term c_j^{-1} joins only places with $|y-y'|<1$; its bound is at most $(2/\theta)e^{-c_f|y-y'|}$. These estimates give entry decay with the same Γ_F^2. Only walks leaving the enlargement change when s changes from one, so their sum is at most $\Gamma_F^2e^{8\bar\kappa}e^{-c_fD_{\mathrm{ex}}}$. Turning a parameter to zero removes every walk crossing that cube. Connectedness of X_ω then makes the covariance a direct sum on the surviving components. At parameters one it is the original inverse. The left parametrix proves the row-twin statements. Local holomorphy and uniform convergence prove these assertions.

The right inverse alone need not preserve the real structure at intermediate parameters. To retain that structure, form the left parametrix as well. On functions supported where all neighbours and blocks remain inside $W_\square$, the full operator and the mirror operator agree. Thus the corresponding columns of $\mathrm{Sl}_\square\mathcal{K}_{j+1}$ are identity columns. Consequently

$$\Big(\sum_\square h_\square\mathrm{Sl}_\square h_\square\Big)\mathcal{K}_{j+1}=1-\mathfrak{N}^{\mathrm{L}},\qquad \mathfrak{N}^{\mathrm{L}}=-\sum_\square h_\square\mathrm{Sl}_\square[h_\square,\mathcal{K}_{j+1}].$$

Expanding its inverse gives

$$\mathcal{W}_\omega^{\mathrm{L}}=\Big[\prod_{i=n}^{1}(-h_{\square_i}\mathrm{Sl}_{\square_i}[h_{\square_i},\mathcal{K}_{j+1}])\Big]h_{\square_0}\mathrm{Sl}_{\square_0}h_{\square_0}.$$

The order in this product runs from n on the left to 1 on the right. These are the same factors as in a right walk, in reverse order. Their supports are still X_ω, and their weighted bounds are the same. The supremum and difference estimates start at the last cube instead of the first; the argument above therefore gives the same constants. Denote by $\Gamma_{j,\mathrm{L}}^{-1}(Z_\circ, s)$ the covariance obtained from these walks.

Write $\mathsf{U}^{\mathfrak{c}}(b) = (\mathsf{U}(b)^*)^{-1}$, $J^{\mathfrak{c}} = \gamma_4 J^\dagger \gamma_4$ and $s^{\mathfrak{c}}(\Delta) = \overline{s(\Delta)}$. Define

$$S_j^{\mathrm{sym}}(Z_\circ, s; \mathsf{U}, J) = \frac{1}{2}\left[\Gamma_j^{-1}(Z_\circ, s; \mathsf{U}, J) + \gamma_4 \Gamma_j^{-1}(Z_\circ, s^{\mathfrak{c}}; \mathsf{U}^{\mathfrak{c}}, J^{\mathfrak{c}})^* \gamma_4\right]. \tag{26}$$

Its second summand is $\Gamma_{j,\mathrm{L}}^{-1}$. Indeed, at a real configuration and a source fixed by the conjugation, $Y \mapsto \gamma_4 Y^* \gamma_4$ fixes the local operators and reverses their products. It exchanges T_A, T_A^+ and the two commutators, hence carries each right walk to its left walk. At a fixed real factor U, write $\mathsf{U} = e^{iX}U$. Both sides of this identity are holomorphic in (s, X, J) on the connected product of the parameter polydisc, the convex complex gauge class and the source ball. They agree on its real points, an open subset of a real form of this space; the holomorphic identity theorem proves equality throughout the domain and continuity gives it on the closed class.

The transpose identity for the chamber operator gives the analogous assertion for charge conjugation. Put $\mathcal{C} = (\sigma_2 \otimes 1_\tau) \otimes \epsilon \otimes 1_{N_f}$, with $\epsilon = i\sigma_2$ on colour; all transposes are in the standard tensor-product bases. Define the linear source transform by $\widehat{J^{\mathfrak{c}}} = \mathcal{C}\widehat{J}^{\mathsf{T}}\mathcal{C}^{-1}$. For physical sources in the span of $\mathfrak{B}$, which are scalar on spin, this reduces to transpose on the internal doublet and flavour. This differs from the antilinear $J^{\mathfrak{c}} = \gamma_4 J^\dagger \gamma_4$ above. On colour the transpose identity uses $\epsilon A^{\mathsf{T}} \epsilon^{-1} = (\operatorname{tr} A)1 - A$, which exchanges q and q^+. Thus charge conjugation reverses the walk factors and exchanges the two parametrices, replacing the source by J^{c}. Their mean satisfies

$$\gamma_4 S_j^{\mathrm{sym}}(Z_\circ, s^{\mathfrak{c}}; \mathsf{U}^{\mathfrak{c}}, J^{\mathfrak{c}})^* \gamma_4 = S_j^{\mathrm{sym}}(Z_\circ, s; \mathsf{U}, J),$$
$$\mathcal{C} S_j^{\mathrm{sym}}(Z_\circ, s; \mathsf{U}, J)^{\mathsf{T}} \mathcal{C}^{-1} = S_j^{\mathrm{sym}}(Z_\circ, s; \mathsf{U}, J^{\mathsf{c}}).$$

Both summands have norm at most Γ_F^2 and the same component restrictions; so does their mean. Each is the whole covariance when $s = 1$. Holomorphy of the second summand in (26) follows because it is the conjugate of a holomorphic function at conjugate arguments.

If $Z_\circ$ extends outside Ω_{j+1}, first intersect its fine cubes with Ω_{j+1} and continue its parameters as one on $Z_\circ^\sharp$. The compression to the integrated places is unchanged by this intersection. The completed parameters and hence the walk sums are the same; if the intersection is empty they act on the zero space. This extends the construction and its component identities to these regions. Parameters are supplied on all big cubes. A walk can leave Ω_{j+1} through an earlier layer, so restricting the parameter set to the top layer would not give the stated component decomposition. □

6.2. **The local fermion normalizer.**

Lemma 6.8 (Rows reaching stored layers)**.** Choose the weight constant C_H at least twice the enlarged whole-background row constant and above the fine-resolvent constant below. On the regularity and source domains of Lem. 6.5, with $j = k - 1$, one has

$$|\mathcal{K}_k(\mathsf{n})^{-1}(v, v')| \le C_H (1 + |x_v - x_{v'}|_1)^{-2} e^{-\alpha_D d_{h,k}(x_v, x_{v'})}, \qquad \mathsf{n} = -a_K J. \tag{27}$$

The same bound holds jointly with the complex gauge class. If a row x of level n lies in the layer $\mathcal{L}_m$, $n \le m$, then at a real regular configuration

$$\sum_P |\mathcal{H}_{(n)}(x; P)| \le \tfrac{1}{2} C_H L^{-(m-n)}.$$

The sum includes stored columns of every level. The row twins obey the same estimate. At the admitted complex configurations and matrix sources the bound holds with C_H in place of $C_H/2$. The localized walk rows, their neighbour differences and their sums with the weights $e^{|\omega|}$ have the

constants C_H and $C_H(\alpha)$ of Prop. 6.7 on these domains as well. On the empty history this gives $\sum_y |\mathsf{H}_{n,k}(x,y)| \le C_H L^{-(k-n)}$ and the difference bound $C_H(\alpha)L^{-(1+\alpha)(k-n)}$. ■

Proof. When these estimates are used at an intermediate depth i, apply them to the history cut after that depth. Its regions $\Omega_1,\dots,\Omega_i$, fine operator, layers, scale function, metric and tower through level i are exactly those of the original history at depth i. Its separation and regularity conditions are inherited. The constants below are uniform in the history and its depth, so this application changes none of them. At depth zero the first-fluctuation estimate of Lem. 5.3 applies at every real configuration; its admitted matrix-source perturbation is the Neumann estimate in Lem. 6.5 and requires no gauge regularity at that first step.

The fine resolvent. We give the point-source argument, since a weighted operator bound alone does not give the power in (27). Retain $\alpha_D \le \delta_*/4$ and the separation floors chosen in Lem. 6.5. Write $\mathsf{s} = \mathsf{s}_k$, $d = d_{h,k}$, and fix a place v' and a unit fibre vector ξ. Put

$$x' = x_{v'}, \qquad s' = \mathsf{s}(x'), \qquad F = \mathcal{K}_k(\mathsf{n})^{-1}\delta_{v'}\xi, \qquad \eta(x) = \delta_* d(x,x').$$

All constants in this argument depend only on the previously chosen operator constants. They are independent of the history, volume, cutoff and flavour multiplicity. Let $t = |x-x'|_1$ and $\rho = |x-x'|_\infty$. The Lipschitz bound on s gives

$$t \le (Rs'+1)(e^{d/R}-1) \le 2s'de^{d/R}, \qquad \max\{\mathsf{s}(x)/s', s'/\mathsf{s}(x)\} \le e^{2d/R}.$$

Consequently, if $\delta R \ge 4$,

$$e^{-\delta d} \le C_\delta s'^2 (1+t)^{-2}, \qquad C_\delta = 4\sup_{u\ge 0}(1+u)^2 e^{-\delta u/2}. \tag{28}$$

Choose the fixed $\varepsilon_0 > 0$ in the local regularity argument small enough for its absorptions below, and reduce ϵ_* so that $\epsilon_* L^2 \varepsilon_0^2 \le \epsilon_{\mathrm{lr}}$. If $\varepsilon_0 s' < 64$, the twisted inverse bound gives

$$|F(x)| \le 2C_S(\mathsf{s}(x)s')^{1/2} e^{-\eta(x)} \le 2C_S s' e^{-15\delta_* d/16}.$$

Apply (28) with $\delta = \delta_*/2$. The remaining exponential has rate at least α_D, and $s' < 64/\varepsilon_0$ bounds the remaining prefactor. This proves the claim in this case.

Suppose now that $l = \varepsilon_0 s' \ge 64$. The separation of the layers puts $Q_{2l}(x')$ inside Ω_1, with its required neighbouring bonds, and gives

$$|U_p - 1| \le \epsilon_* L^2 s'^{-2}, \qquad \tfrac{3}{4}s' \le \mathsf{s} \le \tfrac{5}{4}s'$$

on this cube and on the layer blocks meeting it. The lengths of these blocks lie between $3s'/4$ and $5Ls'/4$. Use the axial gauge rooted at x', extended by the identity outside $Q_{2l}(x')$, so that $u(x') = 1$ and

$$|U^u(b) - 1| \le \epsilon_* L^2 s'^{-2} |z - x'|_1$$

for bonds at z in the cube. After this unitary gauge change the operator is

$$\mathcal{K} = D[U^u] + m_0 + \mathsf{W} - \mathsf{Y}, \qquad \mathsf{Y} = a_K \widehat{J} - (D[\mathsf{U}^u] - D[U^u]).$$

The second term in Y is zero at real links. The penalty kernel on a block of level n is bounded by $5\theta L^{-4n}$, by the complete-average estimates. The finite-range coefficients of Y near x' are bounded by $\upsilon = (L\varrho_\sharp + 3c_{\mathrm{hop}}\alpha_1^*)/s'$.

A local approximation with two terms. Let $G = (D[1]+m_0)^{-1}$ be the free kernel on $\mathbb{Z}^3$ from Lem. 6.6, read in the lifted cutoff cube. Choose cutoffs χ, χ' with gradients at most $4/l$, where $\chi = 1$ on $Q_l(x')$ and vanishes off $Q_{3l/2}(x')$, and $\chi' = 1$ on $Q_{3l/2+1}(x')$ and vanishes off $Q_{2l-1}(x')$. Put

$$\widetilde{g} = \mathsf{Y}\chi G\delta_{v'}\xi, \qquad F^{\mathrm{p}} = \chi G\delta_{v'}\xi + \chi' G\widetilde{g}.$$

The support of $\widetilde{g}$ lies where $\chi' = 1$. Direct multiplication gives $\mathcal{K}F^{\mathrm{p}} = \delta_{v'}\xi + g + g'$, where

$$g = [D[U^u],\chi]G\delta_{v'}\xi + \chi(D[U^u]-D[1])G\delta_{v'}\xi + \mathsf{W}\chi G\delta_{v'}\xi,$$
$$g' = [D[U^u],\chi']G\widetilde{g} + \chi'(D[U^u]-D[1])G\widetilde{g} + \mathsf{W}\chi' G\widetilde{g} - \mathsf{Y}\chi' G\widetilde{g}.$$

In particular the second term in F^{p} cancels the first occurrence of $-\mathsf{Y}\chi G\delta_{v'}\xi$. This cancellation is needed for the weighted norm of the error.

The free bound $|G(z)| \le C(1+|z|)^{-2}$ gives the following bounds for the three terms of g, in their displayed order:

$$Cl^{-3} \text{ on the cutoff shell}, \qquad C\epsilon_* L^2 s'^{-2}(1+|z-x'|)^{-1}, \qquad C\theta\varepsilon_0 s'^{-3}.$$

The last term is supported on the blocks meeting the cutoff; their union has at most $C(Ls')^3$ sites. For it, sum $5\theta L^{-4n}(1+|z-x'|)^{-2}$ over the cutoff cube. For the second approximation use the elementary convolution bound in three dimensions,

$$\sum_y (1+|z-y|)^{-2}(1+|y-x'|)^{-2} \le C(1+|z-x'|)^{-1}.$$

Thus $|G\widetilde{g}(z)| \le Cv(1+|z-x'|)^{-1}$. The four terms of g' are bounded respectively by

$$Cs'^{-1}l^{-2}, \qquad C\epsilon_* L^2 s'^{-3}, \qquad C\theta\varepsilon_0^2 s'^{-3}, \qquad Cs'^{-2}(1+|z-x'|)^{-1},$$

on the second shell, the cutoff cube, its meeting blocks, and the neighbouring cutoff cube. Here and below fixed powers of ε_0, L are included in C. On all these supports $\mathsf{s} \asymp s'$ and η is bounded. Squaring and summing the displayed bounds gives

$$\|\mathsf{s}^{1/2} e^{\eta}(g+g')\|_2 \le Cs'^{-1}.$$

For example, the square of the last term's weighted norm is bounded by

$$Cs'\, s'^{-4} \sum_{|z-x'|_\infty \le 2l} (1+|z-x'|_\infty)^{-2} = O(s'^{-2}).$$

In contrast, the uncancelled $\widetilde{g}$ would have weighted norm of order $vs'^{1/2}$. Also $|F^{\mathrm{p}}(x)| \le C(1+\rho)^{-2}$ on $\rho \le 2l$ and $F^{\mathrm{p}} = 0$ outside this cube.

The remainder. Set $F^{\mathrm{r}} = uF - F^{\mathrm{p}} = -\mathcal{K}^{-1}(g+g')$. The twisted inverse estimate yields

$$\|\mathsf{s}^{-1/2} e^{\eta} F^{\mathrm{r}}\|_2 \le C_{\mathrm{par}} s'^{-1}.$$

Its equation is

$$(D[U^u] + m_0)F^{\mathrm{r}} = -(g+g') - \mathsf{W}F^{\mathrm{r}} + \mathsf{Y}F^{\mathrm{r}}.$$

On a block of level n, Cauchy–Schwarz and the penalty kernel bound give

$$|\mathsf{W}F^{\mathrm{r}}(x)| \le C\theta L^{-5n/2}\|F^{\mathrm{r}}\|_{\ell^2(B)} \le C\theta\mathsf{s}(x)^{-2} e^{-\eta(x)} s'^{-1}.$$

Define

$$v(x) = \begin{cases} (1+\rho)^{-2}, & \rho \le l, \\ (1+l)^{-2}(s'/\mathsf{s}(x))e^{-\eta(x)}, & \rho > l, \end{cases} \qquad A = \sup_x |F^{\mathrm{r}}(x)|/v(x).$$

This supremum is finite on the finite torus. When $\rho \le 32$, or when $\rho > l$ and $\varepsilon_0\mathsf{s}(x) < 64$, the weighted ℓ^2 bound already gives $|F^{\mathrm{r}}(x)| \le Cv(x)$. At every remaining site apply Lem. 6.6 to the real operator $D[U^u] + m_0$ on a cube of half-side

$$l_x = \rho/8 \quad (\rho \le l), \qquad l_x = \varepsilon_0\mathsf{s}(x)/16 \quad (\rho > l).$$

These lengths are at least four. The enlarged cube stays at distance at least $7\rho/10$ from x', and $v \le 3v(x)$ there. The same separation and curvature bounds as above verify the local regularity hypotheses.

For $\rho \le l$, the local ℓ^2 contribution is bounded by $C\rho^{-2}(\rho/s')^{1/2}$. The penalty contribution is $Cl_x s'^{-3}$. The curvature, shell and remaining source errors cost at most $C(\epsilon_* L^2 s'^{-2} + l^{-2} + s'^{-2})$ when their supports meet the cube. Each is at most $Cv(x)$. For $\rho > l$, the first two contributions are bounded by

$$C\big(l_x^{-3/2}\mathsf{s}(x)^{1/2} + l_x\theta\mathsf{s}(x)^{-2}\big)s'^{-1}e^{-\eta(x)} \le C\mathsf{s}(x)^{-1}s'^{-1}e^{-\eta(x)} \le Cv(x).$$

The errors $g+g'$ away from their centre are $O(s'^{-3})$. They contribute only within distance $3Ls'$ of x', where $\mathsf{s}(x) \asymp s'$ and η is bounded, so their contribution has the same bound.

It remains to absorb $\mathsf{Y}F^{\mathrm{r}}$. If n' is the least level of a block meeting the local cube, its contribution is at most

$$3C_{\mathrm{lr}} l_x (L^{1-k}\varrho_\sharp + 3c_{\mathrm{hop}}\alpha_1^* L^{1-n'}) Av(x).$$

Here $l_x \le \varepsilon_0 L^k/8$ and $l_x L^{-n'} \le \varepsilon_0/6$. Choose $C_{\mathrm{lr}}\varepsilon_0 \le 1/8$, $L\varrho_\sharp \le 1/8$ and $c_{\mathrm{hop}}\alpha_1^* L \le 1/64$, reducing the existing radii if necessary. The coefficient is at most $1/2$. Thus $A \le C + A/2$, proving $|F^{\mathrm{r}}(x)| \le Cv(x)$.

For $\rho \le 2l$, $d(x,x') \le 8\varepsilon_0$, so the bounds on $F^{\mathrm{p}}, F^{\mathrm{r}}$ give the asserted kernel estimate. For $\rho > 2l$ the approximation vanishes and

$$|F(x)| \le Cl^{-2} e^{-7\delta_* d(x,x')/8}.$$

Apply (28) once more, with $\delta = \delta_*/2$. Its factor s'^2 cancels l^{-2} up to the fixed ε_0^{-2}, and the remaining rate exceeds α_D. This proves (27). The argument used the complex perturbation and the matrix source together; it does not require a positive real part for their sum.

The background rows. For a column P of level $n_P \ge 1$, its fine background is $\mathcal{K}_k^{-1}(T_{n_P}^k)^*\theta_{n_P}\delta_{y_P}$. The source has norm $L^{-n_P/2}$, is supported in its block and has $\mathsf{s} \le L^{n_P}$ there. The weighted inverse estimate therefore bounds $\|\mathsf{s}^{-1/2}e^{\delta_* d(\cdot,y_P)}\psi\|$ uniformly. For a fine stored column the source is $-D_{A_0 S_0}\delta_P$ near the fine layer, and has the same bound. The local regularity argument used for Prop. 6.7, with the penalty and mass terms absorbed in the weighted supremum, gives $|\psi(x)| \le C\mathsf{s}(x)^{-1}e^{-\delta_* d(x,y_P)/2}$. Passing through the level-n averages yields $C_H^\circ L^{-(n_P-n)}e^{-\alpha_D d(x,y_P)}$ for $n \le n_P$. For $n > n_P$ the preliminary factor is L^{n-n_P}; each intervening layer has distance at least R, and $\alpha_D R \ge 2\log L$ pays this factor from half the exponential. The number of places in a distance ball of radius t is bounded by $C(1+te^{t/R})^3$, and the separation floor makes its sum against the remaining exponential uniformly finite. Thus the full row sum is bounded by $(1+C_\Sigma L^{3/2})C_H^\circ L^{-(m-n)}$. Choose $C_H \ge 2(1+C_\Sigma L^{3/2})C_H^\circ$.

The neighbour and heavy-component bounds apply the same local regularity estimate to the column equation. Its logarithmic modulus is bounded by a power $\alpha < 1$; the transported-average defect has the same gain. Summation over columns uses the preceding distance count. The complex and source perturbation may double these whole-kernel constants. The constants for the local walk rows were already formed on the complex source domain in Prop. 6.7; choose C_H and $C_H(\alpha)$ to include them from the outset. Neither the geometric walk ratio nor A_κ is increased. This proves the assertions. □

In particular a formal background of level $n < m$ in a stored layer, given weight at least $\frac{1}{2}C_H L^{-(m-n)}h_F$, can be read through the whole real background at generator weight h_F with norm at most one. This reserve will be used when a history is finally evaluated.

Proposition 6.9 (Closed-walk expansion)**.** The difference $\log\det\Gamma_j(U;J) - \log\det\Gamma_j(1;J)$ is a sum of local, strictly gauge-invariant scalar terms, vanishing at $U=1$. Its individual array members obey

$$\sup_P |N^{\mathrm{F}}(X;P)| \le C_N N_f e^{-(\kappa+1)d(X)}.$$

This is a polymerwise bound, not the family norm obtained by summing $e^{\kappa d(X)}|N^{\mathrm{F}}(X)|$ through a cube. They are holomorphic on the same complex gauge and source domains and are covariant under the isometries preserving the partitions. At $J=0$ they are real at real configurations. The flat-field part at a nonconstant J is a local source-dependent sum; at $J=0$ it is a vacuum number per block with a thermodynamic limit. The bounds are uniform in the volume, cutoff and history. ■

Proof. For $j \geq 1$ let $c_j = \theta_j + \theta q^* q$ and $P = Lq^+ q$, an idempotent. The Schur determinant identity gives

$$\det(\Gamma_j + y) = \det(c_j + y) \det \mathcal{K}_y / \det \mathcal{K}_j, \quad \mathcal{K}_y = \mathcal{K}_j - T_A^+ (\theta_j - p_y) T_A,$$

$$p_y = \frac{y}{1 + y/\theta_j}(1 - P) + \frac{y + \theta/L}{1 + (y + \theta/L)/\theta_j} P, \qquad N_y = \partial_y \mathcal{K}_y = T_A^+ \theta_j (c_j + y)^{-2} \theta_j T_A.$$

Here $\mathcal{K}_0 = \mathcal{K}_{j+1}$ and $\mathcal{K}_\infty = \mathcal{K}_j$. Since $p_y^2 \leq \theta p_y$ and $p_y \geq p_0$, the weighted energy proof applies uniformly for $y \geq 0$, with LC_S in place of C_S. Integrate the logarithmic derivative and subtract the flat configuration:

$$\log \det \Gamma_j(U) - \log \det \Gamma_j(1) = -\int_0^\infty \{ \mathrm{Tr}(\mathcal{K}_y(U)^{-1} N_y(U)) - \mathrm{Tr}(\mathcal{K}_y(1)^{-1} N_y(1)) \}\, dy.$$

The block-local determinant of $c_j + y$ is configuration independent: its eigenspace dimensions are fixed by $P^2 = P$. Use the local walk expansion for $\mathcal{K}_y^{-1}$, and file each trace walk at the polymer of its visited cubes. The trace closes the walk and is gauge invariant. Before the integral, its rank bound is $\nu(3LM_w)^3$, its two T_A factors cost L^{-j}, and the weighted inverse costs CL^{j+1}. The remaining integral is bounded by $\int_0^\infty \|\partial_y p_y\|\, dy \leq C\theta$. Thus the coefficient bound is $C\nu q_w^{|\omega|}$. The walk count and the sum over trace origins give the individual array bound at rate $\kappa + 1$; one spare unit pays the number of origins in its polymer. Summing this bound through a cube without the $e^{\kappa d(X)}$ weight is allowed by the actual polymer count and the large floor on κ. No weighted family bound is inferred from a single spare unit of the rate.

At $j = 0$ use

$$\log \det \mathcal{K}_1 = \int_0^\infty \mathrm{Tr}\big((1 + y)^{-1} - (\mathcal{K}_1 + y)^{-1}\big)\, dy.$$

After subtracting the flat configuration, the one-cube term is $O((1 + y)^{-2})$; a walk of $n \geq 1$ steps has $n + 1$ resolvents and is integrable. The same closed-walk count applies. The source is read only in visited cubes, with the same bound when both configurations have the same source. At $J = 0$, conjugation by γ_4 exchanges a real kernel with its adjoint; averaging the left and right trace-walk expansions makes every filed coefficient real. Gauge covariance, lattice covariance and holomorphy follow term by term.

At the flat field and $J = 0$ the fluctuation symbol has real part bounded below and is analytic in a fixed momentum strip. Its principal logarithm has modulus at most $C\nu$ per block. The Fourier coefficients therefore decay exponentially. Let $\delta_F > 0$ be a common lower bound for this strip width and the Fourier decay rate, depending only on the earlier operator constants. The antiperiodic momentum sum differs from its Brillouin-zone integral by $C\nu e^{-\delta_F \ell L^{K-j}}$, by summing its nonzero Fourier coefficients at the period multiples. This gives the bounded vacuum number, its thermodynamic limit, and the same uniformity. □

The following local form of the construction is needed before any cluster expansion. Its input is a configuration on a polymer; extending that configuration to a regular field on the whole torus is not part of the construction.

Lemma 6.10 (Operators formed on a polymer)**.** Assume the separation, regularity and complex-field bounds above, with $R \geq \max\{R_0, LM_w\}$, $M \geq 3M_w$ and $c_f \min\{M, D_{\mathrm{ex}}\} \geq C_{\mathrm{dc}} \bar{\kappa}$. Let X be a union of fine cubes. Suppose $\mathsf{U} = e^{i\Xi} U$ is given on its inner bonds, the real field U satisfies the regularity bounds at its inner plaquettes, and $|\Xi(b)| \leq \alpha_1^* L^{-n(b)}$ there. The source is given on its sites and satisfies $\sup_{x \in X} \|L^{j-K} J(x)\| \leq \varrho_\sharp$. At $j = 0$ an arbitrary real configuration is also allowed. Then all walks whose supports lie in X, their background-map terms, and the closed-walk normalizer on X are defined from these data and satisfy the preceding bounds with the same constants.

If $Z_\circ^\sharp \subset X$ and parameters vanish off the big cubes contained in X, the two weakened covariances and $S_j^{\mathrm{sym}}(Z_\circ, s)$ are defined, have norm at most Γ_F^2, and obey their component and symmetry

identities. The comparison with parameters one now means parameters $\mathbf{1}_X$, zero off X. The normalizer can be pointed at its trace places p so that

$$\sum_p N_p(X,\mathsf{U};J) = N^{\mathrm{F}}(X,\mathsf{U};J), \qquad |N_p(X,\mathsf{U};J)| \le C_N N_f e^{-(\kappa+1)d(X)}.$$

The same bound holds for the sum over places sent to one site of the next level. These objects are holomorphic at a fixed real factor and depend on U, not on its representation as $e^{i\Xi}U$. If the data extend to a global configuration, the local and global constructions agree on these terms. The identities with the whole inverse and whole determinant remain assertions on a torus. ■

Proof. A local inverse reads the configuration on $W_\square$ only. Reflect it to its mirror torus and put the identity on the links crossing a mirror. A crossing plaquette is the identity; every other plaquette is copied from $W_\square$. On a reversed bond the copied non-unitary part is $-U(b)^{-1}\Xi(b)U(b)$, with the same norm. The copied regions and source therefore satisfy the hypotheses of the weighted inverse estimate. This constructs the mirror inverse without using any bonds outside X.

Fine cubes are unions of the blocks of level $j+1$, and all lower blocks nest in them. Thus a block meeting a retained walk lies in X. The averages, penalties, commutators and finite-range factors adjacent to the walk read only inner bonds and sites of X. Their bounds are precisely those used in the proof of Prop. 6.7. The products have the same bounds, and the restricted walks are a subset of the walks already counted. The row-difference argument takes place on the mirror torus of its first cube; it also uses no global extension.

When the parameters vanish off X, every walk leaving X has coefficient zero. The weakened covariance is therefore the convergent sum of the walks just constructed. Schur's test and the component argument give its bound and restriction identities. The left walks reverse the same local factors, so the proof of (26) applies on X as well.

For the normalizer, file the trace walks at X and take the trace of the diagonal block at p before integrating in y:

$$N_p(X,\mathsf{U};J) = -\int_0^\infty \sum_{\omega:\widehat{X}_\omega = X} \Big\{ \mathrm{tr}_p[T_A \mathcal{W}_\omega(\mathsf{U},y;J) T_A^+ \partial_y p_y] - (\mathsf{U}\to 1) \Big\}\, dy.$$

At level zero use the diagonal trace of the fine resolvent walk itself. Cyclicity of the trace proves the sum rule. A diagonal block has trace bounded by ν times its operator norm. The places in one next-level block are fewer than those in a threefold walk cube, which is the rank count in Prop. 6.9. The same integral and walk count therefore give both pointed bounds. Reversing the trace walk proves reality at real fields and zero source.

All formulas are holomorphic products and uniformly convergent sums of functions of the links. The principal square root in the normalized average is fixed by the common right half-plane. Consequently two allowed decompositions of the same complex links give the same averages and local inverses, and hence the same objects. Restricting global data leaves each factor and the set of retained walks unchanged. This proves the last assertion as an equality of formulas. □

On the history without large fields, a pair in a local analyticity core has first member $e^{iL^{-(j+1)}A'}U$. Its curvature and complex displacement bounds put it in Lem. 6.10. Thus the local objects are defined on the whole core and are functions of its pair. This is the domain assertion used below when coefficients are evaluated at complex, locally defined backgrounds.

6.3. **The early bound for the gauge coupling coefficient.** The normalizer action coefficient must be bounded before M and κ are chosen. The following finite-matrix form of the four-propagator argument gives that bound without using the later polymer norm.

Proposition 6.11 (The fermion contribution to the coupling)**.** On the history without large fields, let c_k^{F} be the action coefficient of the field-dependent $\log\det \mathcal{K}_k$, in level-k units, and put $c_0^{\mathrm{F}} = 0$.

For $0 \le \mathfrak{m}_k \le 1$,

$$|c_k^{\mathrm{F}}| \le N_f c_2^{\mathrm{F}}, \qquad b_{k+1}^{\mathrm{F}} = c_{k+1}^{\mathrm{F}} - L^{-1} c_k^{\mathrm{F}}, \qquad |b_{k+1}^{\mathrm{F}}| \le (1 + L^{-1}) N_f c_2^{\mathrm{F}},$$

where c_2^{F} depends only on the fixed chamber operator, L and b_F. It is independent of k, K, M, κ and the mass in this range. The coefficients are real and their dependence on the top mass is uniformly Lipschitz on this interval. ■

Proof. Work first at one flavour, in level-k units with $h = L^{-k}$ and one coarse block as Bloch cell. The operator is

$$K = D_h + m + W, \qquad D_h = h^{-1} D, \quad W = \theta_k R^+ R, \quad 0 \le m \le 1, \qquad G = K^{-1}.$$

The codomain of R has dimension eight, including colour; chamber multiplicity enlarges its domain and not its rank. Also $\|W\| \le b_F$ at real fields. The scaled energy inverse estimate gives $\|G\| \le C$. Its proof discards nonnegative mass terms and thus applies throughout the stated mass interval (indeed for every nonnegative mass for this operator bound).

We first justify all jets of the penalty that enter the polarization. Write a fine site as $y + z$, with y its coarse centre and z in a fixed unit cube. Strip the coarse Bloch phase $e^{i\xi \cdot y}$, and for an external wave also strip $e^{ip \cdot y}$. On the cell its gauge amplitude is then $T e^{ip \cdot z_b}$, with bond midpoint z_b of uniformly bounded size. A fixed complex strip of external and internal momenta has a uniform amplitude bound. At the flat field apply the complete-average bound stated before Lem. 6.5 to

$$X_b = h\{t_1 A_{1,b}(p_1) + t_2 A_{2,b}(p_2)\}.$$

A fixed small complex disc in t_1, t_2 is inside its domain, independently of k. The actual complete normalized kernel bound and Cauchy's formula give its first two gauge jets, with up to two needed external or internal momentum jets, uniformly in k. The lattice-normalized row of R has entries $h^{3/2} q_k^c(y, x) \iota^*$. Thus its squared row norm, and the corresponding jet norms, are bounded by $C h^3 \# B = C$. The remaining phase $e^{i\xi \cdot z}$ has bounded derivatives as well. This is why absolute cell coordinates cannot be used in the jet estimate.

Every gauge or momentum derivative of $W = \theta_k R^+ R$ needed here is a finite sum of products of two such maps. Its rank is at most a fixed multiple of eight and its operator and trace norms are uniformly bounded. On complex fields one differentiates the holomorphic row twin R^+, not an antiholomorphic adjoint. Any polarization term containing one of these penalty jets has a uniformly bounded trace: use its trace norm and the operator norms of all remaining factors. The latter remain bounded also after momentum differentiation, by the resolvent identity below.

For the terms without a penalty jet a uniform operator norm is insufficient, because the cell dimension grows as h^{-3}. Put $G_0 = (D_h + m + 1)^{-1}$. In a cell fibre ξ the fine modes are $q_n = \xi + 2\pi n$. Free ellipticity gives

$$\|G_0(\xi)\|_4^4 \le d C_{\mathrm{el}}^4 \sum_n (1 + |\xi + 2\pi n|)^{-4} \le C,$$

where d is the fixed one-flavour fine-site dimension, including chamber multiplicity, and $\|\cdot\|_4$ is the fourth Schatten norm. The finite alias sum is bounded by the infinite three-dimensional sum, uniformly in h and ξ. The resolvent identity is

$$G = G_0 - G_0(W - 1)G, \qquad \|G\|_4 \le \|G_0\|_4 [1 + \|W - 1\| \|G\|] \le C.$$

This is the constant-penalty resolvent expansion in the norm appropriate for the four-propagator count.

Only elementary matrix trace inequalities are needed. Hilbert–Schmidt Cauchy–Schwarz gives

$$\|AB\|_2^2 = \mathrm{Tr}(A^* A B B^*) \le \|A\|_4^2 \|B\|_4^2.$$

Apply it to two adjacent pairs and once more to their trace to obtain $|\mathrm{Tr}(A_1 A_2 A_3 A_4)| \le \prod_i \|A_i\|_4$. The ideal inequality $\|ABC\|_4 \le \|A\| \|B\|_4 \|C\|$ allows any bounded vertices between the four resolvents.

The exact second field variation of $\log\det K$ is $\mathrm{Tr}(GK_{A_1A_2}) - \mathrm{Tr}(GK_{A_1}GK_{A_2})$. Keep its penalty vertices and penalty seagull; they were bounded by rank above. The mixed bare seagull vanishes, because a gauge-dependent hop uses one coordinate direction. In the symmetric momentum assignment the remaining bare bubble is

$$\mathrm{Tr}\{G(\xi - p/2)V_1(\xi)G(\xi + p/2)V_2(\xi)\},$$

where the bare vertices have bounded norm and are independent of the external momentum. More precisely, let $\pi_{\mu\nu}(p)$ be the infinite-volume, volume-normalized second field variation of $\log\det K$ along the waves $e^{ip\cdot b_-}T$ and $e^{-ip\cdot b_-}T$ in directions μ and ν, with $\frac{1}{2}\operatorname{tr}T^2 = 1$. We use the action-coefficient convention $c_k^{\mathrm{F}} = -\partial_{p_1}\partial_{p_2}\pi_{12}(0)$. The midpoint phases used above give the same second derivative, since the polarization and its first jet vanish at zero. For internal derivatives,

$$\partial_i G = -GK_iG, \qquad \partial_{12}G = GK_1GK_2G + GK_2GK_1G - GW_{12}G.$$

Here K_i is bounded and $K_{12} = W_{12}$ has bounded rank, since $D_{h,12} = 0$. Every resulting term is therefore either a four-resolvent trace bounded by the preceding inequality or a bounded-rank trace already treated. Integration over the fixed coarse Brillouin zone gives $|c_k^{\mathrm{F}}| \le c_2^{\mathrm{F}}$. No polymer constant has entered.

The same argument holds on a fixed complex disc about each real $m \in [0,1]$. The scaled inverse Neumann estimate supplies a common disc radius; the constant-mass free resolvent retains its fourth-norm bound after that radius is decreased. Cauchy's estimate now applies to the uniformly bounded coefficient itself, proving the Lipschitz assertion. Holomorphy of matrix entries alone would not have sufficed for it. The γ_4 real structure gives reality, and spectator flavours multiply every trace by N_f.

Finally the exact determinant identity gives $\log\det\Gamma_k = \log\det c_k + \log\det\mathcal{K}_{k+1} - \log\det\mathcal{K}_k$. The first determinant is field independent. Changing from level-k units to level-$k+1$ units multiplies the old action coefficient by L^{-1}, giving the claimed telescope. At $k = 0$, $\Gamma_0 = \mathcal{K}_1$, so the correct convention is $c_0^{\mathrm{F}} = 0$; the unblocked bare determinant, which has an infrared-divergent coefficient at zero mass, does not enter this bound. □

7. The small-field fermion step

We first describe the gauge ingredients required by a local step. The gauge chart has a holomorphic shift system $h_Y(s; P, B)$ on regular complex pairs P, with size $c_h \sup|B|$, supported in its wall components, compatible with restriction, and equal to the actual shift at $s = 1$ on real background pairs. It has a holomorphic site factor $\mathfrak{r} \in \mathrm{SL}(2,\mathbb{C})$ with $|\mathfrak{r} - 1| \le C_{\mathfrak{r}} \sup|B|$. The backgrounds compose up to the residual gauge group. Their real curvatures are at most $\bar{C}_{\mathrm{reg}}\epsilon(\gamma^{-2})L^{-2n}$; we require this to be at most $\epsilon_* L^{-2n}$, and require $\alpha_0 L^2 \le \epsilon_*$, $L\alpha_1 \le \alpha_1^*$. These bounds put a depth-$k+1$ background into the class of Lem. 6.10 at depth $k+1$. Its chart field need only be regular at depth k; for its depth-$k+1$ operators we use it as a complex perturbation of the more regular background. The distinction is needed for the vertex estimate. The pure gauge conditions follow from Lem. 10.19 and Prop. 10.20; the following argument checks the data and the matter arrays to which the single-step results apply.

Lemma 7.1 (The tested block fields)**.** Let the calibration frame and canonical auxiliary data be those of the admitted gauge construction. Assume $\epsilon(\gamma^{-2}) \le \varepsilon_\flat$ and $2\epsilon(\gamma^{-2}) \le \varepsilon_{\mathcal{W}}\alpha_0$, where $\varepsilon_{\mathcal{W}}$ is the radius for the chart sum rule. Every block field W passing the density test at level $k+1$ lies in that sum-rule domain. Its chart, axial background, pivot inverses and fluctuation operator are defined, and its background pairs lie in the local analyticity cores. The set $\mathcal{V}_{k+1}^{\chi}$ of these fields is open.

For every finite-dimensional vector-valued function F for which the tested integrand is bounded and Borel, the scalar chart identity holds coordinate by coordinate:

$$\mathbf{T}\big(\mathfrak{G}_k[\mathsf{t}_k]\chi^{\mathrm{sf}}_{\mathsf{t}_k}\mathbf{1}_{\{\mathcal{M}(V)\in\mathcal{V}^{\chi}_{k+1}\}}F(V)\,dV\big)=\chi_{k+1}(T_\ell)\,\mathsf{I}[F](W)\,dW.$$

The right side is defined as zero off $\mathcal{V}^{\chi}_{k+1}$. This includes the finite Grassmann algebra of the step. ■

Proof. The calibrations decrease and have zero increments. The admitted floor on $\log\gamma^{-2}$ gives $R(Lx)\le LR(x)$ for $x\ge\gamma^{-2}$, where R is the least power of L above $(\log x)^{r_c}$. The prescribed torus side satisfies $L^m\ge LMR(L(g^{-2}+\bar B))$. Therefore

$$MR(\mathsf{t}_i)\le L^{m+K-i-1}.$$

This verifies the frame geometry at every depth. Monotonicity of the test sizes on the final coupling range gives their admitted bounds in that frame.

For each plaquette, the density-test construction assigns a density cube, independently of the configuration, whose test implies $|W_p-1|<2\epsilon(\mathsf{t}_{k+1})$. All these tests pass, so

$$\max_p|W_p-1|<2\epsilon(\mathsf{t}_{k+1})\le\varepsilon_{\mathcal{W}}\alpha_0.$$

The admitted chart theorem at this radius supplies its invertible pivots, axial background and operator B_W; the latter vanishes on block gauge directions. The canonical auxiliary data supply the interaction sum rule at this field, including the rule that localizes an arbitrary admitted bare array. The background bound places each restricted pair in the claimed core. Strictness of the tests makes $\mathcal{V}^{\chi}_{k+1}$ open.

Put $F'=\chi^{\mathrm{sf}}_{\mathsf{t}_k}\mathbf{1}_{\{\mathcal{M}(V)\in\mathcal{V}^{\chi}_{k+1}\}}F$. On the chart over a tested W, its configurations $V_W(B)$ satisfy $\mathcal{M}(V_W(B))=W$ and the small-field test; hence $F'(V_W(B))=F(V_W(B))$. Apply the bounded Borel chart identity to each coordinate of F'. Its pushforward is carried by $\mathcal{V}^{\chi}_{k+1}$, and multiplication by the inverse-image indicator becomes multiplication by $\chi_{k+1}(T_\ell)$ after pushforward. This proves the measure identity with the stated support. □

Proposition 7.2 (The objects of a local step)**.** For the gauge construction of Prop. 10.20, under the displayed operator and radius bounds, the scalar conditioned Gaussian, the chart shift, the site factor, the local gauge vertices and S^{sym}_k form a local step on each admitted pair domain. They are holomorphic on its slices, obey component restrictions at zero parameters, and agree with the actual step at parameters one on a torus. Their joined integral has the scalar absolute-density majorant after the integrand is replaced by its fibre norm. At a tested block field its sum rule and its coefficientwise measure identity hold pointwise as in Lem. 7.1. ■

Proof. The chart and its scalar conditioned integral are supplied by the admitted one-step construction, independently of an old array. The shift bound, regularity of the background and $L\alpha_1\le\alpha_1^*$ put every local first member in Lem. 6.10. That lemma constructs the covariance on the polymer itself, with Gram constant Γ_F; the same restricted data give its component identities. For an even integrand F, the Gaussian contraction bound gives $\|\mu_{S^{\mathrm{sym}}_k}F\|\le\|F\|$. Apply the scalar majorant through each norm-one linear functional on the finite output algebra and take the supremum. Holomorphy follows from the finite Gaussian moment formulas and the locally uniform majorant. At parameters one the covariance is the whole inverse, and the chart identity of Lem. 7.1 applies. An old scalar array is admitted separately by its norm; none of these assertions uses the pure gauge array induction for the matter array. □

For a family $F=(F(X))$ of local terms put

$$\|F\|_{k;a}=\sup_{\square}\sum_{X\supset\square}e^{\kappa d_k(X)}\sup_{P,B}\|F(X;P,B)\|_h.$$

The supremum is over the regular pairs of first radius a and second radius $(\alpha_1/\alpha_0)a$, and the fluctuation radius of the step. The radius system is constant over the levels. It is admissible when $0 < a \le \alpha_0$ and $(1+\beta_s)\delta_k \le s^\star a$. The background legs of level $n \le k$ have weights $h^{(k)}(n) = C_H L^{-(k-n)} h_F$; the fluctuation legs have weight Γ_F. Choose

$$h_F \ge 2A_\kappa \Gamma_F L/(L-1), \qquad c_\psi = 2A_\kappa \Gamma_F / h_F.$$

After a step, a leg of level n splits into a formal new background of weight $C_H L^{-(k+1-n)} h_F$ and a localized fluctuation image of weight $A_\kappa C_H L^{-(k-n)} \Gamma_F$. Thus the split fits the old weight. The stronger displayed floor leaves room also at weight $2\Gamma_F$ on the fluctuation.

Lemma 7.3 (The vertex at one fine field)**.** At the background U of level $k+1$, let $\mathsf{U}' = \mathsf{U}\exp \mathcal{A}(B;\mathsf{U})$ be the fine field of the chart. Rotate the integrated generators by the site factor and shift them by their background. The exponent of the quadratic form and block kernel becomes

$$\bar\Psi_{k+1}\mathbb{D}_{k+1}(\mathsf{U})\Psi_{k+1} + \bar W \Gamma_k(\mathsf{U}) W - \mathcal{V}_k^{\mathrm{F}}.$$

The potential has a local decomposition into bilinear terms, holomorphic in the pair and in B, vanishing at $B = 0$, with

$$\|\mathcal{V}_k^{\mathrm{F}}\|_{k;a} \le C_V h_F^2 \sup|B|.$$

The constant is uniform in a, k, K and the volume. Moreover its normalized fermion integral is exactly

$$\mu_{\Gamma_k(\mathsf{U})^{-1}} e^{\mathcal{V}_k^{\mathrm{F}}} = \frac{\det \Gamma_k(\mathsf{U}')}{\det \Gamma_k(\mathsf{U})} \exp\{-\bar\Psi_{k+1}[\mathfrak{r}^{-1}\mathbb{D}_{k+1}(\mathsf{U}')\mathfrak{r} - \mathbb{D}_{k+1}(\mathsf{U})]\Psi_{k+1}\}.$$

The first and second derivatives of the site factor belong to this identity and are included in the average vertices. ∎

Proof. Gauge covariance carries the quadratic form at the chart field to that at U', with q replaced by $\mathfrak{r}^{-1}q$ and its row twin by $q^+\mathfrak{r}$. The rotation of the integrated generators has Berezin Jacobian $\prod \det \mathfrak{r} \det \mathfrak{r}^{-1} = 1$. The rotation of the outer generators remains in the block kernel. At $B = 0$ the two forms coincide. Completing the square at U leaves no linear fluctuation term; define the vertex as the negative of the difference of the two forms.

One can estimate this difference without differentiating a long product of Schur complements. Let $\mathbb{M}$ be the one-shot matrix on $(\psi, \Psi_k, \Psi_{k+1})$, consisting of $D + m_0$ and the two block kernels. Its ψ block is $d = \mathcal{K}_k$. For the perturbed matrix $\mathbb{M}'$ write $\delta\mathbb{M} = \mathbb{M}' - \mathbb{M}$ and $\mathbb{G} = d^{-1} \oplus 0$. The Schur identity is

$$\mathfrak{Q}' - \mathfrak{Q} = \langle \bar\Phi^{\mathrm{bg}}, \delta\mathbb{M}\Phi^{\mathrm{bg}\prime}\rangle, \qquad \Phi^{\mathrm{bg}\prime} = \sum_{m\ge 0} (-\mathbb{G}\delta\mathbb{M})^m \Phi^{\mathrm{bg}}.$$

Indeed multiplying a block matrix by its background column annuls its ψ component; subtracting the two such products gives the first identity. The two background equations give the second by a Neumann series.

The Dirac part of a vertex at a bond has size $CL^{-k}\sup|B|$. A cube contains $O(L^{3k})$ bare places, and each fine leg has weight $CL^{-k}h_F$. Its bilinear norm is therefore $C\nu h_F^2 \sup|B|$ since $L^{3k-k-2k} = 1$. For an average vertex the product rule gives

$$\|\delta r_k\| + \|\delta q\| \le C\sup|B| \sum_{n<k} L^{n-k} + C\sup|B| \le C_{\mathfrak{r}} \sup|B|.$$

Each such vertex stands beside a residual $\Psi_k - r_k\psi = \theta_k^{-1}\mathbb{D}_k\Psi_k$ or $\Psi_{k+1} - q\Psi_k$, of bounded weight in level-k units. It has the same estimate. The normalized averages, not their unnormalized contour sums, enter this calculation.

Replace every inverse in the series by its local walks. Their sum with the weight for positive polymer growth is $C_K = A_\kappa C_{\sup}$. Thus a link costs $C_G \sup|B|$, where C_G includes the finite-range vertex, the at most two blocks it reads, and the recounting of its cubes. Choose β_0 after C_G so that

$C_G\beta_0 \le 1/2$ and the shifted fields remain in the complex class. The series then converges locally uniformly for $\sup|B| \le \beta_0$.

Apply the admitted shift system to a chain and take its finite differences in the cube parameters. A difference at a cube costs $(e^{\bar\kappa}-1)^{-1} \le e^{-(\bar\kappa-1)}$ by Cauchy's formula. It vanishes unless its wall component meets the chain. File it at the polymer containing the vertices, walks, transport paths and parameter cubes. If C_g^n bounds the number of connected sets of n cubes through a given cube, require

$$x := C_g e^{(\kappa+2)c_0-(\bar\kappa-1)} \le e^{-8}.$$

The sum over parameter components costs at most $\prod_{c\in R}(1+\sum_{n\ge1} x^n) \le e^{|R|/8}$, where R is the set of occupied chain cubes. Its exponential is paid by the geometric factor per link and the weights $e^{|\omega|}$ in the walk sum. This bounds the sum of chain norms with rate $\kappa+2$ for each outer vertex. For a fixed cube $\square$, the two spare units pay the outer vertex location, since

$$d_k(Y) \ge \operatorname{dist}(c,\square)/M, \qquad \sum_c e^{-2\operatorname{dist}(c,\square)/M} \le 37.$$

This proves the summed family bound at rate κ. Schwarz's lemma on $t \mapsto tB$ gives its factor $\sup|B|$. The localization sums to the original vertex at real chart fields because the finite differences telescope. Both sides are holomorphic on the connected fluctuation polydisc, so the identity holds throughout it. This also proves covariance of the total and its independence of the chosen filing.

Finally integrate the completed quadratic form at the perturbed field. Its integrated block is exactly $\Gamma_k(\mathsf{U}')$: $\mathfrak{r}\mathfrak{r}^{-1}$ cancels there. Its outer Schur complement is $\mathfrak{r}^{-1}\mathbb{D}_{k+1}(\mathsf{U}')\mathfrak{r}$. Dividing by the unperturbed determinant proves the stated exact identity. At $k=0$ no fine inverse is eliminated; the same argument is its finite local expression and the series is empty. □

7.1. **The joined cluster expansion.** In this subsection the conditional mean is zero. The history version, with its stored field and conditional mean, is proved in Lems. 9.1 and 9.9. Here are the scalar gauge inputs required for this expansion. They are the precise inputs to be matched to the admitted pure construction. The conditioned integral $\mathfrak{J}(Z,s)$ is linear, normalized at one, holomorphic for $|s(\Delta)| < e^{\bar\kappa}$, agrees with the restricted Gaussian integral at $s=1$, and factorizes on wall components when the parameters off them vanish. Its positive absolute-density majorant satisfies, with the scaled fluctuation $a = \mathfrak{t}_k^{1/2}B$,

$$|\mathfrak{J}(Z,s)f| \le \mathfrak{J}^{\mathrm{abs}}(Z,s)|f|, \quad \mathfrak{J}^{\mathrm{abs}}(Z,s)F \le F_0 e^{c_{\mathrm{Gauss}}(\hat\varepsilon+\rho)N(Z)} \quad (0 \le F(B) \le F_0 e^{\rho\mathfrak{t}_k\|B\|^2}).$$

Here c_{Gauss} is the fixed Gaussian volume constant, and $N(Z) \le 18(LM)^3\#Z$ counts real fluctuation variables. This is an exponential volume bound, not a uniform operator bound in Z. Finite differences in a set $\mathcal{K}$ of parameters cost $(e^{\bar\kappa}-1)^{-\#\mathcal{K}}$. Test defects lie in $[0,1]$ and a marked large-field bond costs $p = e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathfrak{t}_k)^2/2}$.

We use the scalar Mayer and cluster count for at most four origins, each of norm at most $c_{\mathrm{cl}} = w_*$ at its input rate, with output rate $\kappa_+ = (1+4\beta_s)\kappa$. There are two ways a scalar origin can enter this count. An origin of sort Σ_{v} is given on the tested fluctuation domain, and each vertex inserts the tests on its support. An origin of sort Σ_{q} is given at all real fluctuations. The quadratic correction is given on the larger domain, but in the step it must be used with the tests of the first sort. The scalar count reads a potential only through its norm, measurability and quadratic growth; it requires neither scalar gauge invariance nor vanishing at $B=0$. Localization reads the pair on the resulting polymer and commutes with cube isometries. These scalar facts, including their rate and geometry conditions, are the external cluster input verified in Prop. 10.20.

Here is the norm used for one origin, at a fixed admitted pair:

$$\|\mathbb{V}\|_{\star,r} = \sup_{Y,a} \frac{e^{rd_k(Y)}\|\mathbb{V}(Y;a)\|}{1+M^{-3}\displaystyle\sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle}|a(b)|^2}. \tag{29}$$

The set $\mathfrak{X}^{\circ+}\langle Y\rangle$ consists of the positively oriented nonpivot fluctuation bonds read by Y. The variable is $a=\mathsf{t}_k^{1/2}B$: the supremum is over the tested domain for sort Σ_{v} and all real fluctuations for sort Σ_{q}. The numerator uses absolute value for a scalar origin and the fibre norm for an even Grassmann-valued origin. All bounds are uniform in the admitted pair. This is the quadratic-growth norm of the scalar cluster construction in [3, §4.2] and [3, eq. 4.lxxi]; the label Σ_{q} specifies the domain, not polynomial degree. Its input rate is $\lambda_V=(\mathsf{C}_{\mathrm{ex}}/L)\lambda_1+2\lambda_{\mathrm{s}}$. The admitted rate inequalities give $\lambda_V\le(1-2\delta_w)\kappa\le\kappa-1$. Consequently the summed potential norm in Lem. 7.5 dominates $\|\mathbb{V}\|_{\star,\lambda_V}$ on the same domain.

Lemma 7.4 (The quadratic origin and Gaussian normalization)**.** Put $\tau=x_k/\mathsf{t}_k$, with $|\tau-1|<\sigma_\star$. In the scaled fluctuation $a=\mathsf{t}_k^{1/2}B$, let

$$\mathcal{V}^{\mathrm{q}}(\square;a)=-\tfrac12(1-\tau)(a,\widehat{G}_\square a),\qquad \|\mathcal{V}^{\mathrm{q}}\|_{\star,\lambda_V}\le|1-\tau|K_{\mathsf{G}}\le w_*/2.$$

It vanishes on polymers other than single fine cubes. Use its restriction $\mathcal{V}^{\mathrm{q}}|_{\mathrm{v}}$ as a tested origin in the step, and use the unrestricted family only for its untested Gaussian logarithm $G_\tau(X)=E^\circ[\{\mathcal{V}^{\mathrm{q}}\}](X)$. For at most three additional tested origins of sizes w_o,

$$\sum_X\left(E[\{\mathcal{V}^{\mathrm{q}}|_{\mathrm{v}}\}\cup\underline{\mathbb{V}}](X)-G_\tau(X)\right)=\log\mathsf{E}_k\exp\sum_{o,Y}\mathbb{V}^o(Y),$$

where E_k divides the tested integral by the untested integral of $e^{\mathcal{V}^{\mathrm{q}}}$. Polymer by polymer the difference is bounded by

$$\left(K_\Pi\sum_{o\ne\mathrm{q}}w_o+3K_P^\Pi p_*\right)e^{-\kappa_+d(X)},\qquad p_*=\mathsf{C}_{\mathrm{ex}}((L+2)M)^3e^{4\lambda_1}p.$$

These assertions hold also for even Grassmann-valued additional origins, using their fibre norm. ■

Proof. On the set where all tests pass, the restricted and unrestricted quadratic potentials coincide. The sum of their cube gauge-fixing operators is the operator of the chart at a tested field, by Lem. 7.1. The finite Mayer identities and the convergent connected expansion therefore give the displayed identity of totals, with G_τ equal to the logarithm of the divisor.

For the local estimate split the difference into three parts. First remove the additional origins while keeping the tested quadratic origin. Every remaining term is marked by one of the removed origins, and the marked cluster bound is $K_\Pi\sum_{o\ne\mathrm{q}}w_o$. Second change the quadratic origin from the tested to the unrestricted sort, at the same finite threshold. Third remove the threshold from that unrestricted quadratic system. The last part is marked by a failed bond and costs $K_P^\Pi p_*$.

To estimate the second part, expand each test attached to a quadratic vertex by

$$\mathsf{in}_{E'}=\sum_{E''\subset E'}(-1)^{\#E''+}\mathsf{out}_{E''}.$$

This identity may be used at every real fluctuation because the quadratic potential is given there. The summand $E''=\emptyset$ is the unrestricted Mayer summand on the same region. The difference consists of two sums: terms with a nonempty added failed-bond set, and unrestricted terms whose original failed-bond set meets the support of a quadratic vertex. Both have a marked failed bond. An added bond has an endpoint in a vertex support; its enlargement contains a big cube of the original connected region and has tree size zero. Give it weight $pe^{4\lambda_1}$. At most $6((L+2)M)^3$ such bonds can meet a fixed big cube. The correspondence between the vertex set, the original failed bonds and the added failed bonds is injective: the latter are exactly those in the tested vertex supports. The scalar connected count therefore applies to the enlarged set of marks. Each of these two sums costs $K_P^\Pi p_*$, and its absolute majorant also justifies the refinement of the series. Adding the three estimates proves the bound. For Grassmann-valued origins, contraction and duality as in Prop. 7.2 give the same scalar majorant. □

Lemma 7.5 (Grassmann-valued cluster expansion)**.** For a local potential define

$$\|V\|_r = \sup_{Z,P,B} \sup_{\square\subset Z} \sum_{\square\subset Y\subset Z} e^{r d_k(Y)} \|V(Y; P|_Y, B)\|.$$

Here Z ranges over the level-k polymers, including the torus, P over its admitted pairs of radius a, and B over the fluctuation domain. All terms in a sum use the restriction of the same P, B. The fibre norm has weights $h^{(k+1)}$ on the backgrounds and h_W on the fluctuations. The level family norm, with the supremum inside the polymer sum and rate κ, dominates $\|V\|_{\kappa-1}$. Let $\mathcal{V} = \mathcal{V}^{\mathrm{g}} + \mathcal{V}^{\natural}$ be even, local and holomorphic on the pair radii, with

$$|\mathcal{V}^{\mathrm{g}}(Y; B)| \le \varepsilon_V (1 + M^{-3} \sum_{b\subset Y} \mathsf{t}_k |B(b)|^2) e^{-\lambda_V d_k(Y)}, \qquad \|\mathcal{V}^{\natural}\|_{\kappa-1} \le \varepsilon_{\natural}, \quad \varepsilon_V + \varepsilon_{\natural} \le c_{\mathrm{cl}}.$$

The joined gauge and fermion integral is the exponential of a sum of local even terms. After subtracting the q-only gauge part its family $\widetilde{E}$ satisfies

$$\|\widetilde{E}(X)\| \le K'_M (\varepsilon_V + \varepsilon_{\natural} + p) e^{-\kappa_+ d(X)},$$
$$\|\widetilde{E}\|_{k+1;a} \le K_M (\varepsilon_V + \varepsilon_{\natural} + p), \qquad K_M = 4096 K'_M,$$
$$K'_M = \max\{K_\Pi, 3K_P^\Pi \mathsf{C}_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1}\}.$$

The constants do not depend on the radius a or on the dimension of the Grassmann algebra. The logarithm is formed in its even commutative algebra. ∎

Proof. For the symmetrized decoupled covariance $S(Z, s) = S_k^{\mathrm{sym}}(Z, s)$ set

$$\mathfrak{J}_F(Z, s) F = \mathfrak{J}(Z, s)[B \mapsto \mu_{S(Z,s)} F(B)].$$

Its Gram bound and Lem. 5.1 give $\|\mu_S F\| \le \|F\|$ on the closed decoupling polydisc. For every linear functional ℓ of norm at most one,

$$|\ell(\mathfrak{J}_F F)| \le \mathfrak{J}^{\mathrm{abs}} \|F\|.$$

Taking the supremum over ℓ gives this bound in the fibre norm. Its coordinates are holomorphic in s, by the Gaussian moment formulas and the scalar quadratic-growth majorant. The gauge functional factorizes over wall components and the fermion covariance is a direct sum there; Gaussian integration of even elements therefore factorizes as well.

Use the same Mayer indices as in the scalar construction. Submultiplicativity gives $\|e^G - 1\| \le e^{\|G\|} - 1$. At every real B, the norm of a Mayer product is bounded by the corresponding product with potentials replaced by their norms. The positive conditioning majorant bounds its integral; scalar Cauchy estimates, applied through ℓ, bound each parameter bracket. Thus each local activity is bounded by the scalar majorant. Scalar test defects multiply its norm by $1 - \chi$ and have the same tail bound.

To justify the logarithm, let $z(\gamma)$ be the resulting commuting activities and let their scalar majorant satisfy

$$\sum_{\gamma' \nsim \gamma} \|z(\gamma')\| e^{a(\gamma')} \le a(\gamma).$$

The scalar connected-graph bound then bounds absolutely the series

$$S = \sum_{n\ge 1} \frac{1}{n!} \sum_{\gamma_1,\dots,\gamma_n} \rho^T(\gamma_1, \dots, \gamma_n) \prod_i z(\gamma_i).$$

The exponential identity for commuting formal variables is coefficientwise the identity $\Xi = e^S$ for the gas. Absolute convergence makes it an identity in the Banach algebra; $\Xi^{-1} = e^{-S}$. No assertion that a Grassmann-valued quantity is a positive number is used. Regroup the series by its union polymer to obtain the local logarithm. The local logarithm to be retained is

$$\widetilde{E}(X) = E[\{\mathcal{V}^{\mathrm{q}}|_{\mathrm{v}}, \mathcal{V}^{\mathrm{g}}, \mathcal{V}^{\natural}\}](X) - G_\tau(X).$$

Apply Lem. 7.4; in addition to the terms marked by the two nonquadratic origins, it includes the change of sort and the removal of the threshold from the quadratic origin. Its bound has no additive $|1-\tau|$ term.

Finally, polymers of size n through a cube number at most $2048 \cdot 9216^n$. The floor $4\beta_s\kappa \geq 2\log 18432 + 1$ in particular gives

$$\sum_{X \ni \square} e^{-(\kappa_+ - \kappa)d(X)} \leq 4096.$$

This proves the family bound. Uniform majorants give holomorphy. All operations use only data on the polymer, so they preserve locality and covariance. Since the radii enter only through the input bounds, outputs at two radii agree on their common domain. □

Lemma 7.6 (First variation of the expansion)**.** Use the setting and norms of Lem. 7.5, with $\varepsilon_V + \varepsilon_\natural < c_{\mathrm{cl}}$. Let $f = (f(Y))$ be even, local and holomorphic on the same pair and fluctuation domains, vanish at $(B, \bar{W}, W) = 0$, and have finite potential norm $\|f\| = \|f\|_{\kappa-1}$. At each pair, its cluster-origin norm satisfies $\|f\|_{\star,\lambda_V} \leq \|f\|$. On the disc $|\lambda|\|f\| < c_{\mathrm{cl}} - \varepsilon_V - \varepsilon_\natural$, the activity is holomorphic and

$$\|\widetilde{E}[\mathcal{V} + \lambda f](X) - \widetilde{E}[\mathcal{V}](X)\| \leq K_\Pi |\lambda| \|f\|_{\star,\lambda_V} e^{-\kappa_+ d_{k+1}(X)}.$$

Its derivative $E'[\mathcal{V}; f]$ is linear in f and satisfies

$$\|E'[\mathcal{V}; f](X)\| \leq K_\Pi \|f\|_{\star,\lambda_V} e^{-\kappa_+ d_{k+1}(X)}, \qquad \sup_{\square} \sum_{X \supset \square} e^{\kappa d_{k+1}(X)} \|E'[\mathcal{V}; f](X)\| \leq K_M \|f\|.$$

The same difference and derivative bounds hold when λf is added as a fourth tested origin to any admitted system of at most three tested origins, for $|\lambda|\|f\|_{\star,\lambda_V} < w_*$. The derivatives are holomorphic in any additional parameters for which the data are holomorphic with uniform bounds on an open set. ■

Proof. Keep λf as a separate origin. Mark every vertex belonging to it. The terms of the expansion without marked vertices are exactly those of the original system, with the same regions, tests and conditional integrals. Absolute convergence therefore identifies the difference of the expansions with the sum of terms containing a marked vertex. The marked-origin estimate of [3, Prop. 4.17], and its proof, transferred to the even Grassmann algebra by the contraction and duality argument of Prop. 7.2, bounds that sum by

$$K_\Pi |\lambda| \|f\|_{\star,\lambda_V} e^{-\kappa_+ d(X)}.$$

This argument reads only the sizes of the at most four origins. In particular it applies to the system consisting of the tested quadratic origin, $\mathcal{V}^g$, $\mathcal{V}^\natural$ and λf.

On the stated disc the last two origins can be joined into $\mathcal{V}^\natural + \lambda f$. The connected sum is unchanged by joining origins of the same sort, and the subtracted Gaussian family is independent of λ. This proves the first bound. Divide by $|\lambda|$ and let $\lambda \to 0$. Joint holomorphy in two insertion parameters proves linearity of the derivative. The polymer count in Lem. 7.5 gives

$$\sum_{X \supset \square} e^{-(\kappa_+ - \kappa)d_{k+1}(X)} \leq 4096, \qquad 4096 K_\Pi \leq K_M,$$

and proves the family bound. A Cauchy integral on a smaller insertion circle proves holomorphy in the additional parameters. □

7.2. **Relevant extraction and old activities.** The relevant coefficient is extracted from a translation-covariant total. Its individual polymer terms must nevertheless retain gauge invariance: the estimate of an old term will compare it with the trivial pair in a local gauge.

Lemma 7.7 (Joint gauge invariance)**.** At every level the local potentials, the pointed activities and their scalar parts are invariant under the joint action on the pair and the fermion generators. For an SU(2) site map u, writing α_u for the induced automorphism,

$$F(X;P^u)=\alpha_u F(X;P).$$

This holds polymer by polymer and, for a pointed family, pointer by pointer, whenever both pairs belong to its domain. ■

Proof. The block kernel and its completed quadratic form are gauge covariant. Their difference in Lem. 7.3 is therefore jointly invariant. The local inverses, walks and parameter brackets are covariant on their supports. The symmetric covariance preserves this covariance, and Gaussian integration commutes with α_u. Hence each Mayer term and each connected sum has the displayed property. In the normalizer the covariance is under a trace, giving scalar invariance. The operation $n^{-1}N_p$ used for pointing commutes with α_u, since it counts generators at a place. The action is extracted in its plaquette shares, each invariant, and the mass in its local bilinear, also invariant. These observations prove the assertion inductively, starting with the vertex and the normalizers. Taking degree zero gives strict scalar invariance of each term. □

Lemma 7.8 (The local mass coefficient)**.** Suppose a local even family E has norm at most ε and its total at the trivial pair is translation covariant, cubic covariant, and invariant under colour, $\mathrm{U}(N_f)\times\mathrm{U}(N_f)$ and charge conjugation. After reading its legs at their backgrounds, its quadratic kernel $\kappa(u)$ satisfies

$$\sum_u \|\kappa(u)\|_{\mathrm{tr}} e^{c_M|u|}\le C^{\psi}_{am}h_F^{-2}\varepsilon,\quad \sum_u \kappa(u)=\mathsf{cf}_m(E)1,\quad \sum_u u_\mu\kappa(u)=\mathsf{cf}_z(E)\gamma_\mu,$$

where $c_M=\min\{\alpha_D/4,\kappa/(c'M)\}$. In particular $|\mathsf{cf}_m(E)|,|\mathsf{cf}_z(E)|\le C^{\psi}_{am}(\nu h_F^2)^{-1}\varepsilon$. The coefficients are additive, holomorphic, and real at real data with the γ_4 real structure. ■

Proof. The degree-two fibre norm is the weighted trace norm of its matrix kernel: a singular-value decomposition gives one inequality, and testing against matrices of operator norm at most one gives the other. For a term $E(X)$, substitution of the background kernels consequently gives a kernel $\Sigma_X(y,y')$ with

$$\sum_{y,y'}\|\Sigma_X(y,y')\|_{\mathrm{tr}}e^{\alpha_D(d(y,X)+d(y',X))/2}\le Ch_F^{-2}\|E(X)\|.$$

The factors $C_HL^{-(j-n)}$ are precisely the leg weights and cancel. Sum over one cell of the translation-covariant total. The inequality $|y-y'|\le d(y,X)+\operatorname{diam}X+d(y',X)$ and $\operatorname{diam}X\le c'M(1+d_j(X))/2$ pays its exponential with the background and polymer rates. The remaining sums are bounded uniformly, proving the kernel bound.

The common flavour phase rules out $\Psi\Psi$ and $\bar\Psi\bar\Psi$. Colour invariance makes the kernel the identity in colour. Independent flavour rotations on the two τ_3 eigenspaces give $a_+(u)\otimes e_+ + a_-(u)\otimes e_-$, with identity in flavour. The binary octahedral action on spin has a scalar part and one vector part. Its zeroth moment is therefore $\mu_+e_+ + \mu_-e_-$ and its first moments are $a_+\sigma_\mu e_+ + a_-\sigma_\mu e_-$. Inversion has lift $1\otimes\tau_2$, exchanges the eigenspaces and reverses u; hence $\mu_+=\mu_-$ and $a_+=-a_-$. This is the asserted form. The γ_4 conjugation fixes the scalar zeroth moment and the first moment because $\gamma_4\gamma_\mu\gamma_4=-\gamma_\mu$; their coefficients are real. The trace norm of $m1$ is $\nu|m|$. The coefficient formulas and local uniform convergence prove the remaining assertions. □

At the trivial field put $\eta_j=\theta_j/(\theta_j+\mathfrak{m}_j)$ and $\mathfrak{s}_j=\eta_j^2$. The background of a constant block field c is $L^{-j}\eta_j\iota c$. Thus the zeroth moment of $L^{-j}\mathsf{S}_1(\psi^{\mathrm{bg}})$ is η_j^2. At small block momentum the light background symbol is $\lambda(p)=\theta_j(\theta_j+\mathfrak{m}_j+i\gamma\cdot p)^{-1}+O(|p|^2)$. The heavy components and nonzero folded momenta contribute only $O(|p|^2)$. Since the barred map is the row twin, its symbol is also $\lambda(p)$, not $\lambda(p)^*$. The first moment of the mass form is consequently $-2\eta_j^2\gamma_\mu/(\theta_j+\mathfrak{m}_j)$; that of the

bare kinetic form is $\eta_j^2\gamma_\mu$. The zeroth moment of the latter is zero. If $\eta_j \ge c_b > 0$, removing a mass coefficient

$$\mu = -\mathsf{cf}_m(E)/\eta_j^2$$

makes the remaining quadratic zeroth moment vanish, with norm cost at most

$$K_\mu = 1 + c_{\mathsf{S}} C_{am}^{\psi} c_b^{-2}/\nu, \qquad c_{\mathsf{S}} = 48M^3\nu C_H^2.$$

Indeed a cube contains $48M^3L^{3j}$ bare places, each mass bilinear has norm at most $\nu(C_H L^{-j}h_F)^2$, and its coefficient is $L^{-j}\mu$; the powers of L cancel. We do not remove a wave-function coefficient.

Lemma 7.9 (Reading an old activity)**.** Let a level-j family F be holomorphic on the whole pair core, depend on both members only through their pair, and be jointly gauge invariant there. Suppose its Grassmann degree-zero part $F_0(X;P)$ vanishes for every polymer X and every pair P in its core. Suppose also that it is covariant under its cube translations, has norm at most ε_j and has vanishing quadratic zeroth moment at the trivial pair. With $n = k - j$, its local reading at level k has norm at most

$$\omega_n\varepsilon_j, \qquad \omega_n = C_7 L^{n/4}.$$

Its degree at least four and the gauge-dependent part of its quadratic term have norm at most $C_7L^{-n}\varepsilon_j$. The constants are uniform on the constant pair-radius systems. For matrix sources the same conclusion holds whenever the quadratic kernel admits cell units of total mass zero and uniform first absolute moment as in the proof below. ■

Proof. There is no scalar term by hypothesis. A leg read n levels later costs L^{-n}, whereas a cube contains L^{3n} old cells. Thus degree $2p \ge 4$ costs $L^{-(2p-3)n}$. Take n_0 with $2(2A_\kappa C_H)^2 L^{-n_0} \le 1$. At $n \ge n_0$ expand each leg first to level j and then to level k by its local walks. Each stage costs at most $2A_\kappa C_H$; the extra factors beyond the fourth are bounded by a geometric factor $2^{-(2p-4)}$, which also pays polynomials in the number of legs. The finitely many lags below n_0 are absorbed in C_7.

The quadratic comparison needs two powers of L^{-n} from the gauge field. Fix a polymer X at a cell centre z_c, and fix a local or walk term in each of its two legs. Put

$$D = M(1 + d_j(X)), \qquad D' = D + 9M_w \sum_i (|\omega_i'| + 1).$$

Join each column to z_c through its walk cubes and a tree of cubes of X. Such paths have length at most $C_t D'$; one can take $C_t = 9$ for the tree paths. Hold the transported column vectors at z_c fixed and call the resulting coefficient $\mathcal{E}(P)$. Both the coefficient of $F(X)$ and the first-stage walks vary with the pair. By Lem. 6.10, $\mathcal{E}$ is holomorphic on the open set $\mathcal{O}$ where the restricted pair lies in its core and the first member on the walk support has a small logarithm. It is invariant under real site gauges fixing z_c and has a uniform bound $B_{\mathcal{E}}$ there.

This invariance extends along complex gauge paths staying in $\mathcal{O}$. The derivative at zero of $\mathcal{E}(P^{e^{\epsilon\xi}})$ is complex linear in ξ. It vanishes for the real gauge Lie algebra and hence for its complexification. Along a differentiable path of complex gauges fixing z_c, the derivative is therefore zero whenever the transformed pair stays in $\mathcal{O}$.

Let $s_Q = \min\{M, \mathsf{o}LM\}$, using the cube size allowed by the regular-pair core. If $2(C_tD'+M+2) > s_Q L^n$, the trivial bound $2B_{\mathcal{E}}$ is at most $2B_{\mathcal{E}}(8C_t/s_Q)^2 L^{-2n} D'^2$. Otherwise a cube Q about z_c, of side at most s_Q in level-k units, contains the coefficient bonds, the walk supports and their needed neighbours. Its regular real gauge, normalized to be one at z_c, gives in level-j units, with $\eta_j = L^{-j}$,

$$P = (e^{i\eta_j A_Q}, J_Q), \qquad |A_Q| \le C_{\mathrm{cube}} L^{-n}, \quad |\nabla^{\eta_j} A_Q| \le C_{\mathrm{cube}} L^{-2n}, \quad |J_Q| \le C_{\mathrm{cube}} L^{-3n}.$$

These bounds follow by taking the logarithm of the three factors supplied by the shift, the complex displacement and the real regular gauge. The logarithm is Lipschitz on the admitted ball. Converting a covariant difference to a plain difference adds a product of connection and displacement sizes, which has the required L^{-2n}.

Let ρ_A be the logarithmic radius whose pairs lie in $\mathcal{O}$. The radial disc through (A_Q, J_Q) first gives $2B_{\mathcal{E}}C_{\mathrm{cube}}L^{-n}/\rho_A$ by Schwarz's lemma; if its radius is below one use $2B_{\mathcal{E}}$. For large n remove the connection at the centre. Set $A^c_\nu = A_Q(\langle z_c, z_c + \eta_j e_\nu\rangle)$ and

$$g_v(x) = h_3 h_2 h_1, \qquad h_\nu = \exp(iv(x - z_c)_\nu A^c_\nu), \qquad 0 \le v \le 1.$$

The matrices need not commute, and this order is kept. Writing $g^>_\mu, g^<_\mu$ for the products before and after h_μ, the connection on a μ-bond becomes

$$A_{Q,v} = \mathrm{Ad}(g^>_\mu h_\mu)\mathsf{S}(\mathrm{Ad}(g^<_\mu)A_Q, -vA^c_\mu), \qquad e^{i\eta_j \mathsf{S}(A,B)} = e^{i\eta_j A}e^{i\eta_j B}.$$

Each product of the $h_\nu^{\pm 1}$ is bounded by $e^{3C_{\mathrm{cube}}M/2}$. The Lipschitz estimate for S gives a constant C_{lg} such that

$$|A_{Q,v}| \le C_{\mathrm{lg}}L^{-n}, \quad |\nabla^{\eta_j} A_{Q,v}| \le C_{\mathrm{lg}}L^{-2n}, \quad |\,\mathrm{Ad}(g_v)J_Q| \le C_{\mathrm{lg}}L^{-3n}.$$

In the difference estimate, differentiating a conjugating factor contributes the product of two L^{-n} factors. At $v = 1$, $\mathsf{S}(A^c_\mu, -A^c_\mu) = 0$; subtracting the centre value gives $|A_{Q,1}(b)| \le C_{\mathrm{lg}}L^{-2n}|x - z_c|_1$. Choose n_{lg} so $C_{\mathrm{lg}}L^{-n_{\mathrm{lg}}} \le \rho_A$. For $n \ge n_{\mathrm{lg}}$ every transformed pair remains in $\mathcal{O}$. On the bonds read by the term, $|x - z_c|_1 \le 6C_tD'$, so the endpoint admits a radial disc of radius at least $\rho_A/(6C_{\mathrm{lg}}C_tL^{-2n}D')$. The gauge invariance along the path and Schwarz's lemma on this disc give

$$\|\mathcal{E}(P) - \mathcal{E}(1,0)\| \le 2B_{\mathcal{E}}C_{\mathrm{gf}}L^{-2n}D'^2.$$

The trivial bound covers radii below one. The finitely many intermediate lags use the first radial estimate and $L^{-n} \le L^{n_{\mathrm{lg}}}L^{-2n}$.

To sum these bounds, fix each walk simultaneously at all rows of X. A walk term vanishes at rows for which its first cube is not an allowed first cube; for a barred leg use the last cube of its row twin. The rows in a level-j block have the same allowed first cubes, at most C_{nb} of them. Consequently the number of possible first cubes, and likewise of last cubes, is at most

$$N_X = C_{\mathrm{nb}}M^3 \# X \le 8C_{\mathrm{nb}}M^3(1 + d_j(X)).$$

For a fixed first cube the positive weighted walk sum is at most A_κ. Including the one local term, the sum of the suprema of the row bounds for one leg at level $n' \le j$ is therefore at most

$$2C_HL^{-(j-n')}(1 + N_XA_\kappa) \le 2C_HL^{-(j-n')}(1 + 8C_{\mathrm{nb}}M^3A_\kappa)(1 + d_j(X)).$$

Apply this to the two legs in $B_{\mathcal{E}}$. Absolute convergence and bilinearity identify their sum with the quadratic activity. The additional factor $(1 + d_j(X))^2$ must be retained when summing the bounds of its individual holomorphic terms. Together with the path factor D^2, it costs at most

$$M^2 \sup_{d \ge 0}(1 + d)^4 e^{-7\kappa d/8} < \infty.$$

This number and $(1 + 8C_{\mathrm{nb}}M^3A_\kappa)^2$ are included in C_7 and the later old-activity constants, before choosing the fibre weight. Higher-degree terms are evaluated at one pair and retain the ordinary row walk count.

The two legs contribute L^{-2n}, the gauge comparison L^{-2n}, and the cell count L^{3n}, giving $CL^{-n}\varepsilon_j$ per new cube. Enlarge the filing polymer by the at most eight level-k cubes meeting Q. They lie in a cube of side $2M$ and cost the fixed factor $e^{\kappa c_0}$. The polynomials in D' are paid by the old polymer rate and the walk weights $e^{|\omega'|}$. The complex transports are paid by the same rates under $2C_t\alpha_1M \le \kappa$ and $18C_t\alpha_1M_w \le 1$.

It remains to estimate the trivial-pair quadratic form. Write its kernel as units $\beta_u(y, y')$ filed at a cell $c(u)$, with

$$\sum_{u:c(u)=c}\sum_{y,y'} \|\beta_u(y, y')\|(1 + |y - z_c| + |y' - z_c|) \le Ch_F^{-2}\varepsilon_j, \qquad \sum_{u:c(u)=c}\sum_{y,y'} \beta_u(y, y') = 0.$$

At zero sources these are the polymer kernels; translation covariance makes the last sum equal to the vanishing mass $M^3\mathsf{cf}_m(E)1$. At sources the stated cell-unit condition is precisely what is used. For each unit, carry both legs along its paths to z_c and write $\Phi(y) = \Phi(z_c) + [\Phi(y) - \Phi(z_c)]$. The reference value of a first-stage leg is its row sum at the trivial configuration times $\Phi(z_c)$. Its remainder is a sum of transported differences. Their paths use straight bare transports, whereas the kernel estimate uses the iterated block field. At neighbours x, x' one must include both terms in

$$\Phi(x) - \mathsf{U}([x,x'])\Phi(x') = \Phi(x) - (\mathcal{M}^j\mathsf{U})(x,x')\Phi(x') + [(\mathcal{M}^j\mathsf{U})(x,x') - \mathsf{U}([x,x'])]\Phi(x').$$

The complete-average comparison bounds the last transport defect by $2(c_T\epsilon_* L^{-2n} + C_Q\alpha_1 L^{1-n})$. Here the block-centre kernel differs from one by at most $C_Q\alpha_1 L^{-n} + 14\epsilon_*^2 L^{-4n}$, so that comparison of the normalized kernels also compares the two transports. Multiplication by the leg size $2C_H L^{-n}$ gives the bound $4C_H(c_T\epsilon_* + C_Q L\alpha_1)L^{-(1+\alpha)n}$ for every $\alpha < 1$. Thus the extra term has the required Hölder gain. The terms without differences cancel cell by cell before localization. A term with a difference gains $L^{-3n/4}$ by the background estimate with $\alpha = 3/4$; the other leg costs L^{-n}. Its cell bound is $C\varepsilon_j L^{-11n/4}$, hence after L^{3n} cells it is $C\varepsilon_j L^{n/4}$. The first absolute moment pays the path length. Moving a diagonal unit to the cell of its site changes a new cube sum only by units near its boundary; their number $O(L^{2n})$ and leg cost L^{-2n} give $C\varepsilon_j$.

Localize these differences by the walk bounds with their difference gain. For completeness, the exact hull inequality for the tree size is $d_{j+1}(X^{\mathrm{up}}) \le L^{-1}d_j(X)$, where X^{up} is the union of coarse cubes containing a fine cube of X. In coordinates with fine corners in $\mathbb{Z}^3$ and coarse corners in $L\mathbb{Z}^3$, round $Lq + t$, $0 \le t < L$, to Lq if $t \le \sigma$ and to $L(q+1)$ otherwise, choosing $\sigma \in \{0, \dots, L-1\}$ in each coordinate. Every lattice point of a closed coarse cube is mapped to one of its corners. Thus the image of the tree is connected, uses edges of coarse cubes in X^{up}, and meets each of them: a vertex meeting a fine closed cube maps to a corner of its containing coarse cube. A coordinate edge survives for exactly one cut position. The mean number of image edges is at most L^{-1} times the original count; some choice does no worse, and a spanning tree has no more edges. Iterating gives $d_k(X^{\mathrm{up}}) \le L^{-n}d_j(X)$. Add the walk polymers one at a time and use the positive parts of their size increments; size need not be monotone under enlargement. Their weighted sums cost A_κ per leg. The spare old rate $\kappa(1 - L^{-n})d_j(X)$ pays all remaining powers of the diameter, and $e^{|\omega|}$ pays the path-length polynomials and the recounting of cubes reached by a walk. This proves the asserted local family norm.

On a torus the paths include the antiperiodic seam. A nonwrapping piece and its lift have the same cell number, because the seam signs at the two reference legs cancel. The torus cell sum minus its infinite-lattice value is therefore the sum of wrapping pieces minus that of infinite-lattice pieces that do not fit. A piece in either sum has $d_j(Z_p) \ge \ell_j/(2M) - 2$. File a wrapping piece at its coarse hull and a nonfitting piece at the image of its coarse hull; both wrap. Their two remaining legs and the count of cells give

$$C\varepsilon_j L^n \sum_{d \ge \ell_j/(2M)-2} (1+d)^3 e^{-\kappa(1-L^{-n})d} \le C\varepsilon_j,$$

since $\ell_j/(2M) \ge L^n$. On a nonwrapping output polymer every retained term is the image of its lifted term, so the local constructions agree there. The argument at matrix sources uses the stated cell units and their first absolute moments. On a smaller constant-radius system, $C_{\mathrm{cube}}, C_{\mathrm{lg}}$ and ρ_A scale by at most a/α_0; their ratios and the lag threshold are no larger. This proves uniformity on those systems. □

Lemma 7.10 (Changing the Grassmann weight)**.** Let d_λ, $\lambda > 0$, multiply every barred and unbarred generator, including every formal background generator, by λ. For any positive weight w,

$$\|\mathsf{d}_\lambda F\|_w = \|F\|_{\lambda w}.$$

This identity also holds for the family norms. The map d_λ commutes with background substitution, its local and walk terms, transport, gauge transformations, coefficient evaluation and filing. It preserves locality, holomorphy and the cell-unit cancellation used in Lem. 7.9.

Consequently Lem. 7.9 holds at every weight $h > 0$ in place of h_F, with the same C_7. The bound on its quadratic cell units then has h^{-2} in place of h_F^{-2}; the norm of the family and the weights on its later generators are both formed using h. ■

Proof. The map d_λ carries a decomposition in the definition of the fibre norm bijectively to a decomposition whose degree-p coefficient is multiplied by λ^p. Its cost at w is the original cost at λw. Taking infima proves the identity. Taking the sums and suprema defining the family norms preserves it.

Each listed operation acts on coefficients or substitutes linear combinations for generators, and hence commutes with d_λ. Quadratic kernels, moved diagonal terms and cell sums are multiplied by λ^2. Their zero-sum conditions are preserved. Coefficients are only multiplied by constants, which preserves holomorphy and gauge covariance.

For a family measured at h, apply Lem. 7.9 to $\mathsf{d}_{h/h_F}F$ at h_F. Its norm is the norm of F at h, and its quadratic first-moment bound has the required factor

$$(h/h_F)^2 h^{-2} = h_F^{-2}.$$

All terms in that lemma's local decomposition commute with this scaling. Their norms at h_F, including the norms with $C_H h_F$ on later generators, are therefore exactly the corresponding norms at h and $C_H h$ for F. The same argument applies to the difference between torus and infinite-lattice cell sums, and at each fixed pair radius. The coefficient estimate in Lem. 7.8 supplies the required first-moment bound at h: its two background factors cancel against the two leg weights in precisely the same way. □

The physical flow starts with zero mass correction. The estimates in independent coefficient and source coordinates also use nearby data with a nonzero correction at the first step. We record that step explicitly before estimating the full potential. Its mass insertion requires no walk expansion: after the fermion shift it is a bilinear at one bare site.

Lemma 7.11 (The mass insertion at the first step)**.** Take the data of the local step at $k = 0$, at any real or complex action coefficient for which those data are formed. The fine mass may be the mass of the datum or an admitted matrix mass as in Lem. 6.5; form the background maps at that same mass. Let $N(x)$ act on spin, the internal doublet and flavour, and as the identity on colour. Write

$$n_N = \sup_x \|N(x)\|, \qquad \psi^{\mathrm{bg}} = \mathsf{H}_0(\mathsf{U})\Psi_1, \qquad \bar\psi^{\mathrm{bg}} = \bar\Psi_1 \bar{\mathsf{H}}_0(\mathsf{U}).$$

In the local polynomial algebra these backgrounds are formal legs of weight $h^{(1)}(0) = C_H L^{-1} h_F$. Add $-\mathsf{S}_1(N\psi) = -\sum_v \bar\psi(v) N(x_v)\psi(v)$ to the bare exponent. Then the following assertions hold.

(i) The substitutions of the first step give

$$-\mathsf{S}_1(N\psi) = -\mathsf{S}_1(N\psi^{\mathrm{bg}}) + \sum_v \mathcal{V}_0^{\mathrm{s}}[N](v), \tag{30}$$

where

$$\begin{aligned} \mathcal{V}_0^{\mathrm{s}}[N](v) = {} & -\bar\psi^{\mathrm{bg}}(v) N(x_v) W(v) - \bar W(v) N(x_v)\psi^{\mathrm{bg}}(v) \\ & - \bar W(v) N(x_v) W(v). \end{aligned} \tag{31}$$

The retained term is the level-one mass term $-L^{-1}\mathsf{S}_1(LN\mathsf{H}_0(\mathsf{U})\Psi_1)$.

(ii) File the term at v in the level-zero cube containing x_v. The resulting family is even, is supported on single cubes, and vanishes at $\bar W = W = 0$. Its coefficients are linear in N and independent of B, its pivot completion and the background pair. Thus holomorphic dependence of N on parameters gives holomorphic dependence of this family.

(iii) Each term is jointly gauge invariant. Every model symmetry or conjugation that carries the bare insertion with N to the bare insertion with N' carries $\mathcal{V}_0^{\mathrm{s}}[N]$ to $\mathcal{V}_0^{\mathrm{s}}[N']$, with its places transported as well. For constant $N = s1$, this is the covariance of the scalar mass term.

(iv) For either fluctuation weight $h_W = \Gamma_F$ or $2\Gamma_F$, the two summed norms of Lem. 7.5 satisfy

$$\|\mathcal{V}_0^{\mathrm{s}}[N]\|_\kappa, \quad \|\mathcal{V}_0^{\mathrm{s}}[N]\|_{\kappa-1} \le \tfrac{1}{9} c_\psi c_{\mathsf{S}} h_F^2 n_N. \tag{32}$$

The bound holds at every fluctuation and on every admitted pair radius. It also holds with any $h \ge h_F$ in place of h_F, with $c_\psi(h) = 2A_\kappa \Gamma_F / h$ and background weight $C_H L^{-1} h$.

(v) Under a centered Grassmann Gaussian functional μ_S, for any covariance S, the mean of each cube term has degree zero in the remaining generators. The family is independent of B, so it may be used as the mass origin in the four-variable expansion below. With $c_\diamond = c_{\mathrm{cl}}/4$ and the mass-origin constant $c'_\psi \ge c_\psi$, its radius is

$$r_4 = \frac{c_\diamond}{c'_\psi c_{\mathsf{S}} h_F^2 n_N}, \qquad r_4 = \infty \quad \text{if } n_N = 0. \tag{33}$$

For $|z_4| \le r_4$, the family $z_4 \mathcal{V}_0^{\mathrm{s}}[N]$ has norm at most $c_\diamond/9$. On the coefficient polydisc of the local step, at the fine mass of the datum, the cluster estimate of Lem. 7.5 and the mass-origin coefficient estimates in the proof of Prop. 8.6 use, under their remaining hypotheses, n_N in place of $|s_0|$. The window $n_N \le 2\bar{B}_M/\mathsf{t}_0$ gives the sharp mass-origin radius, whereas $n_N \le A\phi(x_0)$ gives the radius on the corresponding coefficient disc. These assertions require the other step data to lie in the domains of those estimates.

(vi) For $N = s1$ at the fine mass of the datum and within the stated coefficient and mass windows, the same first-step substitution also preserves the symmetries needed for scalar mass extraction in Lem. 7.8. The coefficient quadratic in the mass origin has the bound for the $|s|^2$ contribution in the proof of Prop. 8.6. No scalar zeroth-moment conclusion is asserted here for a general matrix N.

■

Proof. At level zero the map from the bare field to itself is the identity, and the map from level one to the bare background is H_0. The site factor is also the identity. Consequently the fermion substitution is simply $\psi = \psi^{\mathrm{bg}} + W$ and $\bar\psi = \bar\psi^{\mathrm{bg}} + \bar W$. The gauge chart changes no coefficient of the bare mass bilinear. Expanding that bilinear gives (30) and (31); the retained normalization follows from $L^{-1}(LN) = N$. This calculation uses neither the action coefficient nor the value of the admitted fine mass.

The three terms at a place read only its two formal background legs, its fluctuation legs and $N(x_v)$. Their coefficients are entries of $N(x_v)$. This proves (ii). Independence of the pair here is a statement about those polynomial coefficients. When a formal leg is subsequently evaluated as $\mathsf{H}_0(\mathsf{U})\Psi_1$, that evaluated leg still depends on U.

For the joint gauge action of Lem. 7.7, unbarred legs at v are multiplied by $u(x_v)^{-1}$ and barred legs by $u(x_v)$. These act on colour alone and commute with $N(x_v)$, so each bilinear in (31) is invariant. The other model operations act in the same way on a formal background leg and a fluctuation leg at the same place, including the conjugation. Applying such an operation to the expansion of the bare bilinear is therefore the expansion of the transformed bilinear. This proves (iii), without requiring a fixed arbitrary N to be invariant under every symmetry.

For the norm estimate, the fibre norm of $\bar a(v) N(x_v) b(v)$ is at most $\nu \|N(x_v)\| h_a h_b$. Indeed its matrix has trace norm at most $\nu\|N(x_v)\|$; a singular-value decomposition and the defining norm in Lem. 5.1 give the bound. There are $48M^3$ places in a level-zero cube. Since its tree size is zero and no walk occurs, neither a polymer weight nor an additional cube count is needed. Thus, for both rates,

$$\|\mathcal{V}_0^{\mathrm{s}}[N]\| \le 48 M^3 \nu n_N (2 C_H L^{-1} h_F h_W + h_W^2). \tag{34}$$

On the other hand,

$$c_\psi c_{\mathsf{S}} h_F^2 = 2A_\kappa C_H^2 (48M^3\nu)\Gamma_F h_F.$$

The standing choices give $\Gamma_F/h_F \le (2A_\kappa)^{-1}$, $A_\kappa \ge 2$, $C_H \ge 2$ and $L \ge 13$. Dividing the right side of (34) by $c_\psi c_{\mathsf{S}} h_F^2 n_N$, first for $h_W = \Gamma_F$ and then for $h_W = 2\Gamma_F$, gives respectively

$$\frac{1}{A_\kappa C_H L} + \frac{\Gamma_F}{2A_\kappa C_H^2 h_F} \le \frac{1}{52} + \frac{1}{64} < \frac{1}{9},$$

and

$$\frac{2}{A_\kappa C_H L} + \frac{2\Gamma_F}{A_\kappa C_H^2 h_F} \le \frac{1}{26} + \frac{1}{16} < \frac{1}{9}.$$

The zero insertion is immediate. Increasing h_F to h only decreases the quotient Γ_F/h, proving the last assertion of (iv). The estimate reads no gauge fluctuation or pair and therefore holds on all their admitted domains.

Centered Gaussian integration kills the terms linear in W or $\bar W$. It contracts the remaining $\bar W N W$ term to a number, proving the assertion about the mean. Multiplication by $|z_4| \le r_4$ in (32) gives $\|z_4 \mathcal{V}_0^{\mathrm{s}}[N]\| \le c_\diamond/9$. The other requirements of the joined expansion are also already verified: this is an even local family, holomorphic in the pair and fluctuation, zero when all fluctuations vanish, and jointly covariant with N. Hence Lem. 7.5 applies with the same mass-origin allowance. The window enters only through $r_4^{-1} = c'_\psi c_{\mathsf{S}} h_F^2 n_N / c_\diamond$.

For the coefficient cancellations used later, keep the fine mass of the datum and the coefficient polydisc specified in (v). At zero gauge and old-activity origins, the mass origin is independent of B. Its coefficients with one mass origin and any power of the scalar gauge origin have no part of degree at least two in the remaining generators. The coefficient with one mass origin and no other origin is its Gaussian mean, which is scalar. The coefficient with one gauge-fluctuation origin and one mass origin vanishes by evenness in B. The coefficient with two mass origins has quadratic part, up to sign,

$$\langle \bar\psi^{\mathrm{bg}}, N\Gamma_0^{-1} N \psi^{\mathrm{bg}} \rangle.$$

The same Cauchy estimate for the local logarithm bounds it by $2K_M c_{\mathrm{cl}} r_4^{-2}$. This proves the first-step coefficient claims in (v).

When $N = s1$, the insertion has the same model symmetries as the original scalar mass. The one-loop coefficient uses no mass origin; the coefficients with one mass origin just considered vanish in degree at least two. Apart from the displayed two-mass coefficient, the remaining mass-dependent coefficients have weighted degree at least three and are bounded by the same origin count. In the sharp window, r_4^{-1} has order $\phi(\mathsf{t}_0)^2$. The scalar moment estimate of Lem. 7.8 applies to the covariant total and bounds the two-mass contribution by the stated constant times $|s|^2$. This is the part of the argument that needs scalar N and the fine mass and coefficient windows specified in (vi). □

The potential of a complete step consists of the gauge vertex and the variations of the old scalar terms, together with

$$\mathcal{V}^\natural = \mathcal{V}_k^{\mathrm{F}} + \mathcal{V}^{\mathrm{s}} + \mathcal{V}_{\ge 2}^{\mathrm{old}}.$$

At $k = 0$ the mass term in this potential is $\mathcal{V}_0^{\mathrm{s}}[N]$ of Lem. 7.11, with $N = s_0 1$ for a scalar mass correction. It vanishes at the physical start $s_0 = 0$; at a nonzero independent mass coordinate it is retained. All terms vanish at $(B, \bar W, W) = 0$, although the fermion part need not vanish at $B = 0$. At the fixed pair radii, with $2a_{\mathrm{an}} = \mathsf{c}_{\mathrm{os}}\alpha_0$, put $\mathsf{s}_a = c_h \delta_k / a_{\mathrm{an}}$. Apply Lem. 7.9 to $E_{\ge 2}^{(j)}$, whose degree-zero part vanishes identically. The preceding bounds, Cauchy's estimate for an old coefficient in the shift, and the fluctuation-leg room give

$$\|\mathcal{V}_k^{\mathrm{F}}\| \le C_V h_F^2 \delta_k, \quad \|\mathcal{V}^{\mathrm{s}}\| \le (c_\psi + C_G \delta_k) c_{\mathsf{S}} h_F^2 |s_k|,$$

$$\|\mathcal{V}_{\geq 2}^{\mathrm{old}}\| \leq (c_\psi + \mathsf{s}_a + C_G\delta_k)\sum_{j\leq k}\omega_{k+1-j}\varepsilon_j.$$

To obtain the factor c_ψ, expand a homogeneous product under $a+b$. If $h_a + h'_b \leq h$ and $h_b \leq h'_b$, its part with at least one b has norm at most $(h_b/h'_b)\|F\|_h$; this is the convexity inequality $(h_a+h_b)^n - h_a^n \leq (h_b/h'_b)[(h_a+h'_b)^n - h_a^n]$. Apply it to the localized W images. The pure scalar localization gives $|\mathcal{V}_0^{\mathrm{old}}| \leq \mathsf{s}_a K_{\mathrm{old}}$ with $K_{\mathrm{old}} \leq C(E_0 + C_N N_f + B_\sharp + 1)$. At radius a, replace s_a for an old fermion coefficient by $(\alpha_0/a)\mathsf{s}_a$; an explicit old scalar family keeps its fixed-radius bound, while its fluctuation family has factor $C_{\mathcal{A}}\delta_k/a$. Admissibility gives $(\alpha_0/a)\mathsf{s}_a \leq 1/9$. Thus these estimates also hold on each admissible constant radius system. They supply the input sizes for Lem. 7.5. Their simultaneous use in the matter induction is verified in Prop. 10.20.

8. The cumulative mass term

We must control the mass terms accumulated over all preceding scales. The three-square identity first gives the sign of each Gaussian slice. We then bound their sum and identify it with the entire second-order contribution of the interacting step. Finally we estimate the cubic remainder and close the forward mass recursion. These are separate parts of the argument; positivity of the slices alone does not bound their sum.

Lemma 8.1 (The three-square identity)**.** Let $m \geq 0$, $a = \mathfrak{a}_k$, $R = R_k(U)$, $P = R^*R$, $\eta = a/(a+m)$ and $\varphi = a(D[U] + m + aP)^{-1}R^*\chi$, whenever the inverse exists. Then

$$\mathrm{Re}\langle\chi, \mathsf{D}^{(k)}[U;m]\chi\rangle - \frac{am}{a+m}\|\chi\|^2 = (a+m)\|R\varphi - \eta\chi\|^2 + m\|(1-P)\varphi\|^2 + \langle\varphi, \Delta\varphi\rangle.$$

The right side is nonnegative and vanishes at the flat field on constant chamber-independent fields. Consequently, for constant chamber-independent χ, the coefficient of order two in a real gauge fluctuation about the flat field, and its mean under every centered Gaussian slice, are nonnegative. ■

Proof. The defining equation is $(D+m)\varphi = aR^*(\chi - R\varphi)$ and $\mathsf{D}^{(k)}\chi = a(\chi - R\varphi)$. Therefore

$$\mathrm{Re}\langle\chi, \mathsf{D}^{(k)}\chi\rangle = a\|\chi - R\varphi\|^2 + m\|\varphi\|^2 + \langle\varphi, \Delta\varphi\rangle.$$

Use $RR^* = 1$ and $\|\varphi\|^2 = \|R\varphi\|^2 + \|(1-P)\varphi\|^2$. Completing the remaining square uses

$$a\|\chi - u\|^2 + m\|u\|^2 = (a+m)\|u - \eta\chi\|^2 + \frac{am}{a+m}\|\chi\|^2.$$

At the flat field $D[1]R^*\chi = 0$ for constant χ, so $\varphi = \eta R^*\chi$ and all three terms vanish. Each square vanishes at zero fluctuation; its quadratic coefficient is a sum of squares of first derivatives. Gaussian averaging preserves this nonnegativity. □

For a bare mass m, put

$$\mathsf{r}_k(U;m)[\chi] = \eta^{-2}\left(\mathrm{Re}\langle\chi, \mathsf{D}^{(k)}[U;m]\chi\rangle - \frac{\mathfrak{a}_k m}{\mathfrak{a}_k + m}\|\chi\|^2\right), \qquad \eta = \mathfrak{a}_k/(\mathfrak{a}_k + m).$$

Let $\Phi_k(A;m)$ be its coefficient of t^2 at $U = e^{itA}$ on constant block fields, per block site and with normalized trace over their fibre. If $\gamma_{k,i}$ is the unit-curvature slice defined below, set

$$\sigma_i(k;m) = \int \Phi_k(A; L^{-k}m)\,\gamma_{k,i}(dA),$$

so the argument of σ_i is in level-k mass units. For the label mass m let $\mathfrak{m}_k = L^{k-K}\mathsf{m}$ and set

$$F(k;m) = \sum_{i<k}\sigma_i(k;m), \qquad G_k(x) = \sum_{i<k}\frac{L^{k-i}}{x_i}\sigma_i(k;\mathfrak{m}_k), \qquad \mu_k^{1\mathrm{L}}(x) = G_k(x) - LG_{k-1}(x), \quad G_0 = 0.$$

The boundedness of F, and the fact that the entire second order of the interacting step is $\mu_k^{1\mathrm{L}}$, are separate estimates; the three-square identity alone does not prove them.

8.1. **A bound for the sum of the gauge slices.** For chamber-wall types $j \in \{0,1,2,3\}$ put $\varrho_j = rt_j$ and $\nabla_j[U] = 1 - R_j[U]$. Thus $\langle f, \Delta f\rangle = \frac{1}{2}\sum_j \varrho_j \|\nabla_j[U]f\|^2$; this chamber difference is distinct from the site-bond difference. Write $\mathbf{Q}_k^s$ for the composite straight gauge average and Q_k' for the composite site mean. The linearized gauge block map has the form

$$\mathbf{Q}_k^{\mathrm{lin}} = \mathbf{Q}_k^s - d\bar{n}_k, \qquad \mathbf{Q}_k^{\mathrm{lin}} d\lambda = d(\lambda|_{L^k\mathbb{Z}^3}).$$

This follows directly by linearizing the contour averages: their first variation on a constant spinor is its contour mean $i\bar{n}_k(A)$. Put $T_k = \ker \mathbf{Q}_k^{\mathrm{lin}}$ and $\mathfrak{G}_k = \{d\lambda : \lambda|_{L^k\mathbb{Z}^3} = 0\}$. The gauge Gaussian slice i has curvature covariance $\mathcal{C}_{(i+1)} = P_{i+1} - P_i$, where P_i projects onto the closed curvatures whose level-i plaquette average vanishes. These are the linearized, zero-background gauge inputs used here. They concern the pure Gaussian construction, not matter estimates.

The quadratic coefficient of the three squares can be written explicitly. Write $P_k^{\mathrm{F}} = R_k^* R_k$ for the fermion projection, distinct from the curvature projection P_k. At the flat field let $\chi_c = R_k^* c$, $\kappa_1 = \partial_t R_k(e^{itA})^* c|_{t=0}$, $V_A = \partial_t D[e^{itA}]|_{t=0}$, and $\nabla_j^{(1)} = \partial_t \nabla_j[e^{itA}]|_{t=0}$. Choose the representative with $\bar{n}_k(A) = 0$ and set $\zeta = \mathcal{K}_k[1;m]^{-1}(V_A\chi_c - m\kappa_1)$. Differentiating the three squares, whose constant terms vanish, gives

$$\Phi_k(A;m) = \left\langle (\mathfrak{a}_k + m)|R_k\zeta|^2 + m|(1-P_k^{\mathrm{F}})\zeta + \kappa_1|^2 + \sum_j \frac{\varrho_j}{2}|\nabla_j^{(1)}\chi_c - \nabla_j\zeta|^2 \right\rangle_c .$$

Here $\langle\cdot\rangle_c$ is normalized trace over constant block spinors per block. For the proof, the exact identity $\varphi_t = \eta(1-\mathcal{K}_t^{-1}D_t)R_t^* c$ gives $R_t\varphi_t - \eta c = -t\eta R\zeta + O(t^2)$, $(1-P_t^{\mathrm{F}})\varphi_t = -t\eta((1-P_k^{\mathrm{F}})\zeta + \kappa_1) + O(t^2)$, and $\nabla_j[U_t]\varphi_t = t\eta(\nabla_j^{(1)}\chi_c - \nabla_j\zeta) + O(t^2)$. Here $P_t^{\mathrm{F}} = R_k(U_t)^* R_k(U_t)$. Dividing the quadratic coefficient by η^2 proves the formula. Gauge covariance makes it invariant under $\mathfrak{G}_k$. Indeed replacing A by $A + d\lambda$ is the first-order gauge transform by $e^{-it\lambda}$; the nonnegative block form is zero at $t = 0$, so its quadratic coefficient depends on this tangent vector alone. Its colour trace is the sum of the same quadratic form in the three colour components.

In a Bloch fibre $\xi \neq 0$, d identifies $T_k/\mathfrak{G}_k$ with the closed curvatures of vanishing level-k average. If two representatives have the same curl, their difference is $d\lambda$; the linearized constraint forces λ to be constant on the coarse sites, and this constant can be subtracted. Conversely choose λ with $\lambda|_{L^k\mathbb{Z}^3} = \bar{n}_k(B)$ for $B \in \ker \mathbf{Q}_k^s$ to put $B + d\lambda$ in T_k. The slice spaces are nested and

$$\sum_{i<k} \mathcal{C}_{(i+1)} = P_k.$$

Consequently $F(k;m)$ is the Gaussian mean of Φ_k on $T_k/\mathfrak{G}_k$ with density $e^{-|dA|^2/2}$. This identity is essential: the sum is bounded as one Gaussian mean, rather than by k separate bounds.

Lemma 8.2 (The constrained covariance)**.** Use level-k units, $h = L^{-k}$, fine modes $q_n = \xi + 2\pi n$, $\hat{q}_{n\mu} = 2h^{-1}\sin(hq_{n\mu}/2)$, and $k_n = \sum_\mu \hat{q}_{n\mu}^2$. At unit spacing put $k_1(t) = 4\sum_\mu \sin^2(t_\mu/2)$. There is a representative A'' of the quotient satisfying $\mathbf{Q}_k^s A'' = 0$ and $(1-\widetilde{P}_k)d^*A'' = 0$, where $\widetilde{P}_k$ is the projection onto block constants. For a nonnegative quadratic observable invariant under $A \mapsto A + d\lambda$, $Q_k'\lambda = 0$, its mean under this representative equals its mean under the covariance

$$\widehat{C} = \widehat{\mathbb{K}}^{-1/2}\{(1-\Pi) + \Upsilon^{-1}|(1-\Pi)w\rangle\langle(1-\Pi)w|\}\widehat{\mathbb{K}}^{-1/2} \leq \Upsilon_0^{-1}\widehat{\mathbb{K}}^{-1}, \quad \Upsilon_0 = (2/\pi)^6,$$

in each colour, where $\widehat{\mathbb{K}}$ acts as k_n. Here Π projects orthogonally onto the three constraint vectors e_μ, w is the unit vector representing the block part of d^*A in the coordinates $a = \widehat{\mathbb{K}}^{1/2}A$, and $\Upsilon = |\Pi w|^2$. These vectors are constructed in the proof. ∎

Proof. Choose first a gauge with $\bar{n}_k = 0$. A further gauge change with $Q_k'\lambda = 0$ uniquely solves $(1-\widetilde{P}_k)d^*d\lambda = -(1-\widetilde{P}_k)d^*A$: its homogeneous equation has energy $|d\lambda|^2 = 0$ and block mean zero, so it has no kernel in the finite Bloch fibre. For $a = \widehat{\mathbb{K}}^{1/2}A$, the three straight-average constraints

are $a \perp e_\mu$, with orthonormal vectors e_μ supported on their coordinate components. The block part of d^*A is $\langle w, a\rangle$ for a unit vector w. The gauge-fixed form is therefore $|a|^2 - |\langle (1-\Pi)w, a\rangle|^2$ on $\mathrm{ran}(1-\Pi)$, where $\Pi = \sum_\mu |e_\mu\rangle\langle e_\mu|$. Inverting this rank-one perturbation gives the displayed covariance with $\Upsilon = |\Pi w|^2$.

If v_n is the symbol of the block mean, put $S = \sum_n v_n^2/k_n$ and $S_\mu = \sum_n v_n^2/(\hat{q}_{n\mu}^2 k_n)$. The straight-average identity $\mathbf{Q}_k^s d = dQ_k'$ gives $|\langle e_\mu, w\rangle|^2 = S^2/S_\mu$. Since $\hat{q}_{n\mu}^2 \geq \hat{q}_{0\mu}^2$,

$$\Upsilon = \sum_\mu S^2/S_\mu \geq Sk_0 \geq v_0^2 = \prod_\mu \left[\frac{\sin(\xi_\mu/2)}{h^{-1}\sin(h\xi_\mu/2)}\right]^2 \geq (2/\pi)^6.$$

The inverse perturbation is at most Υ^{-1}, proving the covariance bound. Every field in the straight-average constraint space splits uniquely as $A'' + d\lambda$ with $Q_k'\lambda = 0$. Its form splits into $|dA''|^2$ plus the independent gauge energy $|(1-\widetilde{P}_k)d^*d\lambda|^2$. Thus an invariant observable has the same mean under the two laws. □

To apply this lemma to Φ_k, replace κ_1 by $(1-P_k^{\mathrm{F}})\kappa_1$ in its second square and in ζ; call the resulting expression Φ_k^{nat}. It agrees with Φ_k when $\bar{n}_k = 0$. Under $Q_k'\lambda = 0$, its ζ changes by $i\lambda\chi_c$ and $(1-P_k^{\mathrm{F}})\kappa_1$ by $-i\lambda\chi_c$. The first square is unchanged because $R_k\lambda\chi_c = 0$; the second and third are unchanged by cancellation. It therefore has the covariance representation just proved.

Proposition 8.3 (Bounded cumulative loop)**.** There is $B_F < \infty$, depending only on L, b_F and the fixed chamber operator, such that $0 \leq F(k;m) \leq B_F$ for all $k \geq 1$ and $0 \leq m \leq 1$ in level-k units. ■

Proof. In a Bloch fibre put $d_n = h^{-1}\widehat{D}(hq_n)$, $g_n(m) = (d_n + m)^{-1}$ and $R = (v_n\iota^*)_n$. The one-shot Schur formula gives

$$D_k(\xi;m) = [\theta_k^{-1} + \sum_n v_n^2 \iota^* g_n(m)\iota]^{-1}.$$

The free estimates give $\|\widehat{D}(p)^{-1}\| \leq c_1/|p|$ and $\|\operatorname{Re}\widehat{D}(p)^{-1}\| \leq c_{\mathrm{re}}^{-1}$. For the latter, the free real part is bounded above by a constant times $|p|^2|\Pi_0 f|^2 + |\Pi_1 f|^2$ near zero: the within-site terms annihilate Π_0 and the remaining hop is a difference. The free ellipticity estimate bounds this by $C|\widehat{D}(p)f|^2$. Multiplying by the inverse gives the assertion; compactness treats the rest of the zone. Also $\|\Pi_1 g_n(m)\| \leq Ch$, by the heavy block inverse and its difference coupling. The gauge vertex matrices are uniformly bounded.

First take $m = 0$. Accretivity gives $\|D_k\| \leq \theta_k$; near zero the light block is $i\gamma\cdot\xi + O(|\xi|^2)$, $v_0^2 \geq \Upsilon_0$, and the remaining modes give a bounded sum. Hence $\|D_k\| \leq C|\xi|$ there. With $X = RD^{-1}$, $R\mathcal{K}^{-1} = \theta_k^{-1}D_kX$. The Gaussian mean of the first square is

$$3\theta_k^{-1}\int \frac{d^3\xi}{(2\pi)^3}\operatorname{Tr}(D_kX\widehat{V}\widehat{C}\widehat{V}^*X^*D_k^*).$$

Its trace before the two D_k factors is bounded by $C\sum_n v_n^2|q_n|^{-2}k_n^{-1} \leq C|\xi|^{-4}$. Thus its integrand is at most $C|\xi|^{-2}$, integrable uniformly in k in three dimensions.

For the third square write $\zeta = \zeta^0 - \zeta^1$, with $\zeta_n^0 = g_nV_n\chi_c$ and $\zeta^1 = gR^*D_kXV\chi_c$. The gradient energy of g_nf is at most $hc_{\mathrm{re}}^{-1}|f|^2$ because

$$h^{-1}\sum_j \frac{\varrho_j}{2}|\nabla_j g_n f|^2 = \langle f, \operatorname{Re} g_n f\rangle.$$

The block correction therefore costs at most Ch times the already bounded first square. For the main part, bound $\widehat{C}$ by $\Upsilon_0^{-1}\widehat{\mathbb{K}}^{-1}$ and unfold the fine modes. The substitution $q = t/h$ gives the

same finite integral for every h:

$$\int_{[-\pi,\pi]^3} \frac{d^3t}{(2\pi)^3 k_1(t)} \sum_j \frac{\varrho_j}{2} \|\widehat{\nabla}_j^{(1)} - \widehat{\nabla}_j \widehat{D}^{-1} \widehat{V}\|_{\mathrm{HS}}^2 \le C(1 + c_{\mathrm{re}}^{-1}) \int \frac{d^3t}{k_1(t)} < \infty.$$

The inequality uses $(D^{-1})^* \Delta D^{-1} = \operatorname{Re} D^{-1}$. Using only $\|D^{-1}\|$ here would lose this bound.

We next control the average variation in the mass-dependent second square. The exact contour recursion is

$$\bar{n}_{j+1}(A) = \bar{n}_j(A)|_{L^{j+1}\mathbb{Z}^3} + \mathsf{k}(\mathbf{Q}_j^s A), \quad (1 - P_k^{\mathrm{F}})\kappa_1 = -i \sum_{j<k} \mathfrak{n}_A^{(j)} \chi_c,$$

where $\mathfrak{n}_A^{(j)}$ is the contour mean of the straight average $\mathbf{Q}_j^s A$ in its own step, minus its block mean. To prove the recursion insert $\mathbf{Q}_j^{\mathrm{lin}} = \mathbf{Q}_j^s - d\bar{n}_j$ and $\mathsf{k}(d\vartheta) = Q'\vartheta - \vartheta|_{L^{j+1}\mathbb{Z}^3}$. In the first variation of the composite fermion average the remaining $\bar{n}_j$ terms telescope, giving the second identity. Each $\mathfrak{n}_A^{(j)}$ has block mean zero.

Under the covariance bound, a link of $\mathbf{Q}_j^s A$ has variance at most $C_Q L^j$ in bare units. Indeed its multiplier obeys

$$|\hat{f}_j(t)| \le L^{-3j} \prod_\nu \min(L^j, \pi/|t_\nu|) \min(L^j, \pi/|t_\mu|),$$

and its variance is at most $\Upsilon_0^{-1} \int |\hat{f}_j(t)|^2 / k_1(t)\, d^3t$. Rescaling $t = L^{-j}s$ bounds this by $C_Q L^j$; the remaining integral of $|s|^{-2} \prod_\nu \min(1, \pi^2/s_\nu^2) \min(1, \pi^2/s_\mu^2)$ is finite. A contour uses at most $3L$ links. Minkowski's inequality therefore bounds the standard deviation of the sum by $CL \sum_{j<k} L^{j/2} \le C_L L^{k/2}$. The bare mean square is at most $C_L L^k$, or C_L in level-k units. This controls $\mathbb{E}|(1 - P_k^{\mathrm{F}})\kappa_1|^2$ uniformly.

Finally $\|\mathcal{K}(m)^{-1}\| \le C_{\mathcal{K}}$ in each fibre, uniformly in $m \ge 0$ and k. The Woodbury formula proves it: nonzero folded modes have a uniform gap, while on the zero folded light mode

$$\mathcal{K}(0)_{00}^{-1} = S_0^{-1} D_k [\theta_k^{-1} + \sum_{n \ne 0} v_n^2 S_n^{-1}],$$

where S_n is the light Schur complement of d_n at mass zero. The product $S_0^{-1} D_k$ is bounded near $\xi = 0$. The heavy blocks are bounded as above. Since $\mathcal{K}(0)^{-1}$ is accretive, $(1 + m\mathcal{K}(0)^{-1})^{-1}$ has norm at most one. The same cancellation and the covariance bound give $\mathbb{E}|\zeta_0|^2 \le C$: the nonzero-mode sum is bounded by $C \sum_{n \ne 0} |q_n|^{-2} k_n^{-1} + C|\xi|^{-2}$ and the zero-mode part by $C|\xi|^{-2}$, all integrable. The resolvent identity gives

$$\zeta(m) - \zeta_0 = -m\mathcal{K}(m)^{-1}[\zeta_0 + (1 - P_k^{\mathrm{F}})\kappa_1].$$

The energy identity for $u = \mathcal{K}(m)^{-1} f$ bounds its gradient and block energy by $\operatorname{Re}\langle u, f\rangle \le C_{\mathcal{K}} |f|^2$. Thus the first and third squares at m are at most twice those at zero plus a bounded multiple of $m^2 \mathbb{E}|\zeta_0 + (1 - P_k^{\mathrm{F}})\kappa_1|^2$; the second is at most $Cm\mathbb{E}(|\zeta_0|^2 + |(1 - P_k^{\mathrm{F}})\kappa_1|^2)$. All are bounded for $0 \le m \le 1$, proving the result. □

8.2. **The whole second order.** For a translation-covariant total P at the trivial pair, let $\|P\|_j$ be the infimum of the norms of its local cube-covariant representatives. This quotient norm is used only for totals; a particular decomposition can have a larger norm. In particular the cancellations below are identities of totals, not identities polymer by polymer.

The infinite-lattice slices and these norms are compatible with the thermodynamic limit. Here is the required argument. In each nonzero Bloch fibre the closedness and vanishing plaquette-average conditions are kernels of finite matrices of trigonometric polynomials. Their orthogonal projections are bounded by one and real analytic outside the common zero set of their maximal minors, a closed null set. The gauge representative constructed above gives a slice covariance $\widehat{C}_A(\xi)$ bounded by $C_{[i]}(1 + |\xi|^{-2})$, with i fixed; its fibres are the same on every torus containing that momentum.

Its zero fibre on a torus is bounded independently of the side. For a coefficient of a quadratic gauge variation of a fermion form, exponential locality of the flat kernels bounds its bond kernel by $Ce^{-c(|b-y|+|b'-y|)}$. Its mean is the pairing of this kernel with the slice covariance. On $|\xi| \geq \delta$ the Riemann sums converge, while the sum and integral over $0 < |\xi| < \delta$ are at most $C\delta$; the zero fibre contributes $O(N^{-3})$. Dominated convergence then gives the same mean on the infinite lattice. Gauge-invariant quadratic forms read only the curl law: two covariances with the same curl image differ by symmetric terms with one gauge-direction factor, which the polarized quadratic form annihilates. These observations justify interchanging the gauge and fermion integrations below.

If torus totals with representatives of norm at most A converge coefficientwise, choose a diagonal subsequence of their representatives on every fixed polymer. Each lies in a bounded finite-dimensional Grassmann space. Fatou's lemma gives a limiting representative of norm at most A; the omitted wrapping polymers are exponentially small. Thus $\|P\|_j \leq A$ in the limit. This proves the norm passages used below without selecting an unbounded representative of the quotient.

Let $\sigma_* > 0$ be the relative coupling radius of the pure one-step expansion in [3, eq. 4.cclvii] and [3, Prop. 4.54], and put $r_x = \frac{1}{4}\min\{\frac{1}{2}, \sigma_*\}$. About a real admissible coupling sequence x°, the full coupling domain is the product of discs $|x_i - x_i^\circ| < r_x x_i^\circ$; the inner domain has radii $\frac{2}{3} r_x x_i^\circ$. The sharp mass windows remain separate hypotheses.

For real $x_i > 0$, let A_i be the centered gauge slice in bare lattice units with covariance $L^{-i} x_i^{-1}$ times that of $\gamma_{k,i}$, and let $\mathbb{E}_i$ denote its Gaussian mean. The slices are independent, and the same slice is used for every $k > i$. At complex x_i, $\mathbb{E}_i$ on the quadratic polynomials below is its continuation linear in x_i^{-1}. Let δ_A and δ_A^2 denote the first and second homogeneous Taylor terms, so δ_A^2 includes the factor $1/2$. At the trivial configuration put

$$\mathscr{T}_k = \sum_{i<k} \left\{ -\mathbb{E}_i \delta_{A_i}^2 \langle \bar{\Psi}, \mathbb{D}_k \Psi \rangle + \tfrac{1}{2} \mathbb{E}_i (\delta_{A_i} \langle \bar{\Psi}, \mathbb{D}_k \Psi \rangle)^2 \right\}.$$

It has degrees two and four. Its last slice is $\mathscr{A}_k$. Write $\mathcal{J}_k$ for Gaussian fermion integration under $\Psi_k = \mathsf{H}_k \Psi_{k+1} + W$ and its row twin.

Lemma 8.4 (Transport of the second order)**.** As polynomials of degrees two and four,

$$\mathscr{T}_{k+1} = (\mathcal{J}_k \mathscr{T}_k)_{\geq 2} + \mathscr{A}_{k+1}.$$

■

Proof. At fixed block average the Schur formula gives

$$\mathbb{D}_{k+1}[\mathbb{D}_k + X] = \mathbb{D}_{k+1} + \bar{\mathsf{H}}_k X \mathsf{H}_k - \bar{\mathsf{H}}_k X \Gamma_k^{-1} X \mathsf{H}_k + O(X^3).$$

For an earlier slice $i < k$, its level-k block field has no linear variation because $A_i \in T_{i+1} \subset T_k$. Its quadratic variation takes values in $\mathfrak{su}(2)$; its mean is an invariant colour vector and hence zero. Thus q may be held fixed inside these means. The displayed formula gives the transported quadratic Taylor term and the additional term $\mathbb{E}_i \bar{\mathsf{H}}_k (\delta_A \mathbb{D}_k) \Gamma_k^{-1} (\delta_A \mathbb{D}_k) \mathsf{H}_k$.

On the other side Wick's rule applied to $\frac{1}{2}\langle \bar{\Psi}, N\Psi \rangle^2$ gives its uncontracted quartic term and exactly this bilinear term. There are two cross contractions, each with coefficient $1/2$. The self-contraction is $-\operatorname{Tr}(N\Gamma_k^{-1}) \langle \bar{\Psi}', \bar{\mathsf{H}}_k N \mathsf{H}_k \Psi' \rangle$; it vanishes since the linear vertex has traceless colour matrices and the flat covariance is the identity in colour. The remaining fully contracted term is constant and is discarded. This proves the formula slice by slice. In finite volume Fubini is algebraic; the exponential fermion kernels and the integrable slice covariance prove absolute convergence at each infinite-lattice coefficient. The new slice is $\mathscr{A}_{k+1}$ by definition. □

Define recursively, with all old terms read at the current level,

$$\widetilde{T}^{(j)} = \mathscr{A}_j + \left(\mathcal{J}_{j-1} \sum_{l<j} T^{(l)} - \sum_{l<j} T^{(l)} \circ (\mathsf{H}_{j-1}, \bar{\mathsf{H}}_{j-1}) \right)_{\geq 2},$$

$$\mu_j^{(2)} = -\mathsf{cf}_m(\widetilde{T}^{(j)})/\eta_j^2, \quad T^{(j)} = \widetilde{T}^{(j)} + L^{-j}\mu_j^{(2)}\mathsf{S}_1(\psi^{\mathrm{bg}}), \quad G_k^{(2)} = \sum_{j\le k} L^{k-j}\mu_j^{(2)}.$$

These expressions include the contractions of earlier four-fermion interactions; retaining only the last slice would give a different mass.

Proposition 8.5 (The sum rule)**.** Each $T^{(j)}$ has quadratic zeroth moment zero at every later level and $\|T^{(j)}\|_j \le C_T h_F^4/\mathsf{t}_{j-1}$. Moreover

$$\sum_{j\le k} T^{(j)} - L^{-k}G_k^{(2)}\mathsf{S}_1(\psi^{\mathrm{bg}}) = (\mathscr{T}_k)_{\ge 2}, \qquad G_k^{(2)} = G_k(x), \quad \mu_k^{(2)} = \mu_k^{1\mathrm{L}}(x).$$

■

Proof. All terms have the symmetries used in Lem. 7.8. The Gaussian curvature slice is invariant under the isometries preserving its level, since its orthogonal projection is defined by centered plaquette averages. The fermion tower and its averages have the same covariance. Thus the zeroth moments are scalar. Their subtraction gives zero at level j. The flat map H_j carries constants to constants, multiplying by $\theta L^{-2}/[\mathfrak{d}_j + \theta/L] > 0$, where $\mathfrak{d}_j = \theta_j\mathfrak{m}_j/(\theta_j + \mathfrak{m}_j)$. It therefore preserves a vanishing zeroth moment at every later level.

The identity holds at $k = 1$. If it holds at k, apply Lem. 8.4. Gaussian integration carries the bare mass form to the next bare background mass form plus a constant. In particular its coefficient becomes $L^{-k} = L\,L^{-(k+1)}$; no new normalization occurs. The recursive definition of $\widetilde{T}^{(k+1)}$ then gives the identity at $k+1$ with $G_{k+1}^{(2)} = LG_k^{(2)} + \mu_{k+1}^{(2)}$.

For the norm, Wick's rule for the new gauge slice and Cauchy's estimate on the vertex give $\|\mathscr{A}_j\|_j \le 4C'C_V^2 h_F^4/\mathsf{t}_{j-1}$. Contractions of earlier quartics are the coefficient linear in the old-activity origin of the joined expansion. Its marked count and the leg-room estimate give

$$K_M c_\psi \sum_{l<j} C_7 L^{(j-l)/4}\|T^{(l)}\|_l.$$

This is a local family bound, obtained with the decoupled covariances; using the whole unlocalized covariance to integrate each polymer would not give such a bound. Put $S_2^\sharp = \sum_{n\ge 1} L^{-3n/4}(1 + n\log L)^{9\mathfrak{p}_0}$ and require $K_\mu K_M c_\psi C_7 S_2^\sharp \le 1/4$. Since $\mathsf{t}_{l-1} = L^{j-l}\mathsf{t}_{j-1}$, extraction and induction close with $C_T = 8K_\mu C'C_V^2$. The torus-to-infinite norm argument above supplies a representative with the same bound.

Finally take quadratic zeroth moments in the sum rule. The left side is $-\eta_k^2 G_k^{(2)}$. In the right side the quartics have no quadratic moment. The mean Hermitian quadratic form on constants is, by the three-square identity, $\eta_k^2\sum_{i<k} L^{k-i}x_i^{-1}\sigma_i(k;\mathfrak{m}_k) = \eta_k^2 G_k(x)$, with the minus sign in $\mathscr{T}_k$. The factor L^{k-i}/x_i combines the slice covariance multiplier L^{-i}/x_i with the factor L^k in $\mathbb{D}_k$. The scalar moment equals this Hermitian moment at real coefficients; linearity in $1/x_i$ extends it to the complex coupling discs. Hence $G_k^{(2)} = G_k(x)$ and taking successive increments proves the mass identity. □

8.3. **The third-order remainder.** The marked scalar expansion also gives a local comparison of tested and untested gauge second moments at real coefficients. For bonds b, b' it writes their difference as $\sum_X c_X(b,b')$, X containing both bonds, with

$$|c_X(b,b')| \le K_{\mathrm{cp}} x_k^{-1} e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_k)^2/2} e^{-\kappa_+ d(X)}.$$

To prove this, put $a = \mathsf{t}_k^{1/2}B$ and insert two linear sources $\sigma_1\mathsf{t}_k^{-1/2}a^a(b) + \sigma_2\mathsf{t}_k^{-1/2}a^{a'}(b')$ in the q origin. These are $\sigma_1 B^a(b) + \sigma_2 B^{a'}(b')$ in unscaled units. The inequality $|a(b)| \le \frac{1}{2}M^{3/2}(1 + M^{-3}|a(b)|^2)$ puts this origin below its allowed norm on the bidisc $|\sigma_i| \le \rho_0 = c_{\mathrm{cl}}\mathsf{t}_k^{1/2}M^{-3/2}$, after assigning half its allowance to the original q potential. The tested logarithm minus the untested Gaussian logarithm is exactly the part marked by test defects, of norm at most $K_P p_* e^{-\kappa_+ d(X)}$ on that bidisc. Its $\sigma_1\sigma_2$

coefficient is the second-moment difference because both first moments vanish by evenness. Cauchy's estimate costs $\rho_0^{-2} \le CM^3 c_{\mathrm{cl}}^{-2} x_k^{-1}$, giving the stated K_{cp}. Locality makes this coefficient zero unless X contains both bonds. This proof uses the admitted marked scalar count and quadratic-growth majorant; a scalar probability tail alone would not give the local comparison.

Put $\phi(t) = t^{-1/2}(1+\log t)^{\mathfrak{p}_0}$ and retain the sharper mass window $|s_i| \le 2\bar{B}_M/\mathsf{t}_i$ throughout the following step. Deform its potential by

$$\mathcal{V}(z) = z_3\mathcal{V}^0(B) + \mathcal{V}_k^{\mathrm{F}}(z_1 B) + z_2\mathcal{V}^{\mathrm{old}}(z_1 B) + z_4\mathcal{V}^{\mathrm{s}}(z_1 B).$$

Here the fluctuation argument is in unscaled units. The weight of $n \in \mathbb{N}_0^4$ is $|n|_w = n_1+2n_2+n_3+2n_4$. With $c_\diamond = c_{\mathrm{cl}}/4$ choose the radii

$$r_1 = \min\{c_\diamond/(C_V h_F^2 \delta_k), \beta_0/(2\delta_k)\}, \quad r_2 = \frac{c_\diamond}{c'_\psi \sum_{j\le k} C_7 L^{(k+1-j)/4}\varepsilon_j},$$

$$r_3 = c_\diamond/(K_0^\sharp \delta_k), \qquad r_4 = c_\diamond/(c'_\psi c_{\mathsf{S}} h_F^2 |s_k|),$$

with infinite radius when its denominator is zero. At $k=0$ there are no old activities, so $r_2 = \infty$. The physical flow has $s_0 = 0$ and therefore $r_4 = \infty$ too. On a coefficient disc with $s_0 \ne 0$, use $\mathcal{V}_0^{\mathrm{s}}[s_0 1]$ from Lem. 7.11 and the displayed finite r_4. The first-step size and origin cancellations remain valid there. The number c'_ψ bounds the displayed leg and shift factors on the enlarged fluctuation disc; it is fixed before the small coupling ceiling. The scalar potential norm in r_3 admits its quadratic Gaussian growth. Replacing it by the supremum over the fluctuation domain would lose two powers of the threshold logarithm.

Proposition 8.6 (Second-order activities and cubic mass error)**.** There are constants C_2, C_3, C'_ρ such that the small-field step preserves

$$\|E_{\ge 2}^{(j)}\|_j \le C_2\phi(\mathsf{t}_{j-1})^2, \qquad E_{\ge 2}^{(j)} = T^{(j)} + R^{(j)}, \quad \|R^{(j)}\|_j \le C_3\phi(\mathsf{t}_{j-1})^3$$

at the trivial pair, the latter as a norm of totals. Its extracted mass satisfies

$$\mu_{k+1} = \mu_{k+1}^{\mathrm{1L}}(x) + \rho_{k+1}, \qquad |\rho_{k+1}| \le C'_\rho \phi(\mathsf{t}_k)^3.$$

The estimates hold at real coupling sequences in their window and at complex sequences in the fixed inner coupling discs, under the same sharp mass windows. They also hold at any fixed weight $h \ge h_F$, with the norm constants formed at h and a sufficiently small coupling ceiling depending on h. ∎

Proof. On $|z_i| \le r_i$, the four potential pieces have sizes at most $c_\diamond$ each. Their local logarithm is holomorphic with norm at most $2K_M c_{\mathrm{cl}}$, including the test tail. If $\widetilde{E}(z) = \sum_n a_n z^n$, Cauchy's estimate gives

$$\|a_n\|_{k+1} \le 2K_M c_{\mathrm{cl}} \prod_i r_i^{-n_i}.$$

Here $r_1^{-1}, r_3^{-1} \le C_r\phi(\mathsf{t}_k)$, $r_2^{-1} \le C_r C_2 \phi(\mathsf{t}_k)^2$ and $r_4^{-1} \le C_r\phi(\mathsf{t}_k)^2$. The last estimate uses the sharp mass window. Choose the coupling ceiling so that every finite $r_i \ge 2$.

There is no fermion coefficient of weight zero or one. The coefficient linear in z_1 vanishes: the first Taylor coefficient of $\mathcal{V}_k^{\mathrm{F}}(z_1 B)$ is linear in B, and the Gaussian, tests and q weight are even. Terms depending only on z_3 are scalar. At $z_1 = 0$ the gauge and fermion integrals factorize, so mixed totals with a positive z_3 power and an old-activity or mass power vanish. The coefficient linear in z_4 is scalar: its single-W terms integrate to zero and its $\bar{W}W$ term is constant. At $k=0$ these are precisely the three terms of (31); Lem. 7.11 supplies the same statement when the independent mass coordinate is nonzero. The coefficient linear in z_2, as a total, is exactly

$$\mathcal{J}_k \sum_{j\le k} E_{\ge 2}^{(j)} - \sum_{j\le k} E_{\ge 2}^{(j)} \circ (\mathsf{H}_k, \bar{\mathsf{H}}_k).$$

The marked expansion of this origin, with Cauchy's estimate in z_2, gives the local bound

$$K_M c_\psi \sum_{j\le k} C_7 L^{(k+1-j)/4}\varepsilon_j.$$

Its normalization cancels the gauge integral, so no test tail is present in this particular coefficient.

We justify this identity also on the infinite lattice, where the old family is defined. For a single old term $v(Y)$ let $L_{X,Y}(v)$ be the coefficient linear in its marked origin in the localized logarithm. On a torus, linearity of the cluster coefficients and the exact Mayer resummation give

$$\sum_X L^T_{X,Y}(v) = \mu_{\Gamma_T^{-1}} v(Y).$$

The term v reads no gauge fluctuation, so its scalar normalization cancels. A summand with output diameter at least R has exponentially small norm, uniformly in the torus. Let its side increase. Nonwrapping local terms agree with their infinite-lattice terms, while the tail tends to zero. The finitely many Gaussian moments in the right side converge with their covariance entries. This proves $\sum_X L_{X,Y}(v) = \mu_{\Gamma^{-1}} v(Y)$ on the infinite lattice. To sum over Y, fix one external coefficient. The exponential bound on the output support makes the sum over X, Y absolutely convergent; the sums may therefore be exchanged. Terms without a difference of external legs cancel through the zero cell sums from Lem. 7.9. This proves the identity for all the finite-norm old families used here, including their cell-unit representation at sources.

The displayed identity is an identity of totals. The expression $\mu_{\Gamma^{-1}} v(Y)$ itself reads the gauge configuration through the whole covariance and is not a local family. The local representative is $\sum_Y L_{X,Y}(v)$, formed with the decoupled covariances. The remainder estimate below uses the quotient norm of totals at exactly this point.

At the trivial pair the coefficient $z_1 z_3$ also vanishes: the expectation of one adjoint colour vector multiplied by a colour-invariant scalar is an invariant vector and is zero. This follows from colour invariance. By the exact identity in Lem. 7.3, the z_1^2 coefficient is $\mathscr{A}_{k+1}$, apart from the tested-Gaussian comparison. The linear determinant vertex has zero colour trace, and the mean of a second-order $\mathfrak{su}(2)$ field inserted through a first derivative is an invariant colour vector; neither adds a term.

To justify the untested measure used here when $x_k \neq \mathtt{t}_k$, write its density as $e^{-x_k H - \mathtt{t}_k G}$ and the comparison Gaussian as $e^{-x_k H - x_k G}$, with H the curvature energy and G the gauge-fixing energy. At the unit coefficient the known curvature covariance is the orthogonal curvature projection. Gaussian elimination in the gauge directions then shows $G_{\min}(dA) = 0$: otherwise the induced curvature form would contain this nonzero extra nonnegative term. Thus both Gaussians have the same curl law. Their quadratic fermion observable is invariant under the residual linear gauge directions, so the covariance argument above makes their means equal. Only the tests change the result. The local second-moment comparison bounds this difference by a total of norm $C_\chi \mathtt{t}_k^{-1} e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathtt{t}_k)^2/2}$. Indeed a polymer has at most $8(1+d(X))$ cubes, so it holds at most $C(LM)^6(1+d(X))^2$ ordered bond pairs. The exponential output rate pays these squares and the leg distances; summing through a cube uses $\sum_{n\ge 0}(n+1)^2 2^{-n} = 12$. The thermodynamic norm passage above preserves the bound. Equivalently one obtains the comparison directly by using the unrestricted origin $\mathcal{V}^{\mathrm{q}} + \zeta\mathcal{V}_1 + \zeta^2\mathcal{V}_2$, where $\mathcal{V}_1, \mathcal{V}_2$ are the first two fermion vertices. In the difference of its tested and untested logarithms every term has a failed-bond mark. Cauchy's estimate in ζ selects its second coefficient and gives the same bound in the quotient norm of totals.

For complex x_k, regard this difference as a holomorphic function with values in the quotient norm of totals. On the full coupling disc it is bounded by $C\phi(\mathtt{t}_k)^2$, and on its real diameter by the last exponentially small bound. Apply the scalar two-constants theorem through bounded linear functionals on that quotient. In the coordinate $(x_k - x_k^\circ)/(r_x x_k^\circ)$, the inner disc has radius $2/3$; there its norm is at most

$$C_\chi \phi(\mathtt{t}_k)^2 e^{-c_{\mathrm{hm}}\gamma_{\mathrm{tail}}\mathfrak{b}(\mathtt{t}_k)^2/2},$$

where $c_{\mathrm{hm}} > 0$ is the minimum harmonic measure of the real diameter on that compact inner disc. A final coupling ceiling makes this at most $C_\chi \phi(\mathsf{t}_k)^3$.

The Cauchy radii now bound all remaining coefficients of weight at least three by

$$\sum_{|n|_w \geq 3} \|a_n\|_{k+1} \leq (C_0 + C_1 C_2)\phi(\mathsf{t}_k)^3.$$

This follows by summing the four geometric series: each weight-one inverse radius costs $C_r\phi$ and each weight-two inverse radius costs $C_r\phi^2$; the exceptional coefficient linear in z_2 is kept separately. The ceiling $C_r^2 C_2 \phi \leq 1$ bounds terms with two or more old-origin factors uniformly by a constant plus a constant times C_2.

For the full family of fermion terms, before the trivial-pair split, weight two contributes at most $C_{(2)}\phi^2$. After mass extraction its bound is

$$K_\mu[C_{(2)} + (C_0 + C_1 C_2)\phi]\phi^2 + K_\mu K_M c_\psi \sum_{j \leq k} C_7 L^{(k+1-j)/4}\varepsilon_j.$$

The geometric calibration relation bounds the last sum by $C_7 S_2^\sharp C_2 \phi(\mathsf{t}_k)^2$; the defining logarithmic factor in $S_2^\sharp$ pays the ratio of the calibration logarithms. Require $K_\mu K_M c_\psi C_7 S_2^\sharp \leq 1/4$, take $C_2 = 4K_\mu(C_{(2)} + C_0)$, and then choose γ so $K_\mu C_1 \phi(\gamma^{-2}) \leq 1/4$. This closes the second-order bound.

At the trivial pair subtract the exact T sum rule. The remaining pre-extraction total is the tested z_1^2 error, the coefficient linear in the old remainders, and the coefficients of weight at least three. The same calculation, using their cubic induction bounds, closes with

$$C_3 = 4K_\mu(C_\chi + C_0 + C_1 C_2).$$

Extract its zeroth moment. Since $\eta_{k+1} \geq c_b$, Lem. 7.8 gives $|\rho_{k+1}| \leq c_b^{-2} C_{am}^\psi h_F^{-2} C_3 \phi(\mathsf{t}_k)^3$; discarding the additional factor $\nu^{-1} \leq 1$ only weakens the bound. This proves the proposition. The squared mass shift has weight four precisely because $|s_k| = O(\mathsf{t}_k^{-1})$; the argument does not assert a cubic error in a wider $O(\phi(\mathsf{t}_k))$ window.

For the assertion at $h \geq h_F$, apply the same expansion to the original potentials with their norms measured at h. The bilinear vertex and mass-member bounds become $C_V h^2 \delta_k$ and $c_{\mathsf{S}} h^2 |s_k|$. For the first-step mass member this follows directly from (32) at weight h, so the change of weight also applies when $s_0 \neq 0$. By Lem. 7.10, the old-reading constant remains C_7. The leg ratios become

$$c_\psi(h) = 2A_\kappa \Gamma_F / h, \qquad c'_\psi(h) = c_\psi(h) + (c_h / a_{\mathrm{an}} + C_G)\, c_\diamond / (C_V h^2).$$

The fluctuation covariance and its Gram bound are unchanged; the larger external weights only improve the leg room. Thus the joined cluster bound has the same K_M and c_{cl}. Form r_1, r_2, r_4 at h; the scalar radius r_3 is unchanged. The preceding coefficient estimates then hold with the norm at h. The explicit second-order family has bound $C_{\mathrm{sl}}(h) C_V^2 h^4 / |x_k|$, where

$$C_{\mathrm{sl}}(h) = K_M M^3 \max\{c_{\mathrm{cl}}^{-1}, 2/(\beta_0 C_V h^2)\}.$$

The tested-Gaussian comparison and its two-constants argument use this bound in the quotient norm at h. Their counting and harmonic-measure constants are unchanged. Form $C_r, C_{(2)}, C_0, C_1, C_\chi$, then C_2, C_3, at h and decrease the coupling ceiling to close the same induction. Since $c_\psi(h) \leq c_\psi(h_F)$, its contraction floor is already satisfied. The finite flow and its extracted numbers are unchanged: only their estimates have been recomputed. □

Proposition 8.7 (Closing the forward mass flow)**.** Put $\overline{B} = LB_\sharp/(L-1)$ and let $x_0 = L^K \mathsf{b}$, $\mathsf{b} \geq \gamma^{-2} + \overline{B}$. Suppose the small-field steps give

$$x_{k+1} = L^{-1} x_k - b_{k+1}, \quad |b_{k+1}| \leq B_\sharp, \qquad s_{k+1} = L s_k + \mu_{k+1}, \quad s_0 = 0,$$

and, whenever the preceding mass windows hold,

$$\mu_j = \mu_j^{\mathrm{1L}}(x) + \rho_j, \qquad |\rho_j| \leq C_\rho \phi_\sharp(x_{j-1})^3, \quad \phi_\sharp(t) = t^{-1/2}(1 + \log t)^{3\mathfrak{p}_0}, \qquad 0 \leq F(k; m) \leq B_F.$$

For the final-level assertion suppose also the calibration bound already given by Prop. 8.6, $|\rho_j| \le C'_\rho \phi_\sharp(\mathsf{t}_{j-1})^3$, with $\mathsf{t}_i = L^{K-i}\mathsf{b}$. For a sufficiently small γ, chosen after these constants,

$$|s_k| \le \frac{3B_F + 1}{x_k} \quad (0 \le k \le K).$$

At the final level,

$$\left| s_K - \frac{F(K;\mathsf{m})}{\mathsf{b}} \right| \le C_s \mathsf{b}^{-3/2} (1 + \log \mathsf{b})^{9\mathfrak{p}_0}.$$

■

Proof. The coupling recursion gives $|x_i - L^{K-i}\mathsf{b}| \le \overline{B} = LB_\sharp/(L-1)$. Choose γ so that all x_i exceed $4\overline{B}$. Then, for $i \le k$, $\frac{1}{2}L^{k-i}x_k \le x_i \le 2L^{k-i}x_k$. The one-loop increments telescope exactly:

$$s_k = \sum_{j \le k} L^{k-j}\mu_j = G_k(x) + \sum_{j \le k} L^{k-j}\rho_j, \qquad |G_k(x)| \le \frac{2B_F}{x_k}.$$

The last inequality uses nonnegativity of the individual σ_i and the bound on their sum. Assume the asserted mass window at levels below k. The remainder estimate is then applicable. A geometric sum gives

$$\sum_{j \le k} L^{k-j} \phi_\sharp(x_{j-1})^3 \le S_3 x_k^{-3/2} (1 + \log x_k)^{9\mathfrak{p}_0},$$

where

$$S_3 = \sup_{x \ge 2} \sum_{n \ge 0} L^n (\tfrac{1}{2}L^{n+1})^{-3/2} \left(\frac{1 + \log(2L^{n+1}x)}{1 + \log x} \right)^{9\mathfrak{p}_0} < \infty.$$

Choose $\gamma^{-2} \ge e^{18\mathfrak{p}_0}$ and $C_\rho S_3 \gamma (1 + 2\log\gamma^{-1})^{9\mathfrak{p}_0} \le \frac{1}{4}$. The sum of the remainders is at most $1/(4x_k)$. Thus $|s_k| \le (2B_F + \frac{1}{4})/x_k \le (3B_F + 1)/x_k$, closing the induction.

At $k = K$,

$$\left| G_K(x) - \frac{F(K;\mathsf{m})}{\mathsf{b}} \right| \le 2\overline{B} B_F \mathsf{b}^{-2},$$

because $|L^{K-i}/x_i - \mathsf{b}^{-1}| \le 2\overline{B}\mathsf{b}^{-2}$. Along the calibrations $\mathsf{t}_{j-1} = L^{K-j+1}\mathsf{b}$ the third-order remainders sum to at most $C'_\rho S_3^\sharp \mathsf{b}^{-3/2}(1 + \log \mathsf{b})^{9\mathfrak{p}_0}$, with

$$S_3^\sharp = \sum_{n \ge 0} L^{-(n+3)/2} (1 + (n+1)\log L)^{9\mathfrak{p}_0} < \infty.$$

Adding the two bounds gives the result with $C_s = 2\overline{B}B_F + C'_\rho S_3^\sharp$. □

8.4. **Scalar extraction, pointing, and the order of constants.** A scalar family used here is strictly gauge invariant term by term. Its first derivative at the trivial configuration is zero because it would be an invariant adjoint colour vector. Differentiating gauge invariance once more shows that its Hessian annihilates exact one-forms in either slot. For a term on X, choose a rooted radial gauge in its bounding box. Each gauge potential is then a sum of curls over a surface of area at most $C\,\mathrm{diam}(X)^2$. Thus its Hessian is $T_X[dA, dA]$, with norm at most a polynomial in $\mathrm{diam}(X)$ times the norm of the term, by Cauchy's estimate. The polymer decay pays this polynomial. The total has a bounded continuous curvature symbol, so its potential symbol begins at order two. Colour and cubic invariance make that quadratic part a multiple of the Wilson-action Hessian. This argument excludes a first-order Chern–Simons density because the individual terms are strictly invariant, rather than invariant up to a difference. It is the scalar classification needed for taking out the gauge-action coefficient; parity is not used for it.

The scalar local extraction has the admitted coefficient bound $|\mathsf{cf}(F)| \le C_{am} r_\alpha^{-2}\|F\|$ on the usable pair radius. Removing its constant and its multiple of the action therefore costs $2 + C_{am} r_\alpha^{-2}$. Applying this to the joined expansion gives

$$\|F_0^{(k+1)}\|_{k+1} \le \tfrac{1}{2} C_w \phi(\mathfrak{t}_k) + \theta_s h_F^2 |s_k|, \qquad |b_{k+1}^{\mathrm{r}}| \le C_b \phi(\mathfrak{t}_k) + \theta_s h_F^2 |s_k|.$$

Indeed the input sizes are $(K_0 + C_V h_F^2)\delta_k$ for the gauge and Dirac vertices, $c'_\psi c_{\mathsf{S}} h_F^2 |s_k|$ for the mass variation, $3\sum_{j\le k} \omega_{k+1-j}\varepsilon_j$ for old fermion terms, and the test tail. Their last sum is at most $3C_7 S_\omega C_2 \phi(\mathfrak{t}_k)^2$, where $S_\omega = \sum_{n\ge 1} L^{-n/4}(1 + n\log L)^{3\mathfrak{p}_0}$. Multiplying by K_M and the extraction factor proves these bounds after C_w, C_b, θ_s are chosen. On a constant smaller radius the coefficient bound costs $(\alpha_0/a)^2$ and the action costs $(a/\alpha_0)^2$; they cancel. Hence the same extraction factor works on all the stated radius systems. The normalizer part is bounded before the polymer constants by Prop. 6.11; the admitted pure normalizer increment is bounded by $(1 + L^{-1})c_{2,\mathrm{loop}}$. Choose $B_\sharp \ge (1 + L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^{\mathrm{F}}) + 1$. The remaining increment is at most one once γ is small, by the displayed scalar extraction estimate and the sharp mass window. This supplies $|b_{k+1}| \le B_\sharp$ in Prop. 8.7, subject to the admitted pure step inputs throughout.

For a homogeneous degree-n term F, $n \ge 2$, let N_p count its legs at place p and put $F_p = n^{-1} N_p F$. Then $\sum_p F_p = F$, and the norm of a sum over any subset of places is at most $\|F\|$. Along the path $\mathcal{V}_t = \mathcal{V}_0 + t\mathcal{V}^{\mathrm{F}}$ define

$$E^\bullet(X, p) = \int_0^1 E'[\mathcal{V}_t; (\mathcal{V}^{\mathrm{F}})_p](X)\, dt.$$

Its sum over p is $E[\mathcal{V}_0 + \mathcal{V}^{\mathrm{F}}](X) - E[\mathcal{V}_0](X)$. Send each place p to its level-$(k+1)$ block site $z(p)$ and write $\widetilde{E}^\bullet(X; z) = \sum_{z(p)=z} E^\bullet(X, p)$. Scalar remainders use the gauge bond pointing, closed fermion normalizers use their diagonal trace pointer, and the field-independent wrapping loops use equal thirds on the three wrapping rows. Quadratic and cubic coefficient extraction commutes with these operations.

The estimates proved above hold for the sum over any set of pointers: its insertion is bounded by the original vertex, and the extracted mass terms have disjoint blocks. In a stored layer, however, different pointers are evaluated at different pairs. We need the sum of their individual norms. The following estimate provides it.

Lemma 8.8 (Summing the pointers)**.** Consider the small-field step from level k to level $k+1$ on $\mathbb{Z}^3$ or on a torus, with the pointing above, the sharp mass window $|s_k| \le 2\bar{B}_M/\mathfrak{t}_k$, and the bounds $\|E_{\ge 2}^{(i)}\|_i \le C_2 \phi(\mathfrak{t}_{i-1})^2$ for $i \le k$. The coupling may be real or in its admitted complex disc; the fine mass is that of the datum. For a pointed family set

$$\|F\|_j^{\mathrm{pt}} = \sup_{\square} \sum_{X \supset \square} e^{\kappa d_j(X)} \sum_{z\in X} \sup_P \|F(X; P; z)\|_{h^{(j)}}.$$

Here P ranges over the regular complex pairs at the fixed radii, separately for each pointed term. Put

$$\varsigma_k = \sum_{i\le k} \omega_{k+1-i} \|E_{\ge 2}^{(i)}\|_i, \qquad \omega_n = C_7 L^{n/4},$$

using the larger of the two family norms on a torus. Let

$$C_2^0 = 2(C_{(2)} + C_0), \qquad \mathfrak{y}_k = C_2^0 \phi(\mathfrak{t}_k)^2 + K_\mu^{-1} c_s h_F^2 |s_k|^2 + K_M c_\psi \varsigma_k,$$

where

$$c_s = 2K_M c_{\mathrm{cl}} (c'_\psi c_{\mathsf{S}} h_F / c_\diamond)^2 \max\{K_\mu, 4c_b^{-2} C_{am}^\psi / \nu\}.$$

Then $\widetilde{E}^{\bullet}_{\geq 2}(X;z) = 0$ unless $z \in X$, and

$$\sup_P \|\widetilde{E}^{\bullet}_{\geq 2}(X;P;z)\| \leq C_E^{-1}\mathfrak{y}_k e^{-\kappa_+ d_{k+1}(X)},$$
$$\|\widetilde{E}^{\bullet}_{\geq 2}\|^{\mathrm{pt}}_{k+1} \leq 8M^3\mathfrak{y}_k,$$
$$\|E^{(k+1)}_{\geq 2}\|^{\mathrm{pt}}_{k+1} \leq (8M^3 + K_\mu - 1)\mathfrak{y}_k \leq C_2^{\mathrm{pt}}\phi(\mathfrak{t}_k)^2, \qquad C_2^{\mathrm{pt}} = 8M^3C_2.$$

The final bound holds under the contraction floor and coupling ceiling

$$\theta_E C_\omega S_\omega \leq \frac{1}{4\sqrt{2}}, \qquad 16c_s h_F^2 \bar{B}_M^2 \gamma^2 \leq C_2,$$

where

$$\theta_E = \sqrt{2}K_\mu K_M \left[c_\psi + (c_h/a_{\mathrm{an}} + C_G)c_{\mathrm{cl}}/(4C_V h_F^2)\right].$$

The earlier families enter only through their unpointed norms. ■

Proof. Only a vertex having a place in X can contribute to $E^{\bullet}(X,p)$. Its assigned block site also lies in X. This proves the support statement. All other terms in the pointing are scalar, except the extracted mass terms, each supported on the single cube containing its pointer.

Use the four-variable deformation and the radii r_i in the proof of Prop. 8.6. A pointed cluster term is the mean of the terms obtained by replacing one vertex by its insertion at z. The cluster majorant is multilinear and increasing in the vertex norms. Since the insertion has no larger norm than that vertex, Lem. 7.5 gives

$$\|\widetilde{E}^{\bullet}(X;P;z;\zeta)\| \leq 2K'_M c_{\mathrm{cl}} e^{-\kappa_+ d_{k+1}(X)} \quad (|\zeta_i| < r_i).$$

The pointed terms are holomorphic on this open polydisc. Indeed, on $|\zeta_i| < \rho r_i$, $\rho < 1$, the three linear origins have total size at most $3\rho c_\diamond$, and the vertex has size at most $c_\diamond$. The potential and its pointing path therefore stay strictly inside the cluster domain. These smaller polydiscs exhaust the stated domain.

Write the part of degree at least two as $\sum_n a^{\bullet}_n(X;z)\zeta^n$. Cauchy's estimate on the radii ρr_i, followed by $\rho \uparrow 1$, gives

$$\|a^{\bullet}_n(X;P;z)\| \leq 2K'_M c_{\mathrm{cl}} r^{-n} e^{-\kappa_+ d_{k+1}(X)}.$$

If r_2 or r_4 is infinite, the corresponding origin vanishes and coefficients with a positive power of that variable vanish. The cancellations used in Prop. 8.6 hold at each pointer: the coefficient $(1,0,0,0)$ is odd in the gauge fluctuation; $(0,0,n_3,0)$ is scalar; and the part of degree at least two of $(0,0,n_3,1)$ vanishes because the mass insertion has only a single fluctuation leg or a fully contracted pair. At $k=0$ this uses Lem. 7.11, including its single-cube filing, so it holds at each pointer for a nonzero scalar mass coordinate as well.

The two weight-two coefficients $(2,0,0,0)$ and $(1,0,1,0)$ are bounded together by

$$C_E^{-1}C_{(2)}\phi^2 e^{-\kappa_+ d(X)},$$

where $\phi = \phi(\mathfrak{t}_k)$ and $K_M = C_E K'_M$. The coefficient $(0,0,0,2)$ is bounded by

$$C_E^{-1}K_\mu^{-1}c_s h_F^2 |s_k|^2 e^{-\kappa_+ d(X)}.$$

Summing the remaining coefficients of weight at least three as in Prop. 8.6 gives

$$C_E^{-1}(C_0 + C_1C_2)\phi^3 e^{-\kappa_+ d(X)} \leq C_E^{-1}(C_{(2)} + \tfrac{3}{2}C_0)\phi^2 e^{-\kappa_+ d(X)}.$$

Here we used $C_2 = 4K_\mu(C_{(2)} + C_0)$, $K_\mu C_1 \phi \leq 1/4$, and $\phi \leq 1/2$.

The coefficient linear in the old origin is kept separately. On the axis $(0,\zeta_2,0,0)$ the potential is $\zeta_2 \mathcal{V}^{\mathrm{old}}_{\geq 2}(0)$. The marked majorant bounds its pointing by $K_\Pi|\zeta_2|\|\mathcal{V}^{\mathrm{old}}_{\geq 2}(0)\|e^{-\kappa_+ d(X)}$ on the disc where this potential has size below c_{cl}. Cauchy's estimate for its first coefficient divides by $|\zeta_2|$. Since

$\|\mathcal{V}^{\mathrm{old}}_{\geq 2}(0)\| \leq c_\psi \varsigma_k$ and $K_\Pi \leq K'_M$, its bound is $C_E^{-1} K_M c_\psi \varsigma_k e^{-\kappa_+ d(X)}$. The preceding estimates prove the first inequality, since $2C_{(2)} + \frac{3}{2}C_0 \leq C_2^0$.

A polymer of tree size d contains at most $8(d+1)$ cubes and hence at most $8M^3 e^d$ pointers. There are at most $2048 \cdot 9216^d$ such polymers through a given cube. The floor $\kappa_+ - \kappa - 1 \geq 2c_E$, with $c_E = \log 18432$, therefore gives

$$\sum_{X \supset \square} e^{\kappa d(X)} \sum_{z \in X} \sup_P \|\widetilde{E}^\bullet_{\geq 2}(X; P; z)\| \leq 8M^3 C_E^{-1} \mathfrak{y}_k \, 2048 \sum_{d \geq 0} (9216 e^{-2c_E})^d \leq 8M^3 \mathfrak{y}_k.$$

The supremum is taken before the pointer sum throughout.

The extracted mass terms stand on disjoint blocks of the sites in a cube. Their norms sum to at most $c_{\mathsf{S}} |\mu_{k+1}| h_F^2$. Mass extraction and $\|\widetilde{E}_{\geq 2}\|_{k+1} \leq \mathfrak{y}_k$ bound this by $(K_\mu - 1)\mathfrak{y}_k$. Thus

$$\|E^{(k+1)}_{\geq 2}\|^{\mathrm{pt}}_{k+1} \leq (8M^3 + K_\mu - 1)\mathfrak{y}_k \leq 8M^3 K_\mu \mathfrak{y}_k.$$

Finally,

$$K_\mu \mathfrak{y}_k \leq \tfrac{1}{2} C_2 \phi^2 + c_s h_F^2 |s_k|^2 + \theta_E \varsigma_k.$$

The mass window gives $|s_k| \leq 2\bar{B}_M \gamma \phi$; the stated ceiling bounds its contribution by $C_2\phi^2/4$. The calibration sum gives $\varsigma_k \leq C_\omega S_\omega C_2 \phi^2$, so the last contribution is at most $C_2\phi^2/(4\sqrt{2})$. Since $1/2 + 1/4 + 1/(4\sqrt{2}) < 1$, the claimed bound follows. The cluster and polymer counts hold on both lattices with the same constants, including both families on a torus. □

The scalar pointing must retain the same quadratic normalization as Lem. 7.4. Let $\mathbb{V}^{\mathrm{int}} = \mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \tau \mathbb{V}^{\mathrm{A}}$ be the gauge vertices and $\mathbb{V}'$ the old scalar variation. Their complex polydisc sizes satisfy

$$n^{\mathrm{int}} \leq \tfrac{7}{2} K_P \delta_k \leq \tfrac{7}{16} w_*, \qquad n' \leq K_\Lambda \mathsf{s}_a N_\star \leq \tfrac{1}{4} w_*.$$

Write $\widehat{g}_b$ for the bond pointing of the quadratic gauge-fixing family. Point its tested and untested logarithms separately, integrating the insertion $-(1-\tau)\widehat{g}_b$ along the path $\sigma \mathcal{V}^{\mathrm{q}}$, $0 \leq \sigma \leq 1$. Call the two terms $G^{\mathrm{v}}(X,b)$ and $G_\tau(X,b)$. Let $E^{\chi}[\emptyset](X,b)$ be the test-only pointing. Finally integrate the marked derivative along the gauge-vertex and old-variation path to form $E^{\bullet,\mathrm{v}}(X,b)$, keeping the tested quadratic origin fixed on this path. Their sum is

$$E^{\chi}[\emptyset](X,b) + G^{\mathrm{v}}(X,b) - G_\tau(X,b) + E^{\bullet,\mathrm{v}}(X,b).$$

Summing over b gives exactly the degree-zero local logarithm, with G_τ subtracted. The four path integrals give the tests alone, the change to the tested quadratic system, the untested quadratic divisor, and the remaining potential; their intermediate endpoints cancel.

The sort-change proof in Lem. 7.4, differentiated by adding an insertion of size K_{G} on a disc of radius w_*/K_{G}, gives

$$|G^{\mathrm{v}} - G_\tau|(X,b) \leq 3|1-\tau| K_{\mathsf{G}} w_*^{-1} K_P^{\Pi} p_* e^{-\kappa_+ d(X)}.$$

The other path derivative has size at most $9(n^{\mathrm{int}} + n')$, so its pointed cluster term is bounded by $9K_\Pi(n^{\mathrm{int}} + n')$ times the same exponential. Send bonds to sites by the tent weights of the chart. Their sum at a site costs at most $3L^3$. The resulting pointing constant is

$$K^{\mathrm{v}}_{\mathbf{M}} = 3L^3 \max\left\{ K_\Pi K_{\mathsf{G}}, (1 + \tfrac{3}{2}\mathsf{C}_{\mathrm{ex}}((L+2)M)^3) K_P^{\Pi} e^{4\lambda_1}, 32 K_\Pi K_P, \ 81 K_\Pi K_\Lambda \right\}.$$

It replaces $K_{\mathbf{M}}$ in the scalar extraction bounds. The Lipschitz estimate in the fourth family splits its change into a change of insertion and a change of system. These have constants $9K_\Pi$ and at most $72K_\Pi$, respectively, when $n^{\mathrm{int}} + n'' \leq w_*$ and $\|\mathbb{V}' - \mathbb{V}''\| \leq w_*/2$. Restrictions and joint gauge transformations commute with every operation, giving local pointed terms on the whole pair cores.

The array supplied to the next step has a degree-zero bound in a supremum over polymers, pointers and configurations, rather than the summed fermion-family norm. At each new level its explicit families give

$$\begin{aligned}&\sup_{X,z} e^{(\kappa+1)d_{k+1}(X)} \sup_{P} |E_0^{(k+1)}(X;P;z)| \\ &\quad \le E_{\mathbf{N}} + C_{\mathsf{n}} + C(C_N + M^3)N_f \\ &\qquad + C(1+L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^{\mathrm{F}}) + CK_{\mathbf{G}}\bar{B}\gamma^2 + C_w\phi(\gamma^{-2}) \le E_0.\end{aligned} \tag{35}$$

The first two terms are the gauge normalizer and its retained one-site constant, the next two are the closed fermion loops and the extracted action coefficient, and the last two are the Gaussian correction and the fluctuation family. For a wrapping row of N/M cubes the free determinant error is at most $C\nu NM^2 e^{-\delta_F N}$. The floor $\delta_F M \ge \kappa + 2$ bounds it by $C\nu M^3$ after multiplication by $e^{(\kappa+1)N/M}$; its pointed shares obey the same bound. The extracted plaquette shares have tree size zero. The pointed fluctuation estimate has rate $\kappa_+ \ge \kappa + 1$. This proves (35) anew at each level.

On the complex coupling disc replace the correction bound by $CK_{\mathbf{G}}(\bar{B}\gamma^2 + r_x)$ and the fluctuation bound by its complex mass-window bound. At a complex fine mass the closed-loop action coefficient is bounded by the Cauchy estimate $CC_{am}r_\alpha^{-2}C_N N_f$. Field-independent wrapping numbers drop out of every background variation and pass outside the linear block map. The common choice of E_0 covers these versions as well. A single spare unit of polymer decay does not imply a summed family bound: only the scalar background variation, of second order in curvature, has the summed bound K_{old} used above.

The choices used in these proofs have the following order. First fix the chamber operator, N_f, L and b_F. Form the free and weighted energy constants, then the complex and source radii, the local supremum constants, M_w and the walk ratios. Choose the admitted pure geometry and polymer constants subject to their conditions and the additional displayed floors on R, M, $\bar{\kappa}$ and κ; then form A_κ. The constants Γ_F, K_M, c_{S}, C_{am}^{ψ}, K_μ and C_7 are then fixed before h_F; homogeneity in h_F cancels from each of them. The cumulative bound B_F also precedes h_F, and sets $\bar{B}_M = 3B_F + 1$. Raise h_F to meet the leg room and

$$h_F \ge 8K_\mu K_M A_\kappa \Gamma_F C_7 S_2^\sharp.$$

Next form $C_r, C_{(2)}, C_0, C_1, C_\chi, c_s$, then C_2, $C_2^{\mathrm{pt}} = 8M^3C_2$, C_T, C_3, C'_ρ and the forward-flow constant. The additional pointer estimate uses $\theta_E C_\omega S_\omega \le 1/(4\sqrt{2})$, imposed when h_F is chosen, and the final ceiling $16c_s h_F^2 \bar{B}_M^2 \gamma^2 \le C_2$. Finally lower γ to meet the finitely many smallness conditions, including

$$\log\gamma^{-2} \ge 18\mathfrak{p}_0, \quad C_r^2 C_2 \phi(\gamma^{-2}) \le 1, \quad K_\mu C_1 \phi(\gamma^{-2}) \le \tfrac{1}{4},$$

$$\sup_{t \ge \gamma^{-2}} e^{-c_{\mathrm{hm}}\gamma_{\mathrm{tail}}\mathfrak{b}(t)^2/4}/\phi(t) \le 1, \quad C'_\rho S_3^\sharp \psi_\sharp(\gamma^{-2}) \le \tfrac{1}{2}.$$

For fixed earlier admissible constants these last ceilings hold for sufficiently small γ. Prop. 10.20 verifies a simultaneous assignment with the pure, history and source conditions.

9. Histories and the common vacuum energy

Let $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$ be the physical torus. We use the admitted pure gauge histories in the calibration frame $\mathsf{t}_j = L^{K-j}\mathsf{b}$. Their tests, coordinates, stored gauge factors and profiles φ_j are unchanged. The running coefficients satisfy $|x_j - \mathsf{t}_j| \le \bar{B}$ and $|s_j| \le \bar{B}_M/x_j$. The weight of the calibration frame is the constant $\mathsf{bw} = \beta = \mathsf{t}_0$. The running coefficients enter the exponent and the Gaussian covariance $x_j^{-1}C_j$; they do not change the tests, the coordinates or the block kernels. The histories and their free-coordinate measures are those of [3, §4.1] and [3, Thm. C]. Write $\mathcal{S} = \mathcal{S}(h)$ for the discrete label of h, distinct from its free evaluation coordinates. The integration regions satisfy $\Omega_j \supset \Lambda_j$, and $Z_j = \Lambda_j^c$. Put $\Omega_0 = T_\ell$, $\mathcal{L}_j = \Omega_j \setminus \Omega_{j+1}$ for $j < k$, and $\mathcal{L}_k = \Omega_k$. For a region

E, $\#_j(E)$ counts its level-j sites, or equivalently its level-j unit cells. The places $\mathfrak{P}_h$ consist of the 48 chambers over each site of $\mathcal{L}_0$ and one coarse place per level-j site of $\mathcal{L}_j$ for $j \geq 1$. The fermions on the layers with $j < k$ are stored; those on the last region are current. The profiles φ_j are the fixed cutoffs of these histories, equal to one on the prescribed inner region and zero on Z_j. Write

$$n_j = \ell^3 L^{3(K-j)}, \quad \nu = 8N_f, \quad \Sigma_{\mathcal{L}}(h) = \sum_{j=0}^{k} \#_j(\mathcal{L}_j), \quad \Sigma_Z(h) = \sum_{j=1}^{k} \#_j(Z_j), \quad \Sigma'(h) = \#\mathfrak{P}_h.$$

There are 48 chamber places per fine site, and $\Sigma_{\mathcal{L}} \leq \Sigma' \leq 48\Sigma_{\mathcal{L}}$. The pure geometry gives $\Sigma_{\mathcal{L}} \leq n_k + L^3\Sigma_Z$. The layer separation is $R_\flat = 8LM$ in the units of the lower of two consecutive levels.

Write $\varepsilon_j^{\mathrm{leg}}$ for the small fermion-origin size at step j, and retain n_j for the number of sites. The extension to a history has two tasks. The gauge expansion must be applied to the actual coefficients and conditional Gaussian of that history. In addition, a fermion term created at an earlier level must remain small when its background fields are read at the present level. We verify these tasks separately. The verification in Prop. 10.20 supplies the gauge charts, conditional structures and coordinate identities for both sequences. Bounds on the matter coefficients are established below.

The exponent of a history has the form

$$-A(\beta_h^F, U) - \langle \bar{\Psi}, \mathbb{D}_h \Psi \rangle - M_h + E_h^{\mathrm{reg}} + E_h^{\mathrm{bd}} - \mathcal{E}_{k,\mathcal{S}}^F, \qquad \beta_h^F = \beta - \sum_{j \leq k} L^j b_j \varphi_j. \tag{36}$$

For a scalar plaquette weight q, the weighted action here is $A(q, U) = \sum_p q(p) a_p(U)$, with the history profiles transferred to plaquettes by the convention of the cited gauge construction. In particular,

$$-A(\beta_h^F, U) = -\beta A(U) + \sum_{j \leq k} L^j b_j A(\varphi_j, U).$$

The last sum is bounded by a constant times $\Sigma_{\mathcal{L}}(h)$ in the scalar estimate below and is included in its layer constant. This permits the history integration estimate to use the fixed frame weight β. Here $s_h = \sum_{j \leq k} L^{-j} \mu_j \varphi_j$ is the mass profile and $M_h = \mathsf{S}_1(s_h \psi^{\mathrm{bg}})$ is its quadratic mass member. On $\Omega_j \setminus \Lambda_j$ its coefficient is $L^{-(j-1)} s_{j-1}$: the entire output on that collar is retained as a boundary term, with no local part extracted there.

9.1. **The conditional expansion with fermion contractions.** The conditional mean near a collar changes with the decoupling parameters. Consequently the local expansion must contract the fermions after each corresponding conditional gauge integral. Using a fixed whole-volume mean in every term would lose the component identities on which the expansion rests.

Lemma 9.1 (Local contractions in the centered expansion)**.** Consider an admitted centered gauge frame at step $k+1$, including the first-step frame, with all the class-size, external-field and Gaussian bounds required by [3, Prop. 5.19] and [3, Prop. 5.20]. Let its values lie in the finite even Grassmann algebra with integrated and external generators, with integrated weights at least Γ_F. Suppose the contraction maps $\mathscr{L}(Z, s; P)$ have the following properties: they fix external elements, are linear over them and have norm at most one; their matrix entries are holomorphic in the decoupling parameters on a neighborhood of the closed unit polydisc and on the pair slices, and are Borel in the real data; they read the pair alone, independently of the integration field, external field and stored coordinates; they factor over wall-components when the separating parameters vanish; their restrictions agree on subdomains; they intertwine gauge transformations and conjugation; and, at parameters one on the torus, they restrict one whole contraction $\mathscr{L}_*$. Then the centered expansion followed by these contractions has the scalar frame's majorants, with the fibre norm in place of absolute value. In particular its cluster family satisfies

$$\|\widetilde{E}(X)\| \leq b_{\mathrm{cs}}^{\mathfrak{F}} e^{-\kappa'' d_{k+1}(X)} \leq \tfrac{1}{2} e^{-\kappa'' d_{k+1}(X)}, \qquad \kappa'' = \kappa_+ + 1 + c_E.$$

The part passing through a specified class of ordinary origins of total size σ is bounded by $K^{\mathrm{lb}}\sigma e^{-\kappa'' d_{k+1}(X)}$, where

$$K^{\mathrm{lb}} = eC_{\mathrm{cs}}(LM)^3 \exp\{C_{\mathrm{cs}}(\bar{c}_P^{+} + \bar{r}_H^{+} + \kappa'')\}.$$

Take the larger of the first-step and positive-level constants in this formula. It is fixed before the fibre weight. The conclusions hold on the admitted external-field box with complex room $r_{\mathrm{bd}} = C_{\mathrm{xt}}\mathfrak{b}(\mathtt{t}_k)$. They include locality on X, joint pair and field holomorphy, Borel dependence, gauge covariance, conjugation, component factorization, and the exponential identity for the whole integral. Each centered member uses the conditional mean of its own decoupled structure. Inside Λ_{k+1} this mean and the external field vanish. ■

Proof. For a Mayer index with region Z, form the scalar conditional integral coefficient by coefficient, with its conditional mean $\mathbf{m}(Z^{\mathrm{I}}, s^{\mathrm{en}})$, and then apply $\mathscr{L}(Z, s; P)$. Denote the resulting function by $f_\iota(s)$. Its finite parameter brackets are the terms of the joined expansion. Use all big cubes as guards, so the parameter set outside $Z^\sharp$ is the full set of cubes of the ambient domain. At parameters supported in a region, both the conditional structure and $\mathscr{L}$ split over its wall-components. The bracket identities and the cancellation of disconnected terms therefore hold in the even algebra, which is commutative.

To estimate a term, apply a linear functional of norm one on the external algebra. The composite of this functional with $\mathscr{L}$ has norm at most one. The scalar contour estimate of Lem. 4.1, at the parameter-dependent center, bounds it by the same majorant. Taking the supremum over the functionals gives the fibre norm bound. Summation over the rooted indices and then over connected clusters is exactly the scalar majorant summation. The version with a marked origin gives its factor $e\max\{K_{\mathrm{or}}, K_{\mathrm{or}}^{+}\} \le K^{\mathrm{lb}}$; this proves the selective bound, rather than a bound by the size of all scalar origins. Projections onto a fixed degree have norm at most one and obey the same estimate.

All finite brackets use values of the members and contractions on their own domains. Their restriction, holomorphy and symmetry properties therefore pass to the terms, and uniform absolute convergence passes them to the sums. At parameters one the conditional structure is the whole structure and its center is the whole conditional mean. Finite Grassmann expansion commutes with the coefficientwise gauge integral; the exponential identity follows by the same finite-volume resummation. On an interior domain there are no external bonds, and every cube meets the integration region. Its indices are then the one-lattice indices, with center zero. Equality with a second frame requires equality of its actual members and contractions; locality alone does not assert that equality. □

For our contractions the covariance is the symmetric weakened covariance of Prop. 6.7, restricted to the integrated fermion places. Parameters outside Ω_{k+1} are retained. The Gram bound gives the contraction norm, and Lem. 6.10 gives the intrinsic domain and restriction properties. The symmetry in (26) is needed here: Wick's rule intertwines Grassmann conjugation with $S \mapsto \gamma_4 S^\dagger \gamma_4$. The unmodified one-sided parametrix would not by itself give that identity at weakened parameters.

Lemma 9.2 (Moving an external field to its stored value)**.** On the admitted pair and external-field boxes, let $F_X(a; P; q)$ be a local family with values in the finite external even algebra. Assume that it reads the field inside X alone, is jointly holomorphic in the pair and field variables, has its stated external legs in X, and satisfies $\|F_X\| \le be^{-rd(X)}$ there. The coordinate change from the field before the strip to the stored field produces local rows whose sum is

$$\sum_X F_X(a_{\mathrm{before}}; P_X; q_X) - \sum_X F_X(a_{\mathrm{st}}; P_X; q_X).$$

Each row is evaluated at its own pair and stored-data restriction. Their norm bound is $\eta_{\mathrm{sr}} be^{-(r-1-c_E)d}$, where $\eta_{\mathrm{sr}} = 2^{14}e^{-\check{R}_{k+1}} < 1/8$ under the chosen floor $\check{R}_{k+1} \ge 12$ and the strip conditions. They

retain holomorphy, locality and the stated symmetries, are supported on made boundary domains, and vanish for the empty history. ■

Proof. The strip construction is a finite difference of values of F_X along the admitted external-field substitution. Its Cauchy contours use the complex room $C_{\mathrm{xt}}\mathfrak{b}(\mathfrak{t}_k)$; the real stored field occupies only the prescribed half of the doubled box. Apply a norm-one functional to the external algebra. Linearity identifies the resulting row with the scalar transfer row. Its contour estimate and the sum over domains give the factor $2^{14}e^{-\check{R}_{k+1}}$ and the loss $1+c_E$ in the rate. Taking the supremum gives the norm estimate. The scalar sum rule is a linear identity, hence holds coefficient by coefficient. Its restriction identities prove the support and locality claims. This argument concerns this external-field row; it is not an assertion that every coordinate identity is supplied by a single transfer estimate. □

9.2. **The running coupling in the calibration frame.** The tests and coordinates are fixed by $\mathfrak{t}_k$, whereas the action has coefficient x_k. The difference changes both the integrated quadratic form and the factor left on the collar. We first keep track of this difference exactly.

Lemma 9.3 (The quadratic form of the actual step)**.** Put $\tau = x_k/\mathfrak{t}_k > 0$, with $|\tau - 1| \le 1/4$. In the fluctuation scaled by $\mathfrak{t}_k^{1/2}$ write $A_1 = G + A^{\mathrm{act}}$ for the calibration quadratic form, where G is its block gauge-fixing form. Let I and E be the integrated and stored collar bonds. Then $G = G_{II} \oplus G_{EE}$, and the actual form is

$$A_\tau = G + \tau A^{\mathrm{act}} = \tau A_1 + (1-\tau)G.$$

Use the conditional Gaussian of τA_1, of covariance $x_k^{-1}C_k$ in the unscaled variable, and put the correction $-\frac{1}{2}(1-\tau)(a, G_{II}a)$ among the tested ordinary origins. Its mean is $-K_1 a'$, where $K_1 = (A_1)_{II}^{-1}(A_1)_{IE}$ and a' is the collar field before the strip. The calibration formed factor on that field is multiplied by

$$e^{R(a')}, \qquad R(b') = -\tfrac{1}{2}(\tau-1)\langle b', (S_1 - G_{EE})b'\rangle, \quad S_1 = (A_1)_{EE} - (A_1)_{EI}(A_1)_{II}^{-1}(A_1)_{IE}. \tag{37}$$

At stored coordinates retain $R(a'_{\mathrm{st}})$ as a scalar boundary term and file $R(a') - R(a'_{\mathrm{st}})$ in its strip rows. The tests, geometric histories and stored gauge factors remain those of the calibration frame. ■

Proof. The action coefficient is one number on the whole current region. Thus τ multiplies all three action blocks, including the mixed and collar blocks. The gauge-fixing form is a sum of terms inside individual coarse blocks. Every such block lies wholly in the integration region or wholly in its collar; hence its mixed block vanishes. Complete the square with τA_1: $a = \widetilde{a} - K_1 a'$. The factor τ cancels from K_1, so every test and vertex reads this same shifted field, and the q-origin is evaluated there as well. The Gaussian leaves

$$\exp\{-\tfrac{1}{2}\tau\langle a', S_1 a'\rangle - \tfrac{1}{2}(1-\tau)\langle a', G_{EE}a'\rangle\}.$$

Factoring out the calibration Schur form proves (37). The retained and rejected localizations of $-\frac{1}{2}(b', S_1 b')$ are identities in b'. Their strip sum rules, together with the transfer rule for the local forms $\frac{1}{2}(b', G_{EE}b')$, give exactly the difference of R at the two collar fields. No collar gauge-fixing term is inserted a second time among the integrated q-origins. □

Lemma 9.4 (Bounds after real calibration)**.** Under the admitted calibration-frame conditions, the conditional structure used in Lem. 9.3 is obtained by replacing its covariance and connecting operator by $\tau^{-1}C$ and $\tau^{1/2}\Gamma$. In the variable $a' = x_k^{1/2}B$, external fields and thresholds are multiplied by $\tau^{1/2}$. The centered quadratic-growth norm increases by at most $4/3$, and the Gaussian test exponent becomes $\tau\gamma_P$. The other geometric frame constants remain those of the calibration frame. The q-origin is confined to the integrated fine cubes. Its untested all-real-fluctuation expansion gives $\mathbf{G}_\tau$; in the actual step it has the tested ordinary sort of Lem. 7.4. ■

Proof. The change of variables is real and positive. Substitution in the covariance, thresholds and conditional mean gives the stated transformations. The only increase in a quadratic-growth denominator is bounded by $\max(\tau,\tau^{-1}) \le 4/3$; the Gaussian exponent scales by τ. The q-origin returns precisely the integrated gauge-fixing difference in Lem. 9.3. Its restriction to integrated cubes follows from the separate collar block there. Applying Lem. 7.4 distinguishes its two sorts and supplies the comparison with $\mathbf{G}_\tau$. These changes of variables prove a statement for real τ; complex-parameter holomorphy is obtained on the stated coefficient domains, not by treating this real Gaussian substitution as a complex change of variables. □

9.3. **The scalar entries and their local variations.** The pure frame accepts a local array and boundary entries with specified bounds. We must supply the actual entries of the matter construction. In particular the coupling extraction has already been removed from the regular activities, and must not be inserted again as a counterterm.

Definition 9.5 (The scalar list of a history)**.** At level k, use calibration $\mathfrak{t}_k$ and zero counterterms in the local array. Its regular entries are the pointed degree-zero activities with their vacuum and action parts removed. Its made boundary entries are the scalar parts of earlier generated terms. Its formed factor family is the unchanged calibration family. If $X(p;j)$ is the pointer cell assigned to a plaquette at level j, set

$$w(p) = \sum_{j\le k} L^{j-k} b_j 1_{\{X(p;j)\not\subset\Lambda_j\}}.$$

With the localizations of the Jacobian, gauge-fixing, action, regular array, old boundary and formed factors denoted respectively by $V^J, V^G, V^A, V^E, V^B, V^\Phi$, the two scalar classes are

$$V^F_{\mathrm{rg}} = V^J + V^G + \tau V^A + V^E - V^w, \qquad V^F_{\mathrm{bd}} = V^B + V^\Phi.$$

Here V^w is the variation of $L^k A(w;U)$ under the shift. The q-origin of Lem. 9.3 is a separate class. ■

Lemma 9.6 (Admission and sum rules for the scalar list)**.** Suppose the earlier history bounds hold and the actual array satisfies (35). Take $B \ge 8M^3E_0$ and the unchanged formed-factor bound $B^\flat_{\mathrm{fr}} = K_{\mathrm{ch}}C^2_{\mathrm{xt}} + 12M^3C_z a_{\mathrm{nf}}$. Then the scalar list has the required local holomorphic entries, pair ties and Borel dependence, with array bound E_0, made-entry bound B and edge bound $8M^3E_0$. Moreover $|w| \le \bar{B}$, and w vanishes on every plaquette whose pointer cell is wholly in the current integration region. Its nonzero plaquettes have a corner in Z_k. The sum of its class potentials is the variation of the actual degree-zero exponent, each part counted once. To state their sizes, put

$$g_k = \mathfrak{t}_k^{-1/2}, \quad y_k = \log \mathfrak{t}_k, \quad \mathfrak{b}_k = \mathfrak{b}(\mathfrak{t}_k), \quad \alpha_{0,k} = C_0 g_k y_k^q, \quad q > \mathfrak{p}_0.$$

Here $\alpha_{0,k}$ is the gauge frame's analytic radius. With $\delta_k = \delta(\mathfrak{t}_k)$, write

$$\vartheta_{0,k} = \frac{\delta_k}{\alpha_{0,k}}, \qquad \mathfrak{s}^\sharp_k[E] = \frac{\delta_k}{\alpha_0}(1 + E + \alpha_0^2 B_\sharp) + M^3 g_k \mathfrak{b}_k^3.$$

These are the quantities in [3, Prop. 4.126], [3, eq. 4.dclxxxv], and [3, eq. 4.dclxxxvi], evaluated at the calibration coupling. The tier bounds are

$$\|V^F_{\mathrm{rg}}\|_{r_V} \le \max(1,\tau) C_{\mathrm{eff}}(\mathfrak{s}^\sharp_k[E_0] + \vartheta_{0,k} g_k^2 y_k^{2q}), \qquad \|V^F_{\mathrm{bd}}\|_{r_V} \le C_{\mathrm{eff}}(B + B^\flat_{\mathrm{fr}})\vartheta_{0,k}.$$

Consequently the effective threshold $y_k \ge Y^F_{\mathrm{eff}}(\epsilon)$ makes these bounds at most $\max(1,\tau)\epsilon$ and ϵ. ■

Proof. The regular entries satisfy the array bound by their construction, with their prescribed pointing. Restriction to an edge gives at most $8M^3$ shares; their norms, rather than a second sum over arbitrary pointers, give the edge bound. The made entries obey their bounds on their full filing domains. The admitted frame's localizations apply to an arbitrary list with precisely these bounds and pair ties; no induction theorem for an unspecified matter array is used.

For the coupling term use $L^k x_k = \beta - \sum_{j\le k} L^j b_j$. Before smoothing the pointer shares its identity is

$$-\beta A(U) + \sum_{j\le k} L^j b_j \sum_{X(p;j)\subset\Lambda_j} a_p(U) = -L^k x_k A(U) - L^k A(w;U).$$

The profiles and their rim shares restore this identity because their weights add to $1_{\{X(p;j)\subset\Lambda_j\}}$: in the interior the profile is one, and on the rim its complement is distributed among shares with sum one. This explains why $-V^w$ replaces the counterterm class. The geometric series in the bounds on b_j gives $|w| \le \bar{B}$. Every excluded pointer has a corner in Z_k; the separation of Z_k from Ω_{k+1} gives $d_k(Y) \ge 7L\check{R}_{k+1} - 2$ whenever its variation on Y is nonzero.

The local sum rules now give respectively the change of the pointed regular array, of $L^k A(w;U)$, of the old boundary entries at unchanged stored data, and of the formed factors. On an interior domain the last three boundary contributions vanish, and V^E is the flat array variation. Applying the admitted localization bounds to this verified list gives the two displayed estimates. Their constants depend on the three entry ceilings; the definition of Y^F_{eff} is the threshold that makes their right sides at most the stated sizes. □

Lemma 9.7 (Variation of an earlier boundary coefficient)**.** Let $g_{j,X}$ be an earlier made or edge family with values in a finite normed space. Suppose it depends on the pair, is Borel in the real and stored data, is holomorphic on the admitted slices of its filing domain, and satisfies $\|g_{j,X}\| \le b^\bullet e^{-\kappa d_j(X)}$. Apply the ordinary or pinned coefficient localization of the history frame, keeping the prescribed stored data fixed. The resulting coefficients are local, analytic on the tiers and Borel there. Their sum is the difference of the values at the shifted and unshifted configurations. Summed over the earlier made and edge families, their bound at each output polymer Y is

$$C^{\mathrm{bd}} \vartheta_{0,k} (b^{\mathrm{mk}} + b^{\mathrm{ed}}) e^{-r_V d_k(Y)}.$$

They vanish unless $d_k(Y) \ge 7L\check{R}_{k+1} - 2$. For a jointly gauge-invariant made term with external legs, depending on the pair, the coefficient at a pinned substitution need not itself be a function of the pair. Its product with the inverse joint action of the levelling function, formed at the same parameters, is a function of the pair and has the joint covariance and tie of the original term. ■

Proof. The coefficient localization is a finite bracket, with coefficients ± 1, of values at substituted triples. The substituted triples do not depend on $g_{j,X}$. Consequently localization commutes with every norm-one linear functional ℓ. The scalar localization estimate uses the analyticity and bound of $\ell \circ g_{j,X}$; it does not require scalar gauge invariance. Its Cauchy radius contributes $C_{\mathrm{sb}} C_{\mathcal{A}} a^\bullet e^{C_{\mathrm{sb}}\bar{\kappa}}$. The domain-count constant $C_B(0)$ includes this factor C_{sb}. The made and edge radii satisfy $a^{\mathrm{mk}} \le c_-^{-1}\vartheta_{0,k}$ and $a^{\mathrm{ed}} \le \vartheta_{0,k}$. Thus the choice

$$C^{\mathrm{bd}} \ge C_{\mathcal{A}} e^{C_{\mathrm{sb}}\bar{\kappa}} C_B(0)(1 + c_-^{-1})$$

gives the stated bound after taking the supremum over ℓ. The bracket sum rule gives the difference of the two endpoint values, at unchanged stored data.

For the assertion about the pair, write

$$\Pi_{j,X}(s,\lambda;\mathfrak{u}) = \theta^{-1}_{u_{s,\lambda}} g_{j,X}\big(T_{X|h'}(s,\lambda;\mathfrak{u});q|_X\big),$$

where θ_v turns the nonstored generators by v and leaves the stored generators fixed. The coefficient and $u_{s,\lambda}$ use the same parameters. At two triples with one pair, let u and $\widetilde{u}$ be their levelling functions. The substituted coefficients are related by $\widetilde{g} = \theta_{\widetilde{u}u^{-1}} g$. For real turns this follows from joint covariance: the connecting gauge transformation is one at the tops and stored sites, so it preserves the stored data. Holomorphy in the turn parameters gives the identity at the pinned turn. Hence $\theta^{-1}_{\widetilde{u}}\widetilde{g} = \theta_u^{-1} g$. This proves pair dependence of Π; the same covariance identity gives its tie.

At fixed real configuration and stored data, normalization to a tame point is linear in the two form coordinates. It preserves the substituted coefficient and the levelling function. At the normalized

point these are locally holomorphic in the form coordinates, the fluctuation and the interpolation parameters. Their composition with normalization is therefore analytic on the tiers. The maps are Borel at fixed parameters, and the finite brackets preserve Borel dependence. These assertions concern the admitted filing domains and tiers. □

Lemma 9.8 (Local bound for the retained collar correction)**.** The term R in (37) has a local family supported on domains meeting stored collar bonds:

$$R_{k+1}(X) = (\tau - 1)\{K_{\mathrm{kept}}(X) + B_{\mathrm{cross}}(X) + Q_X^{\mathrm{gf}}\}, \qquad Q_X^{\mathrm{gf}} = \tfrac{1}{2} \sum_{\square \subset X} (a'_{\mathrm{st}}, \widehat{G}_\square a'_{\mathrm{st}}).$$

The last sum is assigned to a single big collar cube and is zero on other domains. On the full admitted complex pair and stored-field domain,

$$|R_{k+1}(X)| \le r_k e^{-\kappa'' d_{k+1}(X)},$$
$$r_k = \frac{\bar{B}}{\mathfrak{t}_k} C_{\mathrm{xt}}^2 \mathfrak{b}(\mathfrak{t}_k)^2 \{c_{\mathrm{kp}}^{\mathrm{th}} + c_{\mathrm{rj}}^{\mathrm{th}} e^{-(\check{R}_{k+1}-2)} + c_{\mathrm{gf}}\},$$
$$c_{\mathrm{gf}} \le \tfrac{3}{2} c_G (LM)^3, \qquad c_G = 6L^6/r_1^2.$$

Its strip rows have bound $2^{16} e^{-\check{R}_{k+1}} r_k e^{-\kappa_+ d}$. Under the common small-coupling ceiling

$$\log \gamma^{-2} \ge 2\mathfrak{p}_0, \qquad \tfrac{3}{2} \bar{B} \gamma^2 C_{\mathrm{xt}}^2 A_1^2 (\log \gamma^{-2})^{2\mathfrak{p}_0} \{c_{\mathrm{kp}}^{\mathrm{th}} + c_{\mathrm{rj}}^{\mathrm{th}} + \tfrac{3}{2} c_G (LM)^3\} \le 1,$$

the retained family and its rows together are bounded by $e^{-\kappa_+ d}$. ■

Proof. Localize the Schur form in (37) by its kept and rejected families. Its identity in the external field gives the stated local expression. Every fine cube used in the gauge-fixing sum is a source cube of the local Gaussian triple. Its restricted triple is in the class on which $\widehat{G}_\square$ has row bound c_G, also on the complex slices. A big cube has at most $3(LM)^3$ positive bonds; this gives the displayed c_{gf}. The kept and rejected estimates and $|a'_{\mathrm{st}}| \le C_{\mathrm{xt}} \mathfrak{b}(\mathfrak{t}_k)$ give r_k. All terms vanish without a stored collar bond, hence are boundary terms.

The kept and rejected quadratic strip identities and Lem. 9.2 for Q^{gf} give the exact sum $R(a') - R(a'_{\mathrm{st}})$ and the row bound. The strip floor makes the multiplier at most $1/2$. Finally $\mathfrak{b}(t)^2/t = A_1^2 t^{-1} (\log t)^{2\mathfrak{p}_0}$ decreases once $\log t \ge 2\mathfrak{p}_0$. Its value at γ^{-2} and the stated ceiling bound $(3/2) r_k$ by one. □

9.4. **The actual frame and the exact step.**

Lemma 9.9 (The joined frame on a history)**.** Suppose the history induction holds through level k, with earlier fermion-origin sizes $\mathsf{sz}_j^\natural \le 16\varepsilon_\flat/17$. Suppose also that the present size $\mathsf{sz}_k^\natural \le \varepsilon_\flat$ and the scalar-class, q-class and failed-test thresholds of the common choice hold. Then the actual step is the joined centered expansion of Lem. 9.1: it uses covariance $x_k^{-1} C_k$, the calibration tests, the scalar classes of Def. 9.5, the tested q-origin of Lem. 9.3, and the actual fermion potentials contracted with the symmetric weakened covariance. All big cubes are guards, including cubes outside Ω_{k+1}. Its output $E_{\mathrm{lab}}(X)$ obeys

$$\|E_{\mathrm{lab}}(X)\| \le \tfrac{1}{2} e^{-\kappa'' d_{k+1}(X)}, \qquad \|(E_{\mathrm{lab}}(X))_{\ge 2}\| \le K^{\mathrm{lb}} \mathsf{sz}_k^\natural e^{-\kappa'' d_{k+1}(X)}.$$

It is defined on the admitted local pair, external-field and stored-data domains and has the properties in Lem. 9.1. The present-size hypothesis is verified in Prop. 9.14 before this step is applied. ■

Proof. In the scaled variable $\mathfrak{t}_k^{1/2} B$, the integrated covariance is $[\tau (A_1)_{II}]^{-1}$, the inverse block of the reference form τA_1. The tested q-origin supplies the remaining correction $(1-\tau) G_{II}$ to the actual quadratic form, as in Lem. 9.3. The positive real rescaling in Lem. 9.4 has the centered-frame bounds. The density tests place the gauge fields in the local domains by Lem. 7.1; the local vertices and

contractions there are defined by Prop. 7.2 and Lem. 6.10. In particular the contraction parameters outside the integration region have not been discarded.

The actual scalar list is admitted by Lem. 9.6. Its two classes, the class with fermion legs and the q-class give at most four ordinary origins. With $y_k \geq Y^F_{\mathrm{eff}}(2\varepsilon_\flat/5)$ and the q and failed-test ceilings, their amplitudes fit the centered expansion's fixed envelope. Real calibration changes the appropriate quadratic-growth bound by at most $4/3$. Thus the admitted scalar contour estimate applies to these members. The contraction properties were checked after Lem. 9.1, so that lemma gives the whole bound. Setting the fermion class to zero leaves only scalar outputs. Marking that class therefore gives the selective positive-degree bound.

At $k = 0$ no old scalar array or old boundary family is present. The all-field inverse Γ_0^{-1} and the first centered frame give the same construction and the same enlarged constants. The q-only all-real expansion is used solely for the comparison with $\mathbf{G}_\tau$; the actual q-origin here keeps its tests. Borel dependence and the restriction identities follow before summation and persist by the common absolute majorant. □

Lemma 9.10 (Agreement on an interior domain)**.** For $X \subset \Lambda_{k+1}$, the output of Lem. 9.9 is

$$E_{\mathrm{lab}}(X) = \mathbf{G}_\tau(X) + \widetilde{E}_k(X),$$

where $\widetilde{E}_k$ is the small-field output on the same torus. On nonwrapping domains, after the prescribed transport it agrees with the infinite-lattice output. ■

Proof. On X there are no external bonds or stored collar coordinates and the conditional mean is zero. The thresholds are the calibration thresholds. The conditional structure, its parameter extension and its rooted indices are therefore the one-lattice ones. The scalar classes agree term by term: the boundary, formed-factor and missing-weight variations vanish, while the regular array is the same pointed array. The fermion members, including their edge and wrapping assignments, are restrictions of the same local functions. Their weakened covariances agree by the intrinsic restriction identity. Thus every conditional integral and every parameter bracket is the same, not just its support or bound. The small-field definition subtracts the all-real q-family $\mathbf{G}_\tau$; this gives the displayed identity. For nonwrapping domains the local walks, tests and vertices agree with their infinite-lattice counterparts, which proves the last assertion. □

Proposition 9.11 (The exact step with matter)**.** Assume the admitted gauge coordinate identities, the actual scalar and fermion member bounds of Lem. 9.9, and the local strip identities above. Then the gauge step with its fermion contraction is an exact identity of coefficientwise measures. Its conditional quadratic datum is

$$A^F_{\mathsf{d}} = \tau A_{\mathsf{d}} + (1-\tau)\chi^T_E(\widehat{G}_{\mathsf{d}})_{EE}\chi_E,$$

and its scalar normalization changes by

$$c^F_t = -\tfrac{3}{2}\log\tau\,\#I + \log\delta Z^F_k(1),$$

where $\#I$ counts positive integrated gauge bonds. The geometric coordinate maps are unchanged. The fermion Gaussian is normalized inside the step and its determinant is outside; all cluster and strip families occur once. ■

Proof. The II and IE blocks of the displayed datum are τ times the calibration blocks. Its EE block contains the separate collar gauge-fixing correction, as required by Lem. 9.3. The integrated normalization scales by $\tau^{-3\#I/2}$, because the Lie algebra has three real coordinates on each positive bond. The exact fermion Schur identity adds δZ^F_k and the normalized contraction. The cluster exponential of Lem. 9.1 evaluates the resulting integral, and Lems. 9.2 and 9.8 move its external field to stored coordinates with their exact differences. Together with the sum rules of Lems. 9.6 and 9.7, these are precisely the members of the transformed exponent.

The coordinate pushforward identity is applied to each Grassmann coefficient. Split its real and imaginary parts into positive and negative parts; the local norm majorants give their integrability. This justifies the four ordinary measure identities and their recombination. The scalar fermion determinant is continuous and positive at real configurations, so the old coordinate exceptional null sets remain null after multiplication by that density. No positivity or Tonelli assertion is made for a Grassmann-valued integral. □

9.5. **The store and boundary reentry.**

Lemma 9.12 (The stored quadratic factor)**.** Let a Grassmann algebra have N pairs of generators, both of weight h, and let A be their quadratic matrix. Then

$$\|e^{-\bar{\Psi}A\Psi}\|_h \le (1 + h\|A\|^{1/2})^{2N}.$$

Consequently the stored factor of a history is bounded by $e^{c_{\mathrm{store}}\Sigma'(h)}$, where c_{store} may depend on the fixed fibre weight and is independent of the history, cutoff and volume. ■

Proof. The coefficient of degree (p,p) is a minor of A. Its absolute value is at most $\|A\|^p$, since the corresponding matrix of exterior powers has norm at most $\|A\|^p$. Thus the norm is at most $\sum_p \binom{N}{p}^2 h^{2p}\|A\|^p$, which is bounded by the square of $\sum_p \binom{N}{p} h^p \|A\|^{p/2}$. The all-field tower bound $\|\mathbb{D}_h\| \le c_{\mathrm{tw}}$ and $N = \nu\Sigma'(h)$ prove the assertion. This estimates the exponential directly; no bound of the quadratic form by its operator norm times the number of places is required. □

The block Schur identity also holds with stored generators. After integrating the places A_j, its determinant factor is $\delta Z_j^F = \theta^{-\nu \#_{j+1}(\Omega_{j+1})} \det \Gamma_j$, its new quadratic matrix is the Schur complement, and each old background is the new background plus an image of the fluctuation W. These identities are matrix identities before any estimate.

Proposition 9.13 (Generated boundary terms)**.** Under the induction and present-frame hypotheses of Lem. 9.9, the output on a polymer meeting Z_j is a generated boundary term. It is local, even and jointly gauge invariant; it is holomorphic on the admitted pair domains and real under the model's conjugation. At creation its scalar part is bounded by $Be^{-\kappa d_j(X)}$ and its part with fermion legs by

$$(1 + \eta_{\mathrm{sr},j}) K^{\mathrm{lb}} \mathsf{sz}_{j-1}^{\natural} e^{-\kappa_+ d_j(X)}, \qquad \eta_{\mathrm{sr},j} < 1/8.$$

Edge parts from pointed regular activities have the corresponding $4K_M\bar{v}_j$ bound with their pointer factor, where $\bar{v}_j \le b_{j-1}^{\mathrm{blk}}$ is the small-field input size at their birth. The boundary index sets obey $\sum_{X \in \mathsf{lx}_j^{\mathrm{mk}}} e^{-\kappa d_j(X)} \le C_{\mathsf{B}} \#_j(Z_j^{\sim})$. At a stored place of level $m < k$ use weight $w_k(m,x) = (1 + 2^{-(k-m)})h_F$; at a formal background leg of level $n < m$ in that layer use $w_k(n,x) = (\frac{1}{2} + 2^{-(k-m)-1})h^{(m)}(n)$. These weights decrease with k and preserve the bounds at all later steps that have been constructed. ■

Proof. Every output is assigned to a polymer containing the closed enlargement of its integration support. If that polymer is not contained in Λ_j, it meets $Z_j^{\sim}$. The selective estimate of Lem. 9.9 and the strip estimate of Lem. 9.2 give the positive-degree bound. The normalizer, the q-only Gaussian family and the collar correction are scalar; their fixed bounds fit in B. The entropy sum through a cube gives the index-set bound. Restriction of every input to its filed support, followed by the component identities of the conditional construction, gives locality. Gauge covariance, conjugation and holomorphy pass through the finite integrals and the absolutely convergent sums.

For later reentry, retain the room in the weights. At a row in layer m, split an old background into $a + b + c$: the new formal background, the image of W, and its change under the gauge fluctuation. For a formal leg of level $n < m$ the available room is

$$h_b' = w_k(n,x) - w_{k+1}(n,x) = 2^{-(k-m)-2} h^{(m)}(n).$$

The new formal background is not expanded into all stored generators: the whole row bound of Lem. 6.8 gives its admitted reading, while such an expansion here would lose locality and its weight reserve. Only b and c are expanded in walks.

A walk from layer $m < k$ to A_k, read at depth $k+1$, crosses the intermediate layers and the band $\Omega_k \setminus \Omega_{k+1}$. Since $R_b/(3M_w) \geq 8L$, it has at least $8L(k-m)$ steps. Half its available exponential weight pays the length polynomials, and half gives $e^{-4L(k-m)} \leq L^{-4(k-m)}$. Consequently

$$\frac{h_b}{h'_b} \leq 4Lc_\psi(2L^{-4})^{k-m}, \qquad c_\psi = 2A_\kappa \Gamma_F/h_F.$$

The fine shift into layer m is bounded by $c_h L^{-k} L^{-4(k-m)} \sup|B|$. Cauchy's estimate on the link radius $\alpha_1^* L^{-m}$ gives the factor $L^{-(k+1-m)}$ in addition to $2C_{\mathcal{A}}\delta(\mathfrak{t}_k)/\beta_0$. Write $c = c_1 + c_2$, where c_1 changes the background kernel and c_2 reads the changed site factors at its stored columns. A walk reaching a later layer m' loses one crossing band; its Cauchy factor compensates this loss up to L^3. The complex row bound and the room above give

$$\frac{h_{c_1}}{h'_b} \leq \frac{2C_{\mathcal{A}}\delta(\mathfrak{t}_k)}{\beta_0} 8L^4 A_\kappa C_H (2L^{-4})^{k-m},$$
$$\frac{h_{c_2}}{h'_b} \leq \frac{2C_{\mathcal{A}}\delta(\mathfrak{t}_k)}{\beta_0} 6L^5 A_\kappa C_{\mathfrak{r}} \beta_0 (2L^{-4})^{k-m}.$$

For the second estimate use $|\mathfrak{r}_{m'} - 1| \leq C_{\mathfrak{r}} L^{-4(k-m')} \sup|B|$; together with crossing it costs at most L^4 times the decay from m to k. A stored generator itself is split as $\Psi + (\mathfrak{r}_m - 1)\Psi$; its falling weight bounds the second part by the ratio $3C_{\mathfrak{r}}C_{\mathcal{A}}\delta(\mathfrak{t}_k)(2L^{-4})^{k-m}$.

In a made boundary term a nonstored leg also carries the levelling function $u(x)$. The local shift bound is $|u^{\pm 1} - 1| \leq C_{\mathrm{m}} c_h L^{-5(k-m)} \sup|B|$. On the doubled parameter contour, its ratio to the room is at most

$$\frac{2C_{\mathcal{A}}\delta(\mathfrak{t}_k)}{\beta_0} 4C_{\mathrm{m}} c_h \beta_0 (2L^{-5})^{k-m}.$$

Here u is evaluated at the same parameters as the coefficient substitution, as in Lem. 9.7. Coefficient variation itself is estimated by that lemma, separately from these leg changes.

Combining the coefficient and leg changes. Fix a made term with base X at level $j \leq k$. Let g_s be its coefficient at the pinned substituted triple, u_s the levelling function, and c_s the change of its background legs, all formed at the same shift parameters s and at shift amplitude $\lambda = 1$. Thus $u_0 = 1$, $c_0 = 0$, and at $s = \mathbf{1}$ these are the coefficient, levelling function and leg change of the step. Stored generators of h' are left in their places. With $g_s[\ell]$ denoting substitution of the nonstored legs ℓ, put

$$G(s) = g_s[a],$$
$$R^b(s) = g_s[a+b] - g_s[a],$$
$$R^c(s) = g_s[a+b+c_s] - g_s[a+b],$$
$$R^u(s) = g_s[u_s(a+b+c_s)] - g_s[a+b+c_s].$$

The change from the new formal legs to the old reading is

$$G(\mathbf{1}) - G(0) + R^b(\mathbf{1}) + R^c(\mathbf{1}) + R^u(\mathbf{1}).$$

Here $R^c(0) = R^u(0) = 0$; $R^b(0)$ is the term with a leg at W and no dependence on B.

Expand only the parts b, c_s into local terms and walks. For a monomial and a choice ω of these parts, let N_ω be the base $X^\uparrow$ enlarged by the cubes met by their walks. For an edge term also include the blocks read by its stored-leg site factors. The product reads only those components of the shift that meet N_ω, and is consistent when its domain is enlarged. Its brackets $[Y \setminus N_\omega \mid N_\omega]$ are

therefore assigned to $Y \supset N_\omega$. Their sum, together with the value at the base, recovers the product at $s = \mathbf{1}$. The bracket of G gives its difference between $s = \mathbf{1}$ and $s = 0$.

The split above is used to estimate the terms. A separate part of a made term can read the real factor of the triple. For each choice ω, its part from G, R^b or R^c is $g_s[\omega]$, and its part from R^u is $g_s[u_s\omega] - g_s[\omega]$. Their sum is

$$g_s[u_s\omega] = (\theta_{u_s}^{-1} g_s)[\omega] = \Pi_{j,X}(s, 1; \mathfrak{u})[\omega].$$

All parts have the same roots N_ω. Their brackets at each Y thus add to the bracket of $\Pi_{j,X}[\omega]$; the value kept at the base has the same identity because $u_0 = 1$. Lem. 9.7 and covariance of the chosen legs imply that this sum is a function of the pair, with the required tie and joint covariance. Summing these identities at Y proves the corresponding properties of the member of the frame at Y.

For analyticity fix the real configuration U and the stored data on Y. On the normalized, tame slice, the pinned substituted triple and the levelling function are locally holomorphic in the form coordinates, B and s. The canonical substitution radius is at least two, so the levelling function is also holomorphic for $|\lambda| < 2$. The parts b, c_s are locally holomorphic by their local and walk formulas: the moved configuration stays in the admitted complex class for $\sup |B| \le \beta_0$. The normalization preserves the coefficient and levelling function. Composition with it therefore proves analyticity on every tier. The walk sums converge uniformly on the parameter domains, and their finite brackets retain this property. This argument applies to the separate parts used for estimates as well as to their recombination. For an edge term the substituted triple is plain and there is no u_s or R^u; each part already reads the pair.

There is no factor equal to the number of legs. Indeed, if $h_i = h_{a,i} + h'_{b,i}$ and $h_{b,i} \le \rho_i h'_{b,i}$ with the split parts within their rooms, the terms with first split leg i have coefficients $t_i = h'_{b,i} \prod_{l<i} h_{a,l} \prod_{l>i} h_l$. They telescope to $\sum_i t_i \le \prod_i h_i$. Thus the sum of the split terms is at most $\max_i \rho_i$ times the original norm. This permits the walk carrying the split leg to retain its own crossing factor.

Counting the domains of the brackets. For the parts R^c, R^u, put $p_l = h'_{b,l}/h_l$. The nonsplit parts have weights at most $(1 - p_l/2)h_l$. Their telescoping sum is bounded using

$$\sum_i p_i \prod_{l \ne i} (1 - p_l/2) \le 1, \qquad 0 \le p_l \le 1.$$

Indeed, the left side is affine in each p_l; at a vertex with q nonzero coordinates it is $q2^{1-q} \le 1$, and at the zero vertex it is zero. This gives respectively the largest ratio h_c/h'_b or $2h_u/h'_b$ times the coefficient norm. The estimate for R^b uses the preceding telescoping with ratio h_b/h'_b. Each first split leg therefore keeps its own ratio in all three parts.

This count must be made on one parameter contour for each output polymer. Set

$$x = C_g e^{(\kappa+2)c_0} e^{-(\bar\kappa - 1)} \le e^{-8}, \qquad r = 1 + e^{\bar\kappa - 1} < e^{\bar\kappa}.$$

The floor on $\bar\kappa$ gives these inequalities and $r/(r-1) \le e^x$. For a fixed term and Y, let $R_N(s)$ be the sum of its parts from R^b, R^c, R^u with roots $N_\omega = N$. On the common torus $\mathfrak{T}_r(Y) = \{s : |s_D| = r \text{ for every } D \subset Y\}$, Cauchy's formula gives

$$R_N[Y \setminus N \mid N] = \int_{\mathfrak{T}_r(Y)} R_N(s) \prod_{D \in N} \frac{s_D}{s_D - 1} \prod_{D \in Y \setminus N} \frac{1}{s_D - 1} \prod_{D \subset Y} \frac{d\theta_D}{2\pi}.$$

This also includes the base value when $N = Y$. Consequently the norm of the part through the legs at Y is at most

$$\sup_{s \in \mathfrak{T}_r(Y)} \sum_N e^{x\#N} e^{-(\bar\kappa - 1)\#(Y \setminus N)} \|R_N(s)\|.$$

The sum is evaluated at one s. This allows us to apply the algebra norm and telescope the legs before taking the supremum.

Every component of $Y \setminus N$ meets one of at most $26\#N$ neighbouring cubes of N. There are at most C_{g}^{n} connected sets of n cubes through a given cube, and adjoining a cube increases the size by at most c_0. At any rate $0 \le \kappa' \le \kappa + 2$, the resulting domain sum is therefore at most

$$(1-x)^{-26\#N} \le e^{27x\#N}.$$

It follows that $e^{\kappa' d_k(Y)}$ times the preceding norm is bounded by

$$\sup_{s \in \mathfrak{T}_r(Y)} \sum_{N} e^{28x\#N} e^{\kappa' d_k(N)} \|R_N(s)\|.$$

At fixed s, decompose g_s into products of unit vectors in its Grassmann norm, substitute the chosen parts of the legs, and use their walk bounds. Taking the infimum over these decompositions gives the norm of g_s, so the bound retains the ratio of the first split leg. It introduces no separate supremum for each root or monomial.

The remaining factor is paid by the same weight reserves. Indeed,

$$\#N_\omega \le 8\big(d_k(X^{\uparrow}) + 1\big) + 8 \sum_{l} (|\omega_l| + 1), \qquad 224x \le \frac{1}{13}.$$

Thus

$$e^{28x\#N_\omega} \le e^{(d_k(X^{\uparrow})+1)/13} \prod_{l} e^{(|\omega_l|+1)/13}.$$

The walk estimate holds with $e^{3|\omega|/2}$ in place of $e^{|\omega|}$ and the same A_κ; it pays $e^{|\omega_l|/13}$. The preceding bounds, with the floor on h_F and ceiling on the coupling, give

$$\frac{h_b + h_c + 2h_u}{h'_b} \le \frac{1}{8}, \qquad e^{1/13}(h_b + h_c + 2h_u) \le \frac{1}{2} h'_b.$$

Here $h_u = C_{\mathrm{m}} c_h L^{-5(k-m)} \sup |B|\, w_k(n,x)$. The factor two bounds its levelling factor on $|\lambda| < 2$. Hence every expanded leg other than the first split leg retains its weight reserve.

Since $d_k(X^{\uparrow}) \le d_j(X)$, the base factor $e^{d_k(X^{\uparrow})/13}$ is paid by $\kappa_+ - \kappa \ge 2c_E + 1$. The count through a cube is

$$2048 \sum_{m \ge 0} \big(9216 e^{-(2c_E+1)+1/13}\big)^m \le e^{-1/2} C_E, \qquad C_E = 4096.$$

With the edge pointer factor $(d_j(X)+3)^3$ the same sum is at most $2048 \cdot 28 \le e^{-1/2} 94 C_E$. The two factors $e^{1/13}$ from the base and first split leg fit in the spare $e^{1/2}$. For a mass edge the base has size zero and $e^{2/13} < C_E$ suffices. The coefficient variation is still estimated directly by Lem. 9.7, polymer by polymer at rate r_V. The leg contributions use the count below, in the summed norm at rate $\kappa - 1 \ge r_V$.

There are $L^{3(k-j)}$ lower cubes in a level-k cube. A made term reaching layer $m > j$ has $d_j(X) \ge 8L^{m-j} - 2$; an edge term has only $7L - 2$ in its first later layer. Use the uniform lower bound $7L^{m-j} - 2$. The complete count is bounded by $C^{\circ}_{\mathrm{bd}}(c_\psi + 2C_{\mathcal{A}}\delta(\mathsf{t}_k)/\beta_0)$, where

$$\begin{aligned} C^{\circ}_{\mathrm{bd}} &= [9L^4 A_\kappa C_H + (2 + 6L^5 A_\kappa) C_{\mathfrak{r}} \beta_0 + 4 C_{\mathrm{m}} c_h \beta_0] \\ &\quad \times (1 + 376M^3) C_E \left(1 + \sum_{d \ge 0} (2/L)^d\right) \left(2 + \sum_{e \ge 1} L^{3e} e^{-\kappa(1-L^{-1})(7L^e - 2)}\right), \\ C_{\mathrm{bd}} &= C^{\circ}_{\mathrm{bd}} \max\{1, K_M^{-1}, K^{\mathrm{lb}}/K_M\}. \end{aligned}$$

The polynomial $376M^3$ pays edge pointers, since $\sum_{X \ni \square} (d_j(X)+3)^3 e^{-(\kappa_+ - \kappa) d_j(X)} \le 94 C_E$. All these constants precede h_F. Put $\Sigma^{\natural}_k = \sup_{j<k} \mathsf{sz}^{\natural}_j$, $\bar{v}_* = \sup_{j \le k} \bar{v}_j$, and $\bar{\eta} = \sup_j \eta_{\mathrm{sr},j} < 1/8$. Reentry through explicit legs is bounded by

$$C_{\mathrm{bd}} K_M (1 + \bar{\eta}) \left(c_\psi + \frac{2C_{\mathcal{A}}\delta(\mathsf{t}_k)}{\beta_0}\right) \max\{\Sigma^{\natural}_k, \bar{v}_*\}. \tag{38}$$

The mass-profile edges add

$$b_k^{\mathrm{ms}} = C_{\mathrm{bd}} \left(c_\psi + \frac{2C_{\mathcal{A}}\delta(\mathsf{t}_k)}{\beta_0} \right) 3M^3 w_0 \nu C_H^2 h_F^2 \bar{B}_M \gamma^2.$$

The regular terms require their zero moments and are treated in the next proof. Once a new step has been made, unchanged stored and formal legs carry the smaller next weight, so their previously established bounds persist. □

Proposition 9.14 (Closing the matter induction on histories)**.** With the common numerical choices of Prop. 10.20, the actual fermion-origin class at every history step satisfies

$$\mathsf{sz}_k^\natural \le \frac{16}{17}\varepsilon_\flat, \qquad \varepsilon_\flat = (64eL^3 K_{\mathrm{ap}}^{\max})^{-1}.$$

The class size is the supremum of its fibre norm on the admitted tiers and external-field boxes, multiplied by $e^{r_V d_k(Y)}$, with the next weights at its external and stored legs. The history step is defined on these domains, has format (36), and obeys the exact identity of Prop. 9.11. The bounds and locality of the regular and made families persist through the finite induction. ■

Proof. We include earlier frame sizes among the induction hypotheses. They are absent at $k = 0$, where there are no old activities or boundary terms and the all-field inverse supplies the first step. At a later step, complete the fermion square at the new background. The determinant is the normalizer of Prop. 6.9; the current generators become W plus their new background, and the stored quadratic matrix becomes its exact Schur complement. The remaining dependence on the gauge fluctuation and W forms the class whose size we now estimate.

The terms that do not use a boundary. Put $\delta_k = \delta(\mathsf{t}_k)$. The small-field vertex, the current mass member and the old regular activities on one lattice give the bound

$$\begin{aligned} b_k^{\mathrm{blk}} = {} & C_V h_F^2 C_{\mathcal{A}} \delta_k + (c_\psi + C_G C_{\mathcal{A}} \delta_k) c_{\mathsf{S}} h_F^2 \frac{2\bar{B}_M}{\mathsf{t}_k} \\ & + (c_\psi + C_{\mathcal{A}} s_a + C_G C_{\mathcal{A}} \delta_k) C_\omega S_\omega C_2 \phi(\mathsf{t}_k)^2. \end{aligned}$$

Here s_a is the Cauchy shift factor in the old-activity estimate; S_ω is its summed scale weight. These are precisely the previous small-field estimates read at the tier radius $C_{\mathcal{A}}\delta_k$. They control the current-region terms once the old zero-moment cancellation is justified there. For the variation on stored layers, let C_V^F bound the vertex, the mass member and the regular activities by

$$\|\mathcal{V}_B^{\mathrm{F}}(Y; B)\| \le C_V^F h_F^2 |B|_Y e^{-(\kappa-1)d_k(Y)}.$$

Here only the part depending on B is included, and the made boundary terms and edge terms are estimated separately. The vertex count is $L^{3k}L^{-2k}L^{-k} = 1$ per level-k cube. The mass and regular terms use the stored-layer counts below, with C_2^{pt} when their pointers are evaluated separately. Cauchy's estimate on $\sup|B| \le \beta_0$ supplies their factor $|B|_Y$; the shift in $\mathcal{L}_m$ has the additional factor $L^{-4(k-m)}$, which makes the layer sum finite. Thus C_V^F is fixed after C_2^{pt} and C_{lay}, before the final choice of γ, and is uniform in k. The integrated and external collar variations together contribute at most $2C_V^F h_F^2 C_{\mathcal{A}} \delta_k$.

Where an old coefficient can be evaluated. The tiers give a triple at level $t = k + 1$. On $X \subset \Lambda_\ell$, $j \le \ell \le k$, its restriction to level j has a fixed margin τ_{oc} inside the core radii, where

$$\tau_{\mathrm{oc}} = 1 - \frac{(1-\tau_{\mathrm{m}})\theta_{\mathrm{oc}}}{1+\beta_s}, \qquad b_2 = \frac{\tau_{\mathrm{oc}}\alpha_0}{18K^\sharp(1+c_\natural)}.$$

For the local shift define $|B|_X^{\mathrm{sh}}$ as the largest, over the bond and difference-quotient places of X, of $L^{-(m(x)-j)}$ times the exponentially weighted supremum of $|B|$ in the shift estimate. Then

$$|B|_X^{\mathrm{sh}} \le L^{-(\ell-j)} \sup|B|; \qquad |B|_X^{\mathrm{sh}} \le eL^{-160(k-m)} \sup|B|$$

if all those places lie in layers at most $m < k$. The restricted shift and its first difference and source components scale with powers $1, 2, 3$, respectively. Their sizes are thus at most $|\mu| c_h |B|_X^{\mathrm{sh}}$. When this is below b_2, the outer-substitution estimate puts the shifted triple inside the level-j core. Its pair is

$$\left(e^{i\eta_k \mu \mathbf{A}}\mathbf{U},\, L^{-3(t-j)}\mathbf{J} + \mu L^{-3(k-j)}\mathbf{K}\right).$$

Indeed the unshifted triple lies at radii $(1-\tau_{\mathrm{oc}})\alpha_i$, and the outer-substitution bound consumes less than $\tau_{\mathrm{oc}}\alpha_i$. The admitted constants give $128K^{\sharp}(1+c_{\natural})c_{\mathrm{os}} \le 1$ and $128\tau_{\mathrm{oc}} > 23 > 18$, so $c_{\mathrm{os}}\alpha_0 < b_2$. Hence the Cauchy disc $|\mu| c_h \sup|B| \le L^{\ell-j} c_{\mathrm{os}}\alpha_0$ lies in the domain, including lag zero. This verifies coefficient holomorphy and its uniform bound at every shifted pair used below.

Zero moments in the current region. The cancellation is applied to local pieces, not to an unchanged whole-volume background. A piece p consists of a polymer and one local or walk term in each of its two legs; let Z_p contain those three supports. Let m_p be its trivial-background moment for the empty history, and m_p^h its moment along the given history. A missing piece has moment zero. If $Z_p \subset \Lambda_j^{\sim -1}$, all pointed shares of the polymer are present and both local background maps agree there; hence $m_p^h = m_p$. Every other piece is a rim piece. Thus, at a cell c, the moment left by the history is the sum of $m_p^h - m_p$ over those rim pieces. On a torus it also contains the wrapping sum minus the nonfitting infinite-lattice sum already treated in Lem. 7.9.

A different rule decides which terms may use differences of legs. At lags $k - j \ge n_{\mathrm{lag}}$, take a piece whole if $Z_p \not\subset \Omega_k$, or if $Z_p \not\subset \Omega_{k+1}$ and $d_j(Z_p) \ge 8L^{k-j} - 2$; otherwise use the difference decomposition of Lem. 7.9. The first rule, using $\Lambda_j^{\sim -1}$, decides whether its number is unchanged. The second, using the two current regions, decides whether its local estimates have the needed regular scale.

Every piece taken with differences is unchanged. Its support and the local gauge cube used in Lem. 7.9 lie where the smoothed scale is L^k. They lie within distance $8ML^k$ of Ω_{k+1} and hence in Λ_k, since the removed cell layer has width at least $12ML^k$. The core and shifted-core checks above therefore justify the old regular estimate there. The Hölder difference of its background legs supplies the same summable scale factor. The terms with a split stored leg additionally carry the factor below denoted Θ_k'.

To check the algebra, sum only the pieces taken with differences. Their constant terms are $\bar{\Phi}(z_c)(\sum_p' m_p)\Phi(z_c)$. The sum of all empty-history moments is zero on the infinite lattice, or the wrapping-minus- nonfitting sum on a torus. Subtracting the moments of the pieces taken whole leaves exactly the history rim moment above. Each such term is filed on the level-k hull of Z_p, enlarged by the walks of its expanded legs. That hull meets the complement of Ω_{k+1}, so no interior coefficient acquires a dependence on the history. Wrapping terms and the corresponding subtracted empty-history terms are filed on the same wrapping support and cancel there when both are present.

For these whole terms, the two history legs have relative sizes at most L each, and at most $L^{1-(k-j)}$ if $Z_p \subset \Omega_k$. The empty-history comparison legs have size $L^{-(k-j)}$. Count a term through every cell of Z_p, not only its original filing cell. A walk of length n reaches at most $(2n+5)^3$ relevant cubes; put $n_{\mathrm{wk}} = \max_{u \ge 0}(2u+5)^3 e^{-u}$ and $c_{\mathrm{geo}} = 9n_{\mathrm{wk}}^2$. The two walk counts pay these polynomials with their exponential weights. The hull weight leaves $e^{-\kappa(1-L^{-1})d_j(Z_p)}$. A piece leaving Ω_k has $d_j(Z_p) \ge 8L^{k+1-j} - 2$; every other whole piece has $d_j(Z_p) \ge 8L^{k-j} - 2$. The factors $L^{3(k-j)}$ for cells and the two leg sizes are consequently summable. Their bound is $\Theta_k C_\omega c_{\mathrm{rim}} C_2 \phi(\mathfrak{t}_k)^2$, where

$$c_{\mathrm{rim}} = c_{\mathrm{geo}}(1+L^2)\sum_{d \ge 1} L^{3d} e^{-\kappa(1-L^{-1})(8L^d-2)}.$$

Regular terms in stored layers. At a pointer in $\mathcal{L}_m$, retain the formal lower-level legs with the weights of that layer. For large $m - j$, pieces contained in $\Omega_{m-1} \setminus \Omega_{m+2}$ use the difference estimate at the local scale; the remaining pieces are taken whole. The former are in a regular core at some $\check{m} \ge j+2$, so the shifted-domain argument applies with $\ell = \check{m}$. The lower lags are bounded directly

by their finite count. For degree at least four, the extra leg weights supply the power counting; for degree two the Hölder difference is essential.

There is also a cell term in the zeroth moment. Its coefficient $q_j(z)$ has $\sum_{z\in c} q_j(z) = 0$ and commutes with colour. Transport the two fields inside c to z_c by block links and their inverses. Then

$$\sum_{z\in c} \bar\Phi(z) q_j(z) \Phi(z) = \sum_{z\in c} \left[(\bar\Phi(z) - \bar\Phi(z_c)) q_j(z) \Phi(z) + \bar\Phi(z_c) q_j(z) (\Phi(z) - \Phi(z_c)) \right].$$

Each term has one difference. The block-link Hölder estimate applies directly; no replacement by straight transport is made. The two transports have length at most $2M$ each. Their nonunitary parts cost at most $\exp(6MC_Q\alpha_1 L^{-(m-j)}) \le e^{\kappa}$ under the chosen radius ceiling and $m - j \ge n_{\mathrm{lag}} \ge 4$. The term is filed with the cell and the expanded leg and shift supports, hence remains a boundary term.

For a whole piece based at a stored pointer, reaching a later layer $m' > j$ forces $d_j(Z_p) \ge 7L^{m'-j} - 2$; reaching Ω_{k+1} gives the stronger $8L^{k+1-j} - 2$. Use the weighted walk count with $e^{3|\omega|/2}$: one factor $e^{|\omega|}$ pays its reachable cubes, and the remainder pays crossing. This still has constant A_κ, since $e^{1/8}/(1 - e^{-11/8}) < 1.517$ and the local term fits its defining factor two. The individual pointed terms are evaluated at their own pairs. Thus their norms must be summed before the polymer count. By Lem. 8.8, at level j this costs $\epsilon_j^{\mathrm{pt}} = C_2^{\mathrm{pt}} \phi(\mathtt{t}_{j-1})^2$. The bound for these terms is therefore $\Theta_k C_\omega c_{\mathrm{rim}}^{\mathrm{pt}} C_2^{\mathrm{pt}} \phi(\mathtt{t}_k)^2$, with

$$c_{\mathrm{rim}}^{\mathrm{pt}} = \frac{3L}{L-2} c_{\mathrm{geo}} (1 + L^2) \left(1 + \sum_{d\ge 1} L^{3d} e^{-\kappa(1-L^{-1})(7L^d - 2)} \right).$$

Wrapping and nonfitting pieces obey the same count because their size is at least $L^{k-j} - 2$.

Finally, a pointed share whose pointer and filing cell lie on opposite sides of $\partial\Omega_{k+1}$ is taken whole. In a level-k cube at most $6(t+1)L^{2(k-j)}$ lower cells lie within distance $(t+1)M$ of those boundary faces. The residual exponential and this count give at most 39 times the birth norm. The walks of two legs contribute $1 + 2n_{\mathrm{wk}}$. For higher degree, the remaining legs pay their walk count; at lag zero use $2p^2(3/4 + \Theta_k'/4)^{2p-1} < 4.3$ under $\Theta_k' \le 2 \cdot 10^{-3}$. This gives $39(1 + 2n_{\mathrm{wk}}) \Theta_k S_\omega' C_2 \phi(\mathtt{t}_k)^2$, where $S_\omega' = \sum_{d\ge 0} L^{-d} (1 + d \log L)^{2\mathfrak{p}_0}$. These shares are filed with their whole supports, so their domains meet Z_{k+1}.

To collect the estimates, set

$$\begin{aligned} \Theta_k' &= 4Lc_\psi + \frac{2C_{\mathcal{A}}\delta_k}{\beta_0} [9L^4 A_\kappa C_H + (2 + 6L^5 A_\kappa) C_{\mathfrak{r}} \beta_0], \\ \Theta_k &= c_\psi + C_{\mathcal{A}} s_a + C_G C_{\mathcal{A}} \delta_k + \Theta_k'. \end{aligned}$$

The stored-layer difference and finite-lag counts are bounded by a constant C_{lay} fixed before h_F. More explicitly, one can take

$$C_{\mathrm{lay}} = 4M^3 A_\kappa L^{1+\alpha} C(\alpha) C_H \max\{1, C_{\mathrm{leg}}^{\square}, C_\omega\}\, c_{\mathrm{ps}}, \quad c_{\mathrm{ps}} = (2A_\kappa C_H)^2 c_{\mathrm{geo}} L^{3n_{\mathrm{lag}}+12}.$$

The finite-lag factor pays those lags and the extra local or walk terms; the Hölder factor is needed only in the remaining degree-two terms. The cell estimate above is included in $C_{\mathrm{leg}}^{\square}$. Since

$L^{3(k-m)}(2L^{-4})^{k-m} = (2/L)^{k-m}$, the stored-layer sum has the bound

$$\begin{aligned} b_k^{\mathrm{lay}} &= 2C_V^F h_F^2 C_{\mathcal{A}} \delta_k \\ &\quad + \Theta'_k \left(C_\omega S_\omega C_2 + \frac{L}{L-2} C_{\mathrm{lay}} S'_\omega C_2^{\mathrm{pt}} \right) \phi(\mathsf{t}_k)^2 \\ &\quad + \Theta_k \left[C_\omega c_{\mathrm{rim}} C_2 + C_\omega c_{\mathrm{rim}}^{\mathrm{pt}} C_2^{\mathrm{pt}} + 39(1+2n_{\mathrm{wk}}) S'_\omega C_2 \right] \phi(\mathsf{t}_k)^2 \\ &\quad + \frac{L}{L-2} C_{\mathrm{lay}} \Theta'_k w_0 \nu C_H^2 h_F^2 L \bar{B}_M \gamma^2. \end{aligned}$$

Together with the mass-edge bound b_k^{ms} from the preceding proof and b_k^{blk}, this bounds every regular term, vertex and mass edge. Each decreases to zero with γ after the earlier constants and h_F have been fixed.

The two boundary contributions. The old made and edge terms reenter through their explicit legs by (38). The floor $h_F \geq 32 C_{\mathrm{bd}} K_M A_\kappa \Gamma_F$ and ceiling $32 C_{\mathrm{bd}} K_M C_{\mathcal{A}} \delta_k \leq \beta_0$ make this at most $\frac{9}{64} \max\{\Sigma_k^\natural, \bar{v}_*\}$. Their coefficient variation, Lem. 9.7, has bound

$$C^{\mathrm{bd}} \vartheta_{0,k} \left(\tfrac{9}{8} K^{\mathrm{lb}} \Sigma_k^\natural + 108 M^3 K_M \bar{v}_* \right).$$

The factor $108M^3$ is the edge pointer bound $4M^3(d+3)^3 e^{-(\kappa_+ - \kappa)d} \leq 108M^3$. The separate ceiling $16 C^{\mathrm{bd}} \vartheta_{0,k} (\frac{9}{8} K^{\mathrm{lb}} + 108 M^3 K_M) \leq 1$ makes this at most $\frac{1}{16} \max\{\Sigma_k^\natural, \bar{v}_*\}$.

Putting $b_k^{\mathrm{tot}} = b_k^{\mathrm{blk}} + b_k^{\mathrm{lay}} + b_k^{\mathrm{ms}}$, we have proved

$$\mathsf{sz}_k^\natural \leq \frac{13}{64} \max\{\Sigma_k^\natural, \bar{v}_*\} + b_k^{\mathrm{tot}}.$$

At the first step the boundary maximum is absent. Let $\bar{b} = \sup_k b_k^{\mathrm{tot}}$. Since $\bar{v}_j \leq b_{j-1}^{\mathrm{blk}} \leq \bar{b}$, induction gives $\mathsf{sz}_k^\natural \leq 64\bar{b}/51$. The three ceilings $b^{\mathrm{blk}}(\gamma), b^{\mathrm{lay}}(\gamma), b^{\mathrm{ms}}(\gamma) \leq \varepsilon_\natural/4$ therefore give $64\bar{b}/51 \leq 16\varepsilon_\natural/17$. This proves the present-frame size from earlier sizes before using Lem. 9.9.

The scalar entries satisfy Lem. 9.6, including the boundary coefficient variations just checked, and their q and tail classes meet Lem. 9.4. The actual joined frame is consequently admissible. Its interior outputs agree by Lem. 9.10; extract their vacuum, action and mass terms there alone, and file all profile complements on their rim supports. The other outputs, the normalizer outside the interior, and the collar correction are the boundary families of Prop. 9.13. The coordinatewise identity of Prop. 9.11 proves the new format and the exact step. The stored quadratic factor is bounded by Lem. 9.12. This closes the finite induction and preserves every local domain used in it. □

9.6. **Vacuum constants and history masses.** Let e_j be the full vacuum constant per site extracted by the empty history at level j, and set $E_k = \sum_{j \leq k} e_j n_j$. In particular it includes both the gauge and fermion normalizers and the value of the q Gaussian family. The two gauge Gaussian conventions differ by

$$3(L^3-1) \sum_{j<k} n_{j+1} \log(\mathsf{t}_j / x_j).$$

This number is retained in E_k. It is not a bounded error per level-k site. Indeed a block fluctuation has $9(L^3-1)$ real coordinates: $3(L^3-1)$ gauge coordinates and $6(L^3-1)$ transverse coordinates. At the trivial background the action Hessian vanishes on the former, where the gauge-fixing form is $|\lambda|^2$. Scaling the action by τ therefore multiplies the determinant of the full form by $\tau^{6(L^3-1)}$. Equivalently the covariance normalizer contributes $\tau^{-9(L^3-1)/2}$ and the q-Gaussian contributes $\tau^{3(L^3-1)/2}$. Their product is $\tau^{-3(L^3-1)}$, which gives the displayed vacuum correction. The collar term R vanishes at the trivial zero stored field.

Lemma 9.15 (The vacuum constant of a label)**.** There are label-dependent complex numbers $f_j(\mathcal{S})$ such that

$$-\mathcal{E}^F_{k,\mathcal{S}} = E_k - \sum_{j\le k} f_j(\mathcal{S}), \qquad |f_j(\mathcal{S})| \le C^F_\flat(1+\log \mathsf{t}_{j-1})\#_j(Z_j^\sim).$$

The empty-history finite-size remainder is bounded by a constant $C_{\rm fs}$ independent of K and ℓ. It is assigned to the regular scalar sum, with its opposite subtracted from the extracted vacuum constant. Thus E_k contains the infinite-lattice per-site numbers $e_j n_j$. ■

Proof. Define f_j as the number extracted by the empty history on the same torus minus the number extracted by the given history. Write $\mathfrak{f}_k = \sum_{j\le k}\mathfrak{f}^{(j)}$ for the empty-history finite-size number. With the stated assignment the extracted constant is

$$\sum_{j\le k}(e_j n_j + \mathfrak{f}^{(j)} - f_j) - \mathfrak{f}_k = E_k - \sum_{j\le k} f_j.$$

This proves the exact identity. The partial pointer and domain sums defining f_j need not be real; the conjugation identity holds for the full summed energy. The difference has three parts. The missing gauge integral costs its vacuum number, bounded by $C(1+\log \mathsf{t}_{j-1})$, at each cube of Z_j. In the fermion normalizer write the trace as a sum of diagonal place contributions. At a place inside Ω_j its difference from the empty-history contribution requires a walk reaching Ω_j^c and is at most $Ce^{-c_f \operatorname{dist}_j(p,\Omega_j^c)}$. The missing places outside Ω_j have a uniformly bounded free fermion number. Both sums cost $C\#_j(Z_j^\sim)$. Finally a missing regular pair through a pointer z must meet Z_j; the activity and entropy bounds give $CE_0 e^{-(\kappa+1-\log 9216)(\operatorname{dist}_j(z,Z_j)/M-2)}$. Summation over z again costs $C\#_j(Z_j^\sim)$.

A wrapping polymer or trace walk has size at least $\operatorname{side}_j/(2M)-2$. Hence the finite-size number at that level is at most $CE_0\operatorname{side}_j^3 e^{-(\kappa+1-\log 9216)(\operatorname{side}_j/(2M)-2)}$. The sides run through distinct elements of $2L^{\mathbb{N}}$, so the sum is finite. All these numbers depend on the label, not on an evaluation configuration. □

We record the pure history estimate in the form used here. Its hypotheses include the admitted calibration geometry, the failed-test support and curvature conditions, the pivot determinant identities and the retained quadratic lower bound on each stored piece. Suppose a density, scalar or normed, is bounded on its realized free coordinates by

$$e^{\operatorname{Re}(E-\sum_j f_j)+C_E\Sigma_{\mathcal{L}}-A(\mathsf{bw},U)}\prod_{(t,D)}|\mathsf{u}_{t,D}[Q_{t,D}]|,$$

where $\mathsf{bw}\ge(1-a)\beta$, $0\le a<1$, and the piece forms have the corresponding positive retained lower bound. Here $\mathsf{u}_{t,D}$ and $Q_{t,D}$ are the weight and retained quadratic form of a stored component D created at level t. The measure $d\mathscr{M}_h^{\rm fr}$ is the product of Haar probability measures on the free configuration coordinates and Lebesgue measures on the free Lie-algebra coordinates, including the current field, as in [3, eq. 4.cccli]. The count $n_t^{\rm cr}$ is the number of cells created at that level; g_t and $p_0(g_t)$ are its coupling and test threshold in the cited gauge construction. We will also use its nonnegative history weight

$$\Theta^{(k)}(h) = w\operatorname{Dep}_k(h) + c_a\operatorname{Vol}_k(h), \qquad \operatorname{Dep}_k(h) = c_{\rm Dep}\sum_t p_0(g_t)^2 n_t^{\rm cr}(h).$$

Here $0\le w<1$, $c_a, C_{\rm vol}\ge 0$ and $0<c_{\rm Dep}\le c_*$ are fixed. The volume functional is $\operatorname{Vol}_k(h) = C_{\rm vol}\sum_j \check{R}_j\#_j(Z_j(h))$, where $\check{R}_j$ is the coarse-cell radius of the gauge history at level j. Thus the coefficient of a created-cell deposit in $\Theta^{(k)}$ is $c_\Theta = wc_{\rm Dep} < c_*$. The weighted history sum uses the same component estimates, with part of their decay reserved for this weight; see [3, eq. 4.dcccxxi]

and [3, eq. 4.dcccxxii] and their preceding definitions. The component majorant and its product integral [3, Prop. 4.153] and [3, Prop. 4.154] give

$$e^{-\operatorname{Re} E+\operatorname{Re}\sum_j f_j}\int \|g_h\|\,d\mathscr{M}_h^{\mathrm{fr}} \le e^{C_{\mathrm{maj}}(n_k+\Sigma_Z)}\prod_t e^{-(1-a)c_* p_0(g_t)^2 n_t^{\mathrm{cr}}}. \tag{39}$$

This is the cited component estimate, applied to the displayed density: its proof uses the created-cell action allocation, the piece pivot determinants, the positive piece-form bound and Tonelli on the disjoint free coordinates. If also $|f_j| \le C_\flat(1+\log \mathfrak{t}_{j-1})\#_j(Z_j^\sim)$, then for $2\mathfrak{p}_0 > 3r_c+2$ and sufficiently large final threshold, the history summation theorem gives

$$\sum_h e^{-\operatorname{Re} E}\int \|g_h\|\,d\mathscr{M}_h^{\mathrm{fr}} \le e^{(C_{\mathrm{maj}}+2)n_k}. \tag{40}$$

The threshold is formed from $((1-a)c_*, C_{\mathrm{maj}}, C_\flat, \eth)$, $0<\eth<1$, before the cutoff and volume. These are the full inputs to the pure summation interface; the density estimate remains to be verified for the present model.

Theorem D (The upper history bound)**.** Every history has the form (36). Its normed density satisfies

$$\|g_h\| \le e^{-\beta A(U)+C_E^F\Sigma'(h)+E_k-\operatorname{Re}\sum_j f_j}\prod_{(t,D)}|\mathsf{u}_{t,D}[Q_{t,D}]|,$$

and

$$e^{-E_k+\operatorname{Re}\sum_j f_j}\int \|g_h\|\,d\mathscr{M}_h^{\mathrm{fr}} \le e^{C^\dagger_{\mathrm{maj}}(n_k+\Sigma_Z)}\prod_t e^{-c_* p_0(g_t)^2 n_t^{\mathrm{cr}}},\quad C^\dagger_{\mathrm{maj}} = 48L^3C_E^F + C'_*.$$

Its Berezin integral is a positive measure, and the sum of their masses is the partition function. ■□

Proof. The Schur identity, the local conditional expansion and the boundary reentry just proved give the format by induction. Gauge changes, coordinate pushforwards and multiplication by test factors act coefficientwise in the finite-dimensional Grassmann algebra. Joint gauge invariance gives one value on each collared evaluation orbit. The exact bare determinant is nonnegative, so each history's Berezin mass is nonnegative.

The degree-zero regular and boundary sums, together with the coupling-profile shifts, are bounded by $C_5^F\Sigma_{\mathcal{L}}$. To see the profile bound, a level-j cube contains M^3L^{3j} fine sites, its regular plaquette action is at most $C_{\mathrm{reg}}^2L^{-4j}$, and its coefficient is $L^j b_j$; its cost is therefore a constant per cube. A boundary domain has a cube in $Z_j^\sim$, so the entropy bound and $\sum_j \#_j(Z_j^\sim) \le C_Z\Sigma_{\mathcal{L}}$ give its contribution. The wrapping constant is covered by $C_{\mathrm{fs}}\Sigma_{\mathcal{L}}$.

For completeness the same count in all degrees is finite. Each background leg from level j to a layer n costs at most $CL^{1-(n-j)}$. Degree at least four gives net decay $L^{-(n-j)}$ after the cube count, with fixed powers of L in the constant. In degree two, sum the gauge-comparison bounds over the walks at all rows of X, as in the proof of Lem. 7.9. The possible first cubes number at most $N_X = C_{\mathrm{nb}}M^3\#X$. For each leg the sum of its weighted bounds is at most

$$2C_H L^{1-(n-j)}(1+N_X A_\kappa).$$

The two factors $1+N_XA_\kappa$, the squared path length and the complex transports cost at most a fixed multiple of

$$(1+8C_{\mathrm{nb}}M^3A_\kappa)^2 2e^{\kappa/8}M^2\sup_{d\ge 0}(1+d)^4e^{-7\kappa d/8}.$$

The walk weights pay the remaining length polynomials. With $L^{3(n-j)}$ pointers per level-n cube, the scale factors are

$$L^{3(n-j)}L^{2-2(n-j)}L^{2-2(n-j)} = L^4L^{-(n-j)}.$$

These fixed factors enter the constants before h_F is chosen. At the trivial field group the kernel by its total zero moment and use differences of legs; their Hölder exponent 3/4 gives a possible factor

$L^{(n-j)/4}$. The birth size is at most $CL^{-(n-j)}(1+(n-j)\log L)^{2\mathfrak{p}_0}\phi(x_{n-1})^2$, leaving a summable $L^{-3(n-j)/4}$. A missing cancellation near a rim requires a polymer spanning at least RL^{n-j-1} and is covered by its exponential decay. The mass profile costs $w_0\nu C_H^2 h_F^2 L\bar{B}_M\Sigma_{\mathcal{L}}$. Reading formal legs through the whole real background costs at most one per leg by the enlarged row bound. Together with Lem. 9.12 these estimates give a finite C_E^F.

The density estimate uses only degree zero and the stored factor. Before the store is integrated, its scalar coefficient is the positive product of the successive determinants. Equivalently it is $F_h = \det(D+m_0)/\det\mathbb{D}_h > 0$; the Schur identities give this formula, including at arbitrary gauge fields. Taking degree zero of the format gives the scalar history density for $e^{-\beta A}F_h$. Its exponent has the scalar bound just established. Multiply its density by the stored-factor norm bound. The same realized Borel coordinates and piece forms therefore verify every hypothesis of (39), with loss zero and the constant per cube at most $48C_E^F$. This proves both displayed inequalities. □

Theorem E (One vacuum energy in both bounds)**.** With a sufficiently large final threshold,

$$e^{E_k - C_P(1+\log x_k)n_k} \le Z_{K,\ell} \le \sum_h \int \|g_h\|\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{E_k+(C+2)n_k}.$$

The constants are uniform in the cutoff and volume; C_P may depend on the positive lower mass bound. ■□

Proof. The upper bound is (40), with $E = E_k$ and the label error from Lem. 9.15. For the lower bound retain the empty history at the last level. Its mass is not estimated by the free determinant alone: its remaining interacting Grassmann integral must also be bounded below.

Put

$$\Gamma_*^2 = \frac{2}{c_b m_-}, \qquad h_* = \max\{h_F, \Gamma_*\}.$$

Read the old activities at weight $H = 2A_\kappa h_*$ and the level-K generators at weight $2h_*$. Impose $\gamma \le \gamma(H, 5B_s)$ and form the activity constants at this auxiliary weight, using Prop. 8.6 and Lem. 7.10. The activities themselves do not depend on the norm used to estimate them. We shall localize their sum

$$P = \sum_{j\le K}\sum_X E_{\ge 2}^{(j)}(X) = \sum_Y P(Y)$$

on level-K polymers, with

$$\sup_{\square}\sum_{Y\supset\square} e^{\kappa d_K(Y)}\|P(Y)\|_{2h_*} \le \epsilon_P := C_{2,*}\phi(\mathfrak{t}_{K-1})^2. \tag{41}$$

The constant $C_{2,*}$ may depend on m_- through h_*, but is independent of the cutoff and volume.

For degree at least four, and for the activities born at level K, substitute the local and walk parts of every background leg directly. File the resulting term on the level-K hull of X, enlarged successively by its walk polymers. This assignment does not depend on its pointer, so the pointers can first be summed inside the activity. The extra leg weights give the usual factor $L^{-(K-j)}$.

For degree two at level $j < K$, the cancellation must be localized before this substitution. Write $\Phi = \mathsf{H}_{j,K}\Psi_K$ and read each background in two stages, first through $\mathsf{H}_{n,j}$ and then through $\mathsf{H}_{j,K}$. Relative to a direct substitution this costs at most $(A_\kappa C_H)^2$. For a pointed term (z, X), compare its coefficient with the trivial pair, keeping the transports. A piece consists of X and one local or walk term in each of its two level-j legs. Write m_p for its zeroth moment, with both columns transported to z along the paths of the piece. On a torus use the lifted paths, including the seam signs. Expanding each leg as its value at z plus its difference from that value leaves terms with a difference and the term $\bar{\Phi}(z)m_p\Phi(z)$.

File the terms with differences on the hull of the whole piece and its level-K walks. File the term without a difference on the cube containing z and its two level-K walks, independently of X. All such moments at z now belong to the same $P(Y)$ and can be summed there. Their sum is

$q_j(z)$ on the infinite lattice. On a torus it is $q_j(z) + \Sigma^{\mathrm{w}} - \Sigma^{\mathrm{nf}}$, where the last two sums consist of wrapping pieces and the infinite-lattice pieces that do not fit. Each has $d_j(Z_p) \geq \ell_j/(2M) - 2$, with $\ell_j = \ell L^{K-j}$. Thus the torus discrepancy retains exponential decay.

The remaining $q_j(z)$ terms are combined cell by cell. Their coefficients commute with colour and sum to zero in each cell c. Transport the fields to z_c by block links along paths inside c, of length at most $2M$. Then

$$\sum_{z\in c} \bar\Phi(z) q_j(z) \Phi(z) = \sum_{z\in c} \Big[(\bar\Phi(z) - \bar\Phi(z_c)) q_j(z) \Phi(z) + \bar\Phi(z_c) q_j(z) (\Phi(z) - \Phi(z_c)) \Big] .$$

File these terms on the cube containing c, enlarged by the walks of their legs. Every retained term therefore has either a field comparison, a difference, or a wrapping factor. The total P is unchanged.

We give the counts needed for (41). Put $d = K - j$. The local comparison and difference estimates of Lem. 7.9 give the factors L^{-d} and $L^{d/4}$, respectively, per level-K cube. These factors already include the count of L^{3d} level-j cubes. For a piece of degree two let

$$D'_p = M(1 + d_j(X)) + 9M_w \sum_i (|\omega'_i| + 1),$$

where ω'_i are its level-j walks. Recounting the first cubes of those walks costs at most $(2C_H)^2(1 + 8C_{\mathrm{nb}} M^3 A_\kappa)^2 (1 + d_j(X))^2$. The transports in the comparison with the trivial pair cost at most

$$e^{\alpha_1 C_t D'_p/4} \leq e^{\kappa(1+d_j(X))/8} e^{\sum_i (|\omega'_i|+1)/8}$$

under $2C_t\alpha_1 M \leq \kappa$ and $18 C_t \alpha_1 M_w \leq 1$. The comparison also costs D'^2_p, while the difference terms cost at most $(C_t D'_p)^2$. Split the residual rate after taking the level-K hull as

$$e^{-\kappa(1-L^{-d})d_j(X)} = e^{-\kappa'_j d_j(X)} e^{-\kappa'_j d_j(X)}, \qquad \kappa'_j = \tfrac{1}{2}\kappa(1 - L^{-d}) \geq \kappa/4.$$

The first factor pays the polynomial in $d_j(X)$ and the transport exponential, at the finite cost

$$c^{(K)}_{\mathrm{pr}} = 2e^{\kappa/8} C_t^2 M^2 \sup_{u\geq 0} (1+u)^4 e^{-\kappa u/8}.$$

The walk weights pay the factors involving their lengths.

There is a separate count because the degree-two terms are formed at each pointer. By Lem. 8.8 their support contains z, and the pointed form of the unpointed estimate gives

$$\sum_{X\ni z} e^{\kappa d_j(X)} \sup_P \|E^{(j)}_{\geq 2}(X; P; z)\|_H \leq \epsilon_j, \qquad \epsilon_j = C_2 \phi(\mathsf{t}_{j-1})^2.$$

The mass term has this support property too, since it stands on the cube of z. If X meets a level-K cube $\square$, then $\mathrm{dist}(z, \square) \leq M(d_j(X) + 2)$ in level-j units. The second residual factor consequently contributes at most $e^{-\kappa(\mathrm{dist}(z,\square)/M - 2)^+/4}$. There are $M^3 L^{3d}$ pointers inside $\square$ and at most $(2n+3)^3 M^3 L^{3d}$ in the outer shell of width M at distance between nM and $(n+1)M$. Hence their total cost is

$$c^{(K)}_{\mathrm{pt}} L^{3d} \epsilon_j, \qquad c^{(K)}_{\mathrm{pt}} = M^3 \left[1 + \sum_{n\geq 0} (2n+3)^3 e^{-\kappa(n-2)^+/4} \right].$$

The preceding power counts already include $L^{3d}\epsilon_j$. Replacing that count by the pointer sum therefore multiplies them by $c^{(K)}_{\mathrm{pt}}$ alone.

Both stages of a leg can make its term reach additional level-K cubes. A level-j walk ω' followed by a level-K walk ω reaches at most

$$(2|\omega| + 5)^3 [1 + (2|\omega'| + 5)^3] \leq n_{\mathrm{wk}} (1 + 2.3 n_{\mathrm{wk}}) e^{|\omega|} e^{|\omega'|/2}$$

cubes for each cube of the hull. Use the weight $e^{3|\omega'|/2}$ for the first walk. Its weighted sum still has constant A_κ, as proved in the stored-layer count: $e^{|\omega'|}$ pays the length and transport factors,

and $e^{|\omega'|/2}$ pays this new count. The second walk uses $e^{|\omega|}$. The resulting factor for the two legs is $1+2n_{\mathrm{wk}}(1+2.3n_{\mathrm{wk}})$.

The torus discrepancy is filed at the pointer and its two level-K walks. It costs at most

$$c_{\mathrm{wr}}^{(K)}L^{3d}e^{-\kappa(\ell_j/(2M)-2)}\epsilon_j, \qquad c_{\mathrm{wr}}^{(K)}=2(1+2n_{\mathrm{wk}})M^3C_H^2C_{\mathrm{wr}},$$

where C_{wr} is the row-sum constant in the torus part of Lem. 7.9. This is summable in d, since $\ell_j\ge 2L^d$. For the cell terms, the difference of neighboring fields uses the block-link Hölder estimate directly. With $\alpha=3/4$, their count is

$$c_{\mathrm{q}}^{(K)}L^{d/4}\epsilon_j, \qquad c_{\mathrm{q}}^{(K)}=8(1+2n_{\mathrm{wk}})M^4C'^2C_H\max\{2,C(\alpha)\},$$

where C' bounds the weighted row sums of the background kernels. Here M^3 counts the pointers in a cell and $2M$ bounds its transport length. These transports are unitary at the real configurations used for the lower bound.

All displayed geometric constants precede h_F. The calibration estimate

$$\phi(\mathtt{t}_{j-1})^2\le L^{-d}(1+d\log L)^{2\mathfrak{p}_0}\phi(\mathtt{t}_{K-1})^2$$

now completes the sum: its slowest term is bounded by $\sum_{d\ge0}L^{-3d/4}(1+d\log L)^{2\mathfrak{p}_0}<\infty$. This defines a finite $C_{2,*}$ containing the preceding counts and proves (41).

Include the mass member $-M_{h_\emptyset}$ and the bilinear part of P in $-\langle\bar\Psi,p\Psi\rangle$. The mass member has operator norm at most $C_{\mathsf{S}_1}\bar B_M\gamma^2$; impose $C_{\mathsf{S}_1}\bar B_M\gamma^2\le c_bm_-/8$. For the activity bilinear, a row in $\square$ has row sum at most $(2h_*)^{-2}\sum_{Y\supset\square}\|P(Y)\|$, and the same holds for columns. Schur's test therefore bounds its operator norm by $(2h_*)^{-2}\epsilon_P$. Choose the threshold $c_*(m_-)$ in $\phi(\gamma^{-2})\le c_*(m_-)$ so this is at most $c_bm_-/8$. Then $\|p\|\le c_bm_-/4$, and $G_*=D_K+p$ has real part at least $c_bm_-/2$. Its inverse has Gram bound Γ_* and exponential decay. Indeed D_K and the mass member have exponential kernels, and (41) gives rate κ/M for the activity bilinear. Conjugating by $e^{\alpha_*|y-y_0|}$, with $\alpha_*>0$ small enough depending on m_- and these fixed kernel bounds, changes G_* by at most $c_bm_-/4$. The inverse is then bounded by $2\Gamma_*^2e^{-\alpha_*|y-y'|}$, uniformly in the cutoff and volume.

Apply the finite Grassmann forest expansion to the remaining terms of degree at least four, between fixed coarse cubes. The interpolation matrix is positive semidefinite and preserves the covariance's Gram bound; the derivative estimate $\|\partial F\|_{h_*}\le h_*^{-1}\|F\|_{2h_*}$ and the line bound $2e^{-\alpha_*\,\mathrm{dist}}$ give convergence when $\|P\|_{2h_*}C(M_*,\nu)$ is small. Its connected sum has modulus at most ℓ^3. The original integral is nonnegative, and its convergent logarithm is real by conjugation. It is therefore at least $(c_bm_-/2)^{\nu\ell^3}e^{-\ell^3}$.

Restrict the coarse links to the ball of radius $x_K^{-1/2}$. Its action costs at most $24(B_3c_r)^2\ell^3$, and the Haar ball estimate gives volume at least $\exp[-(\frac92\log x_K+3\log7)\ell^3]$. The scalar regular-family bound now gives the last-level empty-history mass at least $e^{E_K-C_P(1+\log x_K)\ell^3}$.

The positive exact history decomposition shows that the empty-history mass at level k dominates that at level K. Finally $|e_j|\le C_e'(1+\log x_{j-1})$ and $\sum_{j>k}n_j\le n_k/(L^3-1)$ give $|E_K-E_k|\le C_e'(1+\log x_k)n_k/(L^3-1)$. Enlarging C_P proves the asserted lower bound with the same E_k as in the upper bound. $\square$

10. The source chain and local source energies

To control the correlations we insert sources before integrating any fields. A source for the action density changes the plaquette weight; a source for a fermion bilinear changes the fine mass matrix. The second change also produces a relevant matrix mass at later scales. We first control this induced mass and the energy extracted from each local step. We then prove that the same energy normalizes the sum of all histories. This distinction matters at complex sources, where a history density need not be positive.

The resulting correlation estimate is Thm. F below. Its constants are independent of both the cutoff and the torus side. The estimate uses locality of the source energy and reflection positivity: the former makes separated sources almost additive, and the latter converts a whole-volume bound into a bound for one cell. Throughout this section we use the ordered choices of constants and the pure gauge construction verified in Prop. 10.20.

The sources are $s = (w, J)$, with w complex Lipschitz and J a bounded field in

$$\mathbb{M} = \mathrm{End}(\mathbb{C}^2_{\mathrm{s}} \otimes \mathbb{C}^2_{\tau} \otimes \mathbb{C}^{N_f}) \otimes 1_{\mathrm{c}}.$$

The physical bilinear sources lie in the specified span of $\mathfrak{B}$, but the intermediate matrix source domain is the entire displayed space. Put $\phi_j = \phi(x_j)$ and choose $n_\pi \geq 1$ with $(1 + 1/n_\pi)^{\mathfrak{p}_0} \leq 4/3$. Set

$$\pi_j = (1 + (K - j)/n_\pi)^{\mathfrak{p}_0}, \quad \|w\|_{\sharp,j} = \max\{\|w\|_\infty, 40M\lambda_j\pi_j \operatorname{Lip} w\}.$$

The fixed physical source norm is $\|w\|_\sharp = \max\{\|w\|_\infty, 40M \operatorname{Lip} w\} = \|w\|_{\sharp,K}$. In fine lattice coordinates let $B_k(x)$ be the ball of radius ML^k centered at x. The corresponding local norm, in physical coordinates, is

$$\|w\|_{\sharp,k;x} = \max\left\{ \sup_{a_K B_k(x)} |w|,\ 40M\lambda_k\pi_k \operatorname{Lip}(w|_{a_K B_k(x)}) \right\}.$$

The geometric estimates used below are

$$\phi_j/\phi_{j+1} \leq q_\phi = 2L^{-1/2}, \quad \sum_{i \leq j} \phi_i \leq \phi_j/(1 - q_\phi), \quad \pi_j/\pi_{j+1} \leq 4/3. \tag{42}$$

We write $S_\uparrow = \sup_k \sum_{j \leq k} \phi_j/\phi_k \leq (1 - q_\phi)^{-1}$. They follow from $|x_j - L^{i-j}x_i| \leq \bar{B}$ and the decreasing range of ϕ, after increasing the final floor.

At pair radius a define

$$\tau_j(w; a) = a x_j^{1/2} 40M\lambda_j \operatorname{Lip} w/\mathsf{b}, \quad a_j^{\mathrm{rad}} = C_{\mathrm{rad}}\delta(\mathsf{t}_j), \quad \mathsf{s}_j(a) = C_{\mathcal{A}}\delta(\mathsf{t}_j)/a.$$

For $i \leq j$,

$$\tau_i(w; a) \leq \sqrt{2}L^{-(j-i)/2}\tau_j(w; a), \qquad \mathsf{s}_j(a) \leq C_{\mathcal{A}}/C_{\mathrm{rad}} \quad (a \geq a_j^{\mathrm{rad}}). \tag{43}$$

The first estimate uses $x_i \leq 2L^{j-i}x_j$ and $\lambda_i = L^{-(j-i)}\lambda_j$. Since $\widehat{\phi}(x) = x^{1/2}\phi(x)$ obeys $\widehat{\phi}(x_j) \leq c_\pi\widehat{\phi}(2\mathsf{b})\pi_j$,

$$\tau_j(w; a_j^{\mathrm{rad}}) \leq 8c_{\mathrm{th}}c_\pi C_{\mathrm{rad}}A_1\varrho\widehat{\phi}(2\mathsf{b})/\mathsf{b} \quad (\|w\|_{\sharp,j} < 8\varrho).$$

This is uniformly small at the final floor. At fixed radius α_0 it is uniformly small on the fixed ball $\|w\|_\sharp < 8\varrho$, where it is at most $16\alpha_0\varrho\phi_j$. It need not be uniformly bounded on the larger level-dependent ball at that fixed radius.

Fix a small operator radius $r_M > 0$, put $r_J = r_M/L$, and let

$$\mathscr{O}_j = \{\|w\|_{\sharp,j} < 4\varrho, \lambda_j\|J\|_\infty < r_J\}, \quad \Xi_j(s) = \lambda_j\left(\pi_j \operatorname{Lip} w/\varrho + L\|J\|_\infty/r_M\right).$$

Then $\Xi_j < 2$, $\mathscr{O}_j \subset \mathscr{O}_i$ for $i \leq j$, and $L\Xi_i \leq (4/3)\Xi_{i+1}$. The coordinates of a step are $\mathfrak{c} = (w, \mathsf{n}, (\sigma_i)_{i \leq j})$ with $\mathsf{n} = -a_K J$. Their domain is

$$\mathscr{D}_j = \{\|w\|_{\sharp,j} < 4\varrho, L^{j+1}\|\mathsf{n}\| < r_M, \|\sigma_i\| < B_\sigma\phi_i\ (i \leq j)\}.$$

At pair radius a replace the condition on w by $\|w\|_\infty < 4\varrho$ and $C_{\mathrm{lin}}\tau_j(w; a) < c_{\mathrm{cl}}/16$; call this $\mathscr{D}_j(a)$. A doubled domain doubles each radius. The final ceilings give $2\mathscr{D}_j \subset 2\mathscr{D}_j(a_j^{\mathrm{rad}})$.

On $2\mathscr{D}_j(a)$ the operator source satisfies $L^{j+1}\|\mathsf{n}\| < 2r_M$. The weighted inverse and walk construction therefore applies through $\mathcal{K}_{j+1}$ and Γ_j, uniformly on the full matrix ball, by Lem. 6.5. In particular it gives holomorphic localized covariances with their Gram bound and backgrounds with the same localized row and Hölder estimates, at the fixed enlarged constants. No positivity of $m_0 + \mathsf{n}$ is used on $\mathscr{O}_j$.

10.1. **The source step and its cell units.** For a matrix kernel k define its symmetric zeroth moment at y by

$$\mathrm{ord}_0[k](y) = \tfrac{1}{2}\sum_{y'}[k(y,y') + k(y',y)].$$

It is the entire filed cell unit, rather than each separate matrix entry, whose total is canceled in the extraction.

Proposition 10.1 (One step at the sources)**.** Suppose the old data are defined on $2\mathscr{D}_i(a)$, $i \le k$, where $a_k^{\mathrm{rad}} \le a \le \alpha_0$. Assume, polymer by polymer and at each pointer, that they read the pair only on its core, are jointly gauge invariant, and are holomorphic in the pair and the independent source coordinates on that core. Their scalar parts are strictly gauge invariant. Assume also that the families, including their pointed scalar parts, agree on common domains when the pair radius is reduced. For the parts with legs, sum the norms of the unextracted activity, the mass member and the individual moved forms; assume this sum is at most $2C_w^{\mathrm{s}}\phi_{i-1}$. After subtracting the explicit normalizers and the Gaussian family, take the maximum of the unpointed scalar fluctuation norm and all its pointed norms. Assume this maximum is at most $2C_w^{\mathrm{s}}\phi_{i-1} + C'_{\mathrm{lin}}\tau_{i-1}(w;a)$. These bounds hold at every admitted reduced radius. Then the next source step is defined and has these bounds at level $k+1$. Its data are holomorphic in the pair and coordinates, and agree when restricted to a smaller admitted pair radius.

Let $k_X(\mathfrak{c})$ be its trivial-background degree-two kernel before local extraction and let $\mathfrak{s}_{k+1}$ be the constant-field mass normalization. Its local increment is

$$\nu_{k+1}(y;\mathfrak{c}) = \mu_{k+1} - \mathfrak{s}_{k+1}^{-1}\sum_X(\mathrm{ord}_0[k_X(\mathfrak{c})](y) - \mathrm{ord}_0[k_X(0)](y)), \tag{44}$$

where the sum runs over polymers of diameter less than a quarter of the torus side. It is holomorphic and, for $\mathfrak{c} \in \mathscr{D}_k$, $\mathfrak{c}' \in 2\mathscr{D}_k$,

$$|\nu_{k+1}(\mathfrak{c})(y) - \nu_{k+1}(\mathfrak{c}')(y)| \le B_\nu\phi_{k+1}\|\mathfrak{c} - \mathfrak{c}'\|_{k,y}, \tag{45}$$

where

$$\|\mathfrak{c} - \mathfrak{c}'\|_{k,y} = \sup_x e^{-\kappa_\sigma|x-y|/L^{k+1}}\max\left\{\frac{\|w - w'\|_{\sharp,k;x}}{\varrho}, \frac{L^{k+1}|\mathsf{n} - \mathsf{n}'|(x)}{r_M}, \max_{i\le k}\frac{|\sigma_i - \sigma_i'|(x)}{B_\sigma\phi_i}\right\}.$$

Here $B_\nu = \alpha_\nu + \beta_\nu B_\sigma$, with α_ν fixed before B_σ and $\beta_\nu \to 0$ as $h_F \to \infty$. ∎

Proof. The relative weight is $v_k = -L^{K-k}w/x_k$. Put its centered quadratic in the q origin, as in Lem. 9.9, and keep its Gaussian family $\mathbf{G}^v$ outside the small family. Its norm is at most $C_G^v(|x_k - \mathsf{t}_k|/\mathsf{t}_k + \sup|v_k|)$. The remaining gauge vertices have size $7K_P\delta(\mathsf{t}_k)$. The weight leaves a linear term. Subtract its constant weight at a base plaquette p_b for each bond b. At a physical critical background this changes nothing, since the derivative of the unweighted action along b vanishes. Define its continuation by that difference formula. The source Taylor estimate on a radius-a pair gains its curvature factor a. Let $\mathrm{sw}_k(v_k)$ be the largest difference of v_k between plaquettes with barycentres at physical ℓ^∞ distance at most $M\lambda_k$. Since $x_k v_k = -\lambda_k^{-1}w$, its swing bound is $x_k\mathrm{sw}_k(v_k) \le \sqrt{3}M\,\mathrm{Lip}\,w$. This bound and the polymer count give $\|V^{\mathrm{lin}}\| \le C_{\mathrm{lin}}\tau_k(w;a)$. This is the reason to retain the radius in the linear size.

At the first step, the physical correction s_0 is zero, but the independent coordinate σ_0 need not be. Form the background at the fine matrix mass $m_0 - a_K\widehat{J}$ and apply Lem. 7.11 with $N = \sigma_0$. The remaining mass potential is $\mathcal{V}_0^{\mathrm{s}}[\sigma_0]$. It is supported on single level-zero cubes, is independent of B in the formal-leg algebra, and has size at most $c_\psi c_{\mathsf{S}} h_F^2\|\sigma_0\|_\infty/9$. It therefore satisfies the same mass-potential bounds used at positive levels. Its covariance is joint with σ_0.

Split the other potential into its scalar part V_0 and its part V_ψ with legs. Localize the old explicit families at their fixed radii; this costs $c_{\mathsf{s}} K^{\mathrm{s}}_{\mathrm{old}} \delta(\mathsf{t}_k)$. Localizing the old scalar fluctuation families costs

$$K^{\mathrm{s}}_{\Lambda} \mathsf{s}_k(a) \sum_{i \le k} L^{-(k-i)} \epsilon^{\mathrm{s},0}_i(a).$$

Choose $C_{\mathrm{rad}} \ge 12 \mathsf{C}_{\mathcal{A}} K^{\mathrm{s}}_{\Lambda} K^{\bullet}_{M}$. Together with the linear term, the gauge vertices and the final ceilings, this gives $\|V_0\| \le c_{\mathrm{cl}}/2$. The source operator bounds and the old-leg estimate give

$$\begin{aligned} \|V_\psi\| &\le C^{\mathrm{s}}_V h_F^2 A_1 \phi_k + c''_\psi c_{\mathsf{S}} h_F^2 (\bar{B}_M + 2B_\sigma)\phi_k \\ &\quad + c''_\psi \sum_{i \le k} \omega_{k+1-i} \epsilon^{\mathrm{s},\psi}_i(a) + c''_\psi \Sigma^{\mathrm{cs}}_k. \end{aligned} \tag{46}$$

Here $\omega_n = C_\omega L^{n/4}$, and

$$c''_\psi = c_\psi + \mathsf{C}_{\mathcal{A}}(\alpha_0/a)\mathsf{s}_a + C_G \mathsf{C}_{\mathcal{A}} \delta_k \le (9/8) c'_\psi.$$

For this bound choose $C_{\mathrm{rad}} \ge 4\mathsf{C}_{\mathcal{A}} c_h \alpha_0 h_F / (a_{\mathrm{an}} A_\kappa \Gamma_F)$, with the original value of Γ_F. Thus the shift is small relative to every admitted radius.

The cell-sum term requires a more precise organization than the identity for the total extracted kernel. Write

$$a_X(y) = \mathrm{ord}_0[k_X(\mathfrak{c})](y) - \mathrm{ord}_0[k_X(0)](y) = \sum_p a_{X,p}(y).$$

Here p records a pair of local or walk terms in the two background legs, and Z_p is its actual supporting polymer. The increment rule includes only X of diameter less than one quarter of the torus side. Although the total extracted kernel can be written as $\sum_X t_X + q_j + r_j$, with

$$t_X = k_X - \delta_{yy'} \mathrm{ord}_0[k_X], \qquad q_j = L^{-j} \mathsf{S}_1(\nu_j \psi^{\mathrm{bg}}) - \mathfrak{s}_j \sum_z \bar{\Psi}(z) \nu_j(z) \Psi(z),$$

this expression does not give the required local summands. We use instead $u_X = k_X - \delta_{yy'} a_X$, filed at the cell of X; on a single cube this includes the old zero-source local kernel $L^{-j} \mu_j \mathsf{S}_1(1_\square \psi^{\mathrm{bg}})|_{\mathsf{n}=0}$. Move $\bar{\Psi}(y) a_{X,p}(y) \Psi(y)$ to the cell containing y. The accompanying local forms are

$$P^\mu_z = L^{-j} \mu_j \big[\mathsf{S}_1(1_z \psi^{\mathrm{bg}}) - \mathsf{S}_1(1_z \psi^{\mathrm{bg}})|_{\mathsf{n}=0} \big], \qquad P_{z;X,p} = -\mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1(a_{X,p}(z) 1_z \psi^{\mathrm{bg}}).$$

The first stands on the cube z and the second on Z_p, pointed at z. Together with the zero-source cube terms already included in u_X, these forms sum to the extracted form $L^{-j} \mathsf{S}_1(\nu_j \psi^{\mathrm{bg}})$. At zero source the moved pieces and these forms vanish, leaving the old polymers. All coefficients are holomorphic on their cores and commute with colour; the summands are jointly gauge invariant. Thus the old-leg estimate applies to each summand on its actual support.

Let $A_j(y,z)$ be the zeroth-moment map of $L^{-j} \mathsf{S}_1(N 1_z \psi^{\mathrm{bg}})$ at $\mathsf{n} = 0$, and let A^{n}_j denote its value at the matrix source. The moment identity for each pair of walks uses the same lifted paths as the local extraction. It gives, for a filed cell c,

$$\begin{aligned} m_c - m_c(0) = \sum_{z \in c} \Big[&\sum_y (A^{\mathsf{n}}_j - A_j)(y,z) \mu_j \\ &+ \sum_{X,p} \Big(1 - \mathfrak{s}_j^{-1} \sum_y A^{\mathsf{n}}_j(y,z) \Big) a_{X,p}(z) + {\sum_{X,p}}' a_{X,p}(z) \Big]. \end{aligned}$$

The prime denotes the polymers omitted by the diameter rule. On the infinite lattice $\sum_y A_j(y,z) = \mathfrak{s}_j$; the displayed expression then reduces to the change of the background map applied to ν_j, and vanishes at $\mathsf{n} = 0$. On a torus the remaining zero-source cell sum is the difference of the filed wrapping-polymer terms.

File a change of a background map, with walk support Y_q, on the level-k hull of $c \cup Y_q$, Z_p, or $Z_p \cup Y_q$, as appropriate. These hulls all contain a cube of c; the outer legs enlarge them in

the usual way. Measure the coefficient at level j with the weight $e^{\kappa L^{-(k-j)}d_j(Y)}(1+d_j(Y))^3$, and denote the result by $\|\cdot\|_{\mathrm{nat}}$. The hull count and the leg estimate give a contribution at most $C_{\mathrm{hl}}L^{k-j}h_F^2\sup_c\|m_c-m_c(0)\|_{\mathrm{nat}}$ per level-k cube. The zero-moment units have the gain $L^{(k-j)/4}$; degree at least four and field dependence have the gain $L^{-(k-j)}$.

For completeness, the estimates for the moved pieces are

$$\sum_{p,y}\|a_{X,p}(y)\|e^{\kappa[d_j(Z_p)-d_j(X)]}\le CA_\kappa^2h_F^{-2}(\|E(X;\mathfrak{c})\|+\|E(X;0)\|).$$

The localized row sums give this inequality. Reaching the filed cell from a cell c' leaves the summable factor $e^{-r(c',c)}$. Consequently, writing $n=k-j\ge1$ and $\epsilon_j=\epsilon_j^{\mathrm{s},\psi}$,

$$\sum_{z\in c,X,p}\|a_{X,p}(z)\|e^{\kappa L^{-n}d_j(Z_p)}(1+d_j(Z_p))^3\le C_{\mathrm{pw}}h_F^{-2}\epsilon_j,\qquad \sum_{Z_p\supset\square,z}\|a_{X,p}(z)\|e^{\kappa d_j(Z_p)}\le C_{\mathrm{pw}}h_F^{-2}\epsilon_j. \tag{47}$$

Here $C_{\mathrm{pw}}=2e^{\kappa/2}S'_\kappa CC_{\mathrm{wk}}A_\kappa^2$, where $S'_\kappa=\sup_d(1+d)^3e^{-5\kappa d/13}$. For the omitted polymers the first bound gains $e^{-c_t\ell/\lambda_j}$. Schwarz's lemma in the matrix source, applied to the walks at rate $\kappa+2$, gives

$$\sum_q\left\|\sum_y(A_j^{\mathfrak{n}}-A_j)_q(y,z)N\right\|e^{\kappa\Delta_q^+}(1+\Delta_q^+)^3\le C_{\mathrm{bg}}^{\mathrm{w}}\frac{L^j\|\mathfrak{n}\|}{r_M}\|N\|,\quad C_{\mathrm{bg}}^{\mathrm{w}}=c_3^2\nu_PA_\kappa^2C_H^2,\quad c_3=\frac{27}{8e}.$$

The torus error in $\mathfrak{s}_j^{-1}\sum_yA_j$ is at most $\eta_{\min}^{-2}C_Ae^{-4c_t\ell/\lambda_j}$. Combining these estimates yields

$$\begin{aligned}\|m_c-m_c(0)\|_{\mathrm{nat}}&\le3M^3C_{\mathrm{bg}}^{\mathrm{w}}\bar B_M\frac{L^j\|\mathfrak{n}\|}{r_M}\phi_j\\&\quad+\left[\eta_{\min}^{-2}\left(C_{\mathrm{bg}}^{\mathrm{w}}\frac{L^j\|\mathfrak{n}\|}{r_M}+C_Ae^{-4c_t\ell/\lambda_j}\right)+e^{-c_t\ell/\lambda_j}\right]C_{\mathrm{pw}}h_F^{-2}\epsilon_j.\end{aligned}$$

At lag zero there is no spare decay for the polynomial factor. Use the second moved-piece bound instead: the same per-cube estimate holds with $C_{\mathrm{hl}}h_F^2$, and with the last omitted-polymer tail replaced by 1. In the source step itself only lags at least n_{lag} enter Σ_k^{cs}; shorter lags are bounded as whole activities. The lag-zero estimate will also be used for the regular sum on a history.

Now $L^{k-j}L^j\|\mathfrak{n}\|/r_M<2/L$. For positive lags the corresponding torus-tail factors are bounded by C_{dx} and C'_{dx}; for all lags use $\max\{1,C'_{\mathrm{dx}}\}$. Summing proves

$$\Sigma_k^{\mathrm{cs}}\le C_0^{\mathrm{cs}}h_F^2\phi_k+c_{\mathrm{cs}}\sum_{j\le k}\omega_{k+1-j}\epsilon_j^{\mathrm{s},\psi}. \tag{48}$$

with

$$C_0^{\mathrm{cs}}=\frac{6C_{\mathrm{hl}}M^3C_{\mathrm{bg}}^{\mathrm{w}}\bar B_M}{L(1-2L^{-1/2})},\qquad c_{\mathrm{cs}}=\frac{C_{\mathrm{hl}}C_{\mathrm{pw}}}{C_\omega}\left[\eta_{\min}^{-2}\left(\frac{2C_{\mathrm{bg}}^{\mathrm{w}}}{L}+C_AC_{\mathrm{dx}}\right)+\max\{1,C'_{\mathrm{dx}}\}\right].$$

Neither a source-chain bound nor spatial regularity of the coordinates σ_i entered this estimate.

By the final ceiling (46) is at most $c_{\mathrm{cl}}/8$. Multiply V_ψ by a complex scalar. The positive-degree output is zero at scalar zero, and its cluster bound holds up to radius $c_{\mathrm{cl}}/(2\|V_\psi\|)$. Schwarz's lemma gives $\|\widetilde E_{\ge2}\|\le5K_M\|V_\psi\|$. Taking out the moment and filing its moved pieces costs $K_\mu^{\mathrm{s}}=K_\mu+\eta_{\min}^{-2}c_{\mathfrak{S}}M^{-3}C_{\mathrm{pw}}$. Put $\theta_E^{\mathrm{s}}=(K_\mu^{\mathrm{s}}/K_\mu)\theta_E=\sqrt2K_\mu^{\mathrm{s}}K_Mc'_\psi$. Then $5K_\mu^{\mathrm{s}}K_Mc''_\psi\le4\theta_E^{\mathrm{s}}$. Consequently

$$\epsilon_{k+1}^{\mathrm{s},\psi}\le C_w^\psi\phi_k+4\theta_E^{\mathrm{s}}\sum_{i\le k}\omega_{k+1-i}[(1+c_{\mathrm{cs}})\epsilon_i^{\mathrm{s},\psi}+c_{\mathfrak{S}}h_F^2(\bar B_M+2B_\sigma)\phi_i].$$

The growing lag is summable against $\phi_{i-1}/\phi_k\le CL^{-(k+1-i)/2}(1+(k+1-i)\log L)^{\mathfrak{p}_0}$. The prescribed floor on h_F closes the leg size.

We next verify the asserted properties at each pointer. Each member of the potential reads the independent coordinates on its filing polymer: the gauge and linear vertices read w, the mass member reads σ_k, and the local inverses read n. At $k=0$ this is the single-cube reading of $\mathcal{V}_0^{\mathrm{s}}[\sigma_0]$ in Lem. 7.11. An old term is filed with the polymer of its coefficient and all walks of its legs. Its local source dependence is therefore retained. Local moments are likewise filed with their actual pieces. The connected expansion then reads only the coordinates on its union polymer.

For a place p, put $\dot V=(V_\psi)_p$, using the share $n^{-1}N_p$ in degree $n\ge 2$; one may also sum these shares over the places of a site. This map preserves locality and evenness, commutes with gauge transformations, and has norm at most one on every polynomial. Define

$$H(X;\mathfrak{c},t,\lambda)=\widetilde{E}[V_0+tV_\psi+\lambda\dot V](X),\qquad E^\bullet(X;\dot V)=\int_0^1\partial_\lambda H(X;\mathfrak{c},t,0)\,dt.$$

For $|t|<9/8$ and $|\lambda|<1$ the potential has size at most

$$\tfrac12 c_{\mathrm{cl}}+(9/8+1)\tfrac18 c_{\mathrm{cl}}=\tfrac{49}{64}c_{\mathrm{cl}}<c_{\mathrm{cl}}.$$

Thus Lems. 7.5 and 7.6 apply on an open set containing the integration segment. The Cauchy formula for $\partial_\lambda H$, followed by the uniformly bounded integral in t, proves holomorphy in $\mathfrak{c}$ and the pair. Both operations preserve the gauge identity of Lem. 7.7. Taking degree zero gives strict scalar invariance. The first-variation bound applies at every pointer, with the norm of V_ψ as its insertion bound.

For the restriction assertion let $a_k^{\mathrm{rad}}\le a'\le a$. The coordinate domains satisfy $2\mathscr{D}_k(a)\subset 2\mathscr{D}_k(a')$. The explicit old families are localized at their fixed radii in both constructions. The old scalar fluctuation arrays, including every pointed member, agree on the radius-a' cores by the inductive hypothesis. Their bracket localization commutes with this restriction, with the same slot and zero counterterms. Hence their fluctuations agree at every admitted B; at $k=0$ this array is empty. The two potentials and their insertions therefore agree at the smaller radius, as do H and $E^\bullet$.

The scalar terms pointed at bonds obey the same restriction identity. The tests, quadratic origin and gauge vertices are defined at the fixed radii. Their paths and insertions depend only on their values at the given pair. The remaining scalar origin is the localized old array just considered, together with the linear source term. It has the same value on the smaller core at every fluctuation, so its paths, insertions and connected sums also agree there. Summing with the fixed tent functions preserves this identity.

For degree zero let D denote the new scalar activity less its explicit Gaussian and normalizer families. The scalar subtraction is exactly

$$\begin{aligned}F_0^{(k+1)}(X;U;z;\mathfrak{c})=D(X;U;z;\mathfrak{c})-D(X;1;z;\mathfrak{c})+D(X;1;z;0)\\-1_{X=\{z\}}\widetilde{e}_{k+1}-\mathbf{L}_0(X;U;z).\end{aligned}$$

Thus $\|F_0\|\le 2\|D\|+\|D(0)\|+M^3|\widetilde{e}_{k+1}|+\|\mathbf{L}_0\|$, where $M^3|\widetilde{e}_{k+1}|\le\|D(0)\|$ by its translation-cell mean. The local action part $\mathbf{L}_0$ is the zero-source one, formed at its fixed radius. Every term in this subtraction is local and holomorphic in the independent coordinates. The evaluations at the trivial pair, which belongs to every radius, preserve agreement on smaller cores; $\widetilde{e}_{k+1}$ and $\mathbf{L}_0$ are independent of that radius. Thus the pointed scalar families have all the properties required in the next step's hypothesis. There is no inverse-square radius loss from extracting a new action coefficient. The scalar recursion is

$$\begin{aligned}\epsilon_{k+1}^{\mathrm{s},0}\le{}&(C_w+C_w^\flat+C_w^\psi)\phi_k+\tfrac{17}{8}\theta_s h_F^2(\bar B_M+2B_\sigma)\phi_k\\&+2\theta_E\sum_{i\le k}\omega_{k+1-i}[(1+c_{\mathrm{cs}})\epsilon_i^{\mathrm{s},\psi}+c_{\mathsf{S}}h_F^2(\bar B_M+2B_\sigma)\phi_i]\\&+\tfrac12 C'_{\mathrm{lin}}\tau_k(w;a)+2K_MK_\Lambda^{\mathrm{s}}\mathsf{s}_k(a)\sum_{i\le k}L^{-(k-i)}\epsilon_i^{\mathrm{s},0}.\end{aligned}$$

The last sum contributes at most $\frac{1}{4}C_w^{\mathrm{s}}\phi_k + \frac{1}{10}C'_{\mathrm{lin}}\tau_k$, by (43) and $2K_M K_\Lambda^{\mathrm{s}}\mathsf{s}_k(a) \le 1/5$. The lag sum of old leg sizes costs at most $\frac{1}{2}C_w^{\mathrm{s}}\phi_k$. Choose

$$C_w^{\mathrm{s}} = C_w + C_w^{\flat} + C_w^{\psi} + 3\theta_s h_F^2(\bar{B}_M + 2B_\sigma) + 4L^{1/2}C_{\mathrm{pc}}S_{1/4}\theta_E^{\mathrm{s}}c_{\mathsf{S}}h_F^2(\bar{B}_M + 2B_\sigma).$$

The scalar size closes, with the linear source term retained at its actual radius-dependent size.

The moment rule (44) is a holomorphic value at the trivial pair, independent of the radius at which it was formed. For its locality estimate, expand its background rows in walks. A term f reads w and the σ_i on its polymer X, and n on X and its walk ω. Extend the difference of w from X by the real and imaginary McShane extensions, cut off at their supremum. Its level norm costs at most $C(1 + d(X))$ times the maximum normalized coordinate difference on $X \cup \omega$. A Cauchy line from $\mathfrak{c}$ inside the doubled domain gives $|f(\mathfrak{c}') - f(\mathfrak{c})| \le 2CM_0(1 + d(X))t$, where M_0 is its term bound. The maximum, rather than a sum over the coordinates, occurs. Choose $\kappa_\sigma \le \min\{\kappa/(4c_{\mathcal{D}}M), 1/(24M_w)\}$. After paying the distance in (45), the remaining polymer factor is $(1 + d(X))e^{-3\kappa d(X)/4}$. A local leg has factor one; a walk leg of t steps has factor $C_{\mathrm{nb}}^{-t}e^{-11t/4}$, since $C_{\mathrm{nb}}q_{\mathrm{sup}} \le e^{-3}$ and the distance bound uses only $e^{-t/4}$. A zero-step walk and the local term are separate terms.

Let S_κ be the supremum of the sum of these products over the polymers and both legs contributing to the symmetric zeroth moment at a fixed site, over all levels, tori and $\mathbb{Z}^3$. We verify that

$$S_\kappa \le 8C_{\mathrm{nb}}(1 - e^{-1/4})^{-1}C_{\mathrm{cnt}}\left[1 + 8(M/M_w + 2)^3(1 - e^{-11/4})^{-1}\right], \tag{49}$$

where

$$C_{\mathrm{cnt}} := \sup_{\square} \sum_{X \ni \square} e^{-\kappa d(X)/2} \le 2048(1 - e^{-2})^{-1}.$$

The last inequality follows from the polymer count in Lem. 7.5, with the existing rate floor $9216e^{-\kappa/2} < e^{-2}$.

The operators at every point of $2\mathscr{D}_k(a_k^{\mathrm{rad}})$ are those of the history without large fields, at its fine mass matrix. Hence there is one layer, with blocks of side one in level units, and no stored place. The walk cubes form one partition into cubes of side M_w. For the leg whose column is the fixed site y, at most C_{nb} walk cubes can meet the block supporting that column. At each first or last cube there are at most C_{nb}^t walks of t steps. The local term and the walk terms therefore have total factor at most $1 + C_{\mathrm{nb}}(1 - e^{-11/4})^{-1}$. Including both terms in the symmetric zeroth moment costs at most $C_{\mathrm{nb}}(1 - e^{-1/4})^{-1}$.

For a fixed such leg, X meets either the block of y or the threefold walk cube at its row. That cube has side $3M_w \le M$ and meets at most eight cubes of side M. For the other leg, count its possible first cubes at the rows of X; a row twin is counted by its last cubes. The polymer has at most $8(1 + d(X))$ cubes of side M. The cutoff of a walk cube extends by $M_w/4$ on each side. If it is nonzero at a row in a given M-cube, the walk cube lies in the concentric cube of side $M + 5M_w/2$. At most $(M/M_w + 2)^3$ disjoint walk cubes can do so. Indeed choose a translate of the grid $M_w\mathbb{Z}^3$ with exactly $(M/M_w + 2)^3$ points in that larger cube; each half-open walk cube contains exactly one grid point. The same count holds on a torus; if its side is smaller than the larger cube, the torus has side M and contains only $(M/M_w)^3$ walk cubes. Thus the other leg contributes at most

$$1 + 8(1 + d(X))(M/M_w + 2)^3(1 - e^{-11/4})^{-1}.$$

The polynomial factors are absorbed by $(1 + d)^2e^{-\kappa d/4} \le 1$, using $\kappa > 8$. Summing first over the leg at y, then the polymers meeting the indicated eight cubes, and finally the other leg proves (49).

Let C' be the fixed constant from the extension and Cauchy bound above, enlarged if necessary so that $C' \ge 2$. The coefficient estimate of Lem. 7.8 now gives

$$c_{\mathrm{inc}} := C'\eta_{\min}^{-2}\max\{1, C_{am}^{\psi}\}e^{\kappa/4 + 1/2}S_\kappa, \qquad B_\nu = 4c_{\mathrm{inc}}h_F^{-2}C_w^{\mathrm{s}}.$$

Here $\eta_{\min}$ is the uniform positive lower bound on the constant-field background factor. All factors in c_{inc} are fixed before h_F. Only the displayed terms in C_w^{s} involve B_σ, and their coefficients are proportional to θ_E. Hence $B_\nu = \alpha_\nu + \beta_\nu B_\sigma$ with the claimed order of choices.

The symmetry induction also distinguishes the two initial data. Along a physical source chain, $s_0 = \sigma_0 = 0$, and the initial density is the bare density at its matrix mass. For independent constant-field coordinates, the bare exponent has the additional term $-\mathsf{S}_1(\sigma_0\psi)$. The density is covariant jointly with its matrix entries; Lem. 7.11 proves the same covariance for the potential left after the first substitution. The exact step integral and the subsequent coefficient operations therefore propagate the total's covariance on both domains. □

Lemma 10.2 (Comparison with the infinite lattice)**.** At constant coordinates and the trivial gauge field, a source step on the torus and on $\mathbb{Z}^3$ with the same old families has, on polymers of diameter less than a quarter of the side, activity difference bounded by

$$K_M e^{-2c_t\ell/\lambda_{k+1}}(\|V\| + e^{-\gamma_{P,k}})e^{-\kappa_+ d(X)}.$$

The corresponding moment difference costs at most $C_{am}^{\psi} h_F^{-2}$ times this bound. ■

Proof. The translation-invariant fine inverse on the torus is

$$\sum_{a\in\mathbb{Z}^3} (-1)^{a_1+a_2+a_3}(\mathcal{K}_j^{\infty})^{-1}(x, x' + \ell L^K a).$$

Exponential decay gives absolute convergence and verifies both inverse identities. The same periodization holds for the fluctuation covariance and background maps by their Schur formulas. Between points closer than a quarter of the side, nonzero images are at least three quarters of the side away; the difference is at most $Ce^{-4c_t\ell/\lambda_{k+1}}$ in the weighted kernel norms, for $c_t \le \alpha_D/16$. Local inverses and supported walks in such a polymer are the same on both lattices. Only these image errors and the tails of background rows remain. The activity is holomorphic on the operator balls with the enlarged constants; the unperturbed data lie inside them. Schwarz's lemma on the complex line between the two sets of data bounds the output difference by its uniform cluster bound times the relative kernel error. Absorb the fixed factor and use the weaker displayed exponential. The fibre coefficient estimate gives the moment assertion. □

Proposition 10.3 (The constant reference chain)**.** For $|c| < 8\varrho$ the infinite-lattice constant chain $\sigma_k^{\mathrm{c}}(c)$ is a scalar, holomorphic in c and real for real c. It obeys

$$|s_k + \sigma_k^{\mathrm{c}}(c)| \le 2\bar{B}_M/\mathsf{t}_k, \qquad |\sigma_k^{\mathrm{c}}(c)| \le B_0\phi_k|c|/\varrho.$$

On the torus its constant-coordinate increment differs from $\mu_{k+1} + \sigma_{k+1}^{\mathrm{c}} - L\sigma_k^{\mathrm{c}}$ by at most $C_{\mathrm{tor}}\phi_{k+1}e^{-c_t\ell/\lambda_{k+1}}|c|/\varrho$. The constants B_0, C_{tor} precede B_σ. ■

Proof. Put $x_i' = x_i - L^{K-i}c$. We first identify this splitting with the holomorphic zero-source flow at bare label $\mathsf{b} - c$. That flow extracts its own coupling increments $\tilde{b}_i$; the source splitting retains b_i and leaves $(\tilde{b}_i - b_i)a_p$ in the scalar activities. Compare the two on finite tori, with the same tests and critical backgrounds. Assuming their mass increments agree through level i, the two exponents at that level differ only by a scalar constant and this redistribution of the action. The next integral is the same. Its fermion-degree-two parts therefore agree; different gauge Gaussian normalizers have degree zero only. At the trivial pair take the zeroth moments. The residual torus moments tend to zero by Lem. 10.2 and the wrapping-polymer tail. The nonzero constant-field normalization then identifies the next mass increments. This proves the identification by induction. Holomorphy is taken from the first splitting. The second contains action multiples and is not asserted to lie in the normalized family space.

Here the complex-coupling flow is controlled by an induction on the fixed disc $|c| < 8\varrho$, with tests and calibrations frozen. Impose $64\varrho \le r_x\gamma^{-2}$ and $26C_w\phi(\gamma^{-2})\gamma^2 \le r_x$. The coupling recursion,

whose fixed one-loop terms do not depend on the coupling, then gives

$$|\tilde{x}_i - x_i| \le \tfrac{2}{3} r_x x_i, \qquad |\tilde{x}_i| \ge \mathsf{t}_i/4.$$

The exact one-loop telescope and its nonnegative slices give $|G_i(\tilde{x})| \le 4B_F/\mathsf{t}_i$. The cubic remainder sum is at most $1/(2\mathsf{t}_i)$ at the final ceiling. Consequently $|\tilde{s}_i| \le (4B_F + 1/2)/\mathsf{t}_i \le 2\bar{B}_M/\mathsf{t}_i$. These bounds keep the next step in its complex coupling and mass domains, closing the induction and justifying composition of the holomorphic steps throughout the disc.

The difference in coupling increments is $|\tilde{b}_i - b_i| \le 2C_w\phi(x_{i-1})$, whence $|\tilde{x}_i - x'_i| \le 4C_w\phi(x_i)$. Its change in the one-loop coefficient is $O(\phi(x_i)/x_i^2)$, absorbed by the cubic remainder after lowering γ. The one-loop mass structure is $\mu_j^{1\mathrm{L}} = G_j - LG_{j-1}$. Therefore

$$\sigma_k^{\mathrm{c}}(c) = G_k(x') - G_k(x) + \sum_{j\le k} L^{k-j}(\mu_j^{\mathrm{rem}}(c) - \mu_j^{\mathrm{rem}}(0)).$$

The nonnegative loop slices have total at most B_F. Since $x_i \ge \mathsf{t}_i/2$ and $|x'_i| \ge \mathsf{t}_i/4$, $|1/x'_i - 1/x_i| \le 8|c|/(\mathsf{bt}_i)$. Thus $|G_k(x') - G_k(x)| \le 8B_F|c|/(\mathsf{bt}_k)$. Schwarz's lemma on the 8ϱ disc and the cubic remainder give

$$\left|\sum_{j\le k} L^{k-j}(\mu_j^{\mathrm{rem}}(c) - \mu_j^{\mathrm{rem}}(0))\right| \le \frac{C_\rho S_3}{2\varrho}\phi_\sharp(x_k)^3|c|.$$

The forward mass-flow bound on this disc gives the stated window; increasing the final floor so that $\phi_\sharp^3 \le \phi$ gives $B_0 = 16B_F + C_\rho S_3$. The identification with the constant-coupling flow gives $\sigma_k^{\mathrm{c}}(c) = \widetilde{s}_k - s_k$ on $\mathbb{Z}^3$. It is therefore a scalar number, holomorphic in c and real at real c, by the reality of that flow. This conclusion does not assert the same reality for individual pointed or polymer pieces.

For the torus comparison, match each polymer of diameter less than a quarter of the side with its translate in $\mathbb{Z}^3$. Define Δ_i by the level-i family norm with

$$\sum_\sigma \|F_\sigma^{\mathrm{tor}}(Z) - F_\sigma^{\infty}(Z)\|$$

in place of the norm of the activity at Z. The summands are the new activity before extraction, each moved form $P_{z;X,p}$ on its own $Z_p = Z$, and the local mass member. The last member has zero difference: both lattices use the same μ_i and the same algebra of formal legs. Let Δ_i^0 be the maximum of the scalar difference norm and the corresponding difference norms at every pointer, including the action terms. At constant coordinates $\mathsf{n} = 0$, the new cell-sum identity has only the error in $\mathfrak{s}_i^{-1}\sum_y A_i(y,z)$ and the omitted large polymers. Keep each summand on its Z_p. The latter polymers cannot enter a new polymer of diameter less than one quarter of the side. The former, together with the residual zero-source torus moment, give a forcing r_i^{c} bounded by $C\phi_i e^{-4c_t\ell/\lambda_i}$, using the moved-piece bound and the constant-chain sizes $\epsilon_i^{\mathrm{c}} \le C\phi_{i-1}$. These constants precede B_σ. Extraction is bounded in this norm. Indeed, if Z_p has diameter less than a quarter of the side, its walks do not wrap around the torus. Their local inverses and moment maps agree on both lattices. Thus the difference of the two forms is the moment of the single difference family E^{dif} before extraction. The piece count in eq. (47) gives

$$\sum_{Z_p \supset \square}\sum_z \|a_{X,p}^{\mathrm{dif}}(z)\| e^{\kappa d_i(Z_p)} \le C_{\mathrm{pw}} h_F^{-2}\|E^{\mathrm{dif}}\|.$$

Multiplying by the norm of a local mass form and using $\mathfrak{s}_i \ge \eta_{\min}^2$ bounds the forms by

$$\eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}}\|E^{\mathrm{dif}}\|.$$

Including the unextracted term therefore costs at most K_μ^{s}, the same factor used for the source-step sizes. The derivative of the new activity with legs in the old leg potential is at most $50K_M/9$ on the admitted ball. Since $\theta_E^{\mathrm{s}} = \sqrt{2}K_\mu^{\mathrm{s}}K_M c'_\psi$, its contribution is at most $4\theta_E^{\mathrm{s}}$ times the old leg norm.

Its derivative in the scalar potential has an additional factor $\|\mathcal{V}_\psi\|/c_{\mathrm{cl}}$, because the part with legs vanishes when $\mathcal{V}_\psi = 0$. These estimates, Lem. 10.2 and the local moment rule give

$$\begin{aligned}\Delta_{k+1} &\le C_{\mathrm{img}} K_M e^{-2c_t\ell/\lambda_{k+1}}\epsilon_{k+1} + 4\theta_E^{\mathrm{s}}\sum_{i\le k}\omega_{k+1-i}\Delta_i \\ &\quad + \epsilon_\gamma \sum_{i\le k} L^{-(k-i)}\Delta_i^0 + C\sum_{i\le k} L^{k+1-i}|r_i^{\mathrm{c}}|, \\ \Delta_{k+1}^0 &\le C' K_M e^{-2c_t\ell/\lambda_{k+1}}\epsilon_{k+1} + C_{\mathrm{dl}}\sum_{i\le k}\omega_{k+1-i}\Delta_i \\ &\quad + \epsilon_\gamma \sum_{i\le k} L^{-(k-i)}\Delta_i^0 + C\sum_{i\le k} L^{k+1-i}|r_i^{\mathrm{c}}|.\end{aligned}$$

Here $\epsilon_{k+1} \le C\phi(\mathfrak{t}_k)$, since the linear term is zero at a constant source; all constants precede B_σ, and $\epsilon_\gamma \to 0$. The lag sums are bounded by $C_\omega S_\omega$ and $S_L = \sum_{n\ge1} L^{1-3n/2}(1+n\log L)^{\mathfrak{p}_0}$. The r_i^{c} terms are summable by their doubly exponential fall in $k-i$. Choose the earlier leg floor so that $4\theta_E^{\mathrm{s}} C_\omega S_\omega \le 2^{-1/2}$ and the final ceiling so that

$$\epsilon_\gamma S_L \le \min\{1/2, (1-2^{-1/2})/(4C_{\mathrm{dl}}C_\omega S_\omega)\}.$$

If C' bounds the forcing terms, the two recursions close with $D = 4C'/(1-2^{-1/2})$ and $D_0 = 2(C' + C_{\mathrm{dl}}C_\omega S_\omega D)$. The large-diameter polymers omitted by the rule have the same exponential bound. The increment difference is holomorphic and zero at $c=0$; Schwarz's lemma supplies its factor $|c|/\varrho$. □

Define the source chain by

$$\sigma_0 = 0, \quad \sigma_{k+1} = L\sigma_k + \nu_{k+1}(\mathfrak{c}_k(s)) - \mu_{k+1}, \quad \mathfrak{c}_k(s) = (w, -a_K J, (\sigma_i(s))_{i\le k}).$$

The increment is read as a constant on its coarse cell. Write $\mathcal{N}_k = -\lambda_k J + \sigma_k$ and $\mathcal{Z}_k = 1 - L^{K-k}\partial_J\sigma_k|_0$ on constant matrix sources.

Lemma 10.4 (The source chain)**.** The chain is formed and holomorphic on $\mathscr{O}_k$, stays in $\mathscr{D}_k$, and satisfies

$$\|\sigma_k - \sigma_k^{\mathrm{c}}(w(\cdot))\| \le C_{\mathrm{var}}\phi_k\Xi_k + 2C_{\mathrm{tor}}C_{\mathcal{L}}\phi_k e^{-c_t\ell/(2\lambda_k)}\|w\|_\infty/\varrho.$$

Moreover

$$\|\sigma_k(0,J)\| \le B_\nu S_\uparrow \phi_k\lambda_k\|J\|/r_M, \qquad \|\mathcal{Z}_k - 1\| \le B_\nu S_\uparrow\phi_k/r_M,$$

and $\|\mathcal{Z}_{k+1} - \mathcal{Z}_k\| \le B_\nu\phi_{k+1}/r_M$. At every site the response is equivariant under the two flavour factors:

$$V\mathcal{Z}_k(J)V^{-1} = \mathcal{Z}_k(VJV^{-1}), \qquad V = e_+\otimes V_1 + e_-\otimes V_2, \quad V_1, V_2 \in \mathrm{U}(N_f).$$

For $S = \mathrm{supp}(s - s')$,

$$|\sigma_k(s)(x) - \sigma_k(s')(x)| \le C_t\phi_k e^{-\kappa_\sigma \operatorname{dist}(x,S)/L^k}.$$

All statements hold uniformly in the volume and cutoff. ■

Proof. Put $q = (4/3)q_\phi < 3/4$, $C_{\mathrm{tl}} = \sup_{t\ge0}(1+t)e^{-\kappa_\sigma t}$, $C_{\mathrm{var}} = 2B_\nu(1 + C_{\mathrm{tl}} + 42M)/(1 - \frac{8}{3}L^{-1/2})$, and $C_{\mathcal{L}} = \sum_{u\ge0}(2L^{1/2})^u e^{-c_t L(L^u-1)}$. Choose

$$B_\sigma \ge 4B_0 + 3C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}}, \qquad B_\sigma \ge 2C_{\mathcal{L}}B_\nu.$$

These inequalities can be met: first make β_ν small by raising h_F, then choose B_σ.

At a point x_0 compare with the reference chain at $c = w(x_0)$. The weighted local difference in w is bounded by $(C_{\mathrm{tl}} + 41M)\varrho\Xi_{k+1}$; that in the fine mass by $r_M\Xi_{k+1}/L$. Cauchy's estimate for the

constant reference chain bounds its variation in w. If D_k is the supremum of the resulting chain difference, the increment locality estimate gives the induction

$$D_k \le C_{\mathrm{var}}\phi_k\Xi_k + T_k, \quad T_k = 2C_{\mathrm{tor}}\sum_{j\le k} L^{k-j}\phi_j e^{-c_t\ell/(2\lambda_j)}\|w\|_\infty/\varrho.$$

Indeed the coefficient of $C_{\mathrm{var}}\phi_{k+1}\Xi_{k+1}$ in the next step is at most $q + B_\nu(C_{\mathrm{tl}} + 41M)/C_{\mathrm{var}} + 4B_\nu/(3LB_\sigma) \le 1$. The remaining forcing terms are at most T_{k+1} by $2C_{\mathcal{L}}B_\nu \le B_\sigma$. The defining series for $C_{\mathcal{L}}$ gives $T_k \le 2C_{\mathrm{tor}}C_{\mathcal{L}}\phi_k e^{-c_t\ell/(2\lambda_k)}\|w\|_\infty/\varrho$. The window follows from $\Xi_k < 2$ and $\|w\|_\infty < 4\varrho$; holomorphy follows inductively by composition.

At $w = 0$ the normalized matrix-source entry dominates the feedback entries. The increment bound is $B_\nu\phi_j\lambda_j\|J\|/r_M$. Summing it with L^{k-j} gives the first matrix bound. Differentiating this holomorphic line at zero gives the two response estimates.

For locality set $a_j(t) = \sup_{\mathrm{dist}(x,S)\ge t}|\sigma_j(s)(x) - \sigma_j(s')(x)|$. The same local estimate gives

$$a_j(t) \le La_{j-1}(t) + B_\nu\phi_j\left[C_{\mathrm{loc}}e^{-\kappa_\sigma t/L^j} + e^{\kappa_\sigma\sqrt{3}}\sup_{u\ge 0} e^{-\kappa_\sigma u/L^j}\max_{i<j}\frac{a_i(t-u)}{B_\sigma\phi_i}\right],$$

with $a_i(t-u) = 2B_\sigma\phi_i$ if $u > t$ and $C_{\mathrm{loc}} = 8e^{\kappa_\sigma\sqrt{3}+\kappa/4}$. For $t \le t_\flat L^j$ the claimed exponential follows from the window by choosing $C_t \ge 2B_\sigma e^{\kappa_\sigma t_\flat}$. For larger t, choose $2L^{1/2}e^{-\kappa_\sigma(L-1)t_\flat} \le 1/4$. The first term then costs at most one quarter of the inductive bound; the forcing term costs one quarter after raising C_t, and the feedback costs at most one half since $e^{2\kappa_\sigma}B_\nu/B_\sigma \le 1/2$. This proves the exponential locality estimate. □

The flavour identity passes through the moment rule, the averages and the covariances; differentiating it at zero gives the displayed equivariance of $\mathcal{Z}_k$. At $J = 0$ it implies, site by site,

$$\sigma_k(x; w, 0) = n_k^0(x) + n_k^3(x)\tau_3, \qquad n_k^0(x), n_k^3(x) \in \mathrm{End}(\mathbb{C}^2_{\mathrm{s}}).$$

These coefficients are spin operators, with no assertion of pointwise reality. The scalar constant reference chain gives, with $\Delta = \sigma_k(x; w, 0) - \sigma_k^{\mathrm{c}}(w(x))$, $n_k^3\tau_3 = (\Delta - \tau_1\Delta\tau_1)/2$. The source-chain estimate therefore bounds $\|n_k^3\|$ by the variation term and the torus error. At constant w only the torus error remains; it vanishes for the constant chain on $\mathbb{Z}^3$. The conjugation used for whole energies and partition functions is $s^\star = (\bar{w}, \gamma_4 J^\dagger\gamma_4)$. Its pseudoscalar signs must be retained, even for real coefficients in the physical basis.

Lemma 10.5 (Windows along arbitrary histories)**.** For the chain obtained using only polymers inside the integration regions of a history, let $S_j = \bigcup_{i\le j}\widetilde{Z}_i$ and $R'_\flat = 7LM$. Then

$$|\sigma_j^h(s)(x) - \sigma_j(s)(x)| \le 8B_\nu\phi_j e^{-\kappa_\sigma\,\mathrm{dist}(x,S_j)/(2L^j)} \quad (\mathrm{dist}(x, S_j) \ge R'_\flat L^j),$$

and throughout Ω_j the difference is at most $16B_\nu\phi_j$. Thus $\|\sigma_j^h\| \le 2B_\sigma\phi_j$. Replacing the full-chain coordinates by history coordinates in the local source energies changes their level-j sum by at most $C'_{\mathfrak{E}}\#_j(\widetilde{Z}_j)$. ■

Proof. Omitted polymers reach a point at distance t only with $e^{-\kappa_\sigma t/L^j}$. The common polymers obey the increment locality estimate, with the second coordinate in its doubled domain. Hence the recurrence for $a_j(t)$ in the preceding proof holds, now taking $a_i(t-u) = 3B_\sigma\phi_i$ below $R'_\flat L^i$. The claimed bound follows with its exponent divided by two: the old-level term costs at most one quarter if $Le^{-\kappa_\sigma R'_\flat} \le 1/4$; the bracket is at most six times the proposed exponential if

$$(C_{\mathrm{loc}} + 3e^{2\kappa_\sigma})e^{-\kappa_\sigma R'_\flat/4} + 8e^{2\kappa_\sigma}B_\nu/B_\sigma \le 6.$$

These are fixed earlier floors. At a point of Ω_j the preceding source difference is already in the exponentially small range by layer separation. One more increment, each bounded by $B_\nu\phi_j/2$, gives $9B_\nu\phi_j \le 16B_\nu\phi_j$. For the energy sum use its coordinate Cauchy estimate on the doubled domain.

Only the σ_i differ, so no Lipschitz extension polynomial is needed. Its decay times the displayed distance exponential is summable over cubes to $C'_{\mathfrak{E}}\#_j(\widetilde{Z}_j)$. □

10.2. **Source energies and normalized partition functions.** Let $\mathcal{P}$ be the closed physical source ball $\|w\|_\sharp \le 2\varrho$, $\|J\|_\infty \le 2\varrho_J$, with J in the complex span of the physical basis $\mathfrak{B}$, where $2\varrho_J \le c_J m_- \le r_J/2$ and $c_J \le 1/2$. It lies strictly inside every $\mathscr{O}_j$. Let $\mathfrak{E}_j(X;s)$ be the trivial-pair value of the new scalar activity before its constant is extracted, including its gauge and fermion normalizers and Gaussian family. Define

$$e_j(X;s) = \mathfrak{E}_j(X;s) - \mathfrak{E}_j(X;0), \qquad E_k(s) = E_k + \sum_{j\le k}\sum_X e_j(X;s), \qquad Z_k^{\mathrm{sub}}[s] = e^{-E_k(s)}Z[s].$$

No coordinate-dependent constant remains in the activities after this extraction. The source-dependent action and first-moment coefficients remain there; only the coupling shift of zero sources is extracted.

Proposition 10.6 (Local source energies)**.** For $j \le K$, $e_j(X;\cdot)$ is holomorphic on $\mathscr{O}_{j-1}$, zero at zero, and

$$|e_j(X;s)| \le C_{\mathfrak{E}} e^{-\kappa d_j(X)}.$$

For $S = \mathrm{supp}(s - s')$,

$$|e_j(X;s) - e_j(X;s')| \le C_{\mathfrak{E}} e^{-\kappa d_j(X)} e^{-\kappa_\sigma \operatorname{dist}(X,S)/L^j}. \tag{50}$$

The distance here is in fine lattice units. Consequently, for any fixed $0 < \kappa_e < \kappa - c_E$,

$$\sup_{j,\square,s}\sum_{X\ni\square} e^{\kappa_e d_j(X)}|e_j(X;s)| \le C_{\mathrm{locE}}.$$

Conjugation holds for the sum at each level:

$$\sum_{X\in\mathcal{D}_j} e_j(X;s^\star) = \overline{\sum_{X\in\mathcal{D}_j} e_j(X;s)}.$$

Consequently $E_k(s^\star) = \overline{E_k(s)}$. No polymerwise conjugation identity is asserted. ■

Proof. The trivial pair belongs to every pair-radius domain. The source step is holomorphic there, and composition with the holomorphic chain proves holomorphy. Its scalar fluctuation contribution is bounded by the source-step size at a_{j-1}^{rad}, uniformly on the level domain. The gauge normalizer is unchanged and cancels in e_j. The Gaussian-family difference is at most $2e^{-\kappa_+ d_j(X)}$ by its explicit fixed bound.

For the fermion normalizer use the exact derivative

$$\log\det\Gamma_{j-1}(\mathsf{n}) - \log\det\Gamma_{j-1}(0) = \int_0^1 \mathrm{tr}\left[\Gamma_{j-1}(t\mathsf{n})^{-1}(\bar{\mathcal{H}}\hat{\mathsf{n}}\mathcal{H})_{AA}(t\mathsf{n})\right]dt.$$

Expand the inverse and both backgrounds in their source walks and file the trace on their hull. The fibre trace costs $8N_f$ a place. The weighted walk sums give $C_N^{\mathsf{n}} N_f L^j \|\mathsf{n}\| e^{-(\kappa+1)d_j(X)} \le 2C_N^{\mathsf{n}} N_f r_M e^{-(\kappa+1)d_j(X)}$. Together these bounds give the first assertion.

For the difference estimate use the coordinate Cauchy argument in the proof of Prop. 10.1. If X misses S, the source entries agree on X. A background walk can see their difference only by reaching S. After half its decay has been reserved for summation, the remaining half gives the factor $e^{-\operatorname{dist}(X,S)/(12\sqrt{3}M_w L^j)}$. The chain entries have the exponential locality of Lem. 10.4. As $\kappa_\sigma \le 1/(24M_w)$, these give (50). If X meets S, use twice the first bound. The entropy sum proves the last displayed estimate. For conjugation, sum the new activities and the localized normalizers before extracting their values. Read on shell, this is the logarithm of the whole step integral. At the trivial pair conjugation carries its scalar part at s to the complex conjugate of that at $s^\star$. The trivial block field passes every density test, so it belongs to the background pairs on which this identity

holds. Subtracting the real value at zero sources gives the asserted identity for the sum of the e_j. The assignment of the terms to individual polymers need not be preserved by conjugation. □

10.3. **Complex densities and the bounds on stored layers.** The local construction gives holomorphic activities. To use them in a history we must also prove an identity of measures on the history's coordinates. We do this coefficient by coefficient in the even Grassmann algebra. This keeps the coordinate changes and their null sets separate from the later estimate of the density.

Lemma 10.7 (A history density at complex sources)**.** Assume the admitted pure gauge construction and majorant settings at level k, and let $s \in \mathcal{P}$ and $h \in \mathcal{H}_k$. Write $\mathrm{Gr}_h^{\mathrm{ev}}$ for the even algebra of the generators retained by h. Suppose that, on the coordinate space $\mathsf{X}_h^{\mathrm{co}\prime}$, the term of h has the form

$$\hat{\tau}_h^{\mathrm{F}}[s] = \mathsf{Stay}_h^{\mathrm{T}} \chi_k(\Omega_k) e^{\mathbf{\Phi}_h^w} \exp A_{k,h}^{\mathrm{F}} \, \mathscr{M}_h.$$

Here the displayed density is defined and Borel off one $\mathscr{M}_h$-null set, is integrable in the fibre norm, and the identity holds for every Grassmann coefficient on every Borel set. Then there is a Borel, fibre-norm integrable density g_h on the free-coordinate space such that

$$\check{\tau}_h^{\mathrm{F}}[s] = g_h \mathscr{M}_h^{\mathrm{fr}}.$$

There is a Borel set $S' \subset \mathsf{v}_h(\mathcal{V}_0[h])$, outside which $\int \|g_h\|_{h_F} \, d\mathscr{M}_h^{\mathrm{fr}} = 0$, with the following property. For $\mathsf{v} \in S'$ and any configuration $U \in \mathcal{V}_0[h]$ with $\mathsf{v}_h(U) = \mathsf{v}$, put $\phi = \xi_h^{\mathrm{co}\prime}(U)$ and $V_n = \mathcal{M}^n U$. Then the two exponents are defined at ϕ and

$$g_h(\mathsf{v}) = \prod_{(t,D) \in \mathrm{Pc}(h)} g_{t,D}(V_{t-1}) e^{\mathbf{\Phi}_h^w(\phi)} \exp A_{k,h}^{\mathrm{F}}(\phi), \qquad g_{t,D} > 0.$$

In particular its fibre norm is bounded by the product of the scalar moduli and the norm of the last exponential. If on S' this bound is at most $\kappa_h \bar{u}_h$, where $\bar{u}_h$ is the product of the pure gauge piece majorants at loss zero, then

$$\int \|g_h\|_{h_F} \, d\mathscr{M}_h^{\mathrm{fr}} \le \kappa_h \exp\left(C'_\star \sum_{t \le k} \#_t(Z_t(h)) \right).$$

For $h_F \ge 1$, Berezin integration is bounded by this norm integral on every Borel set. ■

Proof. Choose the monomial basis (e_I) of $\mathrm{Gr}_h^{\mathrm{ev}}$. The coefficients ς_I of the bare sourced density after the partial block kernels are bounded complex Borel functions of U. Indeed the kernels are defined at every configuration, the bare density is continuous on the compact configuration space, and the partial Berezin integrals are finite sums of products. The pure gauge measure identities therefore apply to each ς_I.

Let f be the density in the hypothesis. Pull its one exceptional set back by $\Pi_h^{\mathrm{le}} \times \mathrm{id}$ and put $g_h = f \circ (\Pi_h^{\mathrm{le}} \times \mathrm{id})$ off this pullback and $g_h = 0$ on it. The coordinate-change identity gives $\check{\tau}_h[\varsigma_I] = (g_h)_I \mathscr{M}_h^{\mathrm{fr}}$ for every I. Since the algebra is finite dimensional, coefficientwise integrability and fibre-norm integrability are equivalent. This justifies the same construction for all coefficients at once.

Write $\rho_0(U) = e^{-\beta A(U)}$ for the bare positive gauge density. Its minimum on the compact configuration space is positive. The source-free positive term has a Borel full-measure support $S'_0 \subset \mathsf{v}_h(\mathcal{V}_0[h])$. For its complement R, the pure gauge domination of a term at a bounded complex density gives

$$|\check{\tau}_h[\varsigma_I]|(R) \le \frac{\sup |\varsigma_I|}{\min \rho_0} \check{\tau}_h[\rho_0](R) = 0.$$

It follows that $\int_R |(g_h)_I| = 0$ for every I, hence that the fibre-norm integral vanishes there. Intersect S'_0 with the open set of points in their tiers and remove the pulled-back null set; call the result S'. The density vanishes outside the tiers because the staying factor does. This proves the asserted support property.

At $\mathsf{v} \in S'$ the coordinate identities identify the pullback with ϕ. The staying factor is the displayed product of positive piece factors, and $\chi_k(\Omega_k) = 1$. Thus both exponents are defined and the stated identity follows. An exponential in the even algebra is invertible: writing $A = A_0 + (A - A_0)$, its scalar factor is e^{A_0} and its remaining factor is a finite nilpotent exponential, with inverse $e^{-(A-A_0)}$. Submultiplicativity now gives the norm bound. The integral of $\bar{u}_h$ is the pure gauge majorant bound, and the complement of S' has zero norm integral, proving the last estimate. Finally $|\mathrm{Ber} F| \le \|F\|_{h_F}$ and the linear Berezin map commutes with the finite-dimensional integral. □

The next three estimates check that the source-dependent pieces fit the same history geometry. The regular terms with fermion legs are larger than at zero source, and the stored background maps have a smaller margin than a generic doubled operator bound would give. Both facts must be accounted for before closing the history induction.

Lemma 10.8 (The sourced regular sum)**.** Let $s \in \mathcal{P}$, $1 \le k \le K$, $h \in \mathcal{H}_k$, and evaluate the regular activities at the history coordinates $\mathfrak{c}^h$ and at any evaluation configuration in the support of its tests. Then

$$\Big\| \sum_{j=1}^{k} \sum_{(z,X) \in \mathsf{Ix}_j^{\mathrm{rs}}(\mathcal{S})} E_{\ge 2}^{(j)}(X; \mathsf{U}, \bar{\Psi}^{\mathrm{bg}}, \Psi^{\mathrm{bg}}; z; \mathfrak{c}_{j-1}^h) \Big\|_{h_F} \le C_{\mathrm{leg}}^{\mathrm{s}} \Sigma_{\mathcal{L}}(h),$$

and the sourced mass member satisfies

$$\|\mathsf{M}_h(s)\|_{h_F} \le w_0 \nu C_H^2 h_F^2 L(\bar{B}_M + 2B_\sigma) \Sigma_{\mathcal{L}}(h).$$

More precisely, for $n \ge 1$, the contributions filed at a cube of the layer $\mathcal{L}_n$ are bounded by $C_{\mathrm{leg}}^{\square,\mathrm{s}} \phi_n$ and $w_0 \nu C_H^2 h_F^2 L(\bar{B}_M + 2B_\sigma)\phi_{n-1}$, respectively. The mass profile vanishes on $\mathcal{L}_0$. The constants are fixed before the lower mass bound and the final coupling ceiling. ■

Proof. Lem. 10.5 places the coordinates in $2\mathscr{D}_{j-1}(\alpha_0)$. There the part with legs has size $2C_w^{\mathrm{s}}\phi_{j-1}$. Its background maps have the localized row and Hölder bounds at the fine mass matrix, with the admitted enlarged constants. Apply the regular-sum argument cell by cell, using the radial gauge centered at the cell where a summand is filed. Its center belongs to the summand's polymer, so the same diameter bound controls the gauge field.

The degree-two term is split into its zero-moment units and the remaining cell sums. The first part has the gain from subtracting the constant leg. At a row of $\mathcal{L}_n$, the Hölder estimate for a localized background row gives the same gain in the lag $n - j$ as in the zero-source regular sum. The degree-at-least-four and field-dependent parts use the corresponding power counting. For the cell sums one uses the estimate proved in the source step, (48); its coefficients are read at the history coordinates, without assuming that the induced mass field varies slowly. The moved pieces are filed at the hulls of their polymers, while the forms taken out at a pointer are filed on the polymer supporting that form.

The differences between these filings and the interior sums occur at the rims. At lag $n - j$ their bound contains

$$L^{n-j} e^{-\kappa(7L^{n-j}-2)}.$$

The zero-source torus cell sums have the additional bound

$$2C_\omega C_{\mathrm{wr}} L^{n-j} e^{-\kappa(L^{n-j+1}-2)}.$$

Both sequences are summable. Combining them with the weighted piece count of the source step and its ϕ_{j-1} size gives $C_{\mathrm{leg}}^{\square,\mathrm{s}}\phi_n$ per layer cube. Since $x_n \ge \frac{1}{2}\gamma^{-2}$ and $\gamma^{-2} \ge e^{18\mathfrak{p}_0}$, this is bounded by a constant independent of the level. Summing the cubes gives the first estimate.

Every generated mass profile is supported in $\Lambda_j \subset \Omega_1$, $j \ge 1$, and hence vanishes on $\mathcal{L}_0$. On $\mathcal{L}_n$, $n \ge 1$, the norm of the mass coefficient is at most $L^{-(n-1)}(\bar{B}_M + 2B_\sigma)\phi_{n-1}$. Count the fine places

in a layer cube and apply the row-sum bound to its two background legs. The factors are precisely $w_0\nu C_H^2 h_F^2 L$, giving the second estimate. □

Lemma 10.9 (Scalar source activities and their scale sum)**.** On $2\mathscr{D}_{j-1}(\alpha_0)$, the scalar activities obey, polymer by polymer and pointer by pointer,

$$\sup_{X,z} e^{(\kappa+1)d_j(X)} \sup |E_0^{(j)}(X;\cdot;z;\mathfrak{c})| \le E_0^{\mathrm{s}} := E_0 + 2 + K_M^{\bullet} c_{\mathrm{cl}}, \qquad K_M^{\bullet} = K_M + C_E K_{\mathbf{M}}^{\mathrm{v}}.$$

Consequently, at the coordinates of a history,

$$\sum_{j\le k} \sum_{(z,X)\in \mathsf{lx}_j^{\mathrm{rs}}(\mathcal{S})} |E_0^{(j)}(X;\mathsf{U};z;\mathfrak{c}_{j-1}^h) - E_0^{(j)}(X;\mathbf{1};z;\mathfrak{c}_{j-1}^h)|$$
$$+ \sum_{j\le k} |L^j b_j A(\varphi_j,\mathsf{U})| \le C_5^{\mathrm{s}} \Sigma_{\mathcal{L}}(h), \qquad C_5^{\mathrm{s}} = C_{\mathrm{m}}^{\mathrm{sc}}(E_0^{\mathrm{s}} + B_\sharp + M^{-3}\mathsf{B}).$$

The scalar edge-part bound also holds with E_0^{s} in place of E_0. These two sourced constants are fixed before the constant C_E^{F} used in the history estimate. ■

Proof. Decompose the scalar part into the gauge normalizer, the closed fermion loops, the Gaussian family at the weight, the scalar fluctuation family, and the local constant taken out. The gauge normalizer and the local source-free subtractions are unchanged. The closed loops at the fine mass matrix have their original pointwise bound: $L^{j-1}\|\mathsf{n}\| < 2r_M/L \le \varrho_*$ puts their local walk expansion in its admitted domain. Their source-dependent values at the trivial pair have already been assigned to the source energy.

Point the Gaussian family along the segment of its quadratic potential. The insertions are $-(1-\tau)\hat{g}_b + \mathcal{V}_b^v$, with $\mathcal{V}_b^v = -\tau T_2[v](z_b, z)$, and their sum is the quadratic potential itself. Its admitted ceiling gives a bound one at each pointer. Comparing it with its zero-source value costs at most two.

For the scalar fluctuation family, insertions at fermion places are bounded by the part with legs of the potential. The cluster majorant is multilinear and nondecreasing in the vertex norms, so these insertions obey the same bound as that part. The bond insertions require the pointed scalar estimate of Lem. 7.4. The gauge and old-activity insertions cost at most 9 times their input norms, followed by the first-variation constant K_Π and the tent count $3L^3$. The resulting scalar constant is $K_M^{\bullet}$. For the linear source term the tent insertion is $\mathcal{V}^{\mathrm{lin}}(Y; P, h_z B)$, $0 \le h_z \le 1$; the term's bound is a sum over its individual bonds and remains $C_{\mathrm{lin}}\tau_{j-1}(w;\alpha_0)$. Schwarz's lemma for the additional origin bounds its first variation by K_Π times this norm.

The scalar recursion of the source step therefore closes for the pointed terms too, with bound

$$2C_w^{\mathrm{s}}\phi_{j-1} + C'_{\mathrm{lin}}\tau_{j-1}(w;\alpha_0).$$

Here the old families in the recursive term are the whole pointed families; no extra pointer sum is inserted. The C_{rad} floor formed with $K_M^{\bullet}$ makes the coefficient of that term at most $1/6$. The two displayed source sizes are each at most $K_M c_{\mathrm{cl}}/2$ by their admission ceilings. The unchanged terms, the Gaussian cost two, and this fluctuation bound give E_0^{s}.

These scalar families are functions of the background pair, holomorphic on its fixed-radius domain and gauge invariant term by term. Their first curvature derivative at the trivial pair vanishes. Thus their variation on a layer cube at lag $n-j$ has the volume factor $L^{3(n-j)}$ and the quadratic curvature gain $L^{-4(n-j)}$. The scalar sum-over-scales estimate applies with E_0^{s}, and $|b_j| \le B_\sharp$ is unchanged. This proves the stated C_5^{s} bound. The scalar edge parts are the omitted pointed sums together with the shares of the old action subtractions; applying their original count to the pointwise bound gives the final assertion. □

Lemma 10.10 (The remaining weight of stored formal legs)**.** Let $s \in \mathcal{P}$ and let a real evaluation configuration satisfy the regular-background hypotheses for $h \in \mathcal{H}_k$. Use the original background

constant C_H, before its enlargement for complex mass matrices, and set

$$c_{\mathsf{M}} = 2600\nu_P/\alpha_D, \qquad C_{\mathrm{bl}} = \nu'(2 + 4/(L-1)), \qquad C_{\mathrm{fl}} = 4C_{\mathrm{bl}}c_{\mathsf{M}}C_H.$$

Assume $C_{\mathrm{fl}}r_M \le L$. For a row x of level i in a stored layer $\mathcal{L}_m$, $i < m < k$, the background map at $\mathsf{n} = -a_K J$ satisfies

$$\sum_P |\mathcal{H}_{(i)}(x;P)| \le (\tfrac{1}{2} + 2^{-(k-m)-1})C_H L^{-(m-i)}.$$

The row twin has the same bound. Hence a formal leg retained at the weight of Prop. 9.13 costs at most one when it is evaluated in Gr_h at weight h_F. ■

Proof. At zero fine source the row sum is at most $\frac{1}{2}C_H L^{-(m-i)}$. To estimate its change, put $\mathsf{s} = \mathsf{s}_k$ and $d = d_h$. The fine background column has the form $\psi_P = \mathcal{K}_k(\mathsf{n})^{-1}g_P$ on A_0, where g_P is independent of n; it is unchanged on S_0. The resolvent identity gives

$$\psi_P(\mathsf{n}) - \psi_P(0) = -\mathcal{K}_k(\mathsf{n})^{-1}\hat{\mathsf{n}}\psi_P(0).$$

The kernel bound (27) and the zero-source background row sum of Lem. 6.8 imply

$$\sum_P |\psi_P(\mathsf{n})(v) - \psi_P(0)(v)| \le 2C_H^2\|\mathsf{n}\|\mathsf{s}(x_v)^{-1}\sum_{v'}(1 + |x_v - x_{v'}|)^{-2}e^{-\alpha_D d(x_v,x_{v'})/4}.$$

The last sum is at most $c_{\mathsf{M}}\mathsf{s}(x_v)$. Indeed, let $\rho = R\mathsf{s}(x_v) + 1 \le 2R\mathsf{s}(x_v)$ and $t = |x_v - x_{v'}|_1$. The layered metric gives $d \ge R\log(1+t/\rho)$. For $t \le \rho$ this is at least $t/(4\mathsf{s}(x_v))$; for $t > \rho$ the exponential is bounded by $(1+t/\rho)^{-p}$ with $p = \alpha_D R/4 \ge 8$. There are at most $6(1+t)^2$ sites at distance t, each with ν_P places. The two sums are bounded by

$$54\nu_P\left(\frac{32\mathsf{s}(x_v)}{\alpha_D} + \frac{16\mathsf{s}(x_v)}{\alpha_D}\right) \le c_{\mathsf{M}}\mathsf{s}(x_v).$$

At level $i \ge 1$ the closed Schur formula expresses the field as $q_i \cdots q_1\psi + \theta_i^{-1}\mathsf{e}_i$. Only ψ changes with n. The composed averages contribute $2\sqrt{48}L^i$, and the part of e_i depending on ψ contributes $4\sqrt{48}L^i/(L-1)$. The factor $\sqrt{48}$ comes from ι^* in the per-chamber supremum norm. Since $\sqrt{48} < \nu' = 8$, their sum is at most $C_{\mathrm{bl}}L^i$; the averages act trivially in flavour. The change of the level i row is consequently at most $2C_{\mathrm{bl}}c_{\mathsf{M}}C_H^2L^i\|\mathsf{n}\|$.

At a row in $\mathcal{L}_m$ the physical source bound gives

$$L^m\|\mathsf{n}\| = \lambda_m\|J\|_\infty \le L^{-(k-m)}r_M/(2L).$$

The change is therefore at most $\frac{1}{4}L^{-(k-m)}C_H L^{-(m-i)}$, which is bounded by $2^{-(k-m)-1}C_H L^{-(m-i)}$. Add the zero-source half. The row twin is estimated in the same way. The stored weights do not increase with the time of evaluation, so the ratio of the resulting row weight to the weight at which a leg was retained is at most one. □

10.4. **The history induction with complex sources.** At a complex source the scalar coefficient of a history is no longer a positive measure. We therefore carry the identity of measures and the estimates in all Grassmann degrees through the induction. Lem. 10.7 then supplies the density and its integrable norm majorant. The gauge tests and the stored variables remain those of the unsourced construction.

Definition 10.11 (The sourced inductive statement)**.** For $s \in \mathcal{P}$, let $\mathsf{I}^{\mathrm{F}}(k)[s]$ mean that every history through level k has, coefficient by coefficient, the measure representation

$$\mathsf{Stay}_h\, 1_h^{\mathrm{T}}\chi_k(\Omega_k)e^{\mathbf{\Phi}_h^w}\exp A_{k,h}^{\mathrm{F}}\,\mathscr{M}_h,$$

on a common full-measure set, with Borel density and one value at all evaluation configurations of each point. The exponent has the same terms and the same places as the unsourced exponent, evaluated at the fine matrix mass and the coordinate chain of h. More precisely, it has the following properties.

(1) The exponent of the formed factors is $\mathbf{\Phi}_h^w = \mathbf{\Phi}_h + \sum_{t,D} K_{t,D}^v$, where the sum is over the kept pieces, $|K_{t,D}^v| \le \gamma_{\mathrm{kp}} S_{t,D}/2$. Its quadratic bound holds with $\gamma^\dagger = \gamma_{\mathrm{kp}}/2$, and its formed-factor bound with $(\mathsf{B}_{\mathrm{fr}}^\flat + 1, \kappa)$.

(2) The action has running weight $\beta_h^F - a_K^{-1} w$. Its difference from the fixed weight $\beta - a_K^{-1} w$ is the source-independent layer term in (36), already included in the scalar bound. For history integration use $\mathsf{bw} = \beta - a_K^{-1} \operatorname{Re} w \ge (1 - \mathsf{a})\beta$, where the fixed loss $\mathsf{a} \in (0, 1/2]$ satisfies the source ceiling $4\varrho \le \mathsf{a}(\gamma^{-2} - \bar{B})$.

(3) The stored quadratic form is $\mathbb{D}_h(\mathsf{n})$. Its row and column sums are bounded by $C_D^{\Sigma,\mathrm{s}}$, and its exponential by $e^{\mathsf{c}_G \Sigma'(h)}$, with $\max\{\mathsf{c}_{\mathrm{tw}}, C_D^{\Sigma,\mathrm{s}}\}$ in c_G.

(4) The mass profile is the sourced history profile of Lem. 10.5; it and the regular sum with legs have the bounds of Lem. 10.8.

(5) The regular activities are those of Prop. 10.1 at $\mathfrak{c}^h$. Extraction keeps each moved form on its actual polymer Z_p, with its own pointer. The pointed sum rule holds, the source-step sizes hold, and the scalar entries and their edge parts satisfy Lem. 10.9.

(6) The made boundary terms and the edge parts satisfy the unsourced boundary estimates with $\mathsf{B}_\natural$ and E_0^{s} in place of B and E_0. No reality is required at a complex source. They include the rejected halves, the reblocked linear term, the sourced strip rows and the forms pointed on the rims. Their dependence on the history has the coordinate tails of Lem. 10.5. At step j the interior leg size obeys $\bar{v}_j^{\mathrm{s}} \le b^{\mathrm{blk,s}} \le c_{\mathrm{cl}}/8$, and the frame leg size obeys $\mathsf{sz}_j^\natural \le (64/51) b^{\mathrm{tot,s}}$. The rim output with legs is bounded by $K^{\mathrm{lb}} \mathsf{sz}_j^\natural e^{-\kappa'' d_j(X)}$; the sourced strip rows add at most $\eta_j^{\mathrm{sr}} K^{\mathrm{lb}} \mathsf{sz}_j^\natural e^{-\kappa_+ d_j(X)}$, where $\eta_j^{\mathrm{sr}} < 1/8$. Formal legs in stored layers have the cost at most one of Lem. 10.10.

(7) The vacuum constant of the label satisfies

$$-\mathcal{E}_{k,\mathcal{S}}^{\mathrm{F}} = E_k(s) - \sum_{n \le k} f_n(\mathcal{S}) + \sum_{n \le k} g_n(\mathcal{S}; s), \qquad |g_n(\mathcal{S}; s)| \le C'_{\mathfrak{E}} \#_n \widetilde{Z}_n.$$

Each g_n is holomorphic and vanishes at zero source.

(8) In all degrees the sum of the mass, regular and boundary terms and the action counterterms is at most $(C_E^{\mathrm{F}} - \mathsf{c}_G)\Sigma'(h)$. The constant includes the scalar scale sum, scalar edge terms and all rim forms specified above.

At level zero this means the bare density with no activity, formed factor or vacuum constant. ■

The next lemma identifies the expansion used at one step. Its estimates use only the bounds at previous steps. The estimates for the newly generated boundary terms, recorded in the proof of Prop. 10.14, are one-step estimates under precisely that hypothesis.

Lemma 10.12 (The centered frame at a source)**.** Use the common constant choices of Prop. 10.20, including the source ceilings. Suppose $\mathsf{I}^{\mathrm{F}}(k)[s]$ holds, $s \in \mathcal{P}$, and consider a step along $h' = (h, \mathsf{o})$ with $\Omega_1(h') \ne \emptyset$. Set $t = k + 1$, $\tau = x_k/\mathsf{t}_k$, $g_k = \mathsf{t}_k^{-1/2}$, and $v' = \tau v_k = -w/\mathsf{b}$. The quadratic weighted term, the linear term and the cross term, read at the integrated fluctuation alone, form one centered class

$$\mathbb{W}^{\mathrm{ce}} = \mathbb{W}^{\mathrm{q}} + \mathbb{W}^{\mathrm{l}} + \mathbb{W}^{\times}.$$

They are analytic and Borel on the tiers, jointly gauge invariant, local on their assigned polymers and complex linear in w; conjugation holds at real w. At the rate $r_V = (1 - 2\delta_{\mathrm{sp}})(\kappa - 2)$, in the centered radial norm,

$$\|\mathbb{W}^{\mathrm{q}} + \mathbb{W}^{\mathrm{l}}\|_{\mathrm{I}\star, r_V}^{\mathrm{rad}} \le \mathsf{k}_\odot(\sigma_s + \varrho'_s g_k) \le \tfrac{1}{2}\mathsf{m}_\odot,$$
$$\|\mathbb{W}^{\times}\|_{\mathrm{I}\star, r_V}^{\mathrm{rad}} \le C_{\mathrm{cr}}^{\mathrm{V}} \mathsf{C}_{\mathcal{A}} A_1 \varrho \mathsf{t}_k^{-1} (\log \mathsf{t}_k)^{\mathfrak{p}_0} \le \tfrac{1}{4} c_M^{\mathrm{O}},$$

where $\sigma_s = 2\varrho/(\mathsf{b} - \bar{B})$, $\varrho'_s = \sqrt{3}\varrho/20$, and $\mathsf{m}_\odot \le c_M^{\mathrm{O}} = w_0^{\mathrm{O}}\varepsilon^{\mathrm{O}}$. At the pair (t_k, x_k) their centered size is at most $c_M^{\mathrm{O}} \le w_0^{\mathrm{O}}$.

The remaining classes are of sort Σ_{v}. Their sizes on the tiers satisfy

$$\mathsf{sz}^{\mathrm{rg}} \le \varepsilon_\flat, \quad \mathsf{sz}^{\mathrm{bd}} \le \tfrac{1}{2}\varepsilon_\flat, \quad \mathsf{sz}^{\mathrm{q}} \le \varepsilon_\flat, \quad \mathsf{sz}^{\natural} \le \tfrac{64}{51} b^{\mathrm{tot,s}} \le \tfrac{16}{17}\varepsilon_\flat.$$

At the first step the boundary class is absent and the regular class together with the additional scalar matter terms has size at most $3\varepsilon_\flat/2$. The centered expansion exists, is holomorphic on the interior of $\mathcal{P}$, and has

$$\varepsilon_{\mathsf{w}}^{\mathfrak{F}+} L^3 \le \frac{5}{32\mathsf{K}_{ap}^{+}}, \qquad \mathsf{b}_{\mathrm{cs}}^{\mathfrak{F}+} \le \frac{9}{32}.$$

Its cluster sums have the locality, invariance, tie and measurability properties of the joined expansion. On a domain $X \not\subset \Lambda_t$ its output after subtracting the weighted Gaussian part is at most $(9/16)e^{-\kappa'' d_t(X)}$; its part with legs is at most $K^{\mathrm{lb}}\mathsf{sz}^{\natural}e^{-\kappa'' d_t(X)}$. Inside Λ_t the expansion agrees term by term with Prop. 10.1, including its pointed outputs. ∎

Proof. The weighted quadratic and linear terms are the centered Taylor members of the weighted action. The subtraction at a base plaquette removes the constant weight in the linear and cross terms. The bounds for the cross term use its conditional mean chains, detailed below. Each chain is assigned to the hull of its footprint and reads the stored field only at its final bond. Thus it is local, and its assignment is independent of the fields. The Taylor terms and these chains are analytic, Borel and gauge covariant, so their assigned sums have the stated properties. The source ceilings give the two displayed norm bounds. Their sum is at most $3c_M^{\mathrm{O}}/4$ in the calibration frame. On passage to the pair the quadratic-growth factor is at most $\max\{1, \tau^{-1}\} \le 4/3$, proving the centered-class bound.

The regular scalar class is the ordinary regular class with the pointed entries at $\mathfrak{c}^h$, the action shares removed with the same b_j, and the counterterms zero. The plaquette weight made from the history is unchanged: it reads the regions and the numbers b_j, not the external source. The boundary scalar class includes the sourced made terms and formed factors. One must apply their estimates origin by origin. The box theorem requiring real entries cannot itself be applied to complex entries. Its constituent estimates require only analyticity, invariance, the tie and the bounds on entries, made terms and formed factors. These hold here with $(E_0^{\mathrm{s}}, \mathsf{B}_\natural, \mathsf{B}_{\mathrm{fr}}^{\flat} + 1)$, by Lem. 10.9 and the inductive statement. In particular an edge part has at most $M^3 \#_{\pi_j} X \le 8M^3 e^{d_j(X)}$ pointers. Its scalar part is therefore at most $8M^3 E_0^{\mathrm{s}} e^{-\kappa d_j(X)}$, which is covered by $\mathsf{B}_\natural$.

The two scalar class estimates, at the calibration couplings, are

$$\max\{1, \tau\} C_{\mathrm{eff}}\big(\mathsf{s}_k^{\sharp}[E_0^{\mathrm{s}}] + \vartheta_{0,k} g_k^2 y_k^{2q}\big), \qquad C_{\mathrm{eff}}(\mathsf{B}_\natural + \mathsf{B}_{\mathrm{fr}}^{\flat} + 1)\vartheta_{0,k}.$$

Since $\max\{1, \tau\} \le 5/4$ and $\mathsf{s}_k^{\sharp}[E]$ increases with E, the doubled-box ceiling makes each at most $\varepsilon_\flat/2$. The weighted Taylor terms of order at least three belong to the regular class. Their norm is bounded by

$$\tfrac{1}{2}\max\{\mathsf{K}^0 (r_P^{\circ})^{-3}, \mathsf{K}_0 r_0^{-3}\}(\mathsf{C}_{\mathcal{A}} + \theta_{\mathrm{sl}})^3 A_1^3 \gamma (1 + 2\log\gamma^{-1})^{3\mathfrak{p}_0} \le \tfrac{1}{2}\varepsilon_\flat.$$

This follows from the third-order Taylor remainder on the original tiers and $\sup|v'| \le 1/8$. The first step has the same estimate on its block-local domains. This proves the regular and boundary bounds. The quadratic class keeps its unsourced bound under the pair comparison. The leg class is bounded by the one-step recurrence below: it reads earlier frame sizes only, and gives $(64/51)b^{\mathrm{tot,s}}$ at the new step.

The boundary members at a complex source have the same analyticity on the tiers as in the proof of Prop. 9.13. Their coefficients are holomorphic on their filing slices; the background maps at $m_0 + \widehat{\mathsf{n}}$ have the bounds and holomorphy of Lem. 6.5 and the localized operator construction; and the shift, substituted triples and levelling functions read no source. The normalization and composition argument therefore applies unchanged. In particular it uses the admitted slices for a

made boundary coefficient. It does not require an extension of that coefficient to every complex link configuration.

The coefficient and leg pieces of a made term are recombined at each polymer and each choice of leg parts as the bracket of $\Pi_{j,X}[\omega]$. This gives the pair dependence of the member of the frame. Edge pieces have that dependence individually. Their products are Borel by the coefficient, background and shift formulas, and uniform convergence preserves Borel dependence in the walk sums. Thus the source frame has the pair, analyticity and measurability properties required by the joined expansion, as well as the sizes just proved.

The noncentered sizes sum to less than $6\varepsilon_\flat$. Using $\varepsilon_\flat = (64eL^3\mathsf{K}_{\mathrm{ap}}^{\max})^{-1}$ and $\varepsilon^{\mathrm{O}} = (16eL^3\mathsf{K}_{\mathrm{ap}}^{+})^{-1}$, the amplitude is at most $3/(32\mathsf{K}_{\mathrm{ap}}^{+}) + 1/(16\mathsf{K}_{\mathrm{ap}}^{+}) = 5/(32\mathsf{K}_{\mathrm{ap}}^{+})$. The large-field contribution is at most $1/(8\mathsf{K}_{\mathrm{ap}}^{\max})$, so the cluster bound is $9/32$. The canonical source-step hypotheses and the pair comparison give the remaining hypotheses of the centered expansion, including those of its first-step version. Lem. 9.1 applies this expansion in the even Grassmann algebra. Its majorant reads the sizes just obtained; the positive Gaussian, its tests and its admissible centers are gauge objects. It supplies the cluster properties and the bound $9/32$ for both the full frame and its weighted Gaussian subframe. Subtracting them gives $9/16$. Marking the class with legs gives the stated K^{lb} bound.

On an interior domain the conditional mean is zero, no external field is read, the boundary class vanishes, and the cross term has no polymer. The centered quadratic and the ordinary quadratic class combine into the weighted quadratic origin; the linear member joins the scalar potential. The origin-joining identity of Prop. 7.2 and the interior identification of Lem. 9.10 give the asserted termwise equality, also at every pointer. □

Proposition 10.13 (Induction at a complex source)**.** With the common constant choices of Prop. 10.20, $\mathsf{I}^{\mathrm{F}}(0)[s]$ holds for every $s \in \mathcal{P}$, and $\mathsf{I}^{\mathrm{F}}(k)[s] \Longrightarrow \mathsf{I}^{\mathrm{F}}(k+1)[s]$ for $0 \le k < K$. ■

Proof. At level zero the exponent is the weighted Wilson action and the fermion form at $m_0 - a_K\widehat{J}$. Its coefficients are polynomials in the links, and its even exponential is a Borel density. If $\Omega_1(h')$ is empty, every later region is empty and no variable is integrated. The term is simply the image of the tested bare density. All profiles vanish, there are no regular or boundary terms, and the label constant cancels the regular vacuum number. The stored-form bound and the bound on that number prove the assertion in this case.

Otherwise fix $h' = (h, \mathsf{o})$. The gauge change of variables, tests and gauge-fixing factors are unchanged. The exact Schur identities at the matrix mass give the partial fermion integral. The background change and completion of the square use the sourced operators, with their doubled bounds. Stored generators are kept at their old places; the other legs acquire the canonical levelling factors. Only the jointly invariant members that read no stored gauge data are thereby read as members at the fine field. Terms with stored gauge data retain that dependence and are treated by the boundary coefficient estimate. That estimate of Lem. 9.7 applies with $\mathsf{B}_\natural$, since it uses analyticity and bounds without a reality condition. The normalizer is δZ_k^{F} at the matrix source.

Lem. 10.12 gives the joined centered expansion, with the first-step version when $k = 0$. The first fermion covariance is defined at every configuration by its positive real-part bound, so no regularity hypothesis is added at that step. Inside Λ_{k+1} the history coordinates lie in the doubled source window, and the expansion is the source step of Prop. 10.1. Take out its vacuum number, action coefficient, source energy, and local mass increment. The increment uses only pieces with $Z_p \subset \Lambda_{k+1}$. Its forms stay on these Z_p: interior pointers enter the regular sum, rim pointers enter the edge parts. Reinsert the mass through the new profile; its complement remains on the rim. At each pointer the extracted form, its reinserted part and its remainder sum to zero.

We spell out the measure identity because it is independent of positivity at a complex source. Transport of the weighted Wilson action over the bare plaquettes gives the sourced change-of-variables identity. The Taylor decomposition at the conditional mean is exactly the sum of the weighted quadratic, linear and cross terms, the term on stored fluctuations, and the terms of order

at least three. Split the stored quadratic term into its kept and rejected halves. The product of this splitting identity and the joined expansion cancels the preintegral value at the conditional mean. The scalar linear term evaluated at the stored field leaves the integral as an exponential and is reblocked into the new boundary term. The sum rule for the sourced strip rows is the one in Lem. 9.2. These are the sourced counterparts of the identities in Prop. 9.11.

All operations are coefficientwise in the finite-dimensional even algebra. In particular the conditioning integral is followed by the Gaussian Berezin integral with the symmetrized sourced decoupling covariance of Prop. 6.7. Its conjugation maps the source to $\gamma_4 J^\dagger \gamma_4$, and its charge conjugation to J^{c}; it is not a positivity assertion at a fixed complex source. For an interchange of integrals, first use Tonelli for the fibre norm, then Fubini for each coefficient. The needed integrability and common null set are supplied by Lem. 10.7. Wherever the scalar argument uses a nonvanishing exponential, use the invertibility of an even exponential. Its scalar coefficient is itself a nonvanishing complex exponential.

The transported source exponent, the canonical shifts and the two halves are Borel. Activity and boundary coefficients are locally uniform limits of finite sums of Borel coefficients and localized covariances. The coordinate fields σ_n^h depend on the source, not on the configuration. The new density is therefore Borel. Evaluation configurations of a point differ by a gauge transformation. The weighted action is invariant plaquette by plaquette, the halves are covariant, and the matrix mass and increments commute with colour. This proves the one-value assertion.

The remaining task is quantitative. The kept-half and store bounds are proved below; Lem. 10.8 bounds the mass and regular terms with legs; Lem. 10.9 bounds the scalar entries and their scale sum; and Lem. 10.10 pays the stored formal legs. The one-step boundary recurrence below uses only preceding frame sizes and preserves $(64/51)b^{\mathrm{tot,s}}$. The rim outputs of Lem. 10.12, the sourced strip bounds and the pointed extraction just described give the boundary clause. Finally the local energy comparison of Lem. 10.5 gives the label errors. These estimates give each clause of Def. 10.11 at level $k+1$ and complete the induction. □

Proposition 10.14 (All histories at the sources)**.** For $s \in \mathcal{P}$ the sourced history measures are complex measures and their sum is the exact bare sourced measure after the block maps. Uniformly for $k \le K$,

$$\sum_h e^{\Theta^{(k)}(h) - \operatorname{Re} E_k(s)} \int \|g_h(s)\| \, d\mathscr{M}_h^{\mathrm{fr}} \le e^{(C+2)n_k}.$$

Thus Z_k^{sub} is holomorphic on the interior of $\mathcal{P}$, continuous on it, and

$$|Z_k^{\mathrm{sub}}[s]| \le e^{(C+2)n_k}, \qquad Z_K^{\mathrm{sub}}[0] \ge e^{-C_P(1+\log x_K)\ell^3}.$$

The factorization is $Z[s] = e^{E_k(s)} Z_k^{\mathrm{sub}}[s]$, and $Z_k^{\mathrm{sub}}[s^\star] = \overline{Z_k^{\mathrm{sub}}[s]}$. ■

Proof. Prop. 10.13 gives the complex measure identity and the full-degree form of every history. The history chain remains in the doubled coordinate window by Lem. 10.5. The physical source ball lies in the fixed-radius domains, so all local norms needed by the histories are the fixed-radius norms of the source step.

We verify the additional scalar payments. Set $a = \frac{1}{2}(1 - c_\Theta/c_*) \in (0, 1/2]$ and impose $4\varrho \le a(\gamma^{-2} - \bar{B})$. Then the real part of the sourced Wilson weight is at least $(1-a)$ times the unsourced weight. The source quadratic on stored fields is decomposed by the pure weighted-form construction into kept and rejected halves. Its kept part obeys $|K_{t,D}^v| \le C_v^{\mathrm{kp}} \sup|v| S_{t,D}$. Impose $4C_v^{\mathrm{kp}}\varrho \le \gamma_{\mathrm{kp}}(\gamma^{-2} - \bar{B})$. The retained real quadratic lower bound then remains at least $\gamma_{\mathrm{kp}} S_{t,D}/2$. Its rejected part is a bounded scalar boundary family. This application uses only the supremum of the weight, the original positive retained forms and the original tiers. More precisely, at later steps the local source forms obey $|q_Y(P; B, B)| \le k_v e^{-r_v d(Y)} \sup|B|^2$, with k_v proportional to $\sup|v|$. Use the original fixed-real-factor analytic slices and their graded-to-layered restrictions. With

$\beta_\upsilon = 8C_E e^{6\delta_\upsilon M} k_\upsilon$, $r_\upsilon - \delta_\upsilon M \ge c_E$, $W_t \ge LM$ and $\delta_\upsilon LM/2 \ge 342(\kappa'+1)$, [3, Prop. 5.36] and [3, Prop. 5.37] give

$$\|\widehat{\Upsilon}\| \le C_\Upsilon \beta_\upsilon, \quad \operatorname{osc} \widehat{\varphi}_X \le b^{\mathrm{s}}_{\mathrm{fr}} \mathfrak{b}_k^2 e^{-\kappa' d(X)}, \quad b^{\mathrm{s}}_{\mathrm{fr}} = c_\upsilon^2 B_f \beta_\upsilon K_{\mathrm{ch}}^\sharp C_{\mathrm{xt}}^2 e^{-\kappa'}.$$

Choose the source ceiling so that $b^{\mathrm{s}}_{\mathrm{fr}} \le 1/4$. The first step has the direct block-local quadratic bound and the analogous coefficient at most one. These are retained formed factors, with their original quadratic tier factor, and not small fluctuation activities. Their later Cauchy variation is precisely the variation of the original formed-factor family with this enlarged fixed coefficient. The source forms meet these hypotheses by their quadratic Taylor bound and original kernel walk rates; their norms are linear in $\sup|v|$.

The linear source term evaluated on the stored field and mean leaves a collar term of size at most

$$C_{\mathrm{lin}} \max\{\alpha_0, 12L^6\} \varrho (1 + 2c^{\mathrm{w}}_{\mathsf{M}}) C_{\mathrm{xt}} A_1 \mathfrak{t}_j^{-1/2} (\log \mathfrak{t}_j)^{\mathfrak{p}_0}.$$

It is below $c_{\mathrm{cl}}/16$ at the final floor. For the source cross term use its weight-difference continuation,

$$X^v = -\tau \sum_b \sum_{Y \ni b} F_{Y,2}^{v - v(p_b)}(\widetilde{z}_b, F_*).$$

Its constant-weight part is zero at the physical center and is not continued. Expand $m = -K_1 a'$ by the conditional-mean resolvent and leg chains. A chain $(b'', \jmath, b_1, S, b')$ reads the pair on its footprint and a' at a stored bond b'. File it at Y joined to its footprint hull, which meets the collar. If its tree length is l, that hull contains at most $125L^3 + 250L^2 l/M$ fine cubes. The chain exponential therefore pays its enlargement and polymer weight under the existing walk-rate floors. The swing $x_j \sup_Y |v - v(p_b)| \le 2\#_{\mathcal{P}_j} Y \sqrt{3} M \operatorname{Lip} w$ and the stored tier bound then give total size

$$C_{\mathrm{cr}} \mathsf{C}_{\mathcal{A}} A_1 \varrho \mathfrak{t}_j^{-1} (\log \mathfrak{t}_j)^{\mathfrak{p}_0} \le c_{\mathrm{cl}}/16.$$

The rejected half has the separate bound

$$C \mathfrak{b}(\mathfrak{t}_{t-1})^2 e^{-(\check{R}_t - 2)} e^{-\kappa'' d_t(X)}.$$

Since $\check{R}_t \ge (\log \mathfrak{t}_t)^{r_c}$ and $r_c \ge 2$, the threshold-square times this suppression is uniformly bounded. Its entropy sum therefore costs a fixed number per rim cube. This, rather than the tier-weighted retained oscillation bound alone, pays the rejected halves. The retained exponents keep their original formed-factor norm with the original tiers. The linear term left outside the integral is reblocked from X to its coarse hull X^{up}. The inequalities $d_{k+1}(X^{\mathrm{up}}) \le L^{-1} d_k(X)$ and $\lambda_V \ge L^{-1}\kappa'' + 2c_E$ give

$$e^{-\lambda_V d_k(X)} \le e^{-\kappa'' d_{k+1}(X^{\mathrm{up}})} e^{-c_E d_k(X)}.$$

The hull count is at most $(L+2)^3 S_{\mathrm{s}}$. The reblocked family is therefore bounded by

$$K_{\mathrm{rb}} c_{\mathrm{cl}} e^{-\kappa'' d_{k+1}(X')}/16, \qquad K_{\mathrm{rb}} = (L+2)^3 S_{\mathrm{s}} e^{4\lambda_1}.$$

It reads the pair and stored fluctuation on X' alone and vanishes unless X' meets the collar.

Stripping produces additional boundary rows. Those from the two halves have their quadratic bound times $C^{\mathrm{tr}}_{\mathrm{fl}} e^{-\check{R}_t}$, uniformly bounded because $\mathfrak{b}(\mathfrak{t}_{t-1})^2 e^{-\check{R}_t} \le e^{-2} C_{\mathfrak{b}}$. Apply Lem. 9.2 to the other two families, the rim cluster sums and the reblocked linear term. They are local, jointly gauge invariant and analytic on its complex box; their bounds there are respectively $9/32$ and $K_{\mathrm{rb}} c_{\mathrm{cl}}/16$, at rate κ''. The rows sum to the difference between the field before the strip and the stored field and vanish away from the strip. Their extra factor is

$$\eta_t^{\mathrm{sr}} = 2^{14} e^{-\check{R}_t} \le 2^{14} e^{-324} < \frac{1}{16}.$$

In positive degree the rim family and its rows are thus bounded by $(9/8) K^{\mathrm{lb}} \mathsf{sz}_t^\natural$. All scalar additions just listed are counted in $\mathsf{B}_\natural$ and C_E^{F}.

The leg recurrence uses only preceding frame sizes. Write $\Theta^{\mathrm{s}}_k = c''_\psi + \Theta'^{\mathrm{s}}_k$ for the split-leg factor at new and stored generators, respectively; $\Theta'^{\mathrm{s}}_k \le 2\cdot 10^{-3}$. At lags below n_{lag} keep each whole pointed activity. At longer lags use the units, moved pieces and forms constructed above. A piece agrees with its one-lattice counterpart only if its actual polymer lies in the regular core. At a cell of a stored layer $\mathcal{L}_m$, take differences only for an unchanged piece contained in $\Omega_{m-1}\setminus\Omega_{m+2}$. Every other piece is taken whole.

The difference pieces have the cell-sum bound of Lem. 10.8. A whole piece reaching layer $m>j$ has $d_j(Z)\ge 7L^{m-j}-2$, so its cells and the unused decay are paid by

$$c^{\mathrm{s}}_{\mathrm{rim}} = c_{\mathrm{geo}}(1+L^2)\sum_{d\ge 1} L^{3d}e^{-\kappa(1-L^{-1})(7L^d-2)}.$$

The two families of moved pieces cost $1+2C_{\mathrm{pw}}$. For a split stored leg the factor $(2L^{-4})^{k-m}$ combines with the cell count to give $(2/L)^{k-m}$, whose sum is at most $L/(L-2)$. For degree $2p\ge 4$ each unsplit leg keeps at most three quarters of its weight; telescoping and the walk count cost at most

$$2p(1+2pn^2_{\mathrm{wk}})(3/4+\Theta'^{\mathrm{s}}_k/4)^{2p-1}\le 8.6n^2_{\mathrm{wk}}\le c_{\mathrm{geo}}.$$

The torus cell sums are filed on wrapping polymers with $d_j(Z)\ge L^{k-j}-2$ and use the convergent bound $\sum_{d\ge1}L^{3d}e^{-\kappa(1-L^{-1})(L^d-2)}\le 1$. Formal stored legs cost at most one by Lem. 10.10.

These estimates give the stored-layer share

$$b^{\mathrm{sl,s}}_k = C^{\mathrm{s}}_{\mathrm{lay}}\frac{L}{L-2}\Big[4Lc^{\mathrm{s}}_\psi + \frac{2\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}_k)}{\beta_0}\big(9L^4A_\kappa C_H + (2+6L^5A_\kappa)C_{\mathfrak{r}}\beta_0\big)\Big]$$
$$\cdot\,(1+w_0\nu C_H^2h_F^2L(\bar B_M+2B_\sigma))\phi_k.$$

Besides the interior share $b^{\mathrm{blk,s}}_k$, the new terms have the shares

$$\begin{aligned}
b^{\mathrm{lay,s}}_k &= 2C_V^{\mathrm{F,s}}h_F^2\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}_k)\\
&\quad+\Theta'^{\mathrm{s}}_k\big[(1+c_{\mathrm{cs}})2C_{\mathrm{pc}}S_{1/4}C^{\mathrm{s}}_w + C_0^{\mathrm{cs}}h_F^2\big]\phi_k\\
&\quad+\Theta^{\mathrm{s}}_k(1+C_\omega(1+2C_{\mathrm{pw}}))c^{\mathrm{s}}_{\mathrm{rim}}2C^{\mathrm{s}}_w\phi_k+\Theta^{\mathrm{s}}_k2C_{\mathrm{pc}}S_{1/4}C^{\mathrm{s}}_w\phi_k\\
&\quad+\frac{L}{L-2}\Theta'^{\mathrm{s}}_kC_\omega(1+2C_{\mathrm{pw}})(2c_{\mathrm{geo}}(1+L^2)+c^{\mathrm{s}}_{\mathrm{rim}})2C^{\mathrm{s}}_w\phi_k+b^{\mathrm{sl,s}}_k,\\
b^{\mathrm{ms,s}}_k &= C^{\mathrm{s}}_{\mathrm{bd}}\left(c^{\mathrm{s}}_\psi+\frac{2\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}_k)}{\beta_0}\right)\Big[3M^3w_0\nu C_H^2h_F^2\left(\bar B_M\gamma^2+\frac{B_\nu}{6}\phi(\tfrac12\gamma^{-2})\right)\\
&\qquad+2\eta_{\min}^{-2}c_{\mathsf{S}}M^{-3}C_{\mathrm{pw}}C^{\mathrm{s}}_w\phi(\tfrac12\gamma^{-2})\Big].
\end{aligned}$$

The mass share includes the rim complement with coefficient $L^{-j}[\varphi_j\mu_j-(1-\varphi_j)(\nu^h_j-\mu_j)]$, bounded by $L^{-j}(|\mu_j|+B_\nu\phi_j/2)$, and the forms $P_{z;X,p}$ on their actual Z_p, of summed norm at most $2\eta_{\min}^{-2}c_{\mathsf{S}}M^{-3}C_{\mathrm{pw}}C^{\mathrm{s}}_w\phi_{j-1}$. Their coefficients read no configuration; their reentry is through their legs alone.

For the forms this uses a loss of one unit of decay. Their known bound sums $\|P_{z;X,p}\|e^{\kappa d_j(Z_p)}$ through a cube. Apply the preceding bracket count at rate $\kappa-1$, with roots containing the hull $Z_p^{\uparrow}$. After the walk factors and the weight room have been used, its remaining factor is

$$e^{(\kappa-1+1/13)d_k(Z_p^{\uparrow})}e^{2/13}=e^{(\kappa-12/13)d_k(Z_p^{\uparrow})}e^{2/13}.$$

At $j=k$ it is bounded by $e^{\kappa d_j(Z_p)}e^{2/13}$. At $j<k$, use $d_k(Z_p^{\uparrow})\le L^{-(k-j)}d_j(Z_p)$ and

$$(\kappa-12/13)L^{-1}+\kappa(1-L^{-1})\le\kappa.$$

The weight of the form therefore also pays the reach factor used in the boundary count. Finally $e^{2/13}<C_E$, already included in $C^{\mathrm{s}}_{\mathrm{bd}}$. This proves the displayed contribution of the forms in the

summed norm at rate $\kappa - 1 \ge r_V$. The made and ordinary edge terms instead have the spare rate κ_+ and use their earlier domain count. Their coefficient and leg contributions remain $1/16$ and $9/64$ times the boundary maximum, respectively; they are recombined before asserting pair dependence of the frame.

Set $\Sigma^{\natural}_{k+1} = \sup_{1\le j\le k} \mathsf{sz}^{\natural}_j$ and $\bar v^{\mathrm{s}} = \sup_{1\le j\le k} \bar v^{\mathrm{s}}_j$, both zero at $k = 0$. The old boundary terms entering through legs contribute at most $\frac{1}{8}\frac{9}{8}\max\{\Sigma^{\natural}_{k+1}, \bar v^{\mathrm{s}}\}$; their coefficient variations contribute at most $\frac{1}{16}$ times the same maximum, by Lem. 9.7 and the coefficient ceiling. Thus

$$\mathsf{sz}^{\natural}_{k+1} \le \frac{13}{64}\max\{\Sigma^{\natural}_{k+1}, \bar v^{\mathrm{s}}\} + b^{\mathrm{tot,s}}_k, \qquad b^{\mathrm{tot,s}}_k = b^{\mathrm{blk,s}}_k + b^{\mathrm{lay,s}}_k + b^{\mathrm{ms,s}}_k.$$

Every share is bounded by its value at γ, and $\bar v^{\mathrm{s}} \le b^{\mathrm{blk,s}} \le b^{\mathrm{tot,s}}$. Since $(13/64)(64/51) + 1 = 64/51$, induction preserves $(64/51)b^{\mathrm{tot,s}}$. Each of the three share ceilings is $\varepsilon_\flat/4$, giving the bound $16\varepsilon_\flat/17$ for the leg class. This verifies the one-step estimate used in Lem. 10.12.

The complex stored tower has uniformly bounded row and column sums by its closed Schur form and the weighted inverse at n; its fine-store entries remain of range one. Lem. 9.12 therefore bounds its exponential with a fixed enlarged constant. The stored mass profile is the convex interpolation of $L^{-n}(s_n + \sigma^h_n)$ and $L^{-(n-1)}(s_{n-1} + \sigma^h_{n-1})$; its norm per layer cube is bounded by the source window. The source cell-unit estimate supplies the needed power counting. For the component integral keep the actual allocation of the piece stocks. At zero sources its majorant retains $e^{-\gamma_{\mathrm{kp}} S_{t,D}/4} e^{-P^G_{t,D}/2}$ and the created cells are paid by

$$\sum_{t,D}\Big(P^W_{t,D} + \tfrac{1}{2}P^G_{t,D} + \tfrac{1}{4}\gamma_{\mathrm{kp}} S_{t,D}\Big) \ge c_* N(h), \qquad N(h) = \sum_t p_0(g_t)^2 n^{\mathrm{cr}}_t(h).$$

Here P^W, P^G are the nonnegative Wilson and gauge-fixing stocks, and $\sum P^W \le \beta A(U)$. The source leaves $e^{-\gamma_{\mathrm{kp}} S_{t,D}/2}$ after the kept halves: one quarter remains in the same majorant and the other quarter, together with half the gauge-fixing stock, pays the created cells. The Wilson stock is at least $(1-a)P^W$. Since $1 - a \le 1$, the payment is at least $(1-a)c_* N(h)$, with the same integrable piece majorant as at zero sources. The enlarged bounded terms proved above only change its fixed cube constant. Consequently the integrated component bound is (39) with rate $(1-a)c_*$ and $E_k(s)$ in place of E_k. This argument does not require the stronger pointwise loss-a property of the retained forms. No positivity of the complex density is asserted.

The extracted constants along h differ from $E_k(s)$ by the missing source polymers, boundary numbers and the coordinate-chain difference of Lem. 10.5. Each costs at most $C'_{\mathfrak{E}} \#_j(\widetilde Z_j)$ and so has the same label-error form as Lem. 9.15. The component estimate has created-cell rate $(1-a)c_* > c_\Theta$. Thus the weight $e^{\Theta^{(k)}(h)}$ is paid by the same created-cell estimate. Apply the pure history summation interface with this rate and error constant $C^F_\flat + C'_{\mathfrak{E}}$, choosing its threshold before the cutoff. This proves the norm sum. Berezin integration has norm at most one at $h_F \ge 1$, so it bounds Z^{sub}_k. Its lower bound at zero is Thm. E. Holomorphy follows from that of the finite sourced partition function and $E_k(s)$. □

Proposition 10.15 (Sources bounded at an intermediate scale)**.** Use the common assignment of Prop. 10.20, including the positive-mass ceilings for the lower bound in Thm. E. For $1 \le k \le K$ put

$$\mathcal{P}^{\#}_k = \{(w, J) : \ \|w\|_\sharp < 2\varrho, \ \lambda_k\|J\|_\infty \le r_J/2\}.$$

The part of $\mathcal{P}$ with $\|w\|_\sharp < 2\varrho$ lies in $\mathcal{P}^{\#}_K$, and $\mathcal{P}^{\#}_K \subset \mathcal{P}^{\#}_k \subset \mathcal{P}^{\#}_j$ for $1 \le j \le k$. The history induction of Prop. 10.13, with the format of Def. 10.11, holds through level k for every $s \in \mathcal{P}^{\#}_k$, with the same constants. In particular,

$$\sum_{h\in\mathcal{H}_k} e^{\Theta^{(k)}(h)} \int \|g_h(s)\|\, d\mathscr{M}^{\mathrm{fr}}_h \le \exp[\mathrm{Re}\, E_k(s) + (C+2)n_k].$$

The factorization and bounds are

$$\begin{aligned} Z[s] = e^{E_k(s)} Z_k^{\mathrm{sub}}[s], \qquad |Z_k^{\mathrm{sub}}[s]| \le e^{(C+2)n_k}, \qquad Z_k^{\mathrm{sub}}[0] \ge e^{-C_P(1+\log x_k)n_k}, \\ \frac{|Z[s]|}{Z[0]} \le \exp[\mathrm{Re}(E_k(s) - E_k(0)) + (C + 2 + C_P(1+\log x_k))n_k]\,. \end{aligned} \tag{51}$$

The function Z_k^{sub} is holomorphic on the interior, continuous on $\mathcal{P}_k^{\#}$, and respects source conjugation. The constant C is independent of the positive mass floor; C_P has the dependence stated in Thm. E. ■

Proof. The physical source ball has $\|J\|_\infty \le 2\varrho_J \le c_J m_- \le r_J/2$. Since $\lambda_j \le \lambda_k$ for $j \le k$, the inclusions follow. Moreover $\mathcal{P}_k^{\#} \subset \mathscr{O}_j \cap \mathscr{O}_j(\alpha_0)$ for $0 \le j \le k$. Thus the local source chains, the history chains and the energies are defined in their fixed-radius coordinate domains. To extend the history bound we must also check the operators and stored terms; the existence of the local chain alone is insufficient.

Write $\mathsf{n} = -a_K J$. The source ceilings give

$$\begin{aligned} L^j\|\mathsf{n}\| \le L^{-(k-j)} r_J/2 \quad (j \le k), \qquad L^{k+1}\|\mathsf{n}\| \le r_M/2, \\ \lambda_j \|J\|_\infty \le r_J/2 = r_M/(2L) \le \varrho_*/4 \le \varrho_\sharp/4, \qquad m_0 + \|\mathsf{n}\| \le 1. \end{aligned} \tag{52}$$

Here $r_J = r_M/L$, $r_M \le L\varrho_*/2$, and $\varrho_* \le \varrho_\sharp$ are the already chosen radii. We check each use of J in the induction up to a level $t \le k$.

At a step from $t < k$, the first bound in (52) puts the fine mass matrix in the source ball of Lem. 6.5. The tower, backgrounds, decoupled covariances and normalizer therefore have their usual doubled bounds. This includes the first step, where $a_K\|J\|_\infty \le L^{-k} r_J/2$. Restricting J to a polymer preserves this ball, so the local conditional contraction and measure identities also apply. The conjugate source remains in the same class because $\|\gamma_4 J^\dagger \gamma_4\| = \|J\|$.

For the sum of old terms on a layer of level $j \le t$, $L^j\|\mathsf{n}\| \le r_M/(2L)$ is inside the cell-unit window of Prop. 10.1. For a stored layer $m < t$ the stronger lag estimate

$$L^m\|\mathsf{n}\| \le L^{-(t-m)} r_M/(2L)$$

retains the gain used in the stored-leg estimate Lem. 10.10. Indeed its change of row sums is at most $\frac{1}{4}L^{-(t-m)}C_H L^{-(m-i)}$, hence at most $2^{-(t-m)-1}C_H L^{-(m-i)}$, under the chosen ceiling $C_{\mathfrak{fl}} r_M \le L$. The sums over layers consequently use the same constants as on $\mathcal{P}$.

The stored tower has the same row and column bounds as well. On its fine layer the entries are bounded by $\mathsf{c}_{\mathrm{disc}} + m_0 + \|\mathsf{n}\| \le \mathsf{c}_{\mathrm{disc}} + 1$; on the other layers use the source operator bounds just checked. The history mass fields are those of Lem. 10.5, with $\|\sigma_j^h\| \le 2B_\sigma\phi_j$. Their layer coefficients are bounded by $L^{-(j-1)}(\bar{B}_M + 2B_\sigma)\phi_{j-1}$, as in the original sourced format.

The normalizer walks and the label errors use the same coordinate balls. They give the same holomorphic vacuum constants and the error $C'_{\mathfrak{E}}\#_j(\widetilde{Z}_j)$. Every remaining source estimate reads only w: its supremum, its swing, the kept and rejected quadratic halves, and the loss of the real action weight. The condition $\|w\|_\sharp < 2\varrho$ is unchanged. The bare sourced density is entire and has continuous coefficients on the compact link space, so the measure operations remain valid.

These checks prove the format by induction on t, at the fixed class $\mathcal{P}_k^{\#}$. The proof of Prop. 10.14 now gives the history bound with the same created-cell rate and layer constants. Berezin integration gives the upper partition bound. The lower bound at zero is Thm. E; division proves the last inequality. Finally $Z_k^{\mathrm{sub}} = e^{-E_k} Z$, with E_k holomorphic on the open set $\mathscr{O}_k$ containing $\mathcal{P}_k^{\#}$. This proves continuity, holomorphy and conjugation. □

10.5. **Separated energies and the cell bound.** For a physical set S let $\#_j^+(S)$ count the level-j M-cubes within distance $M\lambda_j$ of S, and put

$$\mathscr{C}_k(S;\delta) = C_{\mathscr{C}} \sum_{j\le k} \#_j^+(S) e^{-\kappa_E \delta/\lambda_j}.$$

The following estimates are uniform in K:

$$\mathscr{C}_K(T_\ell;\delta) \le C\ell^3(1+\delta^{-3}), \qquad \mathscr{C}_K(S;\delta) \le C[1+\log_L^+(1/\delta)+(R/\delta)^3] \tag{53}$$

if S lies in a ball of radius R. Indeed $\#_j^+(S) \le \min\{n_j/M^3, 27(1+\operatorname{diam} S/(M\lambda_j))^3\}$. The sums follow by splitting at $\lambda_j = \delta$; the fine tail is bounded by $\sum_{u\ge 0} L^{3u} e^{-\kappa_E L^u} < \infty$.

Proposition 10.16 (Separated source energies)**.** Let $s_1,\dots,s_N$ be sources in the ball satisfying

$$\operatorname{dist}(\operatorname{supp} s_i, \operatorname{supp} s_l) \ge \delta > 0 \qquad (i \ne l),$$

and suppose their sum is in the ball. Then

$$\left| E_K(\sum_i s_i) - \sum_i E_K(s_i) + (N-1)E_K(0) \right| \le \sum_i \mathscr{C}_K(\operatorname{supp} s_i;\delta).$$

For a holomorphic source polydisc $s(z) = \sum_i z_i s_i$, with radii ρ_i, if S_1, S_2 are at distance at least d and the supports are pairwise disjoint, then

$$|\partial_1 \cdots \partial_n E_K(s(z))|_{z=0} \le 2^{n-1} \prod_i \rho_i^{-1} \mathscr{C}_K(S_1; d).$$

■

Proof. For each polymer choose a nearest source support. Its difference from the sum of the single-source energies is bounded by the tails from all other supports. Split the polymer decay and use $\operatorname{diam} X \le c_{\mathcal{D}} M\lambda_j(1+d_j(X))$. At least half the separation from another support must be paid by that diameter or its distance to X. Taking $\kappa_E \le \min\{\kappa_\sigma/8, \kappa/(4c_{\mathcal{D}}M)\}$ leaves the factor $e^{-\kappa_E\delta/\lambda_j}$ and a summable polymer weight. Packing the separated supports on fine levels and using the crude cube count on coarse levels gives the first estimate.

For the second apply $\Delta_n = \prod_i (1-P_i)$, where P_i sets $z_i = 0$. Its norm is at most 2^{n-1} times any one single-source difference norm. For each X, $\max\{\operatorname{dist}(X,S_1), \operatorname{dist}(X,S_2)\} \ge (d - \operatorname{diam} X)/2$. Splitting its exponential weights gives $Ce^{-\kappa_E d/\lambda_j} e^{-\kappa d_j(X)/2} e^{-\kappa_\sigma \operatorname{dist}(X,S_1)/(2\lambda_j)}$. Sum through the nearest cube to S_1 and apply Cauchy's estimate in the n variables. This is the stated bound. □

Proposition 10.17 (A source in one cell)**.** Let a board of physical unit cells have its walls through links, and let $s \in \mathcal{P}$ be supported in one cell at distance at least δ from its walls. For the $\mathsf{N} = \ell^3$ products r of wall reflections, let $r_{\mathrm{lin}} s$ be its linear source image, including the spin lift on J, and set

$$\lambda^{\mathfrak{b}}(s) = \mathsf{N}^{-1} \sum_r (E_K(r_{\mathrm{lin}} s) - E_K(0)), \qquad c_V = C + 2 + C_P[1 + \log(L(g^{-2} + B_\sharp))].$$

Then $\lambda^{\mathfrak{b}}$ is holomorphic, and

$$|Z[s]/Z[0]| \le e^{\operatorname{Re}\lambda^{\mathfrak{b}}(s) + c_V + \mathscr{C}_K(\operatorname{supp} s;\delta)}. \tag{54}$$

For sources in distinct cells the corresponding moment bound is the product of these one-cell bounds. ■

Proof. The source exponential is even and lies in its cell algebra. Reflection maps it to the reflected source exponential: in the physical basis $\gamma(\hat n)\Gamma^\dagger\gamma(\hat n) = \Gamma$. Its fold is thus the exponential of the disjointly reflected source. The supremum and Lipschitz source norms are preserved, since the source vanishes in a strip along each wall. The coordinate-link reflection positivity and its chessboard inequality apply.

For the folded source, use Prop. 10.14 and the zero-source floor in Thm. E. Take the Nth root. Its normalization cost is e^{cv}. The separated-energy estimate replaces $\mathsf{N}^{-1}(E_K(s^{\mathrm{fold}}) - E_K(0))$ by the average of its individual reflected energies at cost $\mathscr{C}_K(S;\delta)$. Odd reflection words use the conjugated source, and the source conjugation changes the energy to its complex conjugate; therefore its real part is the real part of the holomorphic linear-image average displayed above. This gives the cell seminorm bound. The multiple-cell chessboard inequality proves the final assertion, and also bounds a single-cell expectation by its seminorm. □

Proposition 10.18 (A base source and a choice of stopping scale)**.** Use the source-energy hypotheses of Prop. 10.6 and the common constant assignment of Prop. 10.20.

(i) Let $s_0, s_0 + s_i, s_0 + \sum_i s_i \in \mathscr{O}_k$, where the supports S_i of the increments s_i are pairwise at distance at least $\delta > 0$. Then

$$\left| E_k(s_0 + \sum_i s_i) - \sum_i E_k(s_0 + s_i) + (n-1)E_k(s_0) \right| \le \sum_i \mathscr{C}_k(S_i;\delta).$$

For $n \ge 2$, suppose instead that the supports are pairwise disjoint, $\mathrm{dist}(S_1, S_2) \ge d > 0$, and $s_0 + \sum_i z_i s_i \in \mathscr{O}_k$ on the polydisc of radii ρ_i. Then

$$\left| \partial_1 \cdots \partial_n E_k(s_0 + \sum_i z_i s_i)|_{z=0} \right| \le 2^{n-1} \prod_i \rho_i^{-1} \mathscr{C}_k(S_1; d).$$

(ii) Suppose $s_0 = (0, J_0)$, $s_i = (0, J_i)$, $\|J_0\|_\infty \le r_J/2$ and $\|J_i\|_\infty \le \varepsilon \le r_J/8$, with the supports S_i pairwise at distance at least δ. The first bound of (i) at $k = K$ holds with

$$\mathscr{C}^\flat(S;\delta) = C_{\mathscr{C}} \sum_{j=1}^{K} \#_j^+(S) e^{-\kappa_E \delta/\lambda_j} \omega_j, \quad \omega_j = \begin{cases} 1, & \lambda_j \le \delta, \\ \lambda_{j-1}/4, & \lambda_j > \delta. \end{cases}$$

For S in a ball of radius R_S,

$$\mathscr{C}^\flat(S;\delta) \le C^\flat_{\mathscr{C}}[1 + (R_S/\delta)^3], \tag{55}$$

uniformly in the cutoff and volume.

(iii) Let a coordinate-link board have cells of physical side $l = qa_K$, where $q \ge 3$ is odd and ℓ/l is an even integer. Write $\mathsf{N} = (\ell/l)^3$ and let s be supported in one cell $\square$, at positive distance from its walls. For the source exponential G_s, let s^{ch} be the source obtained by folding into all cells. Then

$$\|G_s\|_{\mathrm{ch}}^{\mathsf{N}} = Z[s^{\mathrm{ch}}]/Z[0] \ge 0, \qquad |Z[\sum_{\square} s_\square]/Z[0]| \le \prod_{\square} \|G_{s_\square}\|_{\mathrm{ch}}.$$

Under the partition-bound hypotheses of Prop. 10.15, $s \in \mathcal{P}_k^\#$ implies $s^{\mathrm{ch}} \in \mathcal{P}_k^\#$ and

$$\|G_s\|_{\mathrm{ch}} \le \exp\left[\frac{\mathrm{Re}(E_k(s^{\mathrm{ch}}) - E_k(0))}{\mathsf{N}} + (C + 2 + C_P(1 + \log x_k)) \frac{|\square|}{\lambda_k^3} \right].$$

If the distance from the walls is at least δ, put $\lambda_k^\flat(s) = \mathsf{N}^{-1} \sum_r (E_k(r_{\mathrm{lin}} s) - E_k(0))$. Then also

$$\|G_s\|_{\mathrm{ch}} \le \exp\left[\mathrm{Re}\, \lambda_k^\flat(s) + (C + 2 + C_P(1 + \log x_k)) \frac{|\square|}{\lambda_k^3} + \mathscr{C}_k(S;\delta) \right]. \tag{56}$$

The constant $C^\flat_{\mathscr{C}}$ depends only on $C_{\mathscr{C}}, \kappa_E, M, L$ and is chosen before the positive mass floor and the final small-coupling threshold. ■

Proof. For (i), apply the proof of Prop. 10.16 with s_0 added to every argument. More explicitly, the contribution of a polymer is

$$\Delta_j(X) = e_j(X; s_0 + \sum_i s_i) - \sum_i e_j(X; s_0 + s_i) + (n-1)e_j(X; s_0).$$

Choose a nearest support $S_{i(X)}$ and write this as

$$[e_j(X; s_0 + \sum_i s_i) - e_j(X; s_0 + s_{i(X)})] - \sum_{i \neq i(X)} [e_j(X; s_0 + s_i) - e_j(X; s_0)].$$

Each difference is bounded by (50) using the supports other than $S_{i(X)}$. The same fine-scale packing gives $e^{-\kappa_E \delta/\lambda_j}$; at coarser scales use the differences from s_0 and the cube count. The source-independent constant in that proof therefore remains valid. For mixed derivatives apply $\prod_i (1 - P_i)$ to $e_j(X; s_0 + \sum_i z_i s_i)$. Each $1 - P_i$ changes the source only on S_i. The distance splitting in Prop. 10.16, followed by Cauchy's estimate on smaller polydiscs and a radius limit, proves the second assertion.

For (ii), the sum of the increments has norm at most ε, since their supports are disjoint. The fine levels $\lambda_j \le \delta$ have the bound just proved. For a coarser level let J be one increment or their sum and put

$$F(\tau) = e_j(X; (0, J_0 + \tau J)) - e_j(X; (0, J_0)).$$

The domain $\mathscr{O}_{j-1}$ contains this family for

$$|\tau| < T_j = \frac{r_J/\lambda_{j-1} - \|J_0\|_\infty}{\varepsilon} \ge \frac{4}{\lambda_{j-1}}.$$

Here $F(0) = 0$. Its bound from (50) retains the distance from X to the support of J throughout the disc. Schwarz's lemma gives that bound at $\tau = 1$ multiplied by $T_j^{-1} \le \lambda_{j-1}/4$. The coarse-level sum in the proof of (i) therefore acquires exactly ω_j.

To sum it, write $u = K - j$, so $\lambda_j = L^{-u}$, and use $\#_j^+(S) \le 64(1 + 2R_S L^u/M)^3$. On the levels $\lambda_j > \delta$, which occur only if $\delta < 1$, the sum is at most $C_{\mathscr{C}}$ times

$$432 \sum_{u \ge 0} L^{-u-1} + 432 R_S^3 \sum_{L^u < 1/\delta} L^{2u-1} \le 36 + 40(R_S/\delta)^3.$$

The fine levels are bounded by the exponentially convergent sum used in (53). This proves (55); the logarithmic count of coarse levels has been replaced by a geometric sum.

For (iii), G_s is even and belongs to its cell algebra. Its reflections multiply to $G_{s^{\mathrm{ch}}}$. The reflection-positive chessboard argument in the proof of Prop. 10.17 uses only the even number of slabs in each direction, so gives the displayed identity and product bound on this board. The reflected sources have the same supremum and Lipschitz norms; because they vanish near the walls, the norm of their disjoint sum is their maximum. Source conjugation and the spin lifts preserve the operator norm of J. Thus folding preserves $\mathcal{P}_k^{\#}$.

Apply (51) to s^{ch} and take the Nth root. Since $n_k/\mathsf{N} = |\square|/\lambda_k^3$, this gives the first cell estimate. If the support lies δ inside the cell, its distinct reflected images are separated by 2δ. Part (i) replaces their total energy by the sum of their individual energies. The cube count under a reflection costs at most 27, already included in $C_{\mathscr{C}}$. Finally conjugate images have the same real energy as their linear images. This proves (56). □

10.6. **Uniform separated cumulants.** The insertion norms are

$$\|f\|_0 = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}, \qquad \|f\|_\Gamma = \|f\|_\infty.$$

Set

$$\varrho_0 = \varrho, \qquad \varrho_\Gamma = \varrho_J, \qquad \varrho_{\min} = \min\{\varrho, \varrho_J\}.$$

The truncated function is the mixed derivative of $\log Z[\sum_i z_i s_i]$ at zero. Only its germ at zero is used: $Z[0] > 0$ ensures that this germ exists.

Theorem F (Separated cumulant bounds)**.** Let $n \geq 2$, let the supports of f_i be pairwise disjoint with S_1, S_2 at distance at least $d > 0$, and let each support lie in a ball of radius R. Uniformly in the cutoff and volume,

$$|S_n^T((\alpha_i, f_i)_{i=1}^n)| \leq (1 + d^{-1})^3 B(R)^n (n!)^{5+p/2} \prod_i \|f_i\|_{\alpha_i},$$

where

$$B(R) = C_B e^{c_V} (1 + R)^3 / \varrho_{\min}, \qquad p = 432 C_{\mathscr{C}} / \log L.$$

Without the support-radius restriction one also has the torus bound $C_n(d) e^{n c_V \ell^3} \prod_i \|f_i\|_{\alpha_i}$. For the finitely many cutoffs with $L^K < 16n$, a direct bound is

$$[C_B'(1 + N_f/m_-)(1 + R)^3]^n (n!)^5 \prod_i \|f_i\|_\infty.$$

■□

Proof. For any holomorphic counterterm $\Lambda(s)$, define

$$G_\Lambda(z) = e^{-\Lambda(s(z)) + \Lambda(0)} Z[s(z)]/Z[0], \quad G_\Lambda(0) = 1.$$

The exact differentiation formula is

$$S_n^T = \partial_{[n]} \Lambda + \sum_\pi (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \partial_B G_\Lambda.$$

The sum of its absolute combinatorial coefficients is at most $2^n n!$: order the blocks and then order their elements increasingly, and bound the resulting records by permutations with cuts. This formula needs no zero-free complex source neighborhood beyond the germ.

First take $\Lambda = E_K$. On the polydisc with radii $2\varrho_{\alpha_i}/\|f_i\|_{\alpha_i}$, disjoint supports keep the total source in $\mathcal{P}$. The subtracted-partition upper bound and zero-source floor give $|G_\Lambda| \leq e^{c_V \ell^3}$. Cauchy's estimate bounds every block derivative. Prop. 10.16 and (53) bound the mixed energy derivative by $2^{n-1} \prod_i \rho_i^{-1} C \ell^3 (1 + d^{-3})$. These bounds prove the torus assertion.

Suppose now $L^K \geq 16n$. Set $\eta = (16n)^{-1}$. Use the product hat partition of unity at mesh η. Each function splits into at most $(2R/\eta + 3)^3$ terms supported in cubes of side 2η, and each piece has its insertion norm bounded by $(1 + 80M/\eta)\|f_i\|_{\alpha_i}$. Fix one piece from each insertion. In each coordinate choose walls of unit cells at half-integral fine-grid positions, leaving distance η to all these supports. A support forbids at most $4\eta L^K + 1$ positions; there are n supports and $L^K \geq 16n$, so fewer than L^K positions are forbidden. This chooses a valid through-link board for every allowed torus, without a lower bound involving nR on its side.

Group the pieces by their cells and let Λ be the sum of the holomorphic cell energies $\lambda^{\mathfrak{b}}$ for these groups. On a coordinate face indexed by B, the multiple-cell bound gives

$$|G_\Lambda(z_B)| \leq \exp\left[|B| c_V + \sum_{i \in B} \mathscr{C}_K(S_i; \eta)\right].$$

Since each S_i has diameter at most a fixed multiple of η, the refined count gives $\mathscr{C}_K(S_i; \eta) \leq 216 C_{\mathscr{C}}(1 + \log_L(1/\eta)) + C$. Thus the partition derivative sum costs at most $C^n e^{n c_V} (16n)^{np/2} 2^n n! \prod_i \rho_i^{-1}$. The mixed derivative of Λ is zero unless all pieces are in one cell. In that case the reflected energy bound of Prop. 10.16 gives $C(1 + d^{-1})^3 \prod_i \rho_i^{-1}$ times its fixed combinatorial factor. Finally sum over the piece choices. Their number and norm enlargement cost at most $(32nR + 3)^{3n} (1 + 1280 M n)^n$. Using $n^n \leq e^n n!$, these four powers of n and the partition factor give $(n!)^5$, and the energy count gives the additional $(n!)^{p/2}$. Absorb fixed constants into $B(R)$.

For completeness the coarse cutoffs require no renormalization estimate. At every configuration $\|(D+m_0)^{-1}\| \le 1/m_0$. The determinant inequality $|\det(1+T)| \le e^{\|T\|_1}$ therefore gives

$$|Z[s]| \le Z[0] \exp\left[2a_K^{-1}\sum_p |w(p)| + \frac{384N_f}{m_-}\sum_x \|J(x)\|\right].$$

A support in a radius-R ball contains at most $81a_K^{-3}(1+R)^3$ plaquettes and $27a_K^{-3}(1+R)^3$ sites. Choose the independent source radii as the reciprocals of these costs. The moment bound is then e^n, and the partition formula costs at most $2^n n!$ times their product. Since $a_K^{-1} < 16n$, the action cost is bounded by $C(16n)^4(1+R)^3\|f\|_\infty$ and the bilinear cost by $C(N_f/m_-)(16n)^3(1+R)^3\|f\|_\infty$. The same inequality $n^n \le e^n n!$ proves the last displayed estimate. These cases cover every cutoff. □

10.7. **A common choice of constants.** We now choose the constants used in the preceding estimates. The order matters: a bound for the induced mass is not an input to the constants that prove that bound. Throughout, the group is SU(2), the chamber parameter is $r = 1/2$, and $N_f \ge 1$ counts the four-component flavours, each formed from the two conjugate two-component blocks.

The gauge induction requires numerical bounds and also conditions on its radii, lattices and auxiliary objects. We first verify these conditions for the constants chosen in [3, Prop. 4.163]. The subsequent numerical construction will preserve such a choice while imposing the matter bounds.

Lemma 10.19 (Admission of the pure gauge induction)**.** Choose the gauge constants in the order and with the defining formulas of [3, Prop. 4.163], and its proof. Fix the history mollifier: it is smooth, nonnegative, has integral one, is supported in the supremum-norm ball of radius 1/4, and is invariant under signed coordinate permutations. Use the critical backgrounds, coordinate maps, retained quadratic forms and pivot logarithms of that construction, with their specified branches.

Let $x_0, \dots, x_K$ be an admissible coupling sequence as in section 4, with $x_j \ge \gamma^{-2}$, and suppose $MR(x_K) \le L^m$. At every step and every history the gauge construction has its required domains and numerical bounds. Its local auxiliary maps are holomorphic on their prescribed common balls and vanish at zero datum. Its retained exponents are defined on the full prescribed layered spaces, have finite oscillations, and agree exactly on extending a history. Consequently the canonical finite gauge induction applies. All assertions persist after decreasing γ. ■

Proof. We first pass from the common numerical choice to a coupling sequence and a torus. We then check the two conditions involving actual functions: the auxiliary background maps and the retained exponents. This separates conditions needed to define a step from bounds on its output, which are proved by induction.

The common numerical choice. The proof of [3, Prop. 4.163] fixes the exponents $r_c \ge 2$, $\mathfrak{p}_0 \ge 6r_c+1$ and $q \ge 41\mathfrak{p}_0 + 1$, followed by the reserve fractions, L, the response parameters, the decay rates, the geometric lengths and the analytic radii. The constants in a defining formula are evaluated after its arguments have been fixed. The potential bound E_0, the increment bound $B_\sharp$ and the history bounds are then fixed. The threshold γ is chosen last. In particular none of these choices uses K, a particular coupling sequence or the torus.

Here is why the order permits the last choice. Write $F_p(x) = x^{-1/2}(\log x)^p$. After the earlier constants have been fixed, the remaining small quantities are bounded by finite sums of constant multiples of

$$F_p(x), \qquad (\log x)^{-(q-\mathfrak{p}_0)}, \qquad x^a(\log x)^p e^{-c(\log x)^{r_c}}, \qquad e^{-c(\log x)^{2\mathfrak{p}_0}}, \quad c>0.$$

The exponents and coefficients in each occurrence are fixed. These expressions tend to zero as $x \to \infty$; the third uses $r_c \ge 2$. Each required upper bound can therefore be imposed on the whole interval $x \ge \gamma^{-2}$. The lower bounds on $\log \gamma^{-2}$ are finite. Bounded windows, such as those for the reserve fractions, the normalizer radius and the boundary rate, are made nonempty at their earlier

stages of the choice; they are not repaired by the last choice of γ. This is the numerical construction proved in the cited proposition.

For use in the next proposition, the same order allows an additional floor or ceiling only at a stage where its dependencies have already been fixed and its permitted range remains nonempty. All later constants defined by formulas must then be evaluated at the new earlier values. Thus a larger rate or a smaller radius does not mean retaining numerical values obtained from the previous choice. The next proposition checks these requirements for the matter conditions explicitly.

Couplings, radii and room on the torus. Put $y_j = \log x_j$, $\check{R}_j = R(x_j)$, $\epsilon_j = A_0 F_{\mathfrak{p}_0}(x_j)$, $\delta_j = A_1 F_{\mathfrak{p}_0}(x_j)$ and $\mathfrak{b}_j = A_1 y_j^{\mathfrak{p}_0}$. The choice of γ places y_j above all the monotonicity thresholds. Consequently

$$\epsilon_j \le \epsilon(\gamma^{-2}), \qquad \delta_j \le \delta(\gamma^{-2}), \qquad y_j \ge \log \gamma^{-2}, \qquad \mathfrak{b}_j \ge A_1 (\log \gamma^{-2})^{\mathfrak{p}_0} \ge 1.$$

In particular every linear smallness condition for the background, the shift, the averaging logarithm and the coordinate charts holds at every level. Conditions on the relative change of coupling follow from the same bound $|b_j| \le B_\sharp$ used in choosing the uniform regularity constants.

The running radii are

$$\alpha_{\iota,j} = C_\iota F_q(x_j), \qquad \iota = 0, 1.$$

At either stage containing level j one uses these same values. Truncating a coupling sequence therefore restricts the table without changing any previous radius. Since $\log \gamma^{-2} > 2q$, F_q is decreasing on the required interval. The fixed ceiling conditions include

$$(1+\beta_s) C_0 F_q(\gamma^{-2}) \le \min\{\alpha_0, \alpha_2\}, \qquad (1+\beta_s) C_1 F_q(\gamma^{-2}) \le \alpha_1,$$
$$\mathfrak{o} L^3 M B^{(\mathrm{cg})} C_0 F_q(\gamma^{-2}) \le \alpha_0, \qquad a_\flat \le \alpha_0, \qquad \mathfrak{o} L^3 \ge 3.$$

Here $\mathfrak{o}$ and $B^{(\mathrm{cg})}$ are respectively the fixed local-gauge enlargement and its gauge bound in the layered-space construction of [3]. These inequalities place each layered-space reading in its fixed analytic class. They bound the real and complex local potentials, the current, and the reduced towers required by that class; the tower conditions are part of the layered-space definition. They are not inferred from curvature alone.

The adjacent-scale radius inequality is $\check{R}_j \le L^{K-j} \check{R}_K$. Thus

$$M \check{R}_j \le L^{m+K-j}.$$

The lower bounds $\check{R}_j \ge \max\{L^2, \mathfrak{o} L + 1\}$ and $M > 6$ imply

$$L^2 M \le L^{m+K-j}, \qquad \mathfrak{o} L M + 6 < L^{m+K-j}, \qquad m' < m + K - j.$$

The grain, mesh and containing boxes fit at every level, including the last. The same applies to every truncated history. These are the torus-room inequalities following [3, Prop. 4.163]. The larger local charts used elsewhere in this paper require the stronger inequality $LMR(x_K) \le L^m$; this is supplied by $\ell \in \mathfrak{L}(g)$ in section 4, rather than by the weaker room assumption of this lemma.

Gaussian integration and boundary terms. The historical Gaussian uses the covariance and conditional mean of [3, Prop. 4.99]. Denote its fixed collar factor by Θ_{col}. Write its external datum as $a = a' + a''$, where a' is real and belongs to the prescribed tier: it obeys $|a'| < \mathfrak{b}_k \min\{C_{\mathrm{xt}}, (4\Theta_{\mathrm{col}})^{-1}\}$ on the collar, $|a'| < \theta_c \mathfrak{b}_k$ on the rim, and $|a'| < C_{\mathrm{xt}} \mathfrak{b}_k$ elsewhere. The complex increment a'' vanishes on the rim and satisfies $|a''| \le \epsilon_a \mathfrak{b}_k$ elsewhere. Both its real and imaginary parts enter the conditional-mean estimate. The chosen collar width and rim width control the collar and rim contributions to the conditional mean. The remaining contribution has the factor $e^{-\delta_M (W_t - 2)/2}$ and is made small by the running separation. The complex covariance radius controls the imaginary part of the conditional mean. The common choice imposes both the mean bound and the normalizer positivity bound before the Gaussian expansion is applied. In particular the bounds concern the full external datum, not just its restriction to a polymer.

To spell out the quantitative reserve, let $\mathcal{K}_{\mathrm{ap}}$ be the fixed cluster prefactor and let $S_{\mathrm{rg}}(x)$, $S_{\mathrm{bd}}(x)$ and $S_0(x)$ denote the three potential bounds in the proof of [3, Prop. 4.163]: regular and boundary potentials at a later step, and the potential at the first step. Their explicit formulas there have the first two types of decay displayed above. The Gaussian large-field penalty is

$$\Gamma_P(x) = C_{\mathrm{mn}}^{-1}(LM)^{-3}\theta_c^2 A_1^2(\log x)^{2\mathfrak{p}_0}.$$

The common choice gives, uniformly for $x \geq \gamma^{-2}$,

$$S_{\mathrm{rg}}(x), S_{\mathrm{bd}}(x) \leq (12eL^3\mathcal{K}_{\mathrm{ap}})^{-1}, \qquad S_0(x) \leq (6eL^3\mathcal{K}_{\mathrm{ap}})^{-1},$$
$$(LM)^3 e^{-\Gamma_P(x)} \leq (6\mathcal{K}_{\mathrm{ap}})^{-1}.$$

The cluster estimate is therefore at most $1/3$: at a later step its two terms are bounded by $\mathcal{K}_{\mathrm{ap}}eL^3(S_{\mathrm{rg}} + S_{\mathrm{bd}}) \leq 1/6$ and $\mathcal{K}_{\mathrm{ap}}(LM)^3e^{-\Gamma_P} \leq 1/6$. At the first step the first term is instead $\mathcal{K}_{\mathrm{ap}}eL^3S_0 \leq 1/6$. This separately checks the initial step, where no old potential family is available.

For a step $t = k+1$, the rejected forms and the strip expansion have the joint bound

$$\mathfrak{b}_{\mathrm{rj}} + \mathfrak{b}_{\mathrm{sr}} \leq e^{-\check{R}_t}(c_3 + c_4\mathfrak{b}_k^2),$$

where c_3, c_4 are the fixed constants evaluated in the same proof. They are fixed before γ. The flow bound gives $x_k \leq (3L/2)x_t$. With $y = \log x_t \geq \log(2L)$,

$$\mathfrak{b}_k^2 \leq A_1^2(2y)^{2\mathfrak{p}_0}, \qquad \check{R}_t \geq y^{r_c} \geq y, \qquad (2y)^{2\mathfrak{p}_0}e^{-y/2} \leq (8\mathfrak{p}_0/e)^{2\mathfrak{p}_0}.$$

Consequently the preceding bound is at most

$$\gamma\big[c_3 + c_4A_1^2(8\mathfrak{p}_0/e)^{2\mathfrak{p}_0}\big] \leq \tfrac{1}{3}.$$

Together with the cluster bound this leaves the boundary reserve $1 - 2/3$. The additional smallness conditions for the strip coordinate changes are imposed in the same numerical choice; their tails are a fixed constant times $e^{-\check{R}_t}$, compared with the already specified coordinate radius. Hence they hold for every history on the given sequence. No constant is selected separately for a particular background or history.

The auxiliary background maps. An octet $\square$ of level n is the union of eight level $(n-1)$ cubes meeting a vertex. Put $Z = \square^{\sim 5}$ and let $\mathbf{B}_j^{\square}$ be its reduced determining set at truncation $1 \leq j \leq n$. The canonical maps are

$$H_{\square,j}(B) = \mathcal{H}_{\mathbf{B}_j^{\square},1}(B), \qquad \sup|B| < r_{\mathrm{B}}.$$

Here B assigns a complex Lie-algebra element to each determining row. By [3, Lem. 4.42], this map is the holomorphic potential in the normalized mean-Landau gauge, based at the identity configuration. Its exponential, together with the separate normalizing gauge, represents the critical background over the datum e^{iB}. The map vanishes at $B = 0$. Its radius $r_{\mathrm{B}} > 0$ and derivative bounds are independent of height and volume. Uniqueness for the mean-Landau condition and the node normalization makes these maps commute with adjunction, constant SU(2) gauge rotations and translations of lattice charts, as checked before [3, Prop. 4.50].

The radii used to substitute an old interaction are the absolute system at height k:

$$a_{\iota,j} = \alpha_\iota \quad (1 \leq j \leq k), \qquad a_{\iota,k+1} = \frac{\alpha_\iota}{1+\beta_s}, \qquad \iota = 0, 1.$$

Its output radii are $\bar{a}_\iota = (1+\beta_s)a_{\iota,k+1} = \alpha_\iota$, and $\max_\iota \bar{a}_\iota/a_{\iota,j} = 1$ for $j \leq k$. The common choice includes

$$\alpha_0 + \alpha_1 \leq a^\star, \qquad c_{\mathrm{sh}}M\alpha_0 < \alpha_1, \qquad \alpha_1^2 \leq c_\natural\alpha_0, \qquad \frac{(1+\beta_s)\delta(\gamma^{-2})}{\alpha_0} \leq s^\star.$$

These are precisely the shape and shift-size conditions of [3, Def. 4.48]. The slow-growth comparisons in that definition hold for the displayed absolute system. The choices $a^\star \leq a^\star_{\mathrm{oc}}$ and $c_{\mathrm{sh}} \geq c^{\mathrm{oc}}_{\mathrm{sh}}$ are

made when the fixed analytic radii are chosen. This radius system has a different purpose from the running radii of the layered spaces above.

For $k \geq 2$, the octet construction uses

$$b_\iota^{\mathrm{oc}} = \theta_{\mathrm{oc}} a_{\iota,k+1}, \qquad a_{\iota,k}^{\mathrm{oc}} = \frac{b_\iota^{\mathrm{oc}}}{1+\beta_s}, \qquad a_{\iota,j}^{\mathrm{oc}} = \min\{a_{\iota,j}, b_\iota^{\mathrm{oc}}\} \quad (j<k).$$

Here $0 < \beta_s \leq 1/5$ and $(3/2)\theta_{\mathrm{oc}}^2 < 1$. The proof preceding [3, Prop. 4.50] checks the same shape and slow-growth inequalities for this system. Its shift-size ceiling is $(1+\beta_s)s^\star/\theta_{\mathrm{oc}}$, the ceiling used in that substitution.

Let a pair on $X' = (\square^{\sim 5})^\uparrow$ belong to $\mathcal{U}_k^\circ(X'; b_0^{\mathrm{oc}}, b_1^{\mathrm{oc}})$. In particular, its restriction to Z belongs to the window domain: it has the local potential and curvature bounds, and all the reduced real and complex towers exist with the required bounds at every truncation. Put the averaged datum V in the comb gauge used in [3, Prop. 4.47]. With $\eta_j = L^{-j}$ in the rescaled level-j chart, set

$$B_V = i^{-1}\log V, \qquad A_{\square,k} = H_{\square,k}(B_V),$$
$$Q_{\square,j} = i^{-1}\log M_{\mathbf{B}_j^\square}(e^{i\eta_k A_{\square,k}}), \qquad A_{\square,j} = H_{\square,j}(Q_{\square,j}).$$

The reduced averaging map in this formula uses its specified logarithm branch.

We check the domains of these compositions. In [3, Prop. 4.47], take the top window radii $\widetilde{a}_\iota = b_\iota^{\mathrm{oc}}/(1+2\beta_s)$ and the lower radii $a_\iota = a_{\iota,j}^{\mathrm{oc}}$, with $n = k$, $\tau = 1$ and $B' = 0$. Its hypotheses are

$$\widetilde{a}_0 + \widetilde{a}_1 \leq a^\flat, \qquad a_0 + a_1 \leq a^\flat, \qquad 2B_3^2 K_{\mathrm{tw}} M a_0 < a_1, \qquad \widetilde{a}_0 \leq t_{k-j} a_0.$$

The constants B_3, K_{tw} and the radius $a^\flat$ are fixed in that proposition; the radius is chosen after M to satisfy the logarithm, averaging and response ceilings. The octet radius system gives these inequalities. The comparison sequence is $t_\nu = (6/5)\sqrt{1+\nu}$. For $\Theta_s = \sup_{\nu\geq 1} t_\nu L^{-s\nu}$, the remaining hypotheses are

$$\Theta_1 \leq 1, \qquad (1+10\beta_s)\Theta_2 + \beta_s/4 + 1/1000 \leq 1/8,$$
$$(1+3\beta_s)\Theta_3 + 5\beta_s\Theta_2 + \beta_s/8 + 1/500 \leq 1/16.$$

Their suprema occur at $\nu = 1$. The proof of [3, Prop. 4.50] evaluates the last two left sides at $L = 13$, $\beta_s = 1/5$ as less than 0.082 and 0.039, respectively. Thus these hypotheses hold uniformly in $k - j$.

The substitution proposition now proves that B_V and $Q_{\square,j}$ lie inside the prescribed row balls. It gives, on $\square^{\sim 3}$,

$$|A_{\square,j}|, |\nabla A_{\square,j}| < a_1/2, \qquad |\partial e^{i\eta_j A_{\square,j}} - I| < a_0\eta_j^2/8, \qquad |J^{(j)}(e^{i\eta_j A_{\square,j}})| < a_0/16.$$

Each required inner-polymer restriction therefore belongs to the stated analytic core. The restricted real background is identified by criticality and uniqueness: variations preserving the restricted datum extend by the original field, because the boundary bonds are fixed determining bonds. The identities extend to the complex logarithmic domains by holomorphy. The resulting maps and their domains are the canonical maps at every truncation. At heights $k \leq 1$ no octet substitution is required.

Retained factors and their domains. There are two possible retained exponents for a pair (t, X): the kept quadratic form and the sum of external pivot logarithms. Write $t = l + 1$. Their arguments are the pair read in level-l units and the level-l stored field, extended by zero on the other bonds of X. The radius ceilings above put this pair in $\mathscr{O}(X; \alpha_0, \alpha_1, \alpha_2)$ on the whole layered space, including its stored-data tiers. This class membership is the conclusion of [3, Prop. 4.97]. Here $\mathfrak{M}$ is the fixed power of L specifying the cube partition of the layered regions. The local hypotheses are part of the layered-space definition: at each layer the curvature, current, complex potential and first difference obey the graded radii; real local gauges exist on the prescribed cubes. Also, every cube made of level-n $\mathfrak{M}$-cubes, contained in the domain and the level-n region, and of side at most $12\mathfrak{M}$ in level-n units, has a containing embedded cube in the domain with a real gauge of index n and bound $a_\flat$. The history gaps put the tiny cubes about fluctuation bonds inside the preceding layer, and the

torus-room inequalities make them embedded. The radius tests for the cited proposition are exactly the three running-radius ceiling inequalities above, together with $a_\flat \le \alpha_0$. They hold uniformly at every level. Passing from the level-t triple to the level-$(t-1)$ pair rescales A' by L^{-1} and $\mathbf{J}$ by L^{-3}; hence the underlying configuration and its current in bare units are unchanged. The proposition then supplies $\mathscr{O}(X;\alpha_0,\alpha_1,\alpha_2)$ on the full layered domain.

The fixed rate and pivot conditions are

$$\delta_\star^{\mathrm{pl}} LM \ge C_2(\kappa+1), \qquad L\max\{\alpha_0,\alpha_1\} \le a_{\mathrm{nf}}.$$

The common choice satisfies these, with the stronger rate reserve used in subsequent locality estimates.

The construction preceding [3, Prop. 4.144] checks the pivot condition even for a coarse bond crossing the last layer. The two averaging blocks then lie in a cube of side $2\mathfrak{M}$ in the last two layers. Their real and complex potentials are bounded by $L\alpha_0$ and $L\alpha_1$. The pivot estimate at $a = L\max\{\alpha_0,\alpha_1\} \le a_{\mathrm{nf}}$ therefore applies to this bond as well. It gives $|z_c - 6\log L| \le C_z a$ on the specified logarithm branch. Thus neither crossing bonds nor the last layer require restricting the domain of a factor.

The kept quadratic form is holomorphic in the pair, and on the stored-data tier it obeys

$$|\varphi^{\mathrm{kp}}_{t,X}| \le \tfrac{1}{2} K_{\mathrm{ch}} C_{\mathrm{xt}}^2 \mathfrak{b}_l^2 e^{-\kappa(d_t(X)+1)}.$$

For the other exponent there are at most $6M^3$ pivot bonds in X. Its domain is one cube or two adjacent cubes, so $d_t(X) = 0$, and

$$|\varphi^{\mathrm{el}}_{t,X} - 6N_X \log L| \le 6M^3 C_z a_{\mathrm{nf}}, \qquad N_X \le 6M^3.$$

Define

$$\operatorname{osc}\varphi = \sup_{\mathfrak{q}} \sup_{P,P'} |\varphi(P;\mathfrak{q}) - \varphi(P';\mathfrak{q})|,$$

where $\mathfrak{q}$ ranges over the allowed stored-data tiers and P, P' over the corresponding full layered space. At each fixed $\mathfrak{q}$, subtract the values at P and P'. The first estimate is used twice; in the second, the same constant $6N_X \log L$ cancels. Since $\mathfrak{b}_l \ge 1$, this gives

$$\sum_{\nu\in\{\mathrm{kp},\mathrm{el}\}} \operatorname{osc}\varphi^\nu_{t,X} \le B^\flat_{\mathrm{fr}} \mathfrak{b}_l^2 e^{-\kappa d_t(X)}, \qquad B^\flat_{\mathrm{fr}} = K_{\mathrm{ch}} C_{\mathrm{xt}}^2 + 12M^3 C_z a_{\mathrm{nf}},$$

where an absent exponent contributes zero. This is a bound on actual finite oscillations over the specified domains. Each factor is Borel measurable and holomorphic on fixed-real slices; gauge transformations and conjugation preserve its defining formula and branch.

Keep the name (ν,t,X) attached to each factor. Its pair reading and stored-data tier use only the history through step t, and its radii use only $x_0,\dots,x_t$. Extending a history changes none of these. Thus names, domains and exponents are retained exactly. There are at most two names for each (t,X) and finitely many domains on a finite lattice, so the family is finite. This also shows why the argument does not require an inductive bound on the previously collected interaction to define these factors.

The finite induction. The actual-step hypotheses are now available: the regular backgrounds and their coordinate neighborhoods; the common radii and stored-data tiers; the Gaussian and strip bounds; and the auxiliary maps and retained factors on their full domains. Start with Wilson's positive finite-volume measure. Given the integrable density at one level, [3, Prop. 4.165] supplies the next density and its exact integration identity. Its proof makes the two Borel changes of variables, carries the kept quadratic forms and pivot factors with their fixed names, and applies the Gaussian and strip expansions on precisely the domains checked above. Its abstract measure step also supplies integrability and the exceptional null sets; these are not assumptions on every point in a chart image.

If the initial small region of a step is empty, all its integration coordinates and local families are empty. The coordinate changes are then zero-dimensional, and the same identity is evaluation at the retained field. This case is included in the cited step proposition. The first nonempty step uses S_0, as checked above. At each subsequent step the numerical bounds give the same hypotheses for the new regular and boundary terms. The agreement of radii, maps and retained exponents under truncation makes these successive constructions one finite induction, as in [3, Prop. 4.166]. Finally, all coupling-dependent conditions were imposed uniformly for $x \geq \gamma^{-2}$. Decreasing γ therefore preserves every assertion on the resulting class of coupling sequences and tori. □

Proposition 10.20 (A common choice and the finite induction)**.** Fix $b_F > 0$, $0 < m_- \leq m_+ \leq 1/2$ and the action-source radius $\varrho > 0$. The additional numerical conditions in the local, history and source estimates can be satisfied in the ordered choice of constants, independently of the cutoff and volume. At the resulting complete assignment the hypotheses of the pure gauge induction hold for both the calibration and running sequences. There are $B_\sharp > 0$ and $0 < \gamma \leq 1/2$ with the following property. Put $\bar{B} = LB_\sharp/(L-1)$. For $0 < g \leq \gamma$, an admissible torus side $\ell \in \mathfrak{L}(g)$, $K \geq 1$, and

$$\mathsf{b} \in \bar{I}(g) = [g^{-2} + \bar{B},\, L(g^{-2} + B_\sharp) - \bar{B}], \qquad \mathsf{m} \in [m_-, m_+],$$

the finite flow and every history term are defined, with the asserted domains, locality, bounds and exact identities. This includes the complex source step on its stated open coordinate domain, the common vacuum energy in the two partition bounds, and the local source-energy and cumulant bounds. The choice remains admissible, with these conclusions, after decreasing γ. ■

Proof. We first choose the constants in the order of the gauge construction and verify the additional matter inequalities. We then verify the hypotheses of the gauge induction and apply it to the matter objects constructed above. The pure gauge theorem supplies the regular-pair cores, historical Gaussian triples and gauge coordinate maps. The fermion potentials satisfy their own local and history estimates.

The choices before the fibre weight. Choose the pure fixed fractions and integer exponents first: $r_c \geq 2$, $\mathfrak{p}_0 \geq 6r_c + 1$, and the larger auxiliary exponent required in the pure table. Choose δ_{sp}, β_s, the Gaussian reserves and the halo data in their open windows. The mollifier in the history profiles is also fixed here. Choose an odd $L \geq 13$ above the pure and matter floors. The requirement on b_F is only positivity.

Form the free and background operator constants next. The complex gauge and matrix-source radii α_1^* and $\varrho_\sharp$ precede the supremum inverse constant C_{sup}. The latter precedes the supremum walk constant C_q^{sup} and the least power $M_w \geq L$ of L such that

$$C_{\mathrm{nb}} \max(C_q, C_q^{\mathrm{sup}})/M_w \leq e^{-3}.$$

Thus $q_{\mathrm{sup}} = C_q^{\mathrm{sup}}/M_w$ satisfies $C_{\mathrm{nb}} q_{\mathrm{sup}} \leq e^{-3}$. The row-sum constants C_Σ for the background maps and C_D^Σ for the stored tower are fixed before C_H; neither depends on M or N_f. Form also $C_D^{\Sigma,\mathrm{s}}$ from the same row-sum argument with $2C_D$, decay rate $\alpha_D/2$, and the fine-operator bound $\mathsf{c}_{\mathrm{disc}} + 2$. This sourced tower constant also precedes the fibre weight. Here C_H° includes the preliminary background-kernel bound of Prop. 6.7, before summing over all stored places, and the parametrix and fine-resolvent constants in Lem. 6.8. These use the already fixed local regularity constants and radii, including its curvature and absorption ceilings. The row-sum enlargement is $(1 + C_\Sigma L^{3/2})C_H^\circ$. Choose

$$C_H \geq \max\{6C_S(c_{\mathrm{hop}} + b_F),\, 8LC_{\mathrm{sup}},\, 2(1 + C_\Sigma L^{3/2})C_H^\circ,\, C_D(b_F + C_D^\Sigma) + 1,\, 2 + 2C_{\mathrm{nb}}C_H'\}.$$

The single-walk constants C_H' and $C'(\alpha)$ are formed before C_H from the local inverse and source-map bounds. For $0 < \alpha < 1$ choose also

$$C(\alpha) \geq \max\{LC_H/(1-\alpha),\, 5L^2 + 4C_{\mathrm{nb}}C'(\alpha)\}.$$

The radius $\varrho_* = (2C_D C_H^2)^{-1}$ then lies in the earlier source ball and below $(8L^2C_S)^{-1}$. All inverse and walk estimates are formed on that ball; increasing C_H only decreases the radius subsequently used. The free ellipticity and finite four-resolvent calculation give the early coupling constant c_2^F before M and κ. The constants B_F and $\bar{B}_M = 3B_F + 1$ of the mass sum rule have the same earlier placement. In particular, the coupling bound does not use a late normalizer bound.

The centered conditional construction requires a finite enlargement of the pure rate floors. Keep the pure Gaussian map

$$N_{\mathrm{gs}} = 64B_{\mathrm{nm}}^5\gamma_{\mathrm{pos}}^{-2} + c_r + 2, \qquad \bar{c}_P^+ = C_{\mathrm{mn}}^3\lambda_r^{-1}(N_{\mathrm{gs}} + 1).$$

These are fixed before the rates. With $\kappa'' = (1 + 4\beta_s)\kappa + 1 + c_E$ and $\bar{r}_H^+ = \kappa'' + c_E + C_{\mathrm{cs}}(\bar{c}_P^+ + 2)$, require

$$L \geq \frac{2C_{\mathrm{cs}}(1 + 4\beta_s)}{1 - 2\delta_{\mathrm{sp}}},$$

$$\tfrac{1}{2}(1 - 2\delta_{\mathrm{sp}})\kappa \geq 2(1 - 2\delta_{\mathrm{sp}}) + 2c_E + \frac{C_{\mathrm{cs}}}{L}[1 + 2c_E + C_{\mathrm{cs}}(\bar{c}_P^+ + 2)],$$

$$\bar{\kappa} \geq 1 + C_{\mathrm{cs}}\bar{r}_H^+ + \log C_{\mathrm{cs}}, \qquad \bar{\kappa} \geq (\kappa + 2)c_0 + \log C_g + 9.$$

Choose κ and then $\bar{\kappa}$ above these and the original entropy and strip floors. The geometry constant c_0 is independent of both rates and of M.

Choose the remaining pure geometric lengths in their order and choose the power $M = L^{m'}$ last among the lengths. Let C_{rc} be the geometric constant for recounting localized mass-member walks through cubes: the extra norm factor is at most $1 + C_{\mathrm{rc}}M_w/M$. Indeed a walk reaching beyond distance t from its starting cube face has $|\omega| + 1 \geq t/(CM_w)$ and meets at most $C(1 + |\omega|)^3$ cubes; the available $e^{|\omega|}$ walk weight pays this count. Besides the pure requirements, impose

$$8LM \geq \max\{R_0, 1 + 16/\delta_*, 2\alpha_D^{-1}\log L, LM_w\}, \quad M \geq 3M_w,$$

$$c_f M \geq C_{\mathrm{dc}}\max\{\bar{\kappa}, \kappa + 2\}, \qquad \delta_F M \geq \kappa + 2, \qquad M \geq 2C_{\mathrm{rc}}M_w,$$

$$(LM)^3 \geq C_{\mathrm{mn}}^{-1}\bar{c}_P^+,$$

$$\delta_\star^{\mathrm{pl}} M \geq 500L^2(3\kappa + 2).$$

The last inequality is the conditional-mean-chain floor. The pure kept/rejected quadratic-form floors are included at this stage. Each has the form $M \geq C$, $cM \geq C + p\log M$, or $CM^p e^{-cM} \leq \epsilon$ with fixed positive c, ϵ. Hence they hold for all sufficiently large powers of L. The closed enlargement has reach $D_{\mathrm{ex}} = M$ in the lower-level units, and M/L in the upper-level units. It need not contain an additional layer of big cubes. Thus the condition with $\min(M, D_{\mathrm{ex}})$, read in lower-level units, is exactly the displayed floor on M.

The source-tail conditions are compatible with this order. Set

$$\kappa_\sigma = \min\{\kappa/(4c_{\mathcal{D}}M), 1/(24M_w), 1/4\}, \quad R_\flat' = 7LM, \quad C_{\mathrm{loc}} = 8e^{\sqrt{3}\kappa_\sigma + \kappa/4}.$$

Require $\kappa_\sigma R_\flat' \geq \log(4L)$ and $\kappa_\sigma R_\flat' \geq \kappa + 9$. For the first branch these are floors on κ, since $7L > 4c_{\mathcal{D}}$ and $c_{\mathcal{D}} \leq 6$. For the second and third branches they are floors on M. Indeed $c_f = (3LM_w)^{-1}$ and $C_{\mathrm{dc}} \geq 16c_0/L$ already imply $M \geq 48c_0M_w(\kappa + 2)$. Consequently

$$Le^{-\kappa_\sigma R_\flat'} \leq \tfrac{1}{4}, \qquad (C_{\mathrm{loc}} + 3e^{2\kappa_\sigma})e^{-\kappa_\sigma R_\flat'/4} \leq 2.$$

Now reduce the pure analytic radii in their original order. Include $\alpha_0 \leq \epsilon_* L^{-2}$, $2L\alpha_1 \leq \alpha_1^*$, $6C_Q\alpha_1 ML^{-4} \leq \kappa$ and $114M^3\alpha_0^2 \leq 1$. Retain also the tree-path ceilings $2C_t\alpha_1 M \leq \kappa$ and $18C_t\alpha_1 M_w \leq 1$. Here C_t is the fixed geometric tree-path constant. The last two requirements are ceilings on the radius chosen after M, not upper bounds on that earlier length. Include also the conditional mean/error conditions

$$\epsilon_{\mathrm{dc}} \leq (LM)^{-3}, \quad \theta_m + \epsilon_m \leq \vartheta_* < 1, \quad (LM)^3\epsilon_m^2 \leq 1, \quad 1 + \vartheta_* < C_{\mathcal{A}}, \quad \vartheta_* < \theta_{\mathrm{sl}}.$$

The pure proof makes the exponentially small error by the preceding length choices and the remaining errors by these radius choices. Its mean bound uses the actual historical external tiers, not an independent mean parameter.

Before E_0 form $A_\kappa = 2C_{\mathrm{nb}}e^{(\kappa+2)c_0}$, the finite small-lag count used in reading old legs, the normalizer constant C_N, and the cluster constant K_M. They use only the already chosen geometry, rates, operator bounds and analytic radii. They depend on neither E_0 nor $B_\sharp$ nor h_F. The fixed potential ceiling c_{cl} is chosen independently of the array bounds; K_M does not depend on that choice. The degree-zero pointing estimate also gives $K_M^\bullet = K_M + C_E K_M^{\mathrm{v}}$ at this stage.

Three demands on determined pure constants must be imposed on their chosen inputs. The maps themselves are retained. First form the finite matter boundary bound

$$B^F = 24(LM)^3 E_{\mathrm{nf}} + C_N N_f + 3,$$

where E_{nf} is the pure first-step bound. This number depends on neither K_M, K^{lb}, E_0 nor $B_\sharp$. The reserve 3 bounds the sum $1/2+1+1/16+1$: the history output, the pure rejected terms, the matter strip row and the retained Gaussian mismatch. Their bounds follow from the later frame and strip smallness conditions and Lems. 9.2 and 9.8. Choose E_0 above its pure floors, with $8M^3E_0 \ge B^F$, and with the full matter-array floor

$$E_0 \ge E_{\mathbf{N}} + C(C_N + M^3)N_f + C(1+L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + CK_{\mathbf{G}} + C_n + CC_{am}r_\alpha^{-2}C_N N_f + 2.$$

Every quantity on its right was fixed earlier. The original map $B \ge \max(1, B_{\mathbf{T}}, 8M^3E_0)$ then gives $B \ge B^F$. Second impose $\varepsilon_{\mathcal{W}}\alpha_0 \le 2a_Q$; the unchanged map

$$e_{\mathrm{cov}} = \min\{1, c_2^{\mathrm{SO}}, \varepsilon_{\mathcal{W}}\alpha_0/2\}$$

gives $e_{\mathrm{cov}} \le a_Q$. Third choose A_0 after A_1 with $4\pi A_0 \ge c_{\mathrm{sh}}A_1$. Since $C_{\mathrm{xt}} \ge \pi(1 + c_s A_0/A_1)$ with $c_s > 2$, the original map $C_{\mathcal{A}} = \max(3, 2C_{\mathrm{xt}})$ gives $C_{\mathcal{A}} \ge c_{\mathrm{sh}}$. These are respectively a floor on E_0, a ceiling on α_0, and a floor on A_0. They introduce no return to a determined constant.

Choose $B_\sharp$ after E_0. Include both the pure floor $B_\sharp \ge G_\sharp = C_{am}r_\alpha^{-2}E_0$ and

$$B_\sharp \ge (1+L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + 1.$$

Form $C_{\mathfrak{r}}, C_\delta, C_K, C_G$ after E_0 and $B_\sharp$, and then choose the vertex radius β_0 with

$$C_G\beta_0 \le \tfrac{1}{2}, \qquad c_h\beta_0 \le \min\{\epsilon_P, \alpha_1/2, a_{\mathrm{an}}/2\}, \qquad 2C_P\beta_0 \le \min\{r_P^\circ, \alpha_1/2\}.$$

The last inequality supplies the separate pivot estimate at the first step; the positive-level shift bound alone does not supply it. Form now the remaining old-reading and boundary-counting constants. Together with the already fixed A_κ, K_M and Γ_F, the constants K_μ, C_{bd}, C_ω and S_ω precede the fibre weight. Their definitions contain no fibre weight. The cancellation of h^2 between a mass-member norm and its moment extraction is the reason K_μ is independent of h. For the source estimates form the weighted piece constant C_{pw} and the enlarged cell-unit constant $C_{(N)}$ before C_ω, and then c_{cs} and C_{pc}. Their bounds divide out the same two leg weights h^2; none depends on the subsequent choice of h_F. Form also C_{cnt}, C', S_κ and c_{inc} from (49) and the subsequent formula, after M and the kernel bounds and before h_F.

The fibre weight and source constants. Choose h_F sufficiently large. Besides $h_F \ge 2A_\kappa\Gamma_F L/(L-1)$ and $h_F \ge 32C_{\mathrm{bd}}K_M A_\kappa \Gamma_F$ and

$$h_F \ge \frac{9}{4}C_{\mathrm{bd}}K_M A_\kappa \Gamma_F c_{\mathrm{cl}}/\varepsilon_\flat,$$

impose the flat and differentiated contraction floors. Their factor is

$$\theta_E(h) = \sqrt{2}K_\mu K_M\left[\frac{2A_\kappa\Gamma_F}{h} + \left(\frac{c_h}{a_{\mathrm{an}}} + C_G\right)\frac{c_{\mathrm{cl}}}{4C_V h^2}\right] \longrightarrow 0.$$

In particular choose $2\sqrt{2}\theta_E C_\omega S_\omega \le 1/2$. For the source extraction, first form

$$K_\mu^{\mathrm{s}} = K_\mu + c_b^{-2}c_{\mathsf{S}}M^{-3}C_{\mathrm{pw}}, \qquad \theta_E^{\mathrm{s}}(h) = (K_\mu^{\mathrm{s}}/K_\mu)\theta_E(h).$$

The weighted piece constant C_{pw} and all the factors in K^{s}_{μ} precede h_F. The source history uses $\Gamma^{\mathrm{s}}_F = 2\Gamma_F$ and the boundary constant $C^{\mathrm{s}}_{\mathrm{bd}}$ formed with the doubled operator bounds. Require

$$h_F \geq 32 C^{\mathrm{s}}_{\mathrm{bd}} K_M A_\kappa \Gamma^{\mathrm{s}}_F, \qquad 2\sqrt{2}\theta^{\mathrm{s}}_E C_\omega S_\omega \leq \tfrac{1}{2}.$$

Put $D = 1 - 8/(3\sqrt{L}) > 0$ and $P = 1 + C_{\mathrm{tl}} + 42M$. The increment calculation has $B_\nu = \alpha_\nu + \beta_\nu B_\sigma$, where the coefficient β_ν is a fixed multiple of $\theta_E(h_F)$. Require, at this stage,

$$4\theta^{\mathrm{s}}_E C_{\mathrm{pc}} S_{1/4}(1 + c_{\mathrm{cs}}) \leq \tfrac{1}{2}, \qquad 6P\beta_\nu \leq D/2, \qquad 2C_{\mathcal{L}}\beta_\nu \leq \tfrac{1}{2}.$$

The factors multiplying θ_E in these three conditions are already fixed. In particular the cell-sum argument does not use B_ν or B_σ. These are genuine floors on h_F.

Immediately afterwards choose C_{rad}. All its floors are explicit and have fixed right sides: the auxiliary threshold

$$\delta_{\mathrm{ax}} = \min\{1, s^*\alpha_0/(1+\beta_s), \varepsilon_{\mathfrak{h}}/C_{\mathcal{A}}, \delta(e^{2\mathfrak{p}_0})\}$$

is formed from constants preceding h_F.

$$\begin{aligned} C_{\mathrm{rad}} \geq \max\{&(1+\beta_s)/s^*,\ 12C_{\mathcal{A}}K^{\mathrm{s}}_{\Lambda}K^{\bullet}_M,\ 4c_h C_{\mathcal{A}}\alpha_0 h_F/(a_{\mathrm{an}}A_\kappa\Gamma_F),\\ &C_{\mathrm{sf}}\sqrt{L}A_0/(\varepsilon_{\mathcal{W}}A_1),\ 512c_{\mathrm{ins}}(1+c_{\mathrm{cl}}^{-1})C_{\mathcal{A}}K^{\mathrm{s}}_{\Lambda}K_M,\\ &\sqrt{2}L(2+C_{am}r_\alpha^{-2})K^{\bullet}_M K_\Lambda C_{\mathcal{A}}/(\theta_E(h_F)C_\omega),\\ &16(1+s^*K^{\mathrm{s}}_{\Lambda}K_M)(1+\alpha_0/\delta_{\mathrm{ax}})^2\}. \end{aligned}$$

The last two floors come from radius localization and the later insertion estimate. These floors are independent of C_2, C_3, B_σ or a final small-coupling threshold.

Form the zero-source constants in their proved order: $r_x, C_{\mathrm{sl}}, C_r, C_{(2)}, C_0, C_1, C_\chi, c_s, \theta_s$, then $C_2, C_2^{\mathrm{pt}} = 8M^3C_2, C_T, C_3$, then C'_ρ, C_ρ, C_s, and then $C_w, C_b, C_e, B_z, C_\mu$. Here C_3 is the cubic mass remainder constant; later source remainder constants are different constants. The separate mass-member estimates ensure that none of these depends on the eventual source-window radius. Next form $S_3, C^{\psi}_w, B_0, C_{\mathrm{tor}}$. The constant-source comparison uses only zero-source constants and therefore also precedes B_σ.

The two apparent return inequalities for B_σ are

$$B_\sigma \geq 4B_0 + 3C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}}, \qquad B_\sigma \geq 2C_{\mathcal{L}}B_\nu, \qquad C_{\mathrm{var}} = 2PB_\nu/D.$$

Substitute $B_\nu = \alpha_\nu + \beta_\nu B_\sigma$. The preceding floors make each return coefficient at most $1/2$. For example the explicit finite choice

$$B_\sigma \geq \max\left\{2\left(4B_0 + 8C_{\mathrm{tor}}C_{\mathcal{L}} + \frac{6P}{D}\alpha_\nu\right), 4C_{\mathcal{L}}\alpha_\nu\right\}$$

satisfies both. Now form $C^{\mathrm{s}}_w, B_\nu, C_{\mathrm{var}}, C'_{\mathrm{tor}}, C_\sigma$ and C_ν, and finally

$$B_s = \bar{B}_M + \max\{5C_\sigma, 2B_\sigma\}, \qquad A = 5B_s.$$

The general sourced steps use the open set on which their potential has size below c_{cl}. They do not require their order-ϕ leg families to belong to the smaller balls about the normalized zero-source families. The cubic mass decomposition is used only in the constant-coordinate window, with C_ρ fixed before B_σ. Form C^F_V after the stored-layer constants and C_2^{pt}, using the variation estimate in Prop. 9.14. All local-energy, stored-source, history-majorant and later insertion constants are now finite and can be formed.

Allow the remaining constants to depend on the fixed mass interval, and choose the physical bilinear-source radius at this point. More explicitly, put $S_D = \sum_{y\in\mathbb{Z}^3} e^{-\alpha_D|y|/2} < \infty$. The matrix-mass radius, already fixed from the early operator constants, is

$$r_M = \min\left\{1, \frac{1}{4C_{\mathcal{K}}}, \frac{1}{4C_D S_D}, \frac{1}{8e^{3\alpha_D/2}b_F^2 C_{\mathcal{K}}^2 C_D S_D}, \frac{L\varrho_*}{2}, \frac{L}{C_{\mathrm{fl}}}\right\} > 0,$$

where

$$c_{\mathsf{M}} = 2600\nu_P/\alpha_D, \quad C_{\mathrm{bl}} = \nu'\left(2+\frac{4}{L-1}\right), \quad C_{\mathrm{fl}} = 4C_{\mathrm{bl}}c_{\mathsf{M}}C_H.$$

Here $\nu' = 8$, ν_P is the number of fine places over a site, and C_H is the original background constant. Thus the matrix-mass change costs at most one on a stored formal leg. The radius is chosen from earlier constants; the later auxiliary fibre norm does not change it. Fix $0 < c_J \le \min(1/2, \varrho_*, r_M/L)$ and $0 < 2\varrho_J \le c_J m_-$. Thus all physical sources lie inside the full matrix-source operator ball. Put $h_* = \max\{h_F, \sqrt{2/(c_b m_-)}\}$. For the final Berezin bound recompute the norm constants at $H = 2A_\kappa h_*$. This changes only that auxiliary norm estimate and its threshold. The flow and source-chain constants just fixed remain those at h_F.

The final threshold. Choose γ last. Every remaining requirement is a positive upper bound on a function tending to zero with γ, or a finite lower bound on $\log\gamma^{-2}$. This includes the original pure requirements, the background regularity bound $\bar{C}_{\mathrm{reg}}\epsilon(\gamma^{-2}) \le \epsilon_*$, the relative-coefficient bound $4\bar{B}\gamma^2 \le \sigma_*$, and $\gamma^{-2} \ge e^{18\mathfrak{p}_0}$. For the actual density-test fields and covariance classes require

$$2\epsilon(\gamma^{-2}) \le \varepsilon_{\mathcal{W}}\alpha_0, \qquad 2\epsilon(\gamma^{-2}) \le \varepsilon_{\mathcal{W}}\mathsf{a}_H, \qquad \delta(\gamma^{-2}) \le \mathsf{d}_H,$$
$$C_H^{\mathcal{W}}\delta(\gamma^{-2}) \le (1-L^{-2})e_{\mathrm{cov}}.$$

The last inequality closes the step's estimate $\mathsf{m}_k \le L^{-2}\mathsf{m}_{k+1} + C_H^{\mathcal{W}}\delta(\mathsf{t}_k)$ at the same covariance radius. The constant $C_H^{\mathcal{W}}$ belongs to the gauge sum rule and is distinct from the fermion background constant.

The actual history Gaussian is used at the positive real ratio $\tau = x_j/\mathsf{t}_j$, with $|\tau - 1| \le 1/4$. Its construction and retained collar term are those of Lem. 9.4 and Prop. 9.11. Impose the explicit frame thresholds and the three $\varepsilon_\flat/4$ bounds in Lems. 9.6 and 9.7 and Prop. 9.14. In the stored-layer bound $b^{\mathrm{lay}}(\gamma)$ use the displayed formula for b_k^{lay} with $\delta_k = \delta(\gamma^{-2})$, $\phi(\mathsf{t}_k) = \phi(\gamma^{-2})$ and the corresponding $\Theta'(\gamma), \Theta(\gamma)$; its two pointer terms retain the factor C_2^{pt}. Also require $16c_s h_F^2 \bar{B}_M^2\gamma^2 \le C_2$. These are ceilings on γ after all their coefficients have been fixed. The three-share recursion of Prop. 9.14 has factor $64/51$, so these three shares give $\frac{64}{51}\frac{3}{4}\varepsilon_\flat = \frac{16}{17}\varepsilon_\flat < \varepsilon_\flat$. In particular include the boundary-sum ceiling

$$16C^{\mathrm{bd}}\frac{A_1}{C_0}(\log(\tfrac{1}{2}\gamma^{-2}))^{-(q-\mathfrak{p}_0)}\left(\tfrac{9}{8}K^{\mathrm{lb}} + 108M^3K_M\right) \le 1, \qquad q > \mathfrak{p}_0.$$

All quantities except γ have been fixed. The retained Gaussian mismatch requires, in addition,

$$\frac{3}{2}\bar{B}\gamma^2 C_{\mathrm{xt}}^2 A_1^2(\log\gamma^{-2})^{2\mathfrak{p}_0}\left(c_{\mathrm{kp}}^{\mathrm{th}} + c_{\mathrm{rj}}^{\mathrm{th}} + \frac{3}{2}c_{\mathsf{G}}(LM)^3\right) \le 1.$$

The coefficients here are the fixed kept, rejected and gauge quadratic-form constants. This is the domainwise bound of Lem. 9.8, including the strip rows. The effective scalar-frame threshold is $\log\gamma^{-2} \ge Y_{\mathrm{eff}}^{\mathrm{F}}(2\varepsilon_\flat/5)$, formed with $(E_0, B, B_{\mathrm{fr}}^\flat)$ as in Lem. 9.6; the q-class and failed-test bounds of Lem. 9.4 are imposed at the same stage.

The source scalar arrays use the larger budget

$$E_0^{\mathrm{s}} = E_0 + 2 + K_M^\bullet c_{\mathrm{cl}}, \qquad K_M^\bullet = K_M + C_E K_M^{\mathrm{v}}.$$

Their boundary budget is

$$B_\natural = B + 8M^3(E_0^{\mathrm{s}} - E_0) + C_v^{\mathrm{rj}} + C_v^{\mathrm{sr}} + 2 + K_{\mathrm{rb}}c_{\mathrm{cl}}/16.$$

Use the source effective-frame threshold $\log\gamma^{-2} \ge Y_{\mathrm{eff}}(\varepsilon_\flat/(2+2\sqrt{2}))$ formed at the doubled box $\sqrt{2}(E_0^{\mathrm{s}}, B_\natural, B_{\mathrm{fr}}^\flat + 1)$ of Lem. 10.12, and the three source shares, including the formal-leg bound of Lem. 10.10. These are requirements on the actual complex source step; its identity does not follow from positivity at zero sources. Among the source ceilings are

$$C_{\mathrm{rad}}\delta(\gamma^{-2}) \le \alpha_0, \quad 16\varrho \le r_v(\gamma^{-2} - \bar{B}), \quad 64\varrho \le r_x\gamma^{-2},$$
$$32C_{\mathrm{bd}}^{\mathrm{s}}K_M C_{\mathcal{A}}\delta(\gamma^{-2}) \le \beta_0.$$

The source class with fermion legs is bounded by a fixed constant times $\phi(\gamma^{-2}/2)$. The linear collar term is bounded by a constant times $\gamma(1+2\log\gamma^{-1})^{\mathfrak{p}_0}$, and the cross term by a constant times $\gamma^2(1+2\log\gamma^{-1})^{\mathfrak{p}_0}$. The kept quadratic term leaves half the retained positive floor; the rejected term is estimated using $\mathfrak{b}(t)^2 e^{-(\check{R}-2)} \le C_{\mathfrak{b}}$. The precise coefficients and thresholds were fixed in the source-history proof. Each additional condition is therefore satisfied by reducing γ. The constant-coordinate torus comparison also requires

$$C_\psi \phi(\tfrac{1}{2}\gamma^{-2}) \le c_{\mathrm{cl}}/20, \qquad \epsilon_\gamma S_L \le \min\left\{\tfrac{1}{2}, \frac{1-2^{-1/2}}{4C_{\mathrm{dl}}C_\omega S_\omega}\right\}.$$

Its constants are formed before B_σ, and $\epsilon_\gamma \to 0$ as $\gamma \to 0$.

For the common lower bound impose

$$C_{\mathsf{S}_1}\bar{B}_M\gamma^2 \le c_b m_-/8, \qquad \phi(\gamma^{-2}) \le c_*(m_-), \qquad \gamma \le \gamma(2A_\kappa h_*, 5B_s).$$

The first pays the running mass member; the second pays the activity bilinear and the last interacting forest expansion. In choosing $c_*(m_-)$ include the constants $c_{\mathrm{pt}}^{(K)}, c_{\mathrm{pr}}^{(K)}, c_{\mathrm{wr}}^{(K)}, c_{\mathrm{q}}^{(K)}$ in $C_{2,*}$ from (41), formed at the auxiliary weight $2A_\kappa h_*$. The threshold $\gamma(h,A)$ is obtained by taking the infimum of its positive continuous ceilings on the compact rectangle $[h_F,h]\times[1,A]$. It is positive and decreases in h, A. Thus this last choice implies all earlier uses at h_F and radius $5B_s$.

Finally choose the history-sum threshold with the actual matter and source majorants. In the unmarked application take $a = 1/2$ and rate $c_*/2$. More generally, a fixed deposit mark with created-cell coefficient $c_\Theta < c_*$ uses $a = (1-c_\Theta/c_*)/2$ and rate $c = (c_* + c_\Theta)/2 > c_\Theta$. Its deposit dials can be written as $(w', c'_{\mathrm{Dep}}) = (c_\Theta/c, c)$, so that $w' < 1$ and $c'_{\mathrm{Dep}} \le c$. Apply the actual thresholds of [3, Prop. 4.160], with these dials, the enlarged majorant and the logarithmic label-error constant. Their hypotheses hold since $2\mathfrak{p}_0 > 3r_c + 2$ and the remaining threshold is finite. The quarter-stock allocation in Prop. 10.14 gives this integrated rate; it does not assert a stronger pointwise loss-a property. The later insertion ceilings are also scalar envelopes tending to zero. A finite minimum of all these positive thresholds completes the common choice.

Application of the gauge induction. The pure gauge constants are chosen in the order of [3, Prop. 4.163], and its proof. At each stage above, the additional matter conditions impose floors, positive ceilings or nonempty windows on freely chosen constants, using only quantities already fixed. Every new constant defined by a formula is placed after its arguments. The defining formulas for the pure gauge constants are retained. Requirements on earlier constants are imposed at their original stages, before the subsequent constants are evaluated.

In particular, the condition $B \ge B^F$ is obtained by requiring $8M^3E_0 \ge B^F$ in the choice of E_0. The condition $e_{\mathrm{cov}} \le a_Q$ is obtained by imposing $\varepsilon_{\mathcal{W}}\alpha_0 \le 2a_Q$ on α_0. The condition $C_{\mathcal{A}} \ge c_{\mathrm{sh}}$ is obtained from the floor on A_0 checked above. In each case the original formula gives the required value; no constant defined by a formula is chosen independently of its arguments.

Choose the constants successively with these additional bounds. Omitting the matter-only constants then leaves a choice satisfying the pure gauge conditions, since their order, formulas and inequalities have been preserved. Finally, all conditions on γ hold throughout a sufficiently small interval $(0,\gamma_*]$. For the pure conditions this follows from their uniform imposition for $x \ge \gamma_*^{-2}$ in the proof of [3, Prop. 4.163]; for the additional conditions it follows from the positive thresholds and vanishing upper bounds verified above. Taking their finite minimum completes the choice and shows that decreasing γ preserves it.

Lem. 10.19 therefore supplies all conditions of the pure gauge induction at this one restricted assignment, with the canonical data and substitutions of that lemma. Both coupling sequences used here are in its range. The calibration sequence $\mathsf{t}_j = L^{K-j}\mathsf{b}$ has zero increments. At each stage of the running construction, the increments already obtained satisfy $|b_j| \le B_\sharp$. For $\mathsf{b} \in \bar{I}(g)$ the

recurrence $x_{j+1} = L^{-1}x_j - b_{j+1}$ gives

$$\mathsf{t}_j \geq g^{-2} + \bar{B}, \qquad |x_j - \mathsf{t}_j| \leq \bar{B}, \qquad x_j \geq g^{-2}.$$

The allowed torus sides supply the required chart room for both sequences, as in the proof of Lem. 7.1. The gauge induction consequently applies to each finite prefix of either sequence. It is quantified over all such admissible sequences; its use does not presume that a full matter flow has already been constructed.

The matter and source steps. The gauge result supplies the charts, auxiliary systems and coordinate identities. We now verify its applications to the actual fermion and source coefficients using the estimates proved above.

The fermion covariance and its local normalizer are constructed on their intrinsic domains in Prop. 6.7 and Lem. 6.10. The density-test and sum-rule applicability is Lem. 7.1; the small-field induction and mass estimates give the regular activities before any large-field history is used. The histories then use the algebra-valued expansion, the positive real Gaussian rescaling and the agreement on interior domains of Lems. 9.1, 9.4 and 9.10. The degree-zero arrays and the reentering boundary terms obey Lems. 9.6 and 9.7. Consequently Prop. 9.11 applies at every finite step, as an identity of measures in the even Grassmann algebra.

For complex sources, Lem. 10.7 gives the coefficientwise measure identity on one common full-measure set, and Prop. 10.13 closes the source induction with the scalar and formal-leg estimates just used. The common vacuum assignment is preserved by these identities. The final history sum and lower bound therefore give Props. 10.6 and 10.14 and the uniform cumulant estimates. This proves the asserted finite construction. □

The range of bare labels is a full tail, despite its description by reference intervals. Decrease γ so that $\gamma^{-2} \geq 2\bar{B}$. For any $\mathsf{b} \geq \gamma^{-2} + \bar{B}$, take $g = (\mathsf{b} - \bar{B})^{-1/2} \leq \gamma$. Its lower endpoint is b, while its upper endpoint is $L\mathsf{b} - 2\bar{B} \geq \mathsf{b}$. Consequently

$$\bigcup_{0<g\leq\gamma} \bar{I}(g) = [\gamma^{-2} + \bar{B}, \infty).$$

For each such g the admissible sides are cofinal. Thus the volume-first conclusions apply to every sufficiently large fixed b, with the stated positive mass interval. Using the closed intervals is essential at the displayed lower endpoint.

11. The short-distance two-point functions

We keep the physical coupling b fixed and the mass in $[m_-, m_+] \subset (0, 1/2]$. Write $\lambda_j = L^{j-K}$, $\widehat{\phi}(t) = (1 + \log t)^{\mathfrak{p}_0}$ and $\phi(t) = t^{-1/2}\widehat{\phi}(t)$. Only two-point functions of the action density, or of two bilinears, are considered in this section.

For $0 < \vartheta < 1$ and $d > 0$, a test pair consists of nonnegative $f, h \in C^1_c$ supported in balls of radius d, with

$$d \leq |x - y| \leq 10d \quad (x \in \operatorname{supp} f,\ y \in \operatorname{supp} h), \qquad \int f \geq \vartheta d^3 \|f\|_\infty, \quad \operatorname{Lip} f \leq \frac{\|f\|_\infty}{\vartheta d},$$

and the same last two conditions for h. We also require the pair to lie in a common ball of radius $5d$. Put

$$\mathcal{K}_q(f, h) = \iint f(x)h(y)|x - y|^{-q}\, dx\, dy.$$

Then

$$10^{-q}\vartheta^2 d^{6-q}\|f\|_\infty\|h\|_\infty \leq \mathcal{K}_q(f, h) \leq (4\pi/3)^2 d^{6-q}\|f\|_\infty\|h\|_\infty. \tag{57}$$

Proposition 11.1 (The two leading kernels)**.** Every volume-first continuum limit constructed below satisfies, on these test pairs,

$$S_2((0,f),(0,h)) = \frac{9}{16\pi^2}\mathsf{b}^{-2}\mathcal{K}_6(f,h)(1+\epsilon_0),$$

$$S_2((\Gamma,f),(\Gamma',h)) = \frac{1}{2\pi^2}\mathcal{K}_4(f,h)(\delta_{\Gamma\Gamma'}+\epsilon_B).$$

For each fixed ϑ, the errors are bounded by a function $\epsilon(d)$ tending to zero as $d \downarrow 0$. The bound is uniform in the permitted states and in the mass interval. In particular, each displayed field has a nonzero two-point function. ■

11.1. **Reduction to a local source energy.** Let $E_K(s)$ be the source energy, and let $e_j(X;s)$ be its level-j terms, so that $E_K(s) - E_K(0) = \sum_{j,X} e_j(X;s)$. The local source bound gives, for sources supported in the ball of radius $1/4$ about a board-cell centre,

$$\Psi(s) = e^{-\Lambda(s)}\frac{Z(s)}{Z(0)}, \qquad \Psi(0) = 1, \qquad |\Psi(s)| \le B, \quad \Lambda(s) = \frac{1}{\mathsf{N}}\sum_r \{E_K(rs) - E_K(0)\}. \tag{58}$$

Here r runs over the N board reflections, and B is independent of the cutoff and the side. The source domain is the fixed domain of the preceding section. Write $\|F\|_\circ = \|f\|_\sharp/\varrho$ for $F = (0,f)$ and $\|F\|_\circ = \|f\|_\infty/\varrho_J$ for $F = (\Gamma, f)$.

Lemma 11.2 (Packing with signs)**.** For a test pair with $8a_K \le d \le 1/200$, its truncated two-point function differs from a mean of mixed derivatives of E_K, over translates of the pair and their board reflections, by at most

$$Cd^3\|F\|_\circ\|H\|_\circ.$$

The packing has $N \ge (96d)^{-3}$ translates per cell. Its union meets at most $CN\mathsf{N}(d/\lambda)^3$ cubes of side $\lambda > d$. ■

Proof. For separated sources F, H, the source norm of $z_1F + z_2H$ is the maximum of the two norms. This is immediate for the supremum norm. For the Lipschitz norm, put

$$L_* = \max\{\mathrm{Lip}(z_1 f), \mathrm{Lip}(z_2 h)\}.$$

If x, y lie in different supports, let the segment leave the first at p and enter the second at q. The sum of the two function values is bounded by $L_*(|x-p| + |q-y|)$, hence by $L_*|x-y|$. Thus disjoint translates with arbitrary signs retain the same norms.

On the open bidisc $|z_i| < 2/\|F_i\|_\circ$, the function in (58) is bounded by B. Apply Schwarz's lemma on each complex line to $(\Psi - 1)/(B+1)$. On the bidisc smaller by $1/(4B)$, $|\Psi - 1| \le 1/2$. Its logarithm therefore exists there. Apply Cauchy's estimate on smaller circles and let their radii increase to obtain

$$|\partial_1\partial_2 \log\Psi(0)| \le 4B^2\|F\|_\circ\|H\|_\circ.$$

The derivative is the bilinear form $S^T_{2,K} - \partial_1\partial_2\Lambda$. Choose translates F_t, H_t of the pair inside the cell ball, with the balls of radius $5d$ mutually at distance at least d. A lattice grid with mesh between $12d$ and $13d$, in a centred cube of side $1/4$, supplies at least $(96d)^{-3}$ translates. The cube and the balls of radius $5d$ lie in the cell ball, since $\sqrt{3}/8 + 5d < 1/4$. In each coordinate the number of grid points is at least $(52d)^{-1} - 1 \ge (96d)^{-1}$. Average the preceding bilinear form on $\sum_t \varepsilon_t F_t, \sum_t \varepsilon_t H_t$ over independent signs. Only equal indices survive, so the mean over the N pairs has bound $4B^2N^{-1}\|F\|_\circ\|H\|_\circ$. Lattice translation invariance makes the truncated correlation the same for every pair. The mean of $\partial_1\partial_2\Lambda$ remains, since E_K itself depends on the board. This proves the claim. The same grid, repeated by board reflections, meets at most $CN\mathsf{N}(d/\lambda)^3$ cubes of side $\lambda > d$: each cell has a fixed-volume occupied region and $N \ge cd^{-3}$. □

11.2. **The Gaussian and fermion parts of the energy.** At level j use the source domain

$$\|w\|_\infty,\ 40M\lambda_j \operatorname{Lip} w < \frac{\varrho}{\widehat{\phi}(x_j)}, \qquad \lambda_j\|J\|_\infty < R_J, \tag{59}$$

where $R_J = \min\{r_M, \varrho_S, \varrho'_S/L\}$ is below the matrix-mass and free-operator source radii. The steps through level j are made on one constant pair-radius system $a = C_{\mathrm{rad}}\delta(\mathtt{t}_{j-1})$. At earlier levels $i \le j$ the linear gauge-source term has size t_i^{lin}, with

$$t_i^{\mathrm{lin}} \le \sqrt{2}L^{-(j-i)/2}T, \qquad T = t_j^{\mathrm{lin}} < c_{\mathrm{cl}}/16.$$

This term need not be $O(\phi(x_i))$. Its Gaussian contribution must be retained.

The comparison with a Gaussian expansion involves two changes: removing the tests, and changing the sort assigned to a potential that is defined at every real fluctuation. The latter change is not an identity of individual cluster terms. The following bound accounts for it.

Lemma 11.3 (Changing the sort of an all-fluctuation family)**.** Assume the local-step and cluster-expansion hypotheses of Prop. 7.2 and Lem. 7.5, with a region R, its decoupled Gaussian structure, finite threshold $\mathfrak{b}$ and all bond parameters equal to one. Let a base system have only origins of sort Σ_{q}. Let $m \ge 1$ families $\mathbb{W}_r$ be defined at every real fluctuation and satisfy $\|\mathbb{W}_r\|_{\star,\lambda_V} \le w_\star$. If the base has at most $4-m$ origins, denote by E_{v} and E_{q} the cluster sums after adjoining these families with the indicated sort. Then, for $X \subset R$,

$$|E_{\mathrm{v}}(X) - E_{\mathrm{q}}(X)| \le 2K_P^{\Pi} p_\star e^{-\lambda_{\mathrm{out}} d_t(X)}.$$

If the base has at most $3-m$ origins, an additional family $\dot{\mathbb{W}}$ defined at every real fluctuation, with $0 < \|\dot{\mathbb{W}}\|_{\star,\lambda_V} < \infty$, satisfies

$$|E'_{\mathrm{v}}[\dot{\mathbb{W}}](X) - E'_{\mathrm{q}}[\dot{\mathbb{W}}](X)| \le 2K_P^{\Pi} p_\star \frac{\|\dot{\mathbb{W}}\|_{\star,\lambda_V}}{w_\star} e^{-\lambda_{\mathrm{out}} d_t(X)}.$$

The constants are those of the cluster expansion; in its small-field application $w_\star = c_{\mathrm{cl}}$ and $p_\star$ is its large-field bond weight after the bond count. ∎

Proof. For a set $\mathbf{D}$ of vertices let $Y^{\mathbb{W}}(\mathbf{D})$ be the union of the domains belonging to the added families. An index of the system of sort Σ_{v} has its large-field bonds outside $\mathfrak{X}\langle Y^{\mathbb{W}}(\mathbf{D})\rangle$ and has the inner test on that set. Expand this test as

$$\mathsf{in}_{\mathfrak{e}'} = \prod_{b\in\mathfrak{e}'^{+}} (1 - 1_{\{|B(b)|\ge\mathfrak{b}\}}) = \sum_{\mathfrak{e}''\subset\mathfrak{e}'} (-1)^{\#\mathfrak{e}''^{+}} \mathsf{out}_{\mathfrak{e}''}.$$

This is legitimate at every real fluctuation because the potentials are defined there. Conditioning and the decoupling bracket are linear. The terms with $\mathfrak{e}'' = \emptyset$ agree index by index with the terms of the system of sort Σ_{q} whose large-field bonds avoid the added vertices. In the difference, every remaining term therefore has a marked large-field bond lying in an added vertex.

For the terms already present in the system of sort Σ_{q}, the marked-bond estimate gives $K_P^{\Pi} p_\star e^{-\lambda_{\mathrm{out}} d_t(X)}$. The terms created by the test expansion have the same Gaussian majorant: each extra bond carries its usual weight p. Give such a bond the region $b^\sharp$ and the enlarged weight $\widehat{w}(b) = pe^{4\lambda_1}$. The map from an expanded index to its vertices, old bonds and extra bonds is injective: the vertices determine $Y^{\mathbb{W}}$, and membership in its bond set distinguishes the extra bonds from the old ones. The old regions already cover the cluster; the additional bond regions only increase the covering sum. At a coarse cube there are at most $6((L+2)M)^3$ such oriented bonds. Their enlarged weights sum to at most $p_\star$, while all element weights sum to at most twice the number of coarse cubes. Thus the unmarked activity majorant is unchanged, and its marked version has one extra factor $p_\star$:

$$\bar{h}'(\gamma) \le e^{8c_w - 4\lambda_1} e^{-\lambda_2 d_t(\gamma)}, \qquad \bar{z}'(\gamma) \le p_\star e^{8c_w - 4\lambda_1} e^{-\lambda_2 d_t(\gamma)}.$$

The cluster count gives the same bound $K_P^{\Pi} p_\star e^{-\lambda_{\mathrm{out}} d_t(X)}$ for these terms and proves absolute convergence of the expanded sum. Adding the two bounds proves the first assertion.

For an insertion adjoin the extra origin $\epsilon\dot{\mathbb{W}}$. Both systems are holomorphic for $|\epsilon| < w_\star/\|\dot{\mathbb{W}}\|_{\star,\lambda_V}$. Apply the first assertion on each smaller circle and then Cauchy's estimate at zero. Letting the radius increase to the stated one gives the insertion bound. □

Lemma 11.4 (Energy decomposition)**.** On (59),

$$e_j(X; w, J) = e_j^{\mathrm{G}}(X; w) + e_j^{\mathrm{F}}(X; J) + e_j^{\mathrm{H}}(X; w, J) + e_j^{\mathrm{I}}(X; w, J).$$

The sums of the first two terms satisfy the following identities of logarithmic germs at zero sources:

$$\sum_{j,X} e_j^{\mathrm{G}} = \log \frac{Z^{\mathrm{G}}[w]}{Z^{\mathrm{G}}[0]} - \log \frac{Z^{\mathrm{G}}_{\mathrm{last}}[w]}{Z^{\mathrm{G}}_{\mathrm{last}}[0]},$$

$$\sum_{j,X} e_j^{\mathrm{F}} = \log \frac{\det(D[1] + m_0 - a_K \hat{J})}{\det(D[1] + m_0)} - \log \frac{\det \mathbb{D}_K(1; J)}{\det \mathbb{D}_K(1; 0)}.$$

Here Z^{G} is the linearized gauge Gaussian integral, with the constant last-level gauge mode fixed. The fermion identity also defines the analytic continuation of the quotient of determinants by the product of the invertible fluctuation determinants. Separate global logarithms of its numerator and denominator are not required on the enlarged operator-source ball; both are invertible at zero sources. The term e_j^{H} vanishes to fourth order in w, uniformly in J. Moreover

$$|e_j^{\mathrm{I}}(X; s)| \le C_I \phi(x_j) e^{-\kappa d_j(X)},$$

and changing a source outside X gives the same bound multiplied by $\exp[-\kappa_\sigma \operatorname{dist}(X, \operatorname{supp}(s - s'))/\lambda_j]$. ■

Proof. We give the decomposition of the scalar activities, since it is needed to keep the error small on the domains in (59). Let $\mathcal{L}_i$ be the linear source term plus the linear variation obtained by localizing only the fluctuation-dependent b-parts of the earlier Gaussian activities; let q_i be the quadratic variation of those same parts. The pair-dependent c-parts are included in V_i^{rest} below: their absolute localization has the small shift factor $\delta(\mathsf{t}_{i-1})$. Both $\mathcal{L}_i$ and q_i are given at all fluctuations. In the untested Gaussian expansion form the following families at a pair P. Let $\mathcal{V}_i^{v'}(P)$ be the relative-weight quadratic origin of the source step at $\tau = 1$, with $v'_{i-1} = -L^{K-i+1} w/\mathsf{t}_{i-1}$. Write $E^{\circ}_{P,i}[\mathcal{W}](X)$ for the untested Gaussian cluster logarithm with origins $\mathcal{V}_i^{v'}(P), \mathcal{W}$, less its value with only the first origin. All logarithms use the branch at zero sources. Define

$$\mathcal{E}_i(\eta)(X; P) = E^{\circ}_{P,i}[\eta \mathcal{L}_i(P) + q_i(P)](X), \qquad F_i^{\mathrm{G}} = \mathcal{E}_i(0) + \tfrac{1}{2}\mathcal{E}_i''(0).$$

More explicitly put

$$c_i(X; P) = E^{\circ}_{P,i}[q_i(P)](X), \qquad b_i(X; P; \mathcal{L}, \mathcal{L}') = \tfrac{1}{2}\partial_{\eta_1}\partial_{\eta_2} E^{\circ}_{P,i}[\eta_1 \mathcal{L} + \eta_2 \mathcal{L}' + q_i(P)](X)\Big|_{\eta_1=\eta_2=0}.$$

Then $F_i^{\mathrm{G}}(X; P) = c_i(X; P) + b_i(X; P; \mathcal{L}_i(P), \mathcal{L}_i(P))$. These are the c- and b-parts used in the recursive localization. The origin containing the weighted quadratic form is held fixed. These families are formed at the pair itself. Cauchy's estimate in the two linear parameters gives

$$|c_i(X; P)| \le K'_M \|q_i(P)\| e^{-\kappa_+ d_i(X)}, \qquad |b_i(X; P; \mathcal{L}, \mathcal{L}')| \le 8 K'_M c_{\mathrm{cl}}^{-1} \|\mathcal{L}\| \|\mathcal{L}'\| e^{-\kappa_+ d_i(X)}.$$

The bilinear coefficient is bounded for linear terms of any size, using discs of radii $c_{\mathrm{cl}}/(4\|\mathcal{L}\|)$ and $c_{\mathrm{cl}}/(4\|\mathcal{L}'\|)$.

The Gaussian model uses the appropriate jets at the trivial pair. Constant gauge transformations fix that pair and act on the fluctuation by the adjoint representation, which has no invariant vector. Hence $\mathcal{L}_i(1) = 0$. Let $\mathcal{L}_i^{\mathrm{G}}(A; B)$ be the first background-field jet of $\mathcal{L}_i$ at the pair of e^{iA}; it is bilinear

in A, B. Put $Q_i^{\mathrm{G}}(B) = q_i(1; B)$. These jets are independent of the localization radii. At the trivial pair set

$$\mathcal{E}_i^{\mathrm{G}}(\eta)(X; A) = E_i^{\circ}[\eta\mathcal{L}_i^{\mathrm{G}}(A) + Q_i^{\mathrm{G}}](X), \quad F_i^{\mathrm{G,lin}}(X; A) = \mathcal{E}_i^{\mathrm{G}}(0)(X; A) + \tfrac{1}{2}\partial_\eta^2\mathcal{E}_i^{\mathrm{G}}(0)(X; A).$$

Reflection of the fluctuation makes the expansion even in η. The family $F_i^{\mathrm{G,lin}}$ is a polynomial of degree at most two in A, with no linear part. At the trivial pair $F_i^{\mathrm{G}}(X; 1) = c_i(X; 1) = F_i^{\mathrm{G,lin}}(X; 0)$. The pair-dependent bilinear part has no terms of background orders zero or one; its quadratic jet is that of $F_i^{\mathrm{G,lin}}$. The higher jets of the pair-dependent constant part belong to the remainder, not to the Gaussian model.

At the trivial pair the decoupled structure is complete on the torus. The sum over X is therefore the logarithm of the whole Gaussian integral,

$$\sum_X \mathcal{E}_i^{\mathrm{G}}(\eta)(X; A) = \log \int d\mu_i^{\mathrm{G}}(B) e^{\eta\langle l_i(A), B\rangle + Q_i^{\mathrm{G}}(B)},$$

where μ_i^{G} is the normalized Gaussian at the relative weight. This logarithm is quadratic in η. Thus the local terms of orders at least four have zero sum, although they need not vanish polymer by polymer. Completing the square gives

$$\sum_X F_i^{\mathrm{G,lin}}(X; A) = \tfrac{1}{2}\langle l_i(A), \mathcal{G}_i l_i(A)\rangle + \log \mathfrak{Z}_i(w),$$

where $\mathcal{G}_i$ and $\mathfrak{Z}_i$ are the covariance and mass after adding Q_i^{G}. Take the localization sum rule in background/fluctuation bidegrees $(0, 2)$ and $(1, 1)$. It identifies Q_i^{G} and l_i with the old quadratic tree terms transported through the linearized background maps. These terms include the cross terms between different slices. Together with the logarithmic mass of the relative-weight Gaussian origin they give exact successive Gaussian integrations. Their telescoping is the Gaussian identity in the statement, with the constant block mean fixed to remove the harmonic zero mode.

The constructions are made at the pair, before evaluation at the trivial pair. Use the bond tents $h_z(b)$ of [3, eq. 4.cxxvii], so that $\sum_z h_z(b) = 1$. For $\mathcal{W}_\eta = \eta\mathcal{L}_i(P) + q_i(P)$ put

$$\mathcal{W}_{\eta,t}(Y; B) = \mathcal{W}_\eta(Y; tB), \qquad \dot{\mathcal{W}}_{\eta,t,b}(Y; B) = \partial_{t_b}\mathcal{W}_\eta(Y; (t_c B(c))_c)|_{t_c = t}\,.$$

Let $\dot{\mathcal{W}}_{\eta,t,z} = \sum_b h_z(b)\dot{\mathcal{W}}_{\eta,t,b}$. Keeping the first origin fixed, the pointed family is

$$F_{i,\eta}^{\bullet}(X, z) = \int_0^1 E'_{P,i}[\mathcal{W}_{\eta,t}; \dot{\mathcal{W}}_{\eta,t,z}](X)\, dt,$$

where the prime differentiates the second origin. At $t = 0$ that origin is zero, so summing over z gives $\mathcal{E}_i(\eta)$. Its orders zero and two in η give the pointing of F_i^{G}. Thus localization commutes with the split, pointer by pointer. Polarization and $|h_z B| \le |B|$ give the insertion bound $|\eta|\|\mathcal{L}_i(P)\| + 2\|q_i(P)\|$. It follows that

$$|c_i(X; P; z)| \le 2K'_M\|q_i(P)\|e^{-\kappa_+ d_i(X)}, \qquad |F_i^{\mathrm{b}}(X; P; z)| \le 6K'_M c_{\mathrm{cl}}^{-1}\|\mathcal{L}_i(P)\|^2 e^{-\kappa_+ d_i(X)}.$$

These coefficient bounds hold for linear terms of any size; the pointed bilinear part vanishes at the trivial pair. The Gaussian activity has a constant part and a part bilinear in $\mathcal{L}_i$. On radius a their bounds are respectively $CK'_M t_i^{\mathrm{lin}} e^{-\kappa_+ d(X)}$ and $CK'_M c_{\mathrm{cl}}^{-1}(t_i^{\mathrm{lin}})^2 e^{-\kappa_+ d(X)}$. On the fixed radius the latter has the additional factor $(\alpha_0/a)^2$. The scalar localization has lag weight $L^{-(i-1-i')}$ and factor at most $C\,C_{\mathcal{A}}/C_{\mathrm{rad}}$. Since

$$\sum_{i'<i} L^{-(i-1-i')} t_{i'}^{\mathrm{lin}} \le \tfrac{1}{2} t_i^{\mathrm{lin}},$$

the two radius floors in the common constants give $\|\mathcal{L}_i\| \le 17 t_i^{\mathrm{lin}}/16$ and $\|q_i\| \le t_i^{\mathrm{lin}}/16$ on radius a. On the fixed radii the bounds are $\|\mathcal{L}_i\| \le 2(\alpha_0/a) t_i^{\mathrm{lin}}$ and the same bound for q_i. The bilinear families, summed and pointed, are thus at most $10K'_M c_{\mathrm{cl}}^{-1}(t_i^{\mathrm{lin}})^2 e^{-\kappa_+ d_i(X)}$ on radius a and at most

$32K'_M c_{\mathrm{cl}}^{-1}(\alpha_0/a)^2(t_i^{\mathrm{lin}})^2 e^{-\kappa_+ d_i(X)}$ on the fixed radii. The constant-part variation from the trivial pair is at most $K'_M t_i^{\mathrm{lin}} e^{-\kappa_+ d_i(X)}/8$, or twice this at a pointer. This proves inductively the definedness of every Gaussian family used at the next level.

For the higher fluctuation Taylor terms use the absolute localization on a fixed auxiliary disc of radius $\mathsf{C}_{\mathcal{A}}\delta_{\mathrm{ax}}$, where

$$\delta_{\mathrm{ax}} = \min\{1, \mathsf{s}^\star\alpha_0/(1+\beta_s), \varepsilon_{\mathfrak{h}}/\mathsf{C}_{\mathcal{A}}, \delta(e^{2\mathfrak{p}_0})\} = \delta(\mathfrak{t}_{\mathrm{ax}}) > 0$$

is fixed before h_F and γ. It is the same localization on its smaller step disc. Its order-n coefficient costs ρ_i^n, $\rho_i = \delta(\mathfrak{t}_{i-1})/\delta_{\mathrm{ax}} \le 1/2$. The cubic remainder of the bilinear Gaussian activity is therefore bounded by

$$C\rho_i^3(\alpha_0/a)^2 T t_i^{\mathrm{lin}} \le C\phi(x_{i-1}).$$

The constant Gaussian parts have the same bound after localization, by its factor $C\delta(\mathfrak{t}_{i-1})/\alpha_0$. The gauge vertices, explicit closed-loop variations, the mass response and all fermion activities also contribute at most $C\phi(x_{i-1})$. For closed loops use the array bound of Prop. 6.9; their value and first gauge variation vanish at the flat field at every J. An axial gauge on a box containing X and second-order Schwarz give an extra squared curvature threshold; the spare exponential rate pays the box-size polynomial. No through-cube norm at rate κ is inferred from a single spare unit of array decay.

Inductively split the old scalar family, at every pointer, as $F_i^{\mathrm{G}} + F_{i,\ge 4} + F_{i,\mathrm{h}} + F_{i,\mathrm{r}}$, with the Gaussian family taken less its trivial-pair value. Put $K'' = 16c_{\mathrm{ins}}K_M(1+c_{\mathrm{cl}}^{-1})$ and $N_{\mathrm{h}} = K_M(1+c_{\mathrm{cl}})/2$. The inductive bounds are

$$\|F_{i,\mathrm{r}}\|_{i;a} \le C_{\mathrm{r}}^{\mathrm{lev}}\phi(x_{i-1}), \qquad \|F_{i,\ge 4}\|_{i;a} + \|F_{i,\mathrm{h}}\|_{i;a} \le N_{\mathrm{h}}.$$

We carry locality and holomorphy in the independent coordinates through this induction. Write $\mathcal{O}_j^{\mathrm{lev}}$ for the domain (59) and put

$$\mathfrak{C}_j = \left\{\mathfrak{c} = (w, -a_K J, (\sigma_k)_{k<j}) : (w,J) \in \mathcal{O}_j^{\mathrm{lev}}, \quad \|\sigma_k\| < 2B_\sigma\phi(x_k)\ (k<j)\right\}.$$

Here every σ_k is a field on the bare sites with values in $\mathbb{M}$, and its norm is the supremum over those sites. On the finite torus this is an open set in a finite-dimensional complex vector space. At every pointer and fixed point of a core of radius a, each of F_i^{G}, its constant and bilinear parts, $F_{i,\ge 4}$, $F_{i,\mathrm{h}}$ and $F_{i,\mathrm{r}}$ is holomorphic in $\mathfrak{c} \in \mathfrak{C}_j$ and reads these coordinates in its polymer alone. The Gaussian families have this property also on the fixed pair radii. They and $F_{i,\ge 4}$ read only w. Each family is a local, gauge-invariant function of the pair, holomorphic on the cores of radius a. The bounds above hold throughout $\mathfrak{C}_j$.

The first two non-Gaussian families are holomorphic and fourth order in w, uniformly in J and the independent coordinates σ_k. The scalar subtraction includes the zero-source family $\mathbf{d}_i = D(0)|_1 - \widetilde{e}_i$ at each pointer and the zero-source action part $\mathbf{L}_0$: $F_0^{(i)} = \widetilde{E}_0(1,1) + \mathbf{d}_i - \mathbf{L}_0$. Here $\widetilde{E}_0$ denotes degree zero less its trivial-pair value. In particular no new action coefficient is extracted on the shrinking radius. Write the next potential as $\Gamma_i + zV_i^{\mathrm{rest}} + \xi V_i^{\mathrm{h}}$, $\Gamma_i = \mathcal{L}_i + q_i$. Its logarithm is holomorphic for $|z| \le c_{\mathrm{cl}}/(4\|V_i^{\mathrm{rest}}\|)$ and $|\xi| \le c_{\mathrm{cl}}/(4\|V_i^{\mathrm{h}}\|)$. Schwarz's lemma gives changes bounded by $K''\|V_i^{\mathrm{rest}}\|$ and $K''\|V_i^{\mathrm{h}}\|$. This estimate holds at every pointer. The terms from the tests and the quadratic origin do not depend on z, ξ and cancel in these differences; the insertion terms have the uniform bound $2K'_M c_{\mathrm{ins}} c_{\mathrm{cl}} e^{-\kappa_+ d_i(X)}$. At $z = \xi = 0$, expand in $\eta\mathcal{L}_i + q_i$. The quadratic origin throughout this expansion is $\mathcal{V}^{\mathrm{q}} + \mathcal{V}^v$: the q potential together with the quadratic term added by the source weight. Its untested logarithm is the source-dependent $\mathbf{G}^v$ of Prop. 10.1. Thus the pointing formula of Lem. 7.4 uses this origin in both its tested and untested parts, with $\mathbf{G}^v$ as the divisor. The terms of order at least four are fourth order in w and have norm at most $N_{\mathrm{h}}/3$ by Cauchy on $|\eta| = 2$. The terms of orders zero and two differ from F_i^{G} only by the tests, the origin sort and $|x_{i-1}/\mathfrak{t}_{i-1} - 1|$. Lem. 11.3 bounds the change of origin sort, including the pointing insertions. Its all-fluctuation hypothesis holds for the quadratic and linear families above, and there are at most

four origins including an insertion. Removing the tests gives the same marked large-field bound $CK_M e^{-cb_*^2}$. Finally Schwarz's lemma in the coupling disc gives $CK_M|x_{i-1}/\mathsf{t}_{i-1}-1|$. Both errors are $O(\phi(x_{i-1}))$.

More explicitly, with $\bar v$ the supremum of the pointing insertion norm, the four changes at each pointer cost $K^{\mathrm{v}}_{\mathbf{M}}p$ for the tests and the quadratic-origin pointing term, $2K_P^{\Pi}p_\star\bar v/c_{\mathrm{cl}}$ for the sort, $K_P^{\Pi}p_\star\bar v/c_{\mathrm{cl}}$ for removal of tests, and $2K_\Pi\bar v\bar B/(\sigma_\star\mathsf{t}_{i-1})$ for the coupling. The value at the trivial pair adds a factor two; taking the orders zero and two on $|\eta|=2$ adds a factor $5/4$. Since $\bar v\le 9t_i^{\mathrm{lin}}/4<c_{\mathrm{cl}}/3$, $K_P^{\Pi}p_\star\le K'_Mp$, and $p\le\phi(\mathsf{t}_{i-1})^4$, these terms give $C_1^{\mathrm{lev}}\phi(x_{i-1})$ in the pointed family norm. The constant C_1^{lev} does not depend on C_{rad}.

Consequently, if $\varepsilon_i^{\mathrm{r}}=\|F_{i,\mathrm{r}}\|$,

$$\varepsilon_i^{\mathrm{r}}\le C_0^{\mathrm{lev}\prime}\phi(x_{i-1})+\frac{13}{12}K''K_\Lambda^{\mathrm{s}}\frac{\mathsf{C}_{\mathcal{A}}}{C_{\mathrm{rad}}}\max_{i'<i}\varepsilon_{i'}^{\mathrm{r}},\qquad C_0^{\mathrm{lev}\prime}=K''C_0^{\mathrm{lev}}+C_1^{\mathrm{lev}}+C_w+\theta_sh_F^2(\bar B_M+2B_\sigma).$$

The floor $C_{\mathrm{rad}}\ge 512c_{\mathrm{ins}}(1+c_{\mathrm{cl}}^{-1})\mathsf{C}_{\mathcal{A}}K_\Lambda^{\mathrm{s}}K_M$ makes the last coefficient at most $(13/12)(16/512)<1/24$. It is imposed before C_0^{lev} and $C_0^{\mathrm{lev}\prime}$ are formed. Since the earlier $x_{i'-1}$ are larger and ϕ decreases, $C_{\mathrm{r}}^{\mathrm{lev}}=2C_0^{\mathrm{lev}\prime}$ closes the induction. The same coefficient bounds $F_{i,\mathrm{h}}$ by $N_{\mathrm{h}}/24$; together with $N_{\mathrm{h}}/3$ this closes the higher-order bound. Moreover $F_{i,\mathrm{h}}=\int_0^1\partial_\xi\widetilde{E}_0(1,\xi)\,d\xi$ is fourth order in w, because the derivative is linear in V_i^{h}. With $C_{\mathrm{rest}}=C_0^{\mathrm{lev}}+C_{\mathrm{r}}^{\mathrm{lev}}$, the ceiling $8C_{\mathrm{rest}}\phi(\gamma^{-2}/2)\le c_{\mathrm{cl}}$ makes every z disc have radius at least two.

The additional assertions in the induction follow alongside these bounds. By Prop. 10.1, the explicit members of the step and its fermion covariances are holomorphic in the independent coordinates. A walk reads n on the same set on which it reads the configuration. Every expanded leg is assigned to a polymer containing that walk; the mass-coordinate terms read the bare sites of their assigned polymers. Thus a potential member at Y reads the coordinates in Y alone.

For a localized earlier scalar family, the induction gives a holomorphic family of bare arrays with zero counterterms. The parameter version of scalar localization preserves holomorphy at a fixed pair and fluctuation. Its restriction to Y uses only the entries of that array on polymers contained in Y. Two coordinate vectors agreeing on Y give the same restricted array, and hence the same localized member there. Uniform bounds on the field polydisc give joint holomorphy in the fluctuation and coordinates. Taking the first two Taylor terms in the fluctuation preserves these properties.

The cluster expansion at X reads potential members with $Y^\sharp\subset X$ and the fermion covariance decoupled to cubes in X. Those covariances read n in X alone. Hence the expansion is holomorphic in $(\mathfrak{c},z,\xi)$ and in $(\mathfrak{c},\eta)$ on the discs used above, and reads the coordinates in X alone. This also holds at every pointer: the test and quadratic-origin terms read only w locally, and the bond and fermion insertions are derivatives of the same local holomorphic families. The estimates for an inserted origin justify their parameter integrals.

Differences in z,ξ, Cauchy integrals on $|\eta|=2$, and evaluation at the trivial pair preserve coordinate holomorphy and locality. They therefore preserve these properties in the definitions of all four families. The gauge identities hold before these linear operations; their scalar parts are invariant and functions of the pair. This closes the additional clauses of the induction. At step j the same argument applies to the new activity, since its potential uses only the previously constructed families.

At the trivial pair $\mathcal{L}_j=0$: a linear gauge functional there would be an invariant adjoint colour vector. At this pair, compare the tested expansion with origins $\mathcal{V}^{\mathrm{q}}+\mathcal{V}^{v}$ and Q_j^{G} with the untested expansion at coefficient $\tau=1$. Changing both sorts costs $2K_P^{\Pi}p_*$, and removing the tests costs $K_P^{\Pi}p_*$. The subtracted zero-source value costs at most a further $3K_P^{\Pi}p_*$. Changing the coefficient

costs $2K_\Pi\|Q_j^{\mathrm{G}}\|\bar B/(\sigma_\star \mathtt{t}_{j-1})$. Their sum is bounded by

$$\left[6K_P^\Pi p_* + \frac{2K_\Pi\|Q_j^{\mathrm{G}}\|\bar B}{\sigma_\star \mathtt{t}_{j-1}}\right] e^{-\kappa_+ d_j(X)} \le C\phi(x_{j-1})e^{-\kappa_+ d_j(X)}.$$

The Gaussian divisor is compared separately after subtracting its zero-source value. On $|\tau'-1|<\sigma_\star/2$, keep $\tau' v = v'_{j-1}$ fixed. Then $\mathcal{V}^v$ is unchanged, and the norm of $\mathcal{V}^{\mathrm{q}}+\mathcal{V}^v$ is below $c_{\mathrm{cl}}/2$. The pointed Gaussian bound of Prop. 10.1, summed over at most $8M^3 e^{d_j(X)}$ sites, and Schwarz's lemma on this disc give

$$|\mathbf{G}^v(X;1) - \mathbf{G}_\tau(X;1) - \mathbf{G}_j^{v'}(X)| \le 32M^3 C_G^v(1+\sigma_\star^{-1})\frac{\bar B}{\mathtt{t}_{j-1}} e^{-\kappa d_j(X)}.$$

Here $\mathbf{G}_j^{v'}$ is the relative-weight logarithm at $\tau'=1$, and C_G^v is the fixed pointed bound from that proposition. The loss of one decay unit pays the site count. This error belongs to e_j^{I}. The scalar Gaussian constant, together with $\mathbf{G}_j^{v'}$, gives e_j^{G}; the free fermion normalizer gives e_j^{F}, the fourth-order families give e_j^{H}, and the remaining terms give e_j^{I}. The Gaussian sums telescope by exact Gaussian integration. In this identity the linear term is $\mathcal{L}_i^{\mathrm{G}}$, the first background-gauge jet of $\mathcal{L}_i$, and the quadratic term is $Q_i^{\mathrm{G}} = q_i(1)$; the earlier tree terms are generated by $\mathcal{L}_i^{\mathrm{G}}$. The pair-dependent parts have already been assigned to the small remainder. The fermion logarithmic germs telescope by Lem. 5.5. This proves both sum identities.

The tail. Let $s, s' \in \mathcal{O}_j^{\mathrm{lev}}$, $S = \mathrm{supp}(s-s')$, and $D = \mathrm{dist}(X,S)$. If X meets S, the two bounds for e_j^{I} suffice, after doubling their constant. Otherwise $w = w'$ and $J = J'$ on X. The source-chain bound gives, at every bare site of X and for $k<j$,

$$|\sigma_k(s) - \sigma_k(s')| \le C_{\mathrm{t}}\phi(x_k)e^{-\kappa_\sigma D/\lambda_j},$$

since $\lambda_k \le \lambda_j$ and both sources belong to $\mathscr{O}_k$.

We first record what this implies for a local holomorphic function f of the independent coordinates, bounded on $\mathfrak{C}_j$ by $M_0 e^{-\kappa d_j(X)}$. Set

$$\delta_k = (\sigma_k(s) - \sigma_k(s'))1_X, \qquad \theta = \frac{C_{\mathrm{t}}}{B_\sigma} e^{-\kappa_\sigma D/\lambda_j}.$$

Keep $w', -a_K J'$ fixed and replace every $\sigma_k(s')$ by $\sigma_k(s') + \zeta\delta_k$. This affine curve lies in $\mathfrak{C}_j$ for $|\zeta| < 1/\theta$, because

$$\|\sigma_k(s') + \zeta\delta_k\| \le B_\sigma\phi(x_k)(1+|\zeta|\theta) < 2B_\sigma\phi(x_k).$$

When $\theta<1$, its value at $\zeta=1$ agrees with the chain of s on X. Locality identifies the values of f there. Schwarz's lemma applied to the difference from $\zeta=0$ gives $2\theta M_0 e^{-\kappa d_j(X)}$. When $\theta \ge 1$, the bounds at the two chain points give the same inequality. If $\theta = 0$ the coordinate vectors agree on X and the difference vanishes. Consequently

$$|f(\mathfrak{c}_{j-1}(s)) - f(\mathfrak{c}_{j-1}(s'))| \le C_{\mathrm{tail}} M_0 e^{-\kappa d_j(X)} e^{-\kappa_\sigma D/\lambda_j}, \qquad C_{\mathrm{tail}} = 2\max\{1, C_{\mathrm{t}}/B_\sigma\}.$$

Apply this estimate at each z on the disc

$$|z| \le R = \frac{c_{\mathrm{cl}}}{4C_{\mathrm{rest}}\phi(x_{j-1})} \ge 2$$

to $\mathcal{E}(z;\mathfrak{c})$, the degree-zero part at the trivial pair of the step-j activity $\widetilde{E}(z,0)(X)$, before its scalar value is subtracted. The preceding coordinate induction applies on all of $\mathfrak{C}_j$, not only at the coordinate vectors of physical sources. The potential is within c_{cl} of the quadratic origin, so the cluster bound is

$$|\mathcal{E}(z;\mathfrak{c})| \le K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}.$$

It follows that $\Delta(z) = \mathcal{E}(z;\mathfrak{c}_{j-1}(s)) - \mathcal{E}(z;\mathfrak{c}_{j-1}(s'))$ satisfies

$$|\Delta(z)| \le C_{\mathrm{tail}} K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}e^{-\kappa_\sigma D/\lambda_j}.$$

At $z = 0$ the activity has only the Gaussian scalar potential and quadratic origin. These read w in X alone, so $\Delta(0) = 0$.

The definition of the decomposition gives

$$e_j^{\mathrm{I}}(X; s) - e_j^{\mathrm{I}}(X; s') = \Delta(1).$$

Indeed, the gauge normalizer, weighted Gaussian part and e_j^{G} read w in X alone and have equal values at the two sources. The free fermion normalizer cancels with e_j^{F}. The change obtained by turning on ξ is precisely e_j^{H} and is removed in e_j^{I}. A second application of Schwarz's lemma, now in z, therefore gives

$$\begin{aligned} |e_j^{\mathrm{I}}(X; s) - e_j^{\mathrm{I}}(X; s')| &\le R^{-1} \sup_{|z| \le R} |\Delta(z)| \\ &\le 4 C_{\mathrm{rest}} C_{\mathrm{tail}} K_M (1 + c_{\mathrm{cl}}^{-1}) \phi(x_{j-1}) e^{-\kappa d_j(X)} e^{-\kappa_\sigma D/\lambda_j}. \end{aligned}$$

Take C_I to be the larger of this constant and twice the preceding bound constant for e_j^{I}. Since $\phi(x_{j-1}) \le \phi(x_j)$, this proves the asserted tail. The constants C_{tail} and C_I are fixed before m_-. □

11.3. **The free templates and the sum over scales.**

Lemma 11.5 (The templates)**.** For a test pair with $8a_K \le d \le 1$, $20d \le \ell$ and $10m_+ d \le 1$, the Gaussian sum of mixed derivatives is $\frac{9}{16\pi^2}\mathsf{b}^{-2}\mathcal{K}_6$ with error at most

$$C\mathsf{b}^{-2}\left[\frac{a_K}{\vartheta d} + (d/\ell)^3 + d^3\right] \|f\|_\infty \|h\|_\infty.$$

The free fermion sum is $\delta_{\Gamma\Gamma'}(2\pi^2)^{-1}\mathcal{K}_4$ with error at most

$$C\left[\left(\frac{a_K}{\vartheta d} + (m_+ d)^2 + (d/\ell)^2\right) d^2 + d^6 m_-^{-2} + d^4 m_-^{-1}\right] \|f\|_\infty \|h\|_\infty.$$

■

Proof. In the component norm on $\mathfrak{su}(2)$, the quadratic part of $a_p(e^{iA})$ is $\frac{1}{2}|(dA)_p|^2$. The Gaussian weight is therefore $\exp[-\frac{\beta}{2}\sum_p |(dA)_p|^2]$. The flat constrained-curvature identity [3, Prop. 7.3] identifies the curvature covariance per colour as $\beta^{-1}C_K^{\mathrm{free}}$, with C_K^{free} the closed-curvature projection. Its canonical step hypotheses hold by Prop. 10.20. Thus the connected covariance of the quadratic action densities is

$$\frac{\dim \mathfrak{su}(2)}{2}\, \beta^{-2} C_K^{\mathrm{free}}(p, q)^2.$$

The two source derivatives supply a_K^{-2}, and $\beta a_K = \mathsf{b}$. Away from coincidence the continuum kernel of the projection is the dipole matrix $\mathbb{K}_3(x) = (3xx^{\mathsf{T}} - |x|^2 I)/(4\pi|x|^5)$. The squared entries sum to $3/(8\pi^2|x|^6)$. The free case of Prop. A.2, including its lattice-kernel comparison and Riemann-sum estimate, gives error $C[a_K/(\vartheta d) + (d/\ell)^3]$ on the pair. Wick's rule multiplies this square by $\dim \mathfrak{su}(2)/2 = 3/2$, giving $9/(16\pi^2)$. For the scale-one remainder use instead the all-distance bound [3, Prop. 7.5]:

$$|C_{K,s}^{\mathrm{bg}}(p, q)| \le C(a_K/a_s)^3.$$

At $a_s = 1$ it is Ca_K^3. Thus the square of this kernel and its cross terms with the free kernel are bounded, after the mesh factors, by $C(1 + |x - y|^{-3})$. Integrating over the test pair gives a contribution at most $C\mathsf{b}^{-2}d^3\|f\|_\infty\|h\|_\infty$.

For the fermions put $G_0 = (D[1] + m_0)^{-1}$. The free symbol's light Schur complement is $m_0 + i\gamma \cdot p + O(|p|^2)$, with the differentiated bounds proved above. A smooth dyadic partition and integration by parts give, for $n \ne 0$,

$$\iota^* G_0 \iota(n) = \mathscr{G}_{m_0}(n) + O(|n|^{-3}), \qquad |\Pi_1 G_0 \iota(n)| + |\iota^* G_0 \Pi_1(n)| \le C|n|^{-3}, \quad |\Pi_1 G_0 \Pi_1(n)| \le C|n|^{-4},$$

where

$$\mathscr{G}_m(n) = \frac{\gamma(n)(1+m|n|)e^{-m|n|}}{4\pi|n|^3} + \frac{me^{-m|n|}}{4\pi|n|}.$$

Indeed the light error and mixed symbols have degree zero, and the nonanalytic heavy term has degree one; their dyadic Fourier sums have respectively the displayed third and fourth powers. The analytic heavy inverse has exponential decay. These bounds are uniform in $0 \le m_0 \le 1/2$.

The light propagator between individual chambers has factor 48^{-1}, which cancels the 48^2 chamber pairs in the loop. For the reflection-real basis,

$$-\operatorname{tr}[\Gamma\gamma(u)\Gamma'\gamma(-u)] = 8\delta_{\Gamma\Gamma'}|u|^2.$$

This follows from $\operatorname{tr}(t_j t_{j'}) = \delta_{jj'}$, $\tau_3\tau_a\tau_3 = \epsilon_a\tau_a$ and $c_a^2\epsilon_a = 1$. The scalar-vector cross term has zero spin trace. The remaining mass correction is $O((m_+ d)^2)$, and a heavy factor costs $O(a_K/d)$. Thus $-a_K^2 \operatorname{Tr}(\hat{f}\Gamma G_0 \hat{h}\Gamma' G_0)$ has leading coefficient $8/(16\pi^2) = 1/(2\pi^2)$. The Riemann-sum error is $Ca_K/(\vartheta d)$ relative to $d^2\|f\|_\infty\|h\|_\infty$. The antiperiodic image formula and positive m_- give the additional $C(d/\ell)^2$ term.

Finally $\|\mathbb{D}_K(1;0)^{-1}\| \le C/m_-$. Factor the first source derivative into its two background rows. Their squared row sums are $O(a_K^2)$, so its trace norm is at most $Ca_K^3 \sum_x |f(x)| \le Cd^3\|f\|_\infty$. For the second derivative use the resolvent between the separated supports; its $Ca_K^2|x-y|^{-2}$ bound gives trace norm at most $Cd^4\|f\|_\infty\|h\|_\infty$. The two terms in the second logarithmic derivative are therefore bounded by $C(d^6 m_-^{-2} + d^4 m_-^{-1})$. This uses trace norms of the localized factors, with no global volume or rank factor. It is the last-level subtraction in the free fermion sum. □

Lemma 11.6 (The interaction error)**.** For the packing in Lem. 11.2, the mean of the mixed derivatives of $\sum_{j,X} e_j^{\mathrm{I}}$ is bounded, at bilinear indices, by

$$CR_J^{-2}\phi(\mathsf{b}/d)d^2\|f\|_\infty\|h\|_\infty,$$

and at action-density indices by

$$C\left(\frac{40M}{\vartheta\varrho}\right)^2 \phi(\mathsf{b}/d)\widehat{\phi}(\mathsf{b}/d)^2\|f\|_\infty\|h\|_\infty.$$

■

Proof. On (59) the two source radii are

$$r_{\Gamma,j} = \frac{R_J}{\lambda_j\|f\|_\infty}, \qquad r_{0,j} = \frac{\varrho}{\widehat{\phi}(x_j)\max(\|f\|_\infty, 40M\lambda_j \operatorname{Lip} f)}.$$

Cauchy's estimate gives $C\phi(x_j)e^{-\kappa d_j(X)} r_{\alpha,j}^{-1} r_{\alpha',j}^{-1}$. If $\lambda_j \le d$, a mixed derivative must connect the two supports. The tail in Lem. 11.4 gives a factor e^{-cd/λ_j}, and the number of nearby cells is $C(d/\lambda_j)^3$. If $\lambda_j > d$, apply the same independent signs to the packed sources. The packing meets at most $CN\mathsf{N}(d/\lambda_j)^3$ cells; after taking the mean its contribution is again bounded by $C(d/\lambda_j)^3$ times the single-cell Cauchy bound. This volume factor is essential.

At bilinear indices sum

$$C\sum_j \phi(x_j)\lambda_j^2(d/\lambda_j)^3 \begin{cases} e^{-cd/\lambda_j}, & \lambda_j \le d, \\ 1, & \lambda_j > d. \end{cases}$$

Since $x_j \asymp \mathsf{b}/\lambda_j$, both geometric sums are led by $\lambda_j \asymp d$ and are bounded by $C\phi(\mathsf{b}/d)d^2$. For the action density replace λ_j^2/R_J^2 by

$$\varrho^{-2}\widehat{\phi}(x_j)^2 \max(1, 40M\lambda_j/(\vartheta d))^2.$$

Below d the exponential cuts the sum. Above d its terms are bounded by $C\phi(x_j)\widehat{\phi}(x_j)^2 d/\lambda_j$, again led by $\lambda_j \asymp d$. The polynomial logarithms are paid by the geometric ratios. This proves both bounds uniformly in the number of levels. □

Proof of Prop. 11.1. The fourth-order term in Lem. 11.4 has zero mixed derivative at both kinds of indices. The Gaussian part does not read J, and the free fermion part does not read w. Combine Lems. 11.2, 11.5 and 11.6 and divide by (57). On the lattice the additional relative cutoff error is $Ca_K/(\vartheta^3 d)$. On each volume sequence $20d \le \ell$ eventually. First take the volume limit, then the cutoff subsequence at each fixed test pair. These cutoff and image errors vanish. The remaining bounds tend to zero with d, since $\phi(\mathsf{b}/d)\widehat{\phi}(\mathsf{b}/d)^2 \to 0$; the packing remainder is $O(d)$ after division by either template. Taking d small enough gives a positive lower bound at every equal pair of indices. This proves nonvanishing. □

12. Limits taken volume first

At cutoff K choose a limit ν_K of the torus gauge marginals as their physical sides tend to infinity through $\ell = 2L^m$ satisfying the room condition above. Such limits exist by Prop. 2.5. Their local expectations are $\omega_{\nu_K}(F) = \int \langle F\rangle_U \, \nu_K(dU)$. Let S^T_{n,K,ν_K} denote the cumulants of the physical observables defined in the introduction. Bounds that are uniform in the torus side pass to these expectations, and hence to their cumulants, by Prop. 2.5.

Proposition 12.1 (Compactness on separated supports)**.** For any sequence $K_j \to \infty$ and any choices ν_{K_j}, there is a common subsequence on which all the cumulants of order at least two converge on every separated tuple of $C^1_c(\mathbb{R}^3)$ functions. The limits S_n satisfy the bound of Thm. F. At fixed indices they are distributions of locally finite order off the coincidence diagonals. They obey the permutation, translation, cubic, flavour and conjugation identities of the lattice model. ■

Proof. Fix the indices, number of entries and pairwise separated compact support sets, each in a ball of radius R. The bound of Thm. F is uniform in K, the volume and the positions of those balls. It gives equicontinuity in the product of the insertion norms. Choose the support sets to be finite unions of closed rational cubes and choose a countable dense set of C^1 functions in each of them. These spaces are separable both for the C^1 norm used by the action density and for the supremum norm used by a bilinear. A diagonal subsequence makes all the resulting cumulants converge. The uniform multilinear bound extends convergence to every separated tuple: its supports fit inside pairwise separated sets of the indicated countable kind. The same bound passes to the limit. The kernel theorem on products of separated open sets gives distributions there, with agreement on overlaps by uniqueness.

The exact finite-lattice identities pass to the limit. For a real translation v, choose $v_j \in a_{K_j}\mathbb{Z}^3$ converging to v. Translated test functions converge in their insertion norms; for the C^1 norm this uses uniform continuity of their first derivatives. Their supports remain in fixed separated neighborhoods. Equicontinuity and exact lattice translation invariance give invariance under v. □

Let σ be the centered functional of (2). It is the limit of the corresponding centered lattice products. It is useful to keep centering in this formula: no convergence of the uncentered one-point functions is needed.

Proposition 12.2 (Positivity at coordinate and diagonal planes)**.** The functional σ is reflection positive at every coordinate plane and every plane with normal $e_\mu \pm e_\nu$. Reflection acts geometrically on the reflection-real indices. ■

Proof. The plane $P_0 = \{x_1 = x_2\}$ is a mirror of every lattice $a_K\mathbb{Z}^3$. For a finite combination $F = \sum_i z_i X_i$ whose supports lie strictly on one side, the centered lattice products F_K lie in its

even half algebra for all large K. Centering commutes with reflection, since $\omega_{\nu_K}(\Theta_{P_0}G) = \overline{\omega_{\nu_K}(G)}$. Therefore

$$0 \le \omega_{\nu_K}(F_K \Theta_{P_0} F_K) = \sum_{i,l} \bar{z}_i z_l \, \sigma_K(\theta_{P_0} X_i \cdot X_l).$$

Here the lattice positivity is Prop. 3.1; each concatenation is separated across the plane. The partition formula and convergence of the cumulants pass this inequality to σ. Translations and cubic covariance give all the diagonal planes stated. The same argument at $x_3 = 0$ gives the coordinate planes. □

13. Rotations and reconstruction for the centered fields

Let $\mathcal{I} = \{0\} \cup \mathfrak{B}$ be the finite reflection-real index set, with signs $\pi_\alpha \in \{1, -1\}$. For a tuple $X = ((\alpha_i, f_i))_{i=1}^n$ of real $C_c^1(\mathbb{R}^3)$ functions with pairwise disjoint supports, write $\pi(X) = \prod_i \pi_{\alpha_i}$. The geometric action is $rX = ((\alpha_i, f_i \circ r^{-1}))_i$. Let σ be normalized, multilinear and permutation symmetric on these tuples, with

$$\sigma(\varnothing) = 1, \qquad \overline{\sigma(X)} = \pi(X)\sigma(X).$$

It extends complex linearly to the entries and to finite formal combinations of tuples wherever the indicated concatenation is separated. If $F = \sum_i \zeta_i X_i$ and $G = \sum_j \xi_j Y_j$ lie in the positive side of a plane,

$$\sigma(\theta \bar{F} \cdot G) = \sum_{i,j} \bar{\zeta}_i \xi_j \sigma(\theta X_i \cdot Y_j).$$

Reflection acts geometrically and introduces no index sign in this basis. The sign in improper Euclidean covariance is a different operation.

For the centered cumulants of the preceding sections, σ is given by (2). Thus every singleton has value zero. This is a functional of even observables; it is not assumed to be the moment functional of a positive joint probability law for the bilinears.

13.1. **The analytic hypotheses.** Set $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\|\nabla f\|_\infty\}$, $\|f\|_0 = \|f\|_\sharp$ and $\|f\|_\Gamma = \|f\|_\infty$ for $\Gamma \in \mathfrak{B}$. We use the following assumptions on σ.

- (R0) $\sigma((\alpha, f)) = 0$ for every single entry.
- (R1) It is translation invariant and satisfies $\sigma(wX) = \pi(X)^{\delta(w)}\sigma(X)$ for $w \in W_3$, where $\delta(w) = 1$ for an improper map and zero otherwise.
- (R2) For each $n \ge 2$, $d, R > 0$ there is $C_n(d; R)$ such that $|\sigma(X)| \le C_n(d; R) \prod_i \|f_i\|_{\alpha_i}$ whenever the supports are pairwise at distance at least d and each lies in a ball of radius R, independently of the positions of those balls.
- (R3) $\sigma(\theta\bar{F} \cdot F) \ge 0$ for $\theta(x_1, x_2, x_3) = (x_1, x_2, -x_3)$ and every finite combination supported in $x_3 > 0$.
- (R4) In the reflection Hilbert space at this plane, the joint energy and first tangential momentum satisfy $H \ge |P_1|$.
- (R5) There are $0 < c_{\mathrm{I}} \le 1/16$, $C_{\mathrm{I}} < \infty$ and $p_{\mathrm{I}} \ge 0$ such that, for $0 < \eta \le 1$, $|y_3| \le \eta/8$, real $h \in C_c^1(B(y, c_{\mathrm{I}}\eta))$ and combinations F, G supported in $x_3 > \eta/4$,

$$|\sigma(\theta\bar{F} \cdot (\alpha, h) \cdot G)| \le C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} \|h\|_\sharp \|[\tau_{-\eta e_3/4} F]\| \|[\tau_{-\eta e_3/4} G]\|.$$

- (R6) For each fixed α, its two-point form is reflection positive also at $\Sigma_D = \{x_1 + x_3 = 0\}$: for real f_i supported on one open side,

$$\sum_{i,j} \bar{\zeta}_i \zeta_j \sigma((\alpha, f_i \circ \theta_D), (\alpha, f_j)) \ge 0.$$

(R7) In coordinates $y = (x_3, x_1)$, a label is $\lambda = (\alpha, \upsilon)$, $\upsilon \in C^1_c(\mathbb{R})$ real. Let Prof_t consist of C^1_c profiles supported in $[t, 2t] \times [-t, t]$, with $|a| \le 3$ and $\operatorname{Lip} a \le 12/t$. There is $a_\Gamma < 4$ such that, for every label,

$$0 \le \sigma((\alpha, (a \otimes \upsilon) \circ \theta), (\alpha, a \otimes \upsilon)) \le C_\lambda t^{-a_\Gamma}, \qquad a \in \mathsf{Prof}_t, \quad 0 < t \le 1.$$

The Hilbert space in (R4)–(R5) is the completion of the positive-time combinations modulo the null space of $\sigma(\theta\bar{F} \cdot G)$, and $\Omega = [\varnothing]$. Assumptions (R1)–(R3) make its translations $T_t[F] = [\tau_{te_3} F]$ and $\mathcal{U}(v)[F] = [\tau_v F]$, $v \perp e_3$, respectively a strongly continuous self-adjoint contraction semigroup e^{-tH} and a commuting unitary representation. Indeed reflection and translation make T_t symmetric, and repeated Cauchy–Schwarz gives

$$\|T_t x\| \le \|x\|^{1-2^{-j}} \|T_{2^j t} x\|^{2^{-j}}.$$

The position-independent bound in (R2) bounds the limsup of the second factor by one. Null vectors are preserved by the same inequality, and moving-entry continuity follows from (R2). A countable dense set of C^1 tuples proves separability. In particular e^{-tH} is injective.

Proposition 13.1 (The two-plane spectral inequality)**.** Suppose the translation-invariant separated functional has the position bounds and reflection positivity at $e_3^\perp$ and at $(e_3 + \epsilon e_k)^\perp$, $k = 1, 2$, $\epsilon = \pm 1$. Then its joint spectral measure in the e_3 picture is supported on $E \ge |q_k|$. ■

Proof. Put $e_\pm = (e_3 \pm \epsilon e_k)/\sqrt{2}$ and $W = \{x_3 > |x_k|\}$. For a combination F supported in W, let μ_F be its joint spectral measure carried to the right half-plane by $w = (E - i\epsilon q_k)/\sqrt{2}$. Then

$$g(s, s') = \sigma(\theta\bar{F} \cdot \tau_{se_+ + s'e_-} F) = \int e^{-sw - s'\bar{w}}\, d\mu_F(w), \qquad s, s' \ge 0.$$

Let θ_+ be reflection in $e_+^\perp$ and put $F_1 = \theta_+ \theta F$. Both F_1 and F lie on the positive side of this plane. In its reflection picture,

$$g(s, s') = \langle [F_1]_+, T_s^+ \mathcal{U}^+(s'e_-)[F]_+ \rangle_+.$$

For $\varphi \in L^1(0, \infty)$ and its Laplace transform $q(z) = \int_0^\infty e^{-zs} \varphi(s)\, ds$, Fubini and the spectral theorem in these two pictures give

$$\int |q(w)|^2\, d\mu_F(w) \le \|[F_1]_+\|\, \|[F]_+\| \sup_{u \ge 0} |q(u)| \sup_{y \in \mathbb{R}} |q(iy)|.$$

Take $\varphi(s) = s^{a-1} e^{-\eta s}/\Gamma(a)$, where $a = \varepsilon + i\kappa$, $\varepsilon, \eta > 0$. Thus $q(z) = (z + \eta)^{-a}$. Its two suprema are bounded by $\eta^{-\varepsilon}$ and $\eta^{-\varepsilon} e^{|\kappa|\pi/2}$. Fatou, first as $\varepsilon \downarrow 0$ and then as $\eta \downarrow 0$, gives

$$\int_{w \ne 0} e^{2\kappa \arg w}\, d\mu_F(w) \le \|[F_1]_+\|\, \|[F]_+\| e^{|\kappa|\pi/2}.$$

Letting κ tend to $+\infty$ and $-\infty$ excludes every closed sector with $|\arg w| > \pi/4$. This says $E \ge |q_k|$ on the measure of $[F]$. Every positive-time tuple can be translated into W by τ_{re_3} for large r. The bad spectral projection $Q = 1_{\{E < |q_k|\}}(H, P)$ then satisfies $e^{-rH} Q[X] = Q[\tau_{re_3} X] = 0$. The semigroup is injective, so $Q[X] = 0$ on a dense set and hence $Q = 0$. □

Theorem G (Finite-family rotation criterion)**.** A functional with (R0)–(R7) is covariant under every Euclidean isometry:

$$\sigma(rX) = \pi(X)^{\delta(r)} \sigma(X).$$

It is reflection positive at every plane and its joint energy–momentum spectrum is contained in $\{(E, q) : E \ge |q|\}$. ■□

Proof. The slab and point-plate constructions needed here are proved in Props. A.6 and A.7. They use the coordinate reflection pictures, the cone $H \ge |P_\nu|$, (R5), and continuity (R2). Their product identity is recorded in (A.xviii). We give the finite-label rotation argument, including the step in which the sign rule matters.

Pictures and point plates. Transporting reflection positivity by $w \in W_3$ multiplies the coefficient of a tuple by the real sign $\pi(X)^{\delta(w)}$ and does not change positivity. Thus each coordinate plane has a reflection picture. Axis permutations carry (R4) to every ordered pair of distinct axes. Spectral calculus consequently gives the contraction semigroup

$$T(z) = \int e^{-z_3 E + i(z_1 q_1 + z_2 q_2)}\, d\mathscr{E}, \quad \operatorname{Re} z_3 \geq |\operatorname{Im} z_1| + |\operatorname{Im} z_2|,$$

holomorphic in its open tube. For its planar restriction write $d_{\mathrm{Tb}}(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1|$; for the three-dimensional point construction write $d_3(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1| - |\operatorname{Im} z_2|$. The slab operator $B_l(\alpha, h)[G] = [(\alpha, h) \cdot \tau_{le_3} G]$ is defined for $\operatorname{supp} h \subset \{0 < x_3 < l\}$ and is bounded by (R5), with

$$B_l(\alpha, h)^* = B_l(\alpha, \overline{h \circ \theta_{l/2}}).$$

This follows by reflecting, translating by $-le_3$ and using permutation symmetry in its matrix elements. There is no index sign in this adjoint identity. For a profile supported in $|x_3| < d$ put $B_d^{\mathrm{c}}(\alpha, h) = B_{2d}(\alpha, \tau_{de_3} h)$.

Let Υ_λ be the length of the interval hull of $\operatorname{supp} \upsilon$ and put $\wp = p_{\mathrm{I}} + 4$. The point-plate construction gives a holomorphic insertion $B_\lambda(p, q)$ on the planar tubes with

$$\|B_\lambda(p, q)\| \leq C_{\mathrm{plt}}(1 + \Upsilon_\lambda)\|\upsilon\|_{C^1} \min\{1, d_{\mathrm{Tb}}(p), d_{\mathrm{Tb}}(q)\}^{-\wp}.$$

It is useful to recall its mechanism. Choose a nonnegative radial $j \in C_c^1(\mathbb{R}^2)$ of integral one, supported in $|y| \leq 1/2$, and put $j_\delta(y) = \delta^{-2} j(y/\delta)$. Mollify a point plate by j_δ and put $B^\delta(p, q) = T(p - \delta e_3) B_\delta^{\mathrm{c}}(j_\delta \otimes \lambda) T(q - \delta e_3)$. The convolution identity recovers every ordinary slab profile. A holomorphic mean-value formula on a disc of radius proportional to the tube distance, followed by (R5), bounds these operators independently of δ. Cauchy's estimate and symmetric convolution make them Cauchy in norm at positive real times; the shift law extends their limit throughout the tube. This proof uses only the displayed norm of the fixed label, so it applies separately to each α.

Products give kernels $K_n(Z; \lambda_1, \dots, \lambda_n)$, holomorphic on the tubes for every time order and representing σ on time-ordered plates. The quarter turn in the (x_3, x_1) plane is proper and belongs to W_3, so the kernels patch in four frames. For two ordered lists the inner product of their vectors is the kernel with the first list reflected and reversed. This is the labelled Gram identity; no relation between two distinct labels has been used.

Harmonics and their top coefficient. The four-frame kernel is holomorphic on the union $\mathcal{Z}_n$ of the ordering tubes in (A.xix). Its real domain $\mathcal{O}_n$ consists of configurations whose time coordinates are all distinct or whose tangential coordinates are all distinct. Lem. A.9 constructs unique holomorphic coefficients on $\mathcal{Z}_n$ such that

$$K_n(R_{is} Z; \underline{\lambda}) = \sum_{4|m| \leq \wp n} c_m(Z; \underline{\lambda}) e^{-4ms}.$$

The proof first obtains an entire angular function on a nonempty open set of narrow-sector configurations. It then uses a fixed Vandermonde inverse to define the coefficients throughout $\mathcal{Z}_n$ and the identity theorem to obtain the displayed expansion. The same lemma extends these coefficients to real-analytic functions on every distinct-point configuration, with the rotation phase (A.xxi), also for rotations about arbitrary centers. We write c_m for these extensions as well. The sign rule gives

$$\overline{c_m(Z; \underline{\lambda})} = \pi(\underline{\lambda}) c_{-m}(Z; \underline{\lambda})$$

for real configurations. Consequently nonzero extreme harmonics occur in both directions, even for mixed scalar and pseudoscalar labels.

Let $\bar{m}(n)$ be the largest $|m|$ that is nonzero for some n-point configuration and some labels; put $\bar{m}(n) = 0$ if all these harmonics vanish. In particular $\bar{m}(0) = \bar{m}(1) = 0$. Apply Cauchy–Schwarz to the labelled Gram identity at R_{is} and let $s \to +\infty$. Comparing leading exponents gives

$$2\bar{m}(a + b) \leq \bar{m}(2a) + \bar{m}(2b).$$

For $2a$ points multiply each finite Gram matrix by $e^{-4\bar m(2a)s}$ and take its limit. The result, $t_a(\theta S\cup T)$, is positive Hermitian. The rotation phase transports its Schwarz inequality to any line L:

$$|t_a(\theta_L S\cup T)|^2 \le |t_a(\theta_L S\cup S)|\,|t_a(\theta_L T\cup T)|.$$

If some times coincide, perturb them by distinct small amounts and use continuity. This preserves the same top coefficient.

Polygons and halving. Let $\mathcal{N}_a(S)$ mean that t_a vanishes on every labelled $2a$-point configuration containing S. If $S\subset\theta_L S'\cup S'$, S' lies on one side of L and $\#S'\le a$, then $\mathcal{N}_a(S)$ implies both $\mathcal{N}_a(S')$ and $\mathcal{N}_a(\theta_L S')$. For a configuration $X\supset S'$ on that side the diagonal value $t_a(\theta_L X\cup X)$ vanishes. The last Schwarz inequality makes every corresponding mixed value vanish. These mixed configurations form a nonempty open set in the free point positions, with arbitrary fixed labels. Their distinct-point domain is connected, since its removed affine coincidence sets have codimension two. The real-analytic identity theorem proves the assertion for all positions.

If a did not divide $2\bar m(2a)$, a regular $2a$-gon with one fixed label at all vertices would have $t_a=0$: rotation by π/a fixes it but multiplies its top harmonic by a nontrivial phase. Reflect an arc across the perpendicular bisector of its next side. Its first $\lceil j/2\rceil$ vertices and their reflection contain an arc of j vertices. Repeated halving from the full polygon to a singleton therefore gives $\mathcal{N}_a$ for that singleton. Every point is a vertex of such a polygon and the label was arbitrary. This would make t_a identically zero, a contradiction. Thus $a\mid 2\bar m(2a)$.

For completeness put $\nu(a)=\bar m(2a)$. The preceding convexity makes its increments an increasing bounded integer sequence, since $0\le\nu(a)\le \wp a/2$. They are eventually equal to an integer d, and $\nu(a)=da-c$ eventually, where $c\ge 0$ is the sum of their deficits from d. The divisibility forces every sufficiently large a to divide $2c$, hence $c=0$. All increments equal $\nu(1)$, so

$$\bar m(2a)=a\bar m(2),\qquad \bar m(n)\le \tfrac{n}{2}\bar m(2).$$

Equal-index and mixed two-point functions. For one index α its two-point form is real, since $\pi_\alpha^2=1$, and invariant under cubic isometries. Its one-field reflection space maps isometrically into the full space by $[f]\mapsto[(\alpha,f)]$, intertwining translations and energy. The plate measure of Prop. A.4 is positive and carried by the planar cone. Its transform gives the kernel

$$k_{\lambda,r}(\zeta)=\int e^{-r(E\cos\zeta-iq\sin\zeta)}\,d\rho_\lambda(E,q).$$

The profile estimate (A.xi), obtained from (R7), gives growth at most $e^{(4+a_\Gamma)|\operatorname{Im}\zeta|}$. Since $4+a_\Gamma<8$ and its period is $\pi/2$, $k_{\lambda,r}=c_0(r)+2c_1(r)\cos 4\zeta$. At $\zeta=is$ its positive cone measure gives $k_{\lambda,r}(is)\ge 0$; divide by $\cosh 4s$ and let $s\to\infty$ to obtain $c_1\ge 0$. For the opposite sign let ρ_D be the positive spectral measure of the two-point form turned by $\pi/4$, supplied by (R6). Its cone support is not assumed. For the boosts $b_s(E,q)=(E\cosh s+q\sinh s,\,q\cosh s+E\sinh s)$, the three-harmonic identity and uniqueness of the Laplace–Fourier representation give

$$(\cosh 4-1)\rho_D+(b_1)_*\rho_\lambda+(b_{-1})_*\rho_\lambda=(\cosh 4+1)\rho_\lambda.$$

The positive-measure identity (A.xiii) follows from the transform uniqueness proved in Prop. A.4; its derivation is included in Prop. A.5. In particular ρ_D is bounded by a constant times ρ_λ and is supported on the cone. Its continued kernel is $c_0(r)-2c_1(r)\cosh 4s\ge 0$; the same limit gives $c_1\le 0$. Thus the nonconstant harmonics vanish whenever both labels are equal.

It is essential to compare mixed labels by the top-harmonic Schwarz inequality. Real polarization between a scalar and a pseudoscalar would lose their purely imaginary, reflection-odd correlation. If $\bar m(2)>0$, choose a line separating two arbitrary points z,y' with labels λ,λ'. Set $S=\{(\theta_L z,\lambda)\}$ and $T=\{(y',\lambda')\}$ on its same side. Then

$$|t_1((z,\lambda),(y',\lambda'))|^2\le |t_1((z,\lambda),(\theta_L z,\lambda))||t_1((\theta_L y',\lambda'),(y',\lambda'))|=0.$$

This contradicts the definition of the top harmonic. Hence $\bar{m}(2) = 0$ and therefore $\bar{m}(n) = 0$ for every n.

Separated tests and the group. The zero harmonic proves planar rotation invariance first for plates with disjoint planar projections. Convolution by product mollifiers and finite Riemann sums of their translates extend it to all C^1 entries with those projections; (R2) controls the moving-entry limits. To remove the restriction, partition in x_2 into intervals much shorter than the separation. Every intersecting pair of planar projections then has an x_2 gap with a level outside all interval hulls. Translate all entries above that level by se_1. The correlation before and after rotation is holomorphic on a common strip in s, by the e_2 reflection picture and the cone bound. For large real s the number of intersecting pairs decreases. Induction and the identity theorem give equality at $s = 0$. Prop. A.8 gives this separated-test argument in detail. It uses reflection-space spectral measures and the local continuity bound, so it applies to the present finite family.

We have obtained every rotation about e_2. Conjugation with W_3 gives rotations about each axis; these generate SO(3). Together with $-1 \in W_3$ and the sign character this gives the asserted O(3) covariance. Proper isometries transport coordinate reflection positivity to every plane. Spatial rotations act unitarily in the e_3 picture and rotate (P_1, P_2). Their translates of $H \geq |P_1|$ imply $E \geq \sup_\phi |q_1 \cos\phi + q_2 \sin\phi| = |q|$ on the joint spectrum. □

13.2. **Midpoint insertions and shrinking plates on the lattice.** We give the quantitative step needed for (R5) and (R7). Its analytic inputs are the localized source energies and the subtracted partition bound proved in the source section. In the notation used here they are:

(1) $Z[s]/Z[0] = e^{\mathcal{E}(s)} Z_K^{\mathrm{sub}}[s]/Z_K^{\mathrm{sub}}[0]$, $\mathcal{E}(s) = \sum_{j,X} e_j(X; s)$, with each term holomorphic for the source radii 2ϱ and $2\varrho_J$ and vanishing at zero source; the summed energy has the conjugation identity of Prop. 10.6;

(2) at physical cube side $\mathscr{W}_j = ML^j a_K$, write $\hat{d}_j(X) = d_j(X)$ for the polymer tree size; then the local energy majorants obey

$$|e_j(X; s)| \leq \varepsilon_j(X), \qquad \sup_{\square} \sum_{X \supset \square} \varepsilon_j(X) e^{\kappa_e \hat{d}_j(X)} \leq C_{\mathrm{loc},E};$$

(3) changing the source only at distance D from X changes this term by at most $2\varepsilon_j(X) e^{-\kappa_e D/\mathscr{W}_j}$;

(4) uniformly in the source domain,

$$\frac{|Z_K^{\mathrm{sub}}[s]|}{Z_K^{\mathrm{sub}}[0]} \leq \exp(C_0 \ell^3), \quad C_0 = C + 2 + C_P(1 + \log[L(g^{-2} + B_\sharp)]).$$

These estimates follow from Props. 10.6, 10.14 and 10.20. The arguments below do not assume that a fermion bilinear has a positive classical law.

Fix a board with cells $\square_k = o + lk + [-l/2, l/2]^3$, $l = L^{-i}$ and $o \in a_K \mathbb{Z}^3$, whose walls are lattice link mirrors. On a compatible torus let $\mathsf{N} = (\ell/l)^3$. For an even element D of one cell, let $\mathrm{Fold}\, D$ be its product of images in all cells, using the antilinear reflection for orientation-reversing images, and put

$$\|D\|_{\mathrm{ch},\ell} = \omega_\ell(\mathrm{Fold}\, D)^{1/\mathsf{N}}.$$

This root is nonnegative. The relations of the reflection group hold on even elements: parallel reflections compose to a translation, orthogonal reflections differ on a generator by minus one and hence agree on even products, and a period acts trivially on even products despite the antiperiodic generator sign.

Here is the algebraic chessboard argument. In one direction view the torus as a cyclic list of $2N$ slabs. Reflection positivity at a wall and its antipodal wall bounds the squared modulus of an array expectation by the product of the two half-reflected array expectations. For positive constant-array roots, maximize the absolute value of the expectation divided by the product of these roots over the finite set of arrays with the given entries. Half-reflection preserves maximality. A maximizing

array with the longest constant cyclic string must be constant: reflection at an endpoint doubles the string if its length is at most N, and fills the cycle otherwise. The maximum is therefore one. For a zero root perturb each entry by a real ϵ; its folded expectation is a nonnegative polynomial with leading coefficient one, so choose $\epsilon \to 0$ off its finitely many roots. Repeat the argument in the three directions. It proves

$$|\omega_\ell(\prod_k D_k)| \le \prod_k \|D_k\|_{\mathrm{ch},\ell}.$$

The same argument proves fine-to-coarse domination, and invariance of the root under the board reflections and a cell-centre reflection. This is the finite reflection argument of [3, Prop. 5.86], now in the commuting even cell algebra.

For a fixed-cutoff periodic state ω_ν obtained along sides $\ell_r \to \infty$, the bound passes to the limit. It also gives the multiple-reflection inequality

$$|\omega_\nu(D\,\tau_{le_3} F\,\Theta_c F')| \le \liminf_r \|D\|_{\mathrm{ch},\ell_r}\|F\|_{\nu,c}\|F'\|_{\nu,c},$$

for even F, F' above the lower wall c and D in the intervening cell. To prove it without a bounded-operator assumption, put $\widehat{D}X = D\tau_{le_3}X$ and $D^\sharp = \Theta_{c+l/2}D$. Translation invariance and the reflection relations give $\widehat{D}^\sharp$ as its formal adjoint for the positive reflection form. The alternating products yield $s_n = \langle F, (\widehat{D}^\sharp\widehat{D})^n F\rangle$ with $s_n^2 \le s_{n-1}s_{n+1}$, hence $s_1^n \le s_0^{n-1}s_n$. Choose a coarser board containing the finite dependence region of F, and take n along multiples of its side ratio. Fine-to-coarse chessboard domination gives $s_n \le C_F^2(\liminf_r \|D\|_{\mathrm{ch},\ell_r})^{2n}$. The constant C_F is finite at fixed cutoff: if $F = \sum_\Psi f_\Psi \Psi$, Gram's inequality and $\|D_U^{-1}\| \le m_0^{-1}$ bound it by $\sum_\Psi \|f_\Psi\|_\infty m_0^{-\deg\Psi/2}$. Its nth root disappears as $n \to \infty$. This proves the bound for $\widehat{D}F$; one final Schwarz inequality gives the displayed multiple-reflection estimate. It holds along every subsequence of sides.

Lemma 13.2 (A cell normalizer)**.** Let $p = 1$ or 2, and let f_q have disjoint supports in the middle half of one cell, with indices α_q. Put

$$D_z = \exp\sum_{q=1}^{p} z_q \Phi_{\alpha_q}(f_q), \qquad r_* = \min_q 2\varrho_{\alpha_q}/\|f_q\|_{\alpha_q}.$$

Along a subsequence of sides there is a holomorphic scalar $\mathcal{E}^\circ(z)$ on $|z_q| < r_*$ such that

$$|e^{-\mathcal{E}^\circ(z)}| \liminf_r \|D_z\|_{\mathrm{ch},\ell_r} \le Q_{\mathrm{mid}}(l), \quad Q_{\mathrm{mid}}(l) = e^{2C_{\mathrm{loc},E}c_{\mathrm{geo}}+C_0 l^3},$$

where

$$c_{\mathrm{geo}} = (1-L^{-3})^{-1} + \sup_{1<u_0\le L}\sum_{n\ge 0}(u_0L^n)^3 e^{-\kappa_e[u_0L^n/4-2]_+} < \infty.$$

In particular $\bar{Q}_{\mathrm{mid}} = Q_{\mathrm{mid}}(1)$ is uniform for $0 < l \le 1$. ■

Proof. Reflect the cell source into each board cell, conjugating z on orientation-reversing copies. Their sum s_z^{ch} is inside the source domain for $|z_q| < r_*$: disjoint middle supports preserve the maximum supremum and Lipschitz bounds, including for the matrix indices of operator norm at most one. The reflection-real basis gives

$$\|D_z\|_{\mathrm{ch},\ell}^{\mathsf{N}} = Z[s_z^{\mathrm{ch}}]/Z[0].$$

Give each polymer a root cell by the centre of its first cube in a fixed ordering. Let s_z^k be the source in root cell k alone and define

$$\mathcal{E}_\ell^\circ(z) = \mathsf{N}^{-1}\sum_k \sum_{j,X:\,k(X)=k} \widetilde{e}_j(X; s_z^k),$$

where $\widetilde{e} = e$ for a proper cell image and $\widetilde{e} = \bar{e}$ for an improper one. In the latter case the argument has $\bar{z}$, so the conjugated composition is holomorphic in z. The summable local-energy majorants bound this function uniformly in ℓ at fixed K, and

$$\operatorname{Re}\mathcal{E}(s_z^{\mathrm{ch}}) = \mathsf{N}\operatorname{Re}\mathcal{E}_\ell^\circ(z) + \sum_{j,X}\operatorname{Re}[e_j(X; s_z^{\mathrm{ch}}) - e_j(X; s_z^{k(X)})].$$

For $\mathscr{W}_j \geq l$ the last differences sum to at most $2\mathsf{N}C_{\mathrm{loc},E}(l/\mathscr{W}_j)^3$. For $\mathscr{W}_j < l$, a different cell's middle is at distance at least $l/4$ from the root cell. Locality and the polymer diameter bound give, after the size weight is paid, a contribution per root cell at most $2C_{\mathrm{loc},E}(l/\mathscr{W}_j)^3 e^{-\kappa_e[l/(4\mathscr{W}_j)-2]_+}$. The scales form a geometric sequence, so their total divided by N is at most $2C_{\mathrm{loc},E}c_{\mathrm{geo}}$. The subtracted partition bound adds $C_0 l^3$. This proves the asserted bound for $\mathcal{E}_\ell^\circ$. Montel's theorem at fixed cutoff gives a locally uniform subsequential holomorphic limit, with all derivatives. Taking the liminf of the bound proves the lemma. The subsequence may depend on the insertion; the limiting state itself is fixed. □

Proposition 13.3 (Centered midpoint insertion)**.** For real f in the middle of a board cell of side $l \leq 1$,

$$|\omega_\nu([\Phi_\alpha(f) - \omega_\nu\Phi_\alpha(f)]\tau_{le_3}F\,\Theta_c F')| \leq \frac{\bar{Q}_{\mathrm{mid}}}{\varrho_\alpha}\|f\|_\alpha\|F\|_{\nu,c}\|F'\|_{\nu,c}.$$

For a complex f, twice this constant is valid. ■

Proof. Take $p = 1$ in the normalizer lemma and apply multiple reflections along its subsequence. The holomorphic function $\omega_\nu(D_z\tau_{le_3}F\Theta_c F')e^{-\mathcal{E}^\circ(z)}$ has modulus at most $\bar{Q}_{\mathrm{mid}}\|F\|\|F'\|$. Cauchy's estimate on radius $r_* = 2\varrho_\alpha/\|f\|_\alpha$ bounds its derivative at zero. With $F = F' = 1$ it also bounds $|\omega_\nu\Phi_\alpha(f) - \mathcal{E}^{\circ\prime}(0)|$ by $\bar{Q}_{\mathrm{mid}}/r_*$. The coefficient of $\mathcal{E}^{\circ\prime}(0)$ is bounded by $\|F\|\|F'\|$ using the same multiple-reflection estimate with $D = 1$. Adding the two derivatives gives $2\bar{Q}_{\mathrm{mid}}/r_* = \bar{Q}_{\mathrm{mid}}\|f\|_\alpha/\varrho_\alpha$. Split a complex function into real and imaginary parts to obtain its stated bound. □

Proposition 13.4 (Shrinking plates)**.** Let $0 < t \leq 1$ and let real f be supported in $|x_1| \leq t$, $|x_2 - y_2| \leq \Upsilon/2$, $t \leq x_3 \leq 2t$. Let l be the least power of L at least $4\max\{2t, \Upsilon/2\} + 4a_K$. Then

$$|S_2^T[\nu]((\alpha, f\circ\theta),(\alpha, f))| \leq \frac{\|f\|_\alpha^2}{\varrho_\alpha^2}\left[\log 2\,(Q_{\mathrm{mid}}(l)+1)^2 + \frac{C_{\mathrm{loc},E}C_{\mathrm{dep}}(1+\Upsilon+m')}{4t}\right],$$

where $M = L^{m'}$ and C_{dep} depends only on L, κ_e. The side l is at most the least power $\bar{l}(\Upsilon)$ of L exceeding $4\max\{2, \Upsilon/2\} + 4$. ■

Proof. Choose a nearest lattice board centre to $(0, y_2, 0)$; both reflected supports lie in the middle of its cell. Use the normalizer lemma with $p = 2$ and put $\Psi(z) = \omega_\nu(D_z)e^{-\mathcal{E}^\circ(z)}$. Then $\Psi(0) = 1$ and $|\Psi| \leq Q_{\mathrm{mid}}(l)$. Schwarz's lemma on complex lines makes $|\Psi - 1| \leq 1/2$ on the bidisc of radius $r_*/[2(Q_{\mathrm{mid}}+1)]$. There $|\log\Psi| \leq \log 2$, so

$$|\partial_{12}\log\Psi(0)| \leq 4\log 2\,(Q_{\mathrm{mid}}+1)^2/r_*^2.$$

The second cumulant is this derivative plus $\partial_{12}\mathcal{E}^\circ(0)$.

For an energy term of scale $\mathscr{W}_j$, let D_1, D_2 be its distances from the two plate supports. Locality and Cauchy's estimate give its derivative a bound $2r_*^{-2}\varepsilon_j(X)e^{-\kappa_e\max(D_1,D_2)/\mathscr{W}_j}$. The factor two is included in C_{dep} below. The support separation gives $D_1 + D_2 \geq 2t - (\hat{d}_j(X)+4)\mathscr{W}_j$, and $\max(D_1, D_2) \geq D_2/2 + (D_1+D_2)/4$. For a cube $\square \subset X$ also $D_2 \geq \operatorname{dist}(\square, \operatorname{supp} f) - (\hat{d}_j(X)+4)\mathscr{W}_j$. The source-energy size weight pays the resulting $3\kappa_e\hat{d}_j(X)/4$; the fixed factor $e^{3\kappa_e}$ is absorbed in $c(\kappa_e)$ below. Counting cubes near the plate yields

$$|\partial_{12}\mathcal{E}_\ell^\circ(0)| \leq \frac{C_{\mathrm{loc},E}c(\kappa_e)}{r_*^2}\sum_{j\leq K}(2u_j+3)((\Upsilon/t)u_j+3)(u_j+3)e^{-\kappa_e u_j/2}, \qquad u_j = t/\mathscr{W}_j.$$

For $u_j \ge 1$ the geometric sequence is summable after its cubic polynomial. For $u_j < 1$ its terms are at most $20(\Upsilon/\mathscr{W}_j + 3)$. Their first sum is at most $\Upsilon/[t(1 - L^{-1})]$, and their number is at most $\log_L(M/t) + 1 \le (m' + 2)/t$. Thus the last sum is bounded by $C_{\mathrm{dep}}(1 + \Upsilon + m')/t$. Pass to the normalizer subsequence limit, use $r_* = 2\varrho_\alpha/\|f\|_\alpha$, and add the logarithm derivative bound. The side bound follows directly from $t \le 1$ and $a_K \le 1$; it permits boards larger than one for large fixed width. □

13.3. **Checking the hypotheses on a volume-first limit.** For the chamber construction, the earlier compactness argument gives (R0)–(R3), translation continuity and the sign rule. The exact diagonal lattice reflection inequality gives (R6) on the same subsequence. The independent two-plane argument gives (R4). The remaining quantitative inputs are the midpoint insertion estimate and the shrinking-plate estimate; they are needed here with their full lattice uniformities in the cutoff and board size.

For the real midpoint insertion estimate we may use the weaker constant $2\bar{Q}_{\mathrm{mid}}/\varrho_\alpha$, $\varrho_0 = \varrho$ and $\varrho_\Gamma = \varrho_J$. It implies (R5) with

$$c_{\mathrm{I}} = 1/(40L), \qquad p_{\mathrm{I}} = 0, \qquad C_{\mathrm{I}} = 2\bar{Q}_{\mathrm{mid}}/\min\{\varrho, \varrho_J\}.$$

Indeed choose $l = L^{-i}$ with $\eta/(5L) < l \le \eta/5$, and a nearest board centre $y^* \in L^{-(i+2)}\mathbb{Z}^3$. Its lower wall is $c = y_3^* - l/2$. The support of h lies in the middle cube since $c_{\mathrm{I}}\eta + l/(2L^2) < l/4$, and

$$-0.226\eta \le c < c + l \le 0.226\eta < \eta/4.$$

Write $F' = \tau_{2ce_3}F$ and $G'' = \tau_{-le_3}G$. The lattice midpoint estimate applies to $\theta_c\bar{F}' \cdot (\alpha, h) \cdot \tau_{le_3}G''$. Pass to the fixed-subsequence limit. Its two reflection norms become $\|[\tau_{ce_3}F]\|$ and $\|[\tau_{-(c+l)e_3}G]\|$. Both are bounded by the norms in (R5), using the positive-time contractions with times $c + \eta/4$ and $\eta/4 - c - l$. The midpoint constant is uniform for $l \le 1$, so no power of η^{-1} is lost. We use the factor two in the lattice insertion bound.

For a label $\lambda = (\alpha, \upsilon)$ and $a \in \mathsf{Prof}_t$, the plate function $f = a \otimes \upsilon$ obeys

$$\|f\|_\infty \le 3\|\upsilon\|_\infty, \quad \operatorname{Lip} f \le 15\|\upsilon\|_{C^1}/t, \quad \|f\|_\alpha \le 600M\|\upsilon\|_{C^1}/t.$$

The shrinking-plate estimate on the lattices, uniform at a board side bounded in terms of Υ_λ alone, then gives $\sigma(\theta(\alpha, f) \cdot (\alpha, f)) \le C_\lambda t^{-3}$ in the limit. This is (R7) with $a_\Gamma = 3 < 4$. Thus (R0)–(R7) hold and the rotation criterion applies. No estimate at coincident supports is used in this passage.

13.4. **Point bounds and a common field domain.**

Proposition 13.5 (An L^1 point bound)**.** If (R0)–(R7) hold with $p_{\mathrm{I}} = 0$, then there is $C_{\mathrm{pt}} < \infty$ such that, for $0 < \eta \le 1$, real $h \in C_c^1(\{|x_3| < \eta/2\})$ and combinations F, F' supported in $x_3 > \eta$,

$$|\sigma(\theta\bar{F} \cdot (\alpha, h) \cdot F')| \le 16C_{\mathrm{pt}}\eta^{-4}\|h\|_{L^1}\|[\tau_{-\eta e_3}F]\|\|[\tau_{-\eta e_3}F']\|.$$

The same constant works for the finite set of indices. ■

Proof. Prop. A.10 constructs the finite-labelled point family directly from (R5), with $\|B_\alpha(z, z')\| \le C_{\mathrm{pt}} \min(1, d_3(z), d_3(z'))^{-4}$ and the slab identity (A.xxv). This exponent follows from the three-dimensional mean-value volume $O(r^3)$ and the sharp norm $O(r^{-7})$ of its mollified test. For $d = \eta$ and $|x_3| < \eta/2$, both real tube margins are at least $\eta/2$. Integrating the point bound against $|h|$ gives $16C_{\mathrm{pt}}\eta^{-4}\|h\|_1$. Apply the resulting slab operator to the two vectors $[\tau_{-\eta e_3}F]$ and $[\tau_{-\eta e_3}F']$. The common insertion constant and finite label set give the same constant for every index. □

Theorem H (Finite-family reconstruction)**.** Let σ satisfy the preceding hypotheses with $p_{\mathrm{I}} = 0$. Its reflection Hilbert space carries Hermitian operator-valued tempered distributions φ_α, a strongly continuous unitary representation of the proper orthochronous Poincaré group, and a common dense

invariant polynomial domain $\mathcal{D}_0$ containing Ω. The vacuum is cyclic and invariant, translations have spectrum in the closed forward cone, the fields are scalar under the connected Lorentz group, and

$$[\varphi_\alpha(f), \varphi_\beta(g)] = 0 \quad \text{on } \mathcal{D}_0$$

for spacelike separated test supports, for every pair of indices. Their Euclidean continuations agree with σ on all separated C_c^1 tuples. No uniqueness of the translation-invariant vector is asserted. ■□

Proof. The point construction Prop. A.10, followed by its circular-tube extension, applies to the reflection picture of Thm. G. Resolvent regularization gives the forms (A.xxxi), and Prop. A.11 constructs all fields on the common invariant graph domain $\mathcal{D}_\infty$. Their explicit bound is

$$\|(1+H)^j \varphi_\alpha(f)\psi\| \le \|A_{\alpha,l}\| \sum_{k=0}^{l+j} \binom{l+j}{k} \|\partial_t^k f\|_1 \|(1+H)^{2l+j}\psi\|, \qquad l \in \mathbb{N}, \quad l > 4.$$

The adjoint identity holds for each reflection-real index. Prop. A.13 gives all mixed tempered vacuum distributions as tube boundary values. The cyclicity argument in §A.2.9 proves that their polynomial domain $\mathcal{D}_0$ is dense in the original reflection Hilbert space. Prop. A.14 continues proper rotations to boosts and constructs a common strongly continuous Poincaré representation, under which each field is scalar. Prop. A.15 proves mixed locality on $\mathcal{D}_0$. The ordered recovery formulas (A.xxxvi) and (A.xxxvii), and the subsequent change of ordering direction, identify the Euclidean continuations with σ on every separated tuple. The translation spectrum is the already verified circular cone, and Ω is fixed throughout the construction. Vacuum uniqueness would require $\ker H = \mathbb{C}\Omega$; that condition is not used in any of these steps. □

If the finite index space also carries the flavour action of the construction, define it on bilinear tuple classes by

$$\mathsf{V}(V)[((\Gamma_i, f_i))_i] = [((V\Gamma_i V^{-1}, f_i))_i].$$

Action-density entries are fixed, and the formula extends multilinearly in the indices. Its matrices in the reflection-real basis are real orthogonal and it commutes with the semigroup, translations and proper rotations. It consequently commutes with the Poincaré representation and carries φ_Γ to $\varphi_{V\Gamma V^{-1}}$, fixing φ_0. For the reflection p of the first spatial coordinate, $\mathsf{P}[X] = \pi(X)[pX]$ is unitary and

$$\mathsf{P}\varphi_\alpha(f)\mathsf{P}^{-1} = \pi_\alpha \varphi_\alpha(f \circ p).$$

It conjugates the Poincaré action by p and exchanges the two flavour factors. These assertions follow already for slab insertions and pass through the same point and boundary limits.

The reconstruction fixes full tempered Minkowski vacuum distributions. Its Euclidean kernels agree with the limiting correlations at distinct points. For the extension to Schwartz tests flat on every coincidence diagonal, apply the kernel-integration and density arguments in [3, Lem. 8.44], [3, Lem. 8.45], and [3, Prop. 8.46] to each fixed string of labels. With $\wp = 4$, its point bound is $|s_n(x)| \le C^n (2n^2)^{4n} \min(1, \min_{i\neq j} |x_i - x_j|)^{-4n}$. Taylor's formula for a flat test cancels this singular factor; Schwartz decay then makes its product with s_n integrable. The tensor-density result identifies this integral with the Schwinger functional on that test space. This extension asserts neither convergence of lattice contact terms nor uniqueness of an extension across those diagonals. The separated-moment bound with $(n!)^\varkappa$ is compatible with that identification; it is not a substitute for the polynomial insertion estimate used to construct the fields.

Proof of Thm. A. Choose the constants by Prop. 10.20, which verifies the gauge induction for both coupling sequences, and put $\mathsf{b}_0 = \gamma^{-2} + \bar{B}$. For every $\mathsf{b} \ge \mathsf{b}_0$, the choice $g = (\mathsf{b} - \bar{B})^{-1/2} \le \gamma$ places b in the interval of the pure gauge construction; the permitted physical sides contain all sufficiently large $2L^m$. The determinant, inverse and infinite-volume lattice states are supplied by Props. 2.1, 2.5 and 3.1. The block construction, forward flow and history bounds give the source estimates of Prop. 10.14 and Thm. F. These bounds are uniform in the torus side and pass to every ν_K.

Prop. 12.1 gives a common spacing subsequence on every separated tuple, together with clause (i) and the lattice symmetries. Prop. 12.2 gives coordinate and diagonal reflection positivity on that same limit. The reflection-picture construction and Prop. 13.1 give $H \geq |P_1|, |P_2|$. The source energies and common vacuum energy imply the midpoint and plate bounds in Props. 13.3 and 13.4. Their transfer to the continuum verifies (R5) with $p_{\mathrm{I}} = 0$, and (R7) with exponent $3 < 4$; the other properties (R0)–(R7) have just been proved. Thus Thm. G gives full Euclidean covariance, all-plane reflection positivity and the circular spectral cone. This proves clause (ii).

The point bound Prop. 13.5 verifies the final reconstruction input. The finite-family reconstruction Thm. H then gives the common domain, proper Poincaré action, Hermiticity and pairwise locality. The exact flavour and parity identities pass through the same construction. Finally, the cutoff errors in Prop. 11.1 vanish at each fixed test pair after the volume limit. Its positive leading short-distance terms give a nonzero two-point function for every index. These are clauses (iii) and (iv). □

Appendix A. Kernel comparison and reconstruction

We supply the lattice kernel comparison and the reconstruction arguments used above. The real-time argument keeps the finite family of reflection-real labels throughout.

A.1. **The free Maxwell kernel.** For $u \neq 0$ put

$$\mathbb{K}_3(u) = \frac{3\widehat{u}\widehat{u}^{\mathsf{T}} - I}{4\pi|u|^3} = \nabla^2 \frac{1}{4\pi|u|}, \qquad \mathcal{K}_6(f,h) = \iint f(x)h(y)|x-y|^{-6}\,\mathrm{d}x\,\mathrm{d}y. \tag{A.i}$$

On a torus the distance in the integral is its least-lift distance. We will only use pairs supported together in a ball of radius $\ell/4$, where the integral agrees with the integral of their compactly supported lifts. The eigenvalues of $3\widehat{u}\widehat{u}^{\mathsf{T}} - I$ are $2, -1, -1$, so

$$\sum_{\lambda,\lambda'} \mathbb{K}_3(u)^2_{\lambda\lambda'} = \frac{3}{8\pi^2}|u|^{-6}. \tag{A.ii}$$

Represent a plaquette by (z,λ), with λ its dual coordinate direction, choosing the cyclic orientation. Its barycentre is $x(z,\lambda) = z + \frac{1}{2}(e_1 + e_2 + e_3 - e_\lambda)$ in bare units. Relative to the original positive plaquette orientation let the sign be $\epsilon_p \in \{\pm 1\}$. For $q_\mu(k) = e^{ik_\mu} - 1$ define

$$F_{\lambda\lambda'}(k) = \frac{\overline{q_\lambda(k)}\,q_{\lambda'}(k)}{|q(k)|^2}, \qquad P(k) = I - F(k) \quad (k \neq 0), \qquad P(0) = 0.$$

Let Π_∞ and Π_N be the curl projections on $\mathbb{Z}^3$ and $(\mathbb{Z}/N\mathbb{Z})^3$. Their Fourier formulas are

$$\begin{aligned} \Pi_\infty((z,\lambda),(z',\lambda')) &= \delta_{pq} - \int_{[-\pi,\pi]^3} e^{ik\cdot(z-z')} F_{\lambda\lambda'}(k) \frac{\mathrm{d}k}{(2\pi)^3}, \\ \Pi_N((z,\lambda),(z',\lambda')) &= \delta_{pq} - N^{-3}\delta_{\lambda\lambda'} - N^{-3} \sum_{k \in \frac{2\pi}{N}\mathbb{Z}^3 \cap (-\pi,\pi]^3 \setminus \{0\}} e^{ik\cdot(z-z')} F_{\lambda\lambda'}(k). \end{aligned} \tag{A.iii}$$

Indeed, curl has symbol $q\times$, whose range is the Hermitian orthogonal complement of $\bar{q}$. Its orthogonal projection is $P(k)$. On the infinite lattice, cutting out $|k| \leq \varepsilon$ in the right inverse $-|q|^{-2}\bar{q} \times P(k)$ shows that the range of this multiplier is the closure of the curl range. The cutoff tends to the identity on that range in L^2. Fourier inversion proves (A.iii). The finite formula has $P(0) = 0$ because an exact cochain has no constant mode. In particular, $\epsilon_p\epsilon_q C_K^{\mathrm{free}}(p,q) = \Pi_N(p,q)$ after rescaling the bare lattice to spacing one, with $N = \ell/a_K$.

Lemma A.1. There are absolute constants $C'_{\mathrm{lat}}, C''_{\mathrm{lat}}$ such that, for distinct plaquettes on $\mathbb{Z}^3$,

$$|\Pi_\infty(p,q) - \mathbb{K}_3(x(p) - x(q))_{\lambda(p)\lambda(q)}| \leq C'_{\mathrm{lat}}|x(p) - x(q)|^{-5}. \tag{A.iv}$$

If p, q are distinct plaquettes on the N-torus and $\widetilde{p}, \widetilde{q}$ are lifts whose barycentre difference has sup norm at most $N/2$, then

$$|\Pi_N(p,q) - \Pi_\infty(\widetilde{p},\widetilde{q})| \le C''_{\mathrm{lat}} N^{-3}. \tag{A.v}$$

■

Proof. Fix the two dual directions, put $y = z - z'$ and $s = (e_{\lambda'} - e_\lambda)/2$, so that $u = y + s$ is the barycentre difference. Multiplication by the barycentre phase removes the linear lattice error:

$$m(k) = e^{-ik\cdot s} F_{\lambda\lambda'}(k) = \frac{4\sin(k_\lambda/2)\sin(k_{\lambda'}/2)}{4\sum_\mu \sin^2(k_\mu/2)}, \qquad m^0(k) = \frac{k_\lambda k_{\lambda'}}{|k|^2}.$$

Taylor's formula for sine and its derivatives, and $\sum_\mu 4\sin^2(k_\mu/2) \ge 4|k|^2/\pi^2$ on $[-\pi,\pi]^3$, give, for $|\alpha| \le 6$,

$$|\partial^\alpha F(k)| \le C_\alpha |k|^{-|\alpha|}, \qquad |\partial^\alpha (m - m^0)(k)| \le C_\alpha |k|^{2-|\alpha|}.$$

These derivative bounds follow by differentiating the quotient: the numerator of the second difference vanishes to order six before division by $|k|^2 \sum_\mu 4\sin^2(k_\mu/2)$, and differentiating an inverse preserves its corresponding homogeneous bound.

Choose a radial smooth cutoff χ equal to one on $|k| \le 1/2$ and zero on $|k| \ge 1$. Split the Fourier coefficient of F into $(1-\chi)F$, $\chi(m - m^0)$ with phase $e^{ik\cdot u}$, and χm^0 with that phase. The first is a smooth periodic symbol; five integrations by parts give $O(|u|^{-5})$. For the second, use annuli of radius 2^{-j}. Their Fourier integrals are bounded both by $C 2^{-5j}$ and, after six integrations by parts, by $C|u|^{-6} 2^j$. Splitting the sum at $2^j = |u|$ gives $O(|u|^{-5})$.

For the last term put $E(u) = \int e^{ik\cdot u} \chi(k) |k|^{-2} \,\mathrm{d}k/(2\pi)^3$. The heat-kernel identity for $|k|^{-2}$, with Fubini justified by the integrability at zero and the rapid decay of $\check{\chi}$, gives $E = (4\pi|\cdot|)^{-1} * \check{\chi}$. Write $\check{\chi}(v) = \varphi(|v|)$. This function is radial, $4\pi \int_0^\infty \varphi(t) t^2 \,\mathrm{d}t = 1$, and $|\varphi(t)| \le C(1+t)^{-7}$. The mean of $|u - v|^{-1}$ over $|v| = t$ is $1/\max(|u|, t)$, hence

$$E(u) = \frac{1}{4\pi|u|} - D(|u|), \qquad D(r) = \int_r^\infty \varphi(t) t^2 (r^{-1} - t^{-1}) \,\mathrm{d}t.$$

Direct differentiation gives $D'(r) = O(r^{-6})$ and $D''(r) = O(r^{-7})$. The Fourier transform of χm^0 is $-\partial_\lambda \partial_{\lambda'} E$; it therefore equals $-\mathbb{K}_3(u)_{\lambda\lambda'} + O(|u|^{-7})$. The minus sign in (A.iii) proves (A.iv) for $|u| > 4$. For smaller nonzero barycentre differences, $|u| \ge 2^{-1/2}$, and the projection bound by one covers the remaining cases.

For the periodic comparison take $N \ge 4$ and remove the ball $|k| \le N^{-1}$ from F, using a smooth cutoff supported in $|k| \le 2N^{-1}$ and its 2π-periodic translates. Call the resulting smooth periodic symbol F_N. It equals F at every nonzero Fourier grid point and is zero at zero. The derivative bounds above give $\int |\partial_\mu^4 F_N| \le CN$; thus $|\check{F}_N(y')| \le CN |y'|_\infty^{-4}$ for $y' \ne 0$. Its Fourier series is absolutely summable, and summing it over the grid gives the exact identity

$$N^{-3} \sum_{k \ne 0} e^{ik\cdot y} F(k) = \sum_{n \in \mathbb{Z}^3} \check{F}_N(y + Nn).$$

The term $n = 0$ differs from $\check{F}(y)$ by at most CN^{-3}, the volume of the removed ball. Since $|y|_\infty \le N/2 + 1/2$, for $n \ne 0$ one has $|y + Nn|_\infty \ge N|n|_\infty/4$. The other terms are bounded in sum by $CN^{-3} \sum_{n \ne 0} |n|_\infty^{-4} < \infty$. The zero-mode correction in (A.iii) is also $O(N^{-3})$. This proves (A.v); $N \le 3$ follows by increasing the constant and using the projection bound. □

Rescaling the last lemma gives, for a least lift u of the physical barycentre difference of distinct plaquettes,

$$\left|\epsilon_p \epsilon_q C_K^{\mathrm{free}}(p,q) - a_K^3 \mathbb{K}_3(u)_{\lambda(p)\lambda(q)}\right| \le C_{\mathrm{lat}} a_K^3 (a_K^2 |u|^{-5} + \ell^{-3}), \tag{A.vi}$$

and consequently

$$|C_K^{\mathrm{free}}(p,q)| \le (1 + 3C_{\mathrm{lat}}) a_K^3 |u|^{-3}. \tag{A.vii}$$

Here C_{lat} is absolute and $2^{-1/2}a_K \le |u| \le \sqrt{3}\ell/2$.

For a plaquette kernel k, put

$$\mathcal{K}[k](f,h) = \sum_{p,q} f(x(p))h(x(q))k(p,q)^2.$$

There are no additional mesh factors in this sum. A test pair's supports lie together in a ball of radius $5d$, with cross distances in $[d, 10d]$.

Proposition A.2 (The Maxwell comparison)**.** For a test pair (f,h) at scale d and shape ϑ, with $8a_K \le d$ and $20d \le \ell$,

$$\left|\mathcal{K}[k](f,h) - \frac{3}{8\pi^2}\mathcal{K}_6(f,h)\right| \le C_{\mathrm{tm}}\left[\frac{a_K}{\vartheta d} + (d/\ell)^3 + e\right]\|f\|_\infty\|h\|_\infty \tag{A.viii}$$

for $k = C_K^{\mathrm{free}}$ and $e = 0$. The same estimate holds for $k = C_{K,s}^{\mathrm{bg}}$ with

$$t = d/a_s, \qquad e = t^3 e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}},$$

provided $t \ge 1$, or provided $\delta_{\mathrm{tp}}L(L-1)t/\sqrt{3} \ge 3\log L + \log 2$. The constant is independent of the mesh, side, pair, s and ϑ. It is absolute in the free case; in the background case it depends only on the constants of [3, Prop. 7.6]. ■

Proof. Lift both functions to their common ball of radius $5d$. Their cross distances are unchanged. For each of the three dual directions, the barycentres form a translate of $a_K\mathbb{Z}^3$. Disjoint cubes of side a_K about the contributing points lie in a ball of radius $d + \sqrt{3}a_K/2$. Counting all three directions gives at most $18(d/a_K)^3$ contributing plaquettes for each function: $3(4\pi/3)(1+\sqrt{3}/16)^3 < 18$ because $d \ge 8a_K$. If on their product $|k_1 - k_2| \le A_1 a_K^3 d^{-3}$ and $|k_2| \le A_2 a_K^3 d^{-3}$, it follows that

$$|\mathcal{K}[k_1](f,h) - \mathcal{K}[k_2](f,h)| \le 324A_1(2A_2 + A_1)\|f\|_\infty\|h\|_\infty. \tag{A.ix}$$

Use (A.vi) to replace C_K^{free} by $a_K^3\epsilon_p\epsilon_q\mathbb{K}_3(x(p)-x(q))_{\lambda(p)\lambda(q)}$. Equations (A.vi)–(A.ix) bound the error by a constant times $[(a_K/d)^2 + (d/\ell)^3]\|f\|_\infty\|h\|_\infty$.

The resulting sum is a Riemann sum over nine products of barycentre grids. On every product of cubes that contributes, all cross distances are at least $d - 2\sqrt{3}a_K \ge d/2$. The squared dipole entry is bounded by Cd^{-6} and its gradient by Cd^{-7}. The Lipschitz bounds on f, h thus bound the difference between the integrand and its grid value by $Ca_K\vartheta^{-1}d^{-7}\|f\|_\infty\|h\|_\infty$. The union of contributing cubes for either function has volume at most $8d^3$. Integration and summation over the nine directions give error at most $8\cdot 10^4 a_K(\vartheta d)^{-1}\|f\|_\infty\|h\|_\infty$. Equation (A.ii) identifies the integral.

For the background kernel, $C_{K,s}^{\mathrm{bg}} - C_K^{\mathrm{free}} = -C_{K,s}^{\mathrm{fl}}$. All contributing sup distances are at least $d/\sqrt{3}$. The applicable bound in [3, Prop. 7.6] gives $|C_{K,s}^{\mathrm{fl}}(p,q)| \le Ce\, a_K^3 d^{-3}$. Since $t^3e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}$ is bounded for $t > 0$, (A.ix) and (A.vii) add an error at most $Ce\|f\|_\infty\|h\|_\infty$. □

Lemma A.3 (Smooth reflection pairs)**.** For every $0 < \vartheta < 1$ and $d > 0$ there is $f \in C_c^\infty(\mathbb{R}^3)$ supported in $\mathbb{B}(2de_3, d)$ such that $(\theta f, f)$ is a test pair of scale d and shape ϑ, where $\theta(x_1,x_2,x_3) = (x_1,x_2,-x_3)$. ■

Proof. Put $x = 2de_3$ and

$$r_0 = \max\{\vartheta, (3\vartheta/\pi)^{1/3}\}, \qquad r = (1+r_0)/2, \qquad \epsilon = \tfrac{1}{2}\min\{1-r, r-\vartheta\}.$$

Then $r_0 < r < 1$, $0 < \epsilon < r$, $r + \epsilon < 1$, $r - \epsilon > \vartheta$, and $(\pi/3)r^3 > \vartheta$. Convolve $g_r(y) = \max\{0, 1 - |y-x|/(rd)\}$ with a nonnegative smooth mollifier of integral one supported in $\mathbb{B}(0, \epsilon d)$. The resulting function satisfies

$$0 \le f \le 1, \qquad f(x) \ge 1 - \epsilon/r > 0, \qquad \operatorname{Lip} f \le (rd)^{-1},$$
$$\int f(y)\,\mathrm{d}y = (\pi/3)r^3d^3, \qquad \operatorname{supp} f \subset \mathbb{B}(x, (r+\epsilon)d).$$

The integral follows by polar integration of g_r; convolution preserves it. Since $\|f\|_\infty \geq (r-\epsilon)/r > \vartheta/r$, these bounds imply $\operatorname{Lip} f \leq \|f\|_\infty/(\vartheta d)$ and $\int f(y)\,\mathrm{d}y \geq \vartheta\|f\|_\infty d^3$. Reflection preserves the bounds. The supports lie in balls of radius d centered at $\pm x$, so their distance is at least $2d$; it is at most $4d$ because $f(x) > 0$. Their cross distances are at most $6d$, and both supports lie in $\mathbb{B}(0, 3d)$. These verify the remaining test-pair conditions. □

A.2. **Real-time reconstruction.** We prove the domain, continuation and locality assertions used in Thm. H. The labels range over the finite reflection-real basis fixed in the main text. Proper rotations act trivially on these labels. Reflection reality gives the adjoint identity for each individual label; signs under improper maps do not enter that identity. Every product below retains the order of its labels.

We begin with the coordinate reflection pictures constructed from (R1)–(R4). Their time and tangential translations have joint spectral measure $\mathcal{P}$, with $H \geq |P_1|, |P_2|$. Write $T_s = e^{-sH}$. The plate and point constructions below use only these coordinate inequalities. The circular cone and full Euclidean covariance will enter after those inputs to Thm. G have been supplied.

A.2.1. *Positive spectral measures for plates.* The two-point step of Thm. G requires positivity of a measure for a fixed plate label, including after a diagonal turn. We record that construction and the transform uniqueness used to compare the turned and boosted measures. In this paragraph a plate label is $\lambda = (\alpha, \upsilon)$ with $\upsilon \in C_c^1(\mathbb{R}; \mathbb{R})$, and $a \otimes \lambda$ denotes the test $a(x_3, x_1)\upsilon(x_2)$ with index α. Write $Y = [0, \infty) \times \mathbb{R}$ and $V = \{(E, q) : E \geq |q|\}$. For a profile supported in $y^0 > 0$ put

$$\widetilde{a}(E, q) = \int a(y) e^{-y^0 E + iy^1 q}\,\mathrm{d}y.$$

It is bounded by $\|a\|_1 e^{-\epsilon E}$ if $y^0 \geq \epsilon$ on its support, and convolution multiplies these transforms.

Proposition A.4 (The plate measure)**.** Let s be a translation-invariant symmetric two-point form, reflection positive at $y^0 = 0$ and satisfying the local two-point bound (R2). For each fixed real plate label there is a unique positive locally finite measure ρ on Y such that all profile transforms belong to $L^2(\rho)$ and

$$s(\theta(\bar{a} \otimes \lambda), a' \otimes \lambda) = \int \overline{\widetilde{a}}\widetilde{a}'\,\mathrm{d}\rho. \tag{A.x}$$

For $s = \sigma|_2$ it is carried by V and is invariant under $q \mapsto -q$. ■

Proof. Apply the reflection-space construction to single test functions for s, with no distinguished vacuum. The proof used in Prop. 12.2 uses only its local two-point bound and gives a joint spectral measure $\mathcal{P}^s$ on Y. Write $\Psi_a = [a \otimes \lambda]$ and $\mu_{a,a'}(B) = \langle \Psi_a, \mathcal{P}^s(B)\Psi_{a'} \rangle$. Convolution Riemann sums converge in C^1 with support in a fixed positive-time compact set; (R2) gives their norm convergence. The spectral representation identifies the weak limit and hence gives

$$\Psi_{a' * a} = \widetilde{a}'(\mathcal{P}^s)\Psi_a, \qquad |\widetilde{a}'|^2 \mu_{a,a} = |\widetilde{a}|^2 \mu_{a',a'}.$$

On $O_a = \{\widetilde{a} \neq 0\}$ define $\mathrm{d}\rho_a = |\widetilde{a}|^{-2}\mathrm{d}\mu_{a,a}$. The displayed identity makes these measures agree on overlaps. The sets O_a cover Y: at (E_0, q_0) choose $a(y) = \chi(y^0 - 2)\chi(y^1)e^{-iy^1 q_0}$ with a nonnegative small bump χ, whose transform at that point is positive. Compact exhaustion yields a countable subcover. Disjoint Borel pieces of that cover glue the measures to ρ. It is locally finite, since one transform is bounded away from zero near each point. The same identity shows that $\mu_{a,a}$ vanishes on the zeros of $\widetilde{a}$: intersect that zero set with every member of the countable cover. Thus $\mathrm{d}\mu_{a,a} = |\widetilde{a}|^2\mathrm{d}\rho$ everywhere. Polarization proves the corresponding identity for $\mu_{a,a'}$ and yields (A.x).

A finite complex measure on Y is determined by its Laplace–Fourier transform. To verify this, if the transform vanishes, testing at nonnegative integer time gives vanishing against $p(e^{-E})e^{ivq}$. Polynomial approximation on $[0, 1]$ and Fourier uniqueness in q give vanishing against $g(e^{-E})\mathbf{1}_B(q)$. Approximating closed-set indicators and applying the monotone class theorem gives the zero measure. Vanishing against every profile transform has the same consequence by testing the

continuous transform against profiles. If two measures satisfy (A.x), replace a' by $a'' * a$. The resulting identity tests the finite measure $|\widetilde{a}|^2(\rho' - \rho'')$ against every $\widetilde{a}''$. It is zero by this uniqueness. Divide on each O_a and use the countable cover to get $\rho' = \rho''$.

For $s = \sigma|_2$, the one-field space embeds isometrically in the full reflection space by $[f] \mapsto [(\alpha, f)]$, and intertwines translations. The planar spectral cone therefore carries every $|\widetilde{a}|^2\rho$, hence ρ. Reflection of x_1 and uniqueness give $q \mapsto -q$ invariance. This concerns one equal-index two-point form; no joint classical law is used. □

Here is the growth estimate needed for the angular argument. Let $a_0(s) = 1 + \cos(2\pi(s - 3/2))$ on $[1, 2]$, zero elsewhere, and let $b_1 = (3/2)\mathbf{1}_{[-1/3,1/3]}$, $b = b_1 * b_1 * b_1$. For $1/2 \le \lambda' \le 1$ put $a_{t,\lambda'}(y) = a_0(y^0/t)b(y^1/(\lambda' t))$. These are the allowed plate profiles in (R7): $\int a_0 = \int b = 1$, $0 \le a_0 \le 2$, $|a_0'| \le 2\pi$, $0 \le b \le 9/8$, $|b'| \le 9/4$. Their transforms obey

$$|\widetilde{a}_{t,\lambda'}(E, q)|^2 \ge (t^4/4)e^{-4tE}\widehat{b}(\lambda' t q)^2, \qquad \widehat{b}(v) = \left(\frac{\sin(v/3)}{v/3}\right)^3.$$

There is an absolute $c_b > 0$ such that

$$\int_{1/2}^{1} 2\widehat{b}(\lambda' v)^2 \, \mathrm{d}\lambda' \ge c_b(1 + |v|)^{-6}.$$

For $|v| \le 3$ use concavity of sine on $[0, 1]$. For $|v| \ge 3$, the mean of $\sin^6$ on any interval of length at least $1/2$ has a positive infimum, by periodicity and counting full periods. This gives the stated bound after extracting $|v|^{-6}$. On V, use $|q| \le E$ and $(1 + x)^{-6} \ge 6^{-6}e^5e^{-x}$, $x \ge 0$. Tonelli therefore gives an absolute C_b such that

$$\Gamma_\lambda(5t) := \int e^{-5tE} \, \mathrm{d}\rho_\lambda \le C_b t^{-4} \sup_{1/2 \le \lambda' \le 1} \int |\widetilde{a}_{t,\lambda'}|^2 \, \mathrm{d}\rho_\lambda.$$

Condition (R2) makes this finite at every fixed t; (R7) yields

$$\Gamma_\lambda(t) \le C'_\lambda \max(1, t^{-1})^{4+a_\Gamma}. \tag{A.xi}$$

Consequently $k_\lambda(v) = \int_V e^{-v^0E + iv^1q} \, \mathrm{d}\rho_\lambda$ is holomorphic on $\operatorname{Re} v^0 > |\operatorname{Im} v^1|$ and bounded by Γ_λ of that margin. For ordered profiles the spectral formula and Fubini represent s by this kernel. Cubic quarter turns identify the kernels on real overlaps by product approximate identities. They give a real-analytic kernel $\widetilde{k}_\lambda$ on the punctured plane. To cover arbitrary disjoint profile supports, partition into squares of diameter less than one quarter of their distance. Each pair of pieces is ordered along a signed coordinate axis, so the ordered formula after a quarter turn applies and may be summed.

For $r > 0$, the function $k_{\lambda,r}(\zeta) = k_\lambda(R_\zeta(r, 0))$ has margin $re^{-|\operatorname{Im}\zeta|}\cos(\operatorname{Re}\zeta)$ in its initial strip. Quarter-turn continuation makes it entire and $\pi/2$-periodic. Move its real part into $[-\pi/4, \pi/4]$ and use (A.xi); its growth is $O_r(e^{(4+a_\Gamma)|\operatorname{Im}\zeta|})$ with $4 + a_\Gamma < 8$. Shifting the contour of each Fourier coefficient removes every frequency $4m$ with $4|m| > 4 + a_\Gamma$. Reflection and reality identify the remaining coefficients, giving

$$k_{\lambda,r}(\zeta) = c_0(r) + 2c_1(r)\cos 4\zeta, \qquad c_0(r), c_1(r) \in \mathbb{R}. \tag{A.xii}$$

Proposition A.5 (The fourth harmonic)**.** Under the hypotheses of Thm. G, the coefficient $c_1(r)$ in (A.xii) is zero for each fixed real plate label and each $r > 0$. ■

Proof. At imaginary angles its positive cone measure gives

$$k_{\lambda,r}(is) = \int_V e^{-r(E\cosh s + q\sinh s)} \, \mathrm{d}\rho_\lambda \ge 0.$$

Divide (A.xii) by $\cosh 4s$ and let $s \to \infty$ to get $c_1(r) \ge 0$. For the opposite inequality, turn the equal-index two-point form through $\pi/4$. It is reflection positive at the coordinate plane by (R6), so Prop. A.4 constructs a positive measure ρ_D on Y. Its cone support is not assumed at this stage. For

$b_s(E,q) = (E\cosh s + q\sinh s, q\cosh s + E\sinh s)$, the pushed-forward measure $(b_s)_*\rho_\lambda$ has finite Laplace transforms, since b_s preserves V and its energy is at least $e^{-|s|}E$ there. Its real kernel is $k_\lambda(R_{is}v)$. The three harmonics give

$$k_\lambda(R_i v) + k_\lambda(R_{-i}v) = (\cosh 4 + 1)k_\lambda(v) - (\cosh 4 - 1)\widetilde{k}_\lambda(R_{\pi/4}v).$$

Multiply by $\bar{a}(u)a'(u')$ at $v = u' - \theta u$ and integrate. The last term is the turned two-point form; the other terms have the boosted spectral representations. The uniqueness proved in Prop. A.4 yields

$$(\cosh 4 - 1)\rho_D + (b_1)_*\rho_\lambda + (b_{-1})_*\rho_\lambda = (\cosh 4 + 1)\rho_\lambda. \tag{A.xiii}$$

Hence $0 \le \rho_D \le (\cosh 4 + 1)(\cosh 4 - 1)^{-1}\rho_\lambda$. This proves both cone support and finite Laplace transforms for ρ_D. Its real kernel is the turned kernel, by the profile identity; analytic continuation in the angle gives

$$k[\rho_D](R_{is}(r,0)) = c_0(r) - 2c_1(r)\cosh 4s \ge 0.$$

The same limit gives $c_1(r) \le 0$ and proves the assertion. □

A.2.2. *Slabs and point plates before rotation.* The point-plate argument in Thm. G precedes the circular spectral cone. Its semigroup is therefore the one obtained directly from (R4),

$$\mathsf{T}_3(z) = \int e^{-z_3 E + i(z_1 q_1 + z_2 q_2)}\,\mathrm{d}\mathcal{P}, \qquad d_3(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1| - |\operatorname{Im} z_2| > 0.$$

For its adjoint involution write $z^* = (-\bar{z}_1, -\bar{z}_2, \bar{z}_3)$. The separate inequalities $|q_1|, |q_2| \le E$ bound its multiplier by $e^{-d_3(z)E}$. This gives contractivity; on a compact subset of the open tube it also bounds each derivative multiplier $E^j e^{-d_3(z)E}$, proving norm holomorphy by Taylor's formula. On the plane put $d_{\mathrm{Tb}}(\xi) = \operatorname{Re}\xi^0 - |\operatorname{Im}\xi^1|$, $\mathsf{T}(\xi) = \mathsf{T}_3(\xi^1, 0, \xi^0)$, and $n_0 = (1,0)$. Here $\wp = p_{\mathrm{I}} + 4$ may be larger than four; reconstruction below uses $p_{\mathrm{I}} = 0$.

Proposition A.6 (Slab insertions)**.** For $l > 0$ and $h \in C_c^1(\{0 < x_3 < l\}; \mathbb{C})$, there is a unique bounded operator

$$B_l(\alpha, h)[F] = [(\alpha, h) \cdot \tau_{le_3} F].$$

It is complex linear in h, is bounded by $C(K,l)\|h\|_\sharp$ on every fixed compact support K in the slab, and satisfies slab localization, tangential translation covariance, and

$$B_l(\alpha, h)^* = B_l(\alpha, \overline{h \circ \theta_{l/2}}).$$

If h is real, supported in $B(z, c_{\mathrm{I}}\eta)$, with $\eta/8 \le z_3 \le 3\eta/8$ and $0 < \eta \le 1$, then

$$\|B_{\eta/2}(\alpha, h)\| \le C_{\mathrm{I}}\eta^{-p_{\mathrm{I}}}\|h\|_\sharp. \tag{A.xiv}$$

■

Proof. For this small support, put $h_0 = \tau_{-\eta e_3/4}h$ and apply (R5) to h_0 between $\tau_{\eta e_3/4}F$ and $\tau_{\eta e_3/4}G$. Translation invariance makes the form $\sigma(\theta\bar{F} \cdot (\alpha, h) \cdot \tau_{\eta e_3/2}G)$, bounded by the right side of (A.xiv) times $\|[F]\|\|[G]\|$. It vanishes on null classes and gives a bounded operator by Riesz representation. Pairing with all classes identifies its action as asserted.

For general compact K, let $\delta = \min_{x\in K}\min(x_3, l - x_3) > 0$, $\eta = \min(1, 4\delta)$ and $b = c_{\mathrm{I}}\eta/2$. Use the partition

$$\chi_{b,k}(x) = \prod_{\mu=1}^{3} \chi(x_\mu/b - k_\mu), \qquad \chi(s) = \begin{cases} \cos^2(\pi s/2), & |s| \le 1, \\ 0, & |s| > 1. \end{cases}$$

It sums to one and has Lipschitz constant at most $\sqrt{3}\pi/(2b)$. Only finitely many pieces $h_k = \chi_{b,k}h$ meet K. Let c_k be the projection of $bk_3 - \eta/4$ onto $[0, l - \eta/2]$. The bounds $b \le \eta/32$ and $\eta/4 \le \delta$

put the center of $\tau_{-c_k e_3} h_k$ between $7\eta/32$ and $9\eta/32$; its support has radius $\sqrt{3}b < c_{\mathrm{I}}\eta$. Thus the small-piece construction applies, and

$$B_l(\alpha, h) = \sum_k T_{c_k} B_{\eta/2}(\alpha, \tau_{-c_k e_3} h_k) T_{l-c_k-\eta/2}$$

has the prescribed value on classes. The product estimate $\|h_k\|_\sharp \le (1+40\sqrt{3}\pi M/(c_{\mathrm{I}}\eta))\|h\|_\sharp$ gives the compact-support bound; splitting real and imaginary parts costs at most two. The action on dense classes makes the construction independent of the partition. Translating the class identity proves localization and tangential covariance. Reflection at $l/2$, followed by permutation and conjugation, proves the adjoint identity. Iterating the class identity represents any ordered slab tuple as the product of its successive slab insertions. □

Write $B_d^{\mathrm{c}}(\alpha, h) = B_{2d}(\alpha, \tau_{de_3} h)$. For the fixed real plate label $\lambda = (\alpha, \upsilon)$, let Υ_λ be the length of the interval hull of $\operatorname{supp}\upsilon$. Choose a nonnegative profile $m \in C_c^1([0,\infty))$, constant near zero and zero for $s \ge 1$, with $2\pi\int_0^1 m(s)s\,\mathrm{d}s = 1$. Its radial extension is a C^1 weight on the unit disc. Put $m_0 = \sup m$, $m_1 = \sup|m'|$, $m_2 = \sqrt{2}m_0m_1 + 2m_0^2$, $m_3 = m_0^3 + \sqrt{3}m_0^2m_1$, and define, for $N = 2, 3$,

$$a_{r,t}^{(N)}(x) = \prod_{i=1}^{N} r^{-2} m(r^{-1}\sqrt{x_i^2 + t_i^2}).$$

The iterated disc mean-value formula is

$$u(0) = \int_{(-r,r)^N} \int a_{r,t}^{(N)}(x) u(x + it)\,\mathrm{d}x\,\mathrm{d}t. \tag{A.xv}$$

It holds when u is holomorphic near the closed polydisc; polar integration proves it one variable at a time. The weight has supremum at most $m_0^N r^{-2N}$ and Lipschitz constant at most $\sqrt{N}m_0^{N-1}m_1 r^{-2N-1}$.

Proposition A.7 (Point plates)**.** There is a unique norm-holomorphic family $B_\lambda(\xi, \xi')$ on the planar tube squared, with the shift and adjoint laws, such that

$$\|B_\lambda(\xi, \xi')\| \le C_{\mathrm{plt}}(1 + \Upsilon_\lambda)\|\upsilon\|_{C^1} \min(1, d_{\mathrm{Tb}}(\xi), d_{\mathrm{Tb}}(\xi'))^{-\wp} \tag{A.xvi}$$

and

$$B_d^{\mathrm{c}}(a \otimes \lambda) = \int a(y) B_\lambda(\,\mathrm{d}n_0 + y,\, \mathrm{d}n_0 - y)\,\mathrm{d}y \quad (\operatorname{supp} a \subset \{|y^0| < d\}). \tag{A.xvii}$$

The adjoint is $B_\lambda(\xi'^*, \xi^*)$, where $\xi^* = (\bar{\xi}^0, -\bar{\xi}^1)$. The constant is common to the finite index set. ■

Proof. Take a radial planar mollifier j_δ of mass one supported in $|y| \le \delta/2$ and put

$$A_\delta = B_\delta^{\mathrm{c}}(j_\delta \otimes \lambda), \qquad B^\delta(\xi, \xi') = \mathsf{T}(\xi - \delta n_0) A_\delta \mathsf{T}(\xi' - \delta n_0).$$

It is holomorphic beyond margin δ, self-adjoint at equal positive-time arguments, and has the shift and adjoint laws. Localization gives

$$\int a(y) B^\delta(\,\mathrm{d}n_0 + y,\, \mathrm{d}n_0 - y)\,\mathrm{d}y = B_d^{\mathrm{c}}((a * j_\delta) \otimes \lambda).$$

This follows from convolution Riemann sums in C^1 on a common compact slab support.

Put $\eta = \min(1, d_{\mathrm{Tb}}(\xi), d_{\mathrm{Tb}}(\xi'))$ and $r = c_{\mathrm{I}}\eta/8$. Apply the two-variable mean-value formula (A.xv) to $\langle v, B^\delta(\xi + z, \xi' - z)v'\rangle$ for $\delta \le r$, using $a_{r,t}^{(2)}$. The polydisc of radius $2r$ remains in the tube. Move the imaginary shifts into contractions; the inner integral is the centered insertion at margin $\eta/4$ of $(a_{r,t}^{(2)} * j_\delta) \otimes \lambda$. Its planar factor is supported in $[-2r, 2r]^2$, with supremum $O(r^{-4})$ and Lipschitz constant $O(r^{-5})$. Partition the transverse profile by $\upsilon_k(s) = \chi(s/(4r) - k)\upsilon(s)$. At most

$3(1+\Upsilon_\lambda)/(c_{\mathrm{I}}\eta)$ pieces are nonzero. Each insertion has support in $B(4rke_2, c_{\mathrm{I}}\eta)$ and sharp norm bounded by $40Mm_2r^{-5}\|v\|_{C^1}$. Equation (A.xiv), followed by the integral over $(-r,r)^2$, gives

$$\|B^\delta(\xi,\xi')\| \le (2r)^2 \frac{3(1+\Upsilon_\lambda)}{c_{\mathrm{I}}\eta} C_{\mathrm{I}}\eta^{-p_{\mathrm{I}}} 40Mm_2r^{-5}\|v\|_{C^1}.$$

This is (A.xvi), uniformly in δ.

At $(\epsilon n_0, \epsilon n_0)$ the map $z \mapsto B^\delta(\epsilon n_0 + z, \epsilon n_0 - z)$ is bounded on the polydisc of radius $\epsilon/4$ by $b_\epsilon = C_{\mathrm{plt}}(1+\Upsilon_\lambda)\|v\|_{C^1}(\frac{1}{2}\min(1,\epsilon))^{-\wp}$. Cauchy's derivative estimate and the convolution identity, used in both orders, give

$$\|B^\delta(\epsilon n_0, \epsilon n_0) - B^{\delta'}(\epsilon n_0, \epsilon n_0)\| \le 4b_\epsilon(\delta+\delta')/\epsilon.$$

Thus the norm limit exists. Sandwich it between $\mathsf{T}(\xi - \epsilon n_0)$ and $\mathsf{T}(\xi' - \epsilon n_0)$ to define $B_\lambda(\xi,\xi')$. This is the locally uniform limit of the mollified families and is independent of ϵ. Choosing a smaller ϵ locally proves holomorphy; bounds, shifts and adjoints pass to the limit. Convolution convergence in C^1 proves (A.xvii). That formula identifies all matrix elements at equal positive-time arguments; the shift law gives uniqueness everywhere. □

For clarity, no extra product theorem is needed for the plate kernels in Thm. G. At an interior junction the shift law gives, for positive-margin κ_1, κ_2,

$$\begin{aligned} &B_\lambda(\xi,\kappa_1)\mathsf{T}(\kappa_0)B_{\lambda'}(\kappa_2,\xi') \\ &\quad = B_\lambda(\xi,\epsilon n_0)\mathsf{T}(\kappa_1+\kappa_0+\kappa_2-2\epsilon n_0)B_{\lambda'}(\epsilon n_0,\xi') \end{aligned} \tag{A.xviii}$$

when ϵ does not exceed their two margins. Iterating this identity makes the ordered product depend only on the consecutive point differences. The slab class identity and (A.xvii), integrated successively, show that its vacuum component represents every ordered plate tuple. Taking the adjoint of the first ordered vector reverses its labels and reflects its points; (A.xviii) joins it to the second vector. This is the labelled Gram identity used in Thm. G. The quarter turn in the plane is proper, so equality of the ordered kernel integrals on real overlaps and product approximate identities identify the four frame kernels. Their overlaps are convex with an open real part; the identity theorem identifies the holomorphic pieces there. These are precisely the point-plate products used in the harmonic argument.

Proposition A.8 (Removing intersecting planar projections)**.** If the planar rotation identity holds for n-tuples whose supports have pairwise disjoint planar projections, it holds for every separated n-tuple. ■

Proof. Suppose two groups lie in $x_2 < c-\delta$ and $x_2 > c+\delta$. In the e_2 reflection picture their relative translation by su, $u \perp e_2$, $|u| = 1$, is a matrix element

$$\Phi(s) = \int e^{-2\delta E + is(u_1q_1+u_3q_3)}\,\mathrm{d}m(E,q).$$

The coordinate cone makes it holomorphic for $|\operatorname{Im} s| < \sqrt{2}\delta$. Let A_j be the planar projections and I_j the x_2 interval hulls. First assume that each intersecting pair of projections has a level separating its two intervals and outside every I_j. Induct on the number of intersecting pairs. Translate all entries above a chosen separating level by se_1. For large s the projections on opposite sides are disjoint, and that number decreases. The interval hulls are unchanged. Before and after rotation, the correlations are holomorphic on a common strip: the rotated relative translation is sR_te_1, still perpendicular to e_2. Induction and the identity theorem give the rotation identity also at $s = 0$.

For a general tuple of separation d, choose $0 < \epsilon < d/n$ and partition its tests in x_2 with $\chi(2x_2/\epsilon - k)$. Each interval hull has length at most ϵ. If two nonzero pieces have intersecting planar projections, their heights over that projection differ by at least d. Their interval gap is at least $d - 2\epsilon > (n-2)\epsilon$, so the other $n-2$ hulls cannot cover it. The first case applies to every product. The partition commutes with the planar rotation; summing proves the assertion. □

A.2.3. *Angular coefficients on the four-frame domain.* Let $\mathsf{Tb} = \{\xi : d_{\mathrm{Tb}}(\xi) > 0\}$. For a permutation $\mathfrak{s}$ and a quarter-turn frame $\upsilon \in \frac{\pi}{2}\mathbb{Z}$, define

$$\mathcal{Z}(\mathfrak{s}, \upsilon) = \{Z : z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)} \in R_\upsilon \mathsf{Tb},\ a < n\}, \qquad \mathcal{Z}_n = \bigcup_{\mathfrak{s},\upsilon} \mathcal{Z}(\mathfrak{s}, \upsilon). \tag{A.xix}$$

The point-plate products on these tubes glue to the holomorphic kernel $K_n(Z; \underline{\lambda})$ constructed above. The real part of its domain is

$$\mathcal{O}_n = \{X : \text{all } x_a^0 \text{ are distinct, or all } x_a^1 \text{ are distinct}\}.$$

Each tube is convex and contains the real part of every point in it. Their overlap identifications were proved above from product tests and the identity theorem. The complement of $\mathcal{O}_n$ is the finite union of codimension-two subspaces $\{x_a^0 = x_b^0,\ x_k^1 = x_l^1\}$, $a \neq b$, $k \neq l$. Consequently $\mathcal{O}_n$ is connected; every tube is connected and meets it, so $\mathcal{Z}_n$ is connected. The difference inequalities show that this domain is stable under common complex translations, permutations, quarter turns, conjugation, and R_{is} for real s. For the last assertion, in the light-cone coordinates $\xi^\pm = \xi^0 \pm i\xi^1$, R_{is} multiplies the two coordinates by positive real numbers. Kernels are invariant under the first three operations, with points and labels permuted together. Their conjugation law includes the factor $\pi(\underline{\lambda})$ from the finite-label sign rule.

Lemma A.9 (Angular coefficients)**.** Put $\mathcal{M}_n = \{m \in \mathbb{Z} : 4|m| \leq \wp n\}$. For a fixed word of $n \geq 2$ labels there are unique holomorphic functions c_m on $\mathcal{Z}_n$, zero for $m \notin \mathcal{M}_n$, with

$$K_n(R_{is}Z; \underline{\lambda}) = \sum_{m \in \mathcal{M}_n} c_m(Z; \underline{\lambda}) e^{-4ms}. \tag{A.xx}$$

They inherit permutation and common-translation invariance, and satisfy

$$c_m(R_{is}Z) = e^{-4ms} c_m(Z), \qquad \overline{c_m(X; \underline{\lambda})} = \pi(\underline{\lambda}) c_{-m}(X; \underline{\lambda}) \quad (X \in \mathcal{O}_n).$$

They extend uniquely to real-analytic functions $\widetilde{c}_m$ on configurations of distinct planar points, with

$$\widetilde{c}_m(\upsilon + R_t(X - \upsilon); \underline{\lambda}) = e^{4imt} \widetilde{c}_m(X; \underline{\lambda}). \tag{A.xxi}$$

Here the same vector υ is subtracted from every point. ■

Proof. First take a real configuration X whose consecutive differences make angles of modulus less than $\pi/8$ with n_0. For $\zeta = t + is$ choose a quarter-turn frame υ with $|t - \upsilon| \leq \pi/4$. All differences of $R_\zeta X$ lie in $R_\upsilon \mathsf{Tb}$, with margins at least

$$\min_a |x_{a+1} - x_a| \cos(3\pi/8) e^{-|s|}.$$

Thus $h_X(\zeta) = K_n(R_\zeta X)$ is entire, $\pi/2$-periodic, and bounded by $C_X e^{\wp n|s|}$. Shifting its Fourier contour vertically removes the coefficient of $e^{4im\zeta}$ when $4|m| > \wp n$. Fourier uniqueness on the real circle gives the finite expansion, valid throughout the complex plane.

For arbitrary $Z \in \mathcal{Z}_n$, put $\mu = \#\mathcal{M}_n$ and invert the Vandermonde matrix $V_{lm} = e^{-4ml}$, $0 \leq l < \mu$. With $W = V^{-1}$, define

$$c_m(Z) = \sum_{l=0}^{\mu-1} W_{ml} K_n(R_{il} Z).$$

These functions are holomorphic, since $\mathcal{Z}_n$ is stable under imaginary rotations. On each tube, permuted quarter-turned narrow-sector configurations of the preceding type form a nonempty real open set. There the formula recovers the Fourier coefficients. The holomorphic identity theorem on that tube proves (A.xx) everywhere. The same Vandermonde comparison proves uniqueness, the imaginary-rotation identity, and the permutation, translation and conjugation identities.

We prove the real phase law without assuming that $\mathcal{O}_n$ contains an entire real rotation orbit. For a real point X in one tube, write its successive differences in that frame as $r_a n_{\vartheta_a}$, $|\vartheta_a| < \pi/2$. For

$$t \in I_X = \bigcap_a (-\pi/2 - \vartheta_a, \pi/2 - \vartheta_a),$$

the full strip $R_{t+is}X$ remains in that tube. Analytically continue (A.xx) in the rotation parameter and compare its coefficients. This gives $c_m(R_t X) = e^{4imt} c_m(X)$ on I_X.

Now take configurations with distinct points and no perpendicular pair of distinct difference vectors. They form a rotation-invariant dense open set contained in $\mathcal{O}_n$. For density, the product of squared differences and of the relevant scalar products is a polynomial nonzero at $((a, a^2))_{a=1}^n$; on the line segment from any configuration to this one it has only finitely many zeros. Along each such rotation orbit, the preceding local identity makes $t \mapsto e^{-4imt} c_m(R_t X)$ locally constant on $\mathbb{R}$, hence constant. Continuity extends the identity to every $X, R_t X \in \mathcal{O}_n$.

For any configuration of distinct points, all but finitely many angles modulo π give distinct time coordinates. Choose such a t and put

$$\widetilde{c}_m(X) = e^{-4imt} c_m(R_t X).$$

The preceding phase law makes this independent of t. Near each configuration one may keep t fixed, which gives a holomorphic local extension and hence real analyticity. Translation invariance and this definition prove (A.xxi), including rotations about arbitrary centers. □

A.2.4. *Point insertions.* From here on assume $p_{\mathrm{I}} = 0$, as in Prop. 13.5 and Thm. H. We first give the point construction with its exponent 4. This is the three-dimensional mollifier construction behind the point-kernel method of [3, Lem. 8.28]; the exponent here is obtained from the present insertion estimate (R5). This construction needs only the coordinate cone. In its proof we use T_3 and d_3 from the preceding point-plate construction, before the conclusion of Thm. G. Let $B_d^{\mathrm{c}}(\alpha, h)$ denote the centered slab insertion at margin d supplied by (R5). In matrix elements it is the insertion of (α, h) between vectors translated by d. Translation covariance gives its shift law, and reflection reality gives $B_d^{\mathrm{c}}(\alpha, h)^* = B_d^{\mathrm{c}}(\alpha, \theta\bar{h})$.

Proposition A.10 (Point insertions)**.** For each label there is a unique norm-holomorphic family $B_\alpha(z, z')$ on $\{d_3 > 0\}^2$ with

$$\|B_\alpha(z, z')\| \le C_{\mathrm{pt}} \min(1, d_3(z), d_3(z'))^{-4}, \tag{A.xxii}$$

$$\mathsf{T}_3(w) B_\alpha(z, z') \mathsf{T}_3(w') = B_\alpha(w + z, z' + w'), \tag{A.xxiii}$$

$$B_\alpha(z, z')^* = B_\alpha(z'^*, z^*), \tag{A.xxiv}$$

$$B_d^{\mathrm{c}}(\alpha, h) = \int h(x) B_\alpha(\,\mathrm{d}e_3 + x,\, \mathrm{d}e_3 - x)\,\mathrm{d}x. \tag{A.xxv}$$

The shifts are in the closed tube, and the last identity holds for $h \in C_c^1(\{|x_3| < d\}; \mathbb{C})$. The constant is common to the finite label set. ■

Proof. Choose a nonnegative radial C^1 mollifier j_δ of mass one supported in $|x| \le \delta/2$, and put

$$A_{\alpha,\delta} = B_\delta^{\mathrm{c}}(\alpha, j_\delta), \qquad B_\alpha^\delta(z, z') = \mathsf{T}_3(z - \delta e_3) A_{\alpha,\delta} \mathsf{T}_3(z' - \delta e_3).$$

This is defined when both margins are at least δ and is holomorphic when they exceed δ. Radiality and the slab adjoint relation make $A_{\alpha,\delta}$ self-adjoint. The shift and adjoint laws follow. Slab localization gives

$$B_\alpha^\delta(\,\mathrm{d}e_3 + y,\, \mathrm{d}e_3 - y) = B_d^{\mathrm{c}}(\alpha, \tau_y j_\delta), \qquad \int h(y) B_\alpha^\delta(\,\mathrm{d}e_3 + y,\, \mathrm{d}e_3 - y)\,\mathrm{d}y = B_d^{\mathrm{c}}(\alpha, h * j_\delta)$$

whenever the support has distance greater than δ from the slab boundary. Convolution Riemann sums converge in C^1 with a common compact support, and (R5) permits this limit.

Here is the uniform estimate that removes the mollifier. Use the weight $a^{(3)}_{r,t}$ in (A.xv). It has supremum $O(r^{-6})$ and Lipschitz constant $O(r^{-7})$. Set $\eta = \min(1, d_3(z), d_3(z'))$ and $r = c_{\mathrm{I}}\eta/8$. For $\delta \le r$, apply this formula to $\langle v, B^{\delta}_{\alpha}(z+w, z'-w)v'\rangle$. The polydisc of radius $2r$ loses at most $6r$ in tube margin. Move its imaginary shifts into the contraction factors with arguments $z - \eta e_3/4 + it$ and $z' - \eta e_3/4 - it$. The remaining integral in x is the centered slab insertion of $\gamma_t = a^{(3)}_{r,t} * j_\delta$ at margin $\eta/4$. Its support lies in $[-2r, 2r]^3 \subset B(0, c_{\mathrm{I}}\eta)$, and its sharp norm is bounded by $40Mm_3r^{-7}$, with m_3 depending only on the chosen weight. Consequently (R5) gives

$$\|B^{\delta}_{\alpha}(z, z')\| \le (2r)^3 C_{\mathrm{I}} 40 M m_3 r^{-7} = C_{\mathrm{pt}}\eta^{-4}.$$

The same constants work for all labels.

Fix $\epsilon > 0$. On the polydisc of radius $\epsilon/6$, $w \mapsto B^{\delta}_{\alpha}(\epsilon e_3 + w, \epsilon e_3 - w)$ is bounded by $b_\epsilon = C_{\mathrm{pt}}(\frac{1}{2}\min(1,\epsilon))^{-4}$ for all sufficiently small δ. Cauchy's derivative estimate bounds its change at a real y, $|y| \le \epsilon/12$, by $12b_\epsilon|y|/\epsilon$. Integrate against $j_{\delta'}$. The convolution identity, applied also with δ and δ' interchanged, gives

$$\|B^{\delta}_{\alpha}(\epsilon e_3, \epsilon e_3) - B^{\delta'}_{\alpha}(\epsilon e_3, \epsilon e_3)\| \le 6b_\epsilon(\delta + \delta')/\epsilon.$$

Thus these operators have a norm limit $B_{\alpha,[\epsilon]}$. For $\epsilon \le \min(d_3(z), d_3(z'))$ define

$$B_\alpha(z, z') = \mathsf{T}_3(z - \epsilon e_3) B_{\alpha,[\epsilon]} \mathsf{T}_3(z' - \epsilon e_3).$$

This is the locally uniform limit of B^{δ}_{α}, so is independent of ϵ. Choosing a smaller margin locally proves norm holomorphy. The bound, shifts and adjoints pass to the limit. Uniform norm convergence on a slab support, together with $h * j_\delta \to h$ in C^1, proves (A.xxv). That integral identity determines every matrix element at $(de_3 + y, de_3 - y)$, hence at (de_3, de_3); the shift law then determines the family everywhere. This proves uniqueness. □

A.2.5. *Point forms and energy powers.* We now use the conclusion of Thm. G: the measure $\mathcal{P}$ is carried by $E \ge |q|$. Define the circular tube and its semigroup by

$$d_\circ(z) = \operatorname{Re} z_3 - |\operatorname{Im}(z_1, z_2)|, \qquad \mathrm{Tb}_\circ = \{d_\circ > 0\}, \qquad \mathcal{T}(z) = \int e^{-z_3 E + i(z_1, z_2)\cdot q}\, \mathrm{d}\mathcal{P}. \tag{A.xxvi}$$

The multipliers have modulus at most $e^{-d_\circ(z)E}$. Thus these operators are contractions on the closed tube, strongly continuous there and norm holomorphic in its interior. On every smaller open tube the same exponential bounds all derivatives of the multipliers. Dominated convergence against scalar spectral measures proves boundary continuity. They have the semigroup and adjoint laws, with $z^* = (-\bar z_1, -\bar z_2, \bar z_3)$, and fix Ω. Their interior ranges lie in $\mathcal{D}_\infty = \bigcap_{j\ge 0} \operatorname{dom} H^j$. For Minkowski coordinates $x = (t, \vec x)$ define

$$\iota x = (\vec x, -it), \qquad U(x) = \mathcal{T}(\iota x) = e^{itH}\mathcal{U}(\vec x). \tag{A.xxvii}$$

The map ι carries $\mathbb{R}^{1+2} + iV_+$ onto $\mathrm{Tb}_\circ$, where $V_+ = \{t > |\vec x|\}$.

After Thm. G, the same positive-time point operators extend to the circular tube. For $\epsilon \le \min(1, d_\circ(z), d_\circ(z'))$ set

$$B_\alpha(z, z') = \mathcal{T}(z - \epsilon e_3) B_\alpha(\epsilon e_3, \epsilon e_3) \mathcal{T}(z' - \epsilon e_3).$$

The positive-time shift law makes this independent of ϵ and equal to the preceding family where both are defined. The same formula proves norm holomorphy, the shift and adjoint identities, and the bound $C_{\mathrm{pt}} \min(1, d_\circ(z), d_\circ(z'))^{-4}$. Thus the construction used below retains exponent four on the full circular tube.

Put $B_\alpha(s, s') = B_\alpha(se_3, s'e_3)$ and $R_H = (1 + H)^{-1}$. For real $l > 4$ define

$$A_{\alpha,l} = \Gamma(l)^{-2} \int_0^\infty \int_0^\infty (ss')^{l-1} e^{-s-s'} B_\alpha(s, s')\, \mathrm{d}s\, \mathrm{d}s'. \tag{A.xxviii}$$

This norm integral is self-adjoint and

$$\|A_{\alpha,l}\| \le C_{\mathrm{pt}} \left(1 + \frac{2\Gamma(l-4)}{\Gamma(l)}\right). \tag{A.xxix}$$

Indeed, the point bound is at most $C_{\mathrm{pt}}(1 + s^{-4} + s'^{-4})$. The shift law and the Gamma integral for R_H^a imply

$$R_H^a A_{\alpha,l} R_H^a = A_{\alpha,l+a}, \qquad T_s A_{\alpha,l} T_{s'} = R_H^l B_\alpha(s,s') R_H^l. \tag{A.xxx}$$

For the first identity insert the two integrals for R_H^a and set $s = r + t$, $s' = r' + t'$. The inner integrals are $s^{a+l-1}\Gamma(a)\Gamma(l)/\Gamma(a+l)$. For the second, use $B_\alpha(r+s, r'+s') = T_r B_\alpha(s,s') T_{r'}$. The integrable bound above justifies both applications of Fubini. It follows that

$$q_\alpha(\psi,\psi') = \langle (1+H)^l \psi, A_{\alpha,l}(1+H)^l \psi' \rangle \tag{A.xxxi}$$

is independent of the admissible l on the nested graph domains. On $\mathcal{D}_\infty$ it is Hermitian and obeys

$$q_\alpha(T_s\psi, T_{s'}\psi') = \langle \psi, B_\alpha(s,s')\psi' \rangle. \tag{A.xxxii}$$

To obtain the last identity, apply the second equality in (A.xxx) at $(s/2, s'/2)$ and cancel the resolvents against the bounded operators $(1+H)^l T_{s/2}$ and $(1+H)^l T_{s'/2}$.

A.2.6. *Ordered kernels and their common continuation.* For a label word $\underline{\alpha} = (\alpha_1, \dots, \alpha_n)$, define

$$\mathbf{W}_{\underline{\alpha}}(\zeta) = \langle \Omega, B_{\alpha_1}(\eta,\eta) \mathcal{T}(\zeta_1 - 2\eta e_3) B_{\alpha_2}(\eta,\eta) \cdots \mathcal{T}(\zeta_{n-1} - 2\eta e_3) B_{\alpha_n}(\eta,\eta)\Omega \rangle, \tag{A.xxxiii}$$

where $0 < \eta \le \min(1, \frac{1}{2}\min_j d_\circ(\zeta_j))$. For $z_1, z_{j+1} - z_j \in \mathrm{Tb}_\circ$, similarly put

$$\Psi_{\underline{\alpha}}(z) = \mathcal{T}(z_1 - \eta e_3) B_{\alpha_1}(\eta,\eta) \mathcal{T}(z_2 - z_1 - 2\eta e_3) \cdots \mathcal{T}(z_n - z_{n-1} - 2\eta e_3) B_{\alpha_n}(\eta,\eta)\Omega, \tag{A.xxxiv}$$

with the additional restriction $\eta \le d_\circ(z_1)$. For $n = 1$ omit all differences. The shift law makes these definitions independent of η: replace each point insertion by $T_{\eta-\eta'} B_\alpha(\eta',\eta') T_{\eta-\eta'}$ and absorb the padding into the adjacent semigroups. The end factors fix Ω. Locally taking a strictly smaller fixed margin proves holomorphy, in norm for Ψ, and gives

$$|\mathbf{W}_{\underline{\alpha}}(\zeta)| \le (C_{\mathrm{pt}}\eta^{-4})^n, \qquad \|\Psi_{\underline{\alpha}}(z)\| \le (C_{\mathrm{pt}}\eta^{-4})^n. \tag{A.xxxv}$$

Here the differences have margins at least 2η and, for the second bound, the first point has margin at least η.

For completeness, these kernels recover the original ordered tuples. If h_j lies in the slab $|x_3 - t_j| < \eta/2$ with $t_{j+1} - t_j \ge 2\eta$, put $g_j = \tau_{-t_j e_3} h_j$ and $s_j = t_{j+1} - t_j - 2\eta$. The slab class identity, applied from right to left, gives

$$\sigma((\alpha_j, h_j)_j) = \langle \Omega, B^{\mathrm{c}}_\eta(\alpha_1, g_1) T_{s_1} \cdots T_{s_{n-1}} B^{\mathrm{c}}_\eta(\alpha_n, g_n)\Omega \rangle.$$

If $t_1 \ge \eta$, the same product preceded by $T_{t_1-\eta}$ is the tuple class. In each slab integral split the point margin in half. Consecutive semigroups then combine into the differences in (A.xxxiii) and (A.xxxiv). Their positive margins and the point bounds justify iterated integration and give

$$\sigma((\alpha_j, h_j)_j) = \int \mathbf{W}_{\underline{\alpha}}(x_2 - x_1, \dots, x_n - x_{n-1}) \prod_j h_j(x_j)\, \mathrm{d}x, \tag{A.xxxvi}$$

$$\langle \psi, [(\alpha_j, h_j)_j] \rangle = \int \langle \psi, \Psi_{\underline{\alpha}}(x) \rangle \prod_j h_j(x_j)\, \mathrm{d}x. \tag{A.xxxvii}$$

To change the ordering direction, for $u \in S^2$ put

$$d^u_\circ(z) = \mathrm{Re}(u \cdot z) - |\,\mathrm{Im}(z - (u\cdot z)u)|, \qquad \mathrm{Tb}^{u,\pi}_n = \{z : d^u_\circ(z_{\pi(j+1)} - z_{\pi(j)}) > 0\}.$$

Choose $R \in \mathrm{SO}(3)$ with $Re_3 = u$. On this convex tube define $\mathbf{S}_{\underline{\alpha}}$ by applying (A.xxxiii) to the differences $R^{-1}(z_{\pi(j+1)} - z_{\pi(j)})$, with the labels permuted by the same π. These pieces agree and define one holomorphic function on $\mathrm{Tb}^{\mathrm{all}}_n = \bigcup_{u,\pi} \mathrm{Tb}^{u,\pi}_n$. Here are the two identifications needed

for that assertion. At real configurations in two ordering regions, (A.xxxvi) gives the same value against every product of tests in sufficiently small balls. Product approximate identities identify the continuous kernels there. A nonempty complex overlap is convex and contains the real part of each point, so has a nonempty open real part. The identity theorem therefore identifies the two holomorphic pieces on the whole overlap. It also proves independence of R and simultaneous symmetry of points and labels. Moreover,

$$|\mathbf{S}_{\underline{\alpha}}(z)| \le (C_{\mathrm{pt}}\eta^{-4})^n \quad \text{if } d^u_\circ(z_{\pi(j+1)} - z_{\pi(j)}) \ge 2\eta. \tag{A.xxxviii}$$

These kernels cover all distinct real points. If their least distance is δ, the spherical zone $|u \cdot (x_i - x_j)| < |x_i - x_j|/n^2$ has area $4\pi/n^2$. The union over pairs has area less than 2π. Hence some direction orders the points with gaps at least δ/n^2. For separated test supports choose $\eta = \min(\delta/(4n^2), 1)$ and partition each test into pieces supported in balls of radius $\eta/4$. Each product is ordered in one such direction. Summing (A.xxxvi) proves the kernel formula for every separated tuple. Thus the restriction $\mathsf{s}_{\underline{\alpha}}$ of $\mathbf{S}_{\underline{\alpha}}$ to real points is the Euclidean kernel of σ, including configurations with equal third coordinates.

A.2.7. *Fields on the common graph domain.* Fix an integer $l > 4$ and write $N_H = 1 + H$. Set

$$c_{\alpha,N}(f) = \|A_{\alpha,l}\| \sum_{k=0}^{N} \binom{N}{k} \|\partial_t^k f\|_1.$$

We use the graph seminorms $\|N_H^j \psi\|$ on $\mathcal{D}_\infty$. The projections $P_M = \mathcal{P}(\{E \le M\})$ have ranges in this domain and converge strongly to the identity. In particular,

$$|\langle N_H^j \psi, \chi\rangle| \le C\|\psi\| \quad (\psi \in \mathcal{D}_\infty) \quad \Longrightarrow \quad \chi \in \operatorname{dom} H^j, \quad \|N_H^j \chi\| \le C. \tag{A.xxxix}$$

Indeed, use $\psi = N_H^j P_M \chi$ in the inequality. It bounds $\int_{E\le M}(1+E)^{2j}\,\mathrm{d}m_\chi$ by C^2; monotone convergence proves the claim.

Proposition A.11 (Smeared fields)**.** There is a unique complex linear map $f \mapsto \varphi_\alpha(f)$ from $\mathcal{S}(\mathbb{R}^{1+2})$ to operators $\mathcal{D}_\infty \to \mathcal{D}_\infty$ satisfying

$$\langle \psi, \varphi_\alpha(f)\psi'\rangle = \int f(x) q_\alpha(U(-x)\psi, U(-x)\psi')\,\mathrm{d}x. \tag{A.xl}$$

It obeys, for every $j \ge 0$,

$$\|N_H^j \varphi_\alpha(f)\psi\| \le c_{\alpha,l+j}(f)\|N_H^{2l+j}\psi\|. \tag{A.xli}$$

On $\mathcal{D}_\infty$ its adjoint, translation and energy identities are

$$\varphi_\alpha(f)^* \supset \varphi_\alpha(\bar f), \qquad U(a)\varphi_\alpha(f)U(-a) = \varphi_\alpha(f(\cdot - a)), \qquad [H, \varphi_\alpha(f)] = i\varphi_\alpha(\partial_t f). \tag{A.xlii}$$

■

Proof. The form $q_{\alpha,f}$ on the right of (A.xl) has bounded continuous matrix-element factor, by (A.xxxi), and satisfies

$$|q_{\alpha,f}(\psi,\psi')| \le \|A_{\alpha,l}\|\|f\|_1 \|N_H^l\psi\|\|N_H^l\psi'\|.$$

To obtain an operator, move the energy powers off its first vector. The spectral theorem differentiates $e^{-itH}\psi$ in norm on $\operatorname{dom} H$. Differentiating both vectors and integrating by parts in time gives

$$q_{\alpha,f}(H\psi,\psi') - q_{\alpha,f}(\psi, H\psi') = i q_{\alpha,\partial_t f}(\psi,\psi').$$

Schwartz decay removes the boundary terms, since the matrix element and its time derivative are bounded. Iteration yields

$$q_{\alpha,f}(N_H^N\chi,\psi') = \sum_{k=0}^{N}\binom{N}{k} i^k q_{\alpha,\partial_t^k f}(\chi, N_H^{N-k}\psi'). \tag{A.xliii}$$

Choose $\chi = R_H^l \psi$ and $N = l + j$. The preceding form bound gives

$$|q_{\alpha,f}(N_H^j \psi, \psi')| \le c_{\alpha,l+j}(f) \|\psi\| \|N_H^{2l+j} \psi'\|.$$

For $j = 0$, Riesz representation defines $\varphi_\alpha(f)\psi'$. For every j, (A.xxxix) puts it in $\operatorname{dom} H^j$ and proves (A.xli). Thus it lies in the common domain $\mathcal{D}_\infty$. The defining form proves linearity, independence of l, and the adjoint identity. Changing variables proves translation covariance. The integrated derivative identity proves the commutator after testing against dense graph-domain vectors. □

Two continuity facts will control products and their boundary values. First, for $w, w' \in \mathrm{Tb}_\circ$ and $0 < s \le \min(1, d_\circ(w), d_\circ(w'))$, the regularized field is

$$\mathcal{T}(w)\varphi_\alpha(f)\mathcal{T}(w') = \int f(x)\mathcal{T}(w - se_3 + \iota x) B_\alpha(s,s) \mathcal{T}(w' - se_3 - \iota x)\, \mathrm{d}x. \tag{A.xliv}$$

To prove this matrix-element identity, the positive margins put both vectors in $\mathcal{D}_\infty$. Apply (A.xl), split off T_s on each side, and use (A.xxxii). The remaining factors are contractions: adding ιx does not change a tube margin. The integrand, without f, is strongly continuous and bounded by $C_{\mathrm{pt}} s^{-4}$. For $w_1, \dots, w_n \in \mathrm{Tb}_\circ$ put $z_k(x) = \sum_{r \le k} w_r + \iota x_k$. Iteration gives

$$\begin{aligned} &\langle \psi, \mathcal{T}(w_1)\varphi_{\alpha_1}(f_1) \cdots \mathcal{T}(w_n)\varphi_{\alpha_n}(f_n)\Omega \rangle \\ &\qquad = \int \prod_k f_k(x_k) \langle \psi, \Psi_{\underline{\alpha}}(z(x)) \rangle \, \mathrm{d}x. \end{aligned} \tag{A.xlv}$$

Indeed, split each intermediate semigroup in half, choose $s \le \min(1, d_\circ(w_1), \frac{1}{2} \min_{k \ge 2} d_\circ(w_k))$, and append $T_s \Omega = \Omega$. Every field is now between positive-margin factors. Formula (A.xliv) and its uniform bound justify Fubini; consecutive arguments combine into those of (A.xxxiv).

Second, if all L^1 time-derivative norms of $f_k^\epsilon - f_k^0$ tend to zero and contractive joint spectral multipliers C_k^ϵ converge strongly to C_k^0, then

$$C_1^\epsilon \varphi_{\alpha_1}(f_1^\epsilon) \cdots C_n^\epsilon \varphi_{\alpha_n}(f_n^\epsilon)\Omega \longrightarrow C_1^0 \varphi_{\alpha_1}(f_1^0) \cdots C_n^0 \varphi_{\alpha_n}(f_n^0)\Omega \tag{A.xlvi}$$

in every graph norm. Work from right to left: if $\chi^\epsilon \to \chi^0$ in all graph norms, then (A.xli) bounds the graph norm of the field difference by

$$c_{\alpha,l+j}(f^\epsilon - f^0) \|N_H^{2l+j} \chi^\epsilon\| + c_{\alpha,l+j}(f^0) \|N_H^{2l+j}(\chi^\epsilon - \chi^0)\|,$$

which tends to zero. Spectral multipliers commute with graph powers, and their uniform bound and strong convergence preserve this convergence. Induction starts at Ω.

A.2.8. *Uniform smearing at the boundary.* The next estimate controls passage through a test integral. Pointwise polynomial growth alone would not justify that step.

Lemma A.12. Suppose $\mathbf{F}$ is holomorphic on an open set containing $z(x;\eta) = (\iota x_j + \eta h_j e_3)_{j=1}^n$ for $x \in K$, $0 < \eta \le \eta_0 \le 1$, where K is compact, and $|\mathbf{F}(z(x;\eta))| \le b\eta^{-p}$. For $g \in C_c^\infty$ supported in K write $I_g(\eta) = \int g(x)\mathbf{F}(z(x;\eta))\, \mathrm{d}x$ and $D_h = \sum_j h_j \partial_{t_j}$. If the integer $N \ge p + 1/2$, then

$$\sup_{0 < \eta \le \eta_0} |I_g(\eta)| \le 2^N \eta_0^{-(N-1/2)} b \sum_{r=0}^N \|D_h^r g\|_1. \tag{A.xlvii}$$

The limit as $\eta \downarrow 0$ exists. ■

Proof. The Cauchy–Riemann equations and integration by parts give $I_g' = I_{-iD_h g}$. For $J_r = I_{(-iD_h)^r g}$ we have $J_r' = J_{r+1}$ and $|J_r(\eta)| \le b\|D_h^r g\|_1 \eta^{-(N-1/2)}$. Integrate successively from η_0. Each integration of η^{-a}, $a \ge 3/2$, lowers the exponent by one at cost at most two; after $N - 1$ integrations the exponent is $1/2$, whose final integral costs at most two. The endpoint values give the same bound with the corresponding derivative norms. This proves (A.xlvii). Apply it also to $-iD_h g$. It bounds I_g' down to zero, so I_g is Lipschitz there and has a limit. □

Proposition A.13 (Vacuum boundary distributions)**.** For each label word there is a unique tempered distribution $\mathcal{W}_{\underline{\alpha}}$ on $\mathbb{R}^{3n}$ such that

$$\mathcal{W}_{\underline{\alpha}}(f_1 \otimes \cdots \otimes f_n) = \langle \Omega, \varphi_{\alpha_1}(f_1) \cdots \varphi_{\alpha_n}(f_n)\Omega \rangle. \tag{A.xlviii}$$

For any $a_1, \dots, a_{n-1} \in V_+$ it satisfies

$$\begin{aligned} &\mathcal{W}_{\underline{\alpha}}(f_1 \otimes \cdots \otimes f_n) \\ &\quad = \lim_{\epsilon \downarrow 0} \int \prod_j f_j(x_j) \mathbf{W}_{\underline{\alpha}}(\iota(x_2 - x_1 + i\epsilon a_1), \dots, \iota(x_n - x_{n-1} + i\epsilon a_{n-1}))\, \mathrm{d}x. \end{aligned} \tag{A.xlix}$$

■

Proof. Take the vacuum component in (A.xlv) with $w_1 = e_3$ and $w_{j+1} = \iota(i\epsilon a_j)$. The integral in (A.xlix) becomes

$$\langle \Omega, \varphi_{\alpha_1}(f_1)\mathcal{T}(w_2)\varphi_{\alpha_2}(f_2) \cdots \mathcal{T}(w_n)\varphi_{\alpha_n}(f_n)\Omega \rangle.$$

All spectral contractions converge strongly to the identity. Equation (A.xlvi) proves the boundary formula on Schwartz tensors.

For general compact smooth tests use Lem. A.12 with $h_j = 2j$, $\eta_0 = 1$, $b = C_{\mathrm{pt}}^n$, $p = 4n$ and $N = 4n + 1$. Consecutive margins are 2η, so (A.xxxv) applies. Let $|F|_m$ denote the Schwartz seminorm given by the supremum of derivatives of order at most m, multiplied by $(1 + \sum_j |x_j|)^m$. The elementary integral bound $\|\partial^\beta F\|_1 \le \pi^{2n} |F|_{|\beta|+4n}$ follows by comparison with $\prod_j (1 + |x_j|^2)^{-2}$. Since $\sum_j h_j = n(n+1)$, the resulting limit functional obeys

$$|\mathcal{L}(F)| \le 2^N (N+1) C_{\mathrm{pt}}^n (n(n+1))^N \pi^{2n} |F|_{N+4n}.$$

Cutoffs $\chi(\cdot/R)F$ converge in every Schwartz seminorm, so this extends the functional continuously to all Schwartz tests. It agrees with (A.xlviii) first on compact tensors by the boundary formula and then on Schwartz tensors by (A.xlvi). Finite sums of compact smooth tensors are dense for each fixed seminorm: cut off, convolve with a product mollifier, and approximate the compact convolution integral by Riemann sums of its translates. All derivatives up to that order converge uniformly on a fixed compact set. This proves uniqueness of the tempered extension. For $n = 1$ the same assertion is simply the constant $\mathbf{W}_{\alpha_1}$ times the integral of the test. □

A.2.9. *Cyclicity.* Let $\mathcal{D}_0$ be the span of Ω and the finite labelled field words on Ω. It lies in $\mathcal{D}_\infty$ and is invariant under all fields and translations. We prove its density, including the step from ordered to arbitrary positive-time tuples. If $\psi \perp \mathcal{D}_0$, translation covariance gives

$$\langle \psi, U(a_1)\varphi_{\alpha_1}(f_1) \cdots U(a_n)\varphi_{\alpha_n}(f_n)\Omega \rangle = 0.$$

Replace these translations successively by $\mathcal{T}(w_j)$, $w_j \in \mathrm{Tb}_\circ$. For one replacement move preceding regularized factors to the first vector using the adjoint identities. The first positive-margin semigroup puts that vector in $\mathcal{D}_\infty$, so all subsequent field moves are legitimate. The remaining expression is $\langle \chi, U(a)\chi' \rangle$. Its vanishing for all a makes the Fourier transform of the finite joint spectral measure $m_{\chi,\chi'}$ zero. Fourier uniqueness gives $m_{\chi,\chi'} = 0$, hence also $\langle \chi, \mathcal{T}(w)\chi' \rangle = 0$. Induction gives zero for every fully regularized word. Formula (A.xlv) and product approximate identities therefore give $\langle \psi, \Psi_{\underline{\alpha}}(z) \rangle = 0$. By (A.xxxvii), ψ is orthogonal to every ordered positive-time tuple class.

Fix an arbitrary separated positive-time tuple $X = ((\alpha_i, f_i))_{i=1}^n$. Partition the tests by nonnegative $\chi_{h,k}$ supported in cubes of side $2h$ about hk, summing to one. Call a product good if all center heights differ by at least $4h/c_0$, where $c_0 = 1/2$. For small h it is ordered after a permutation, with $\eta = 2h/c_0$ and first height at least η, so its class is orthogonal to ψ. Let ξ_h be the sum of the remaining classes. On the product of the original compact positive supports and their negative

reflected supports all points are distinct, and the continuous Euclidean kernel is bounded by some M_X. Its kernel formula gives

$$\|\xi_h\|^2 \le M_X \left(\int \prod_i |f_i(x_i)| \sum_{\mathbf{k}\,\mathrm{bad}} \prod_i \chi_{h,k_i}(x_i)\,\mathrm{d}x \right)^2 . \tag{A.l}$$

The sum is at most one and vanishes unless two heights differ by less than $(4/c_0 + 2)h$. Its integral is $O_X(h)$: integrate one of these heights over an interval of that length, bound that test by its supremum on its fixed spatial support, and integrate all remaining tests in L^1. Thus $\xi_h \to 0$, and $\langle \psi, [X] \rangle = 0$. Tuple classes are dense in the reflection Hilbert space, so $\psi = 0$. This proves density of $\mathcal{D}_0$. The translation spectrum is already in $\overline{V_+}$: in the convention $U(x) = \int e^{ip\cdot x}\,\mathrm{d}\mathsf{E}(p)$, E is the image of $\mathcal{P}$ under $(E, q) \mapsto (E, -q)$. No vacuum-uniqueness condition has been used.

A.2.10. *Boosts and the Lorentz representation.* Every element of $\mathcal{L} = \mathrm{SO}^+(1, 2)$ is a product of spatial rotations and the boosts

$$B(s)(t, x_1, x_2) = (t \cosh s + x_1 \sinh s, x_1 \cosh s + t \sinh s, x_2).$$

Indeed, a spatial rotation followed by a boost sends e_0 to the future unit vector Λe_0. Their inverse followed by Λ fixes e_0, so acts by $\mathrm{SO}(2)$ on its spatial orthogonal complement.

Proposition A.14 (Lorentz covariance)**.** There is a unique strongly continuous unitary representation $U(\Lambda)$ of $\mathcal{L}$, fixing Ω, such that

$$U(\Lambda)\mathcal{D}_0 = \mathcal{D}_0, \qquad U(\Lambda)\varphi_\alpha(f)U(\Lambda)^{-1}\psi = \varphi_\alpha(f \circ \Lambda^{-1})\psi \quad (\psi \in \mathcal{D}_0). \tag{A.li}$$

Together with translations it gives a strongly continuous representation of the proper orthochronous Poincaré group. Its spatial rotations agree with those of the reflection picture. ∎

Proof. We first prove, for each fixed label word,

$$\mathbf{W}_{\underline{\alpha}}(\widehat{\Lambda}\zeta_1, \dots, \widehat{\Lambda}\zeta_{n-1}) = \mathbf{W}_{\underline{\alpha}}(\zeta_1, \dots, \zeta_{n-1}), \qquad \widehat{\Lambda} = \iota\Lambda\iota^{-1}. \tag{A.lii}$$

Both sides are holomorphic on the connected product of circular tubes. It suffices to compare them for real differences ξ_j of positive third coordinate, where $\mathbf{W}$ is the Euclidean kernel. Spatial-rotation invariance is already known. For a boost use the complex rotation

$$r_z(y_1, y_2, y_3) = (y_1 \cos z + y_3 \sin z, y_2, y_3 \cos z - y_1 \sin z). \tag{A.liii}$$

It obeys $r_{is} = \widehat{B(s)}$. For the fixed real positive-time differences, the open set where all $r_z\xi_j$ remain in the tube contains the imaginary axis: real boosts preserve the forward tube. A connected neighborhood of the segment from 0 to is remains in this set. For small real z, the rotation is Euclidean and preserves the ordering; hence $\mathbf{W}_{\underline{\alpha}}(r_z\xi_1, \dots, r_z\xi_{n-1})$ is constant there. The one-variable identity theorem continues that equality to is. The identity theorem in all differences, followed by the rotation–boost factorization, proves (A.lii).

In the boundary formula (A.xlix), change variables $x_j = \Lambda y_j$. The transformed approach vectors $\Lambda^{-1}a_j$ remain in V_+, so that formula and (A.lii) give

$$\mathcal{W}_{\underline{\alpha}}(F \circ (\Lambda^{-1})^{\times n}) = \mathcal{W}_{\underline{\alpha}}(F). \tag{A.liv}$$

This holds first on tensors and then on all Schwartz tests by their density and continuity under invertible linear changes. Define on polynomial vectors

$$U(\Lambda)\varphi_{\alpha_1}(f_1) \cdots \varphi_{\alpha_n}(f_n)\Omega = \varphi_{\alpha_1}(f_1 \circ \Lambda^{-1}) \cdots \varphi_{\alpha_n}(f_n \circ \Lambda^{-1})\Omega.$$

Their inner products are vacuum distributions, by moving the first word across with the adjoint identity, reversing its order and conjugating its tests. Equation (A.liv) preserves every such Gram matrix. Thus the rule respects all linear relations and is isometric, with inverse given by Λ^{-1}. Density extends it to a unitary. The group law, covariance and uniqueness follow on the spanning words.

As $\Lambda \to 1$, all time derivatives of $f \circ \Lambda^{-1} - f$ tend to zero in L^1. The chain rule gives finite sums of derivatives of f after linear changes, and Schwartz decay gives one integrable majorant for Λ near 1. Equation (A.xlvi) proves strong continuity on polynomial vectors; unitarity and density extend it to $\mathcal{H}$. On those same words, $U(\Lambda)U(a)U(\Lambda)^{-1} = U(\Lambda a)$. Consequently $U(a,\Lambda) = U(a)U(\Lambda)$ has the Poincaré group law and the stated continuity. Finally, the original spatial rotations commute with H and with each real-time point insertion, hence with $A_{\alpha,l}$, and preserve q_α. They therefore give the same field transformation under (A.xl). Both representations fix Ω, so density of polynomial words identifies them. □

A.2.11. *Mixed locality.*

Proposition A.15 (Locality)**.** For any two labels and spacelike separated supports of $f, h \in \mathcal{S}(\mathbb{R}^{1+2})$,

$$\varphi_\alpha(f)\varphi_\beta(h)\psi = \varphi_\beta(h)\varphi_\alpha(f)\psi \qquad (\psi \in \mathcal{D}_0). \tag{A.lv}$$

■

Proof. First take all tests compactly supported. After testing against polynomial vectors and using Hermiticity, it suffices to interchange adjacent entries $k, k+1$ of a vacuum distribution whose supports are spacelike separated. Set $K = \prod_j \operatorname{supp} f_j$, omitting the trivial case of a zero entry, and write $\xi = \vec{x}_{k+1} - \vec{x}_k$, $\tau = t_{k+1} - t_k$. Put

$$\delta = \min_K(|\xi| - |\tau|) > 0, \qquad A = 4/\delta, \qquad \eta_0 = \min((2A)^{-1}, 1/2),$$

$$h_* = 2A \max_{x\in K,\, j<n} (|\vec{x}_{j+1} - \vec{x}_j| + |t_{j+1} - t_j|) + 6.$$

For $-1 \le \lambda \le 1$ let

$$\begin{gathered} h_k^\lambda = -\lambda/2, \qquad h_{k+1}^\lambda = \lambda/2, \\ h_j^\lambda = \tfrac{1}{2} + (j-k-1)h_* \quad (j > k+1), \qquad h_j^\lambda = -\tfrac{1}{2} - (k-j)h_* \quad (j < k), \\ z_j(x;\lambda,\eta) = \iota x_j + \eta h_j^\lambda e_3, \qquad 0 < \eta \le \eta_0. \end{gathered} \tag{A.lvi}$$

The heights of the two selected entries pass through each other. To keep the analytic kernel defined, use the direction $u_\theta = \cos\theta\, e_3 + \sin\theta(\xi/|\xi|, 0)$ with $\sin\theta = A\eta$. Then $\cos\theta \ge \sqrt{3}/2$ and

$$\begin{aligned} d_\circ^{u_\theta}(z_{j+1} - z_j) &= \eta(h_{j+1}^\lambda - h_j^\lambda)\cos\theta \\ &\quad + A\eta \frac{\xi}{|\xi|} \cdot (\vec{x}_{j+1} - \vec{x}_j) - A\eta|t_{j+1} - t_j|. \end{aligned} \tag{A.lvii}$$

For $j = k$ this is at least $-\eta + A\eta\delta = 3\eta$. For every other j the height gap is at least h_*, and its definition gives the same lower bound. Thus all configurations remain in $\mathrm{Tb}_n^{\mathrm{all}}$, and (A.xxxviii) gives uniformly

$$|\mathbf{S}_{\underline{\alpha}}(z(x;\lambda,\eta))| \le (C_{\mathrm{pt}}\eta^{-4})^n. \tag{A.lviii}$$

The witnessing direction depends on x and η; the glued holomorphic function is the same in every tube.

For smooth g supported in K define $I_g(\lambda,\eta) = \int g(x)\mathbf{S}_{\underline{\alpha}}(z(x;\lambda,\eta))\,\mathrm{d}x$. Holomorphy, integration by parts and compactness at every positive height give

$$\partial_\eta I_g = I_{-iD_\lambda g}, \qquad \partial_\lambda I_g = \eta I_{-iD' g}, \qquad D_\lambda = \sum_j h_j^\lambda \partial_{t_j}, \quad D' = \tfrac{1}{2}(\partial_{t_{k+1}} - \partial_{t_k}). \tag{A.lix}$$

Apply Lem. A.12 uniformly in λ, with $N = 4n+1$. It gives

$$|I_g(\lambda,\eta)| \le Q(g) := 2^N \eta_0^{-(N-1/2)} C_{\mathrm{pt}}^n \sum_{r=0}^{N} \sup_{|\lambda|\le 1} \|D_\lambda^r g\|_1. \tag{A.lx}$$

This is finite because the coefficients of D_λ are uniformly bounded. Take $g = \bigotimes_j f_j$. At $\lambda = 1$ the heights increase in the original order, so (A.xlix) identifies the limit with the original vacuum distribution. At $\lambda = -1$ they increase in the exchanged order. The simultaneous permutation symmetry of points and labels in $\mathbf{S}$ identifies this limit with the distribution having entries $k, k+1$ exchanged. But (A.lix) and (A.lx) give

$$|I_g(1,\eta) - I_g(-1,\eta)| \le 2\eta Q(D'g) \longrightarrow 0.$$

This proves the adjacent interchange for compact tests. Cut off the other Schwartz entries and use graph convergence (A.xlvi) to pass to both products. Density of the polynomial test vectors proves the operator equality for compact f, h. Finally cut off f, h inside their original supports. Spacelike separation is preserved, and the same graph convergence gives (A.lv) for the original Schwartz tests. This argument keeps every label throughout, and therefore proves all mixed commutators. □

Appendix B. Index of notation and conventions

The entries follow the order of the construction. The last column gives the definition or the passage immediately preceding the cited result.

Notation	Meaning and location
W_3, $v = (x, g)$, $v \cdot s$	The signed permutation group, a chamber at the lattice site x, and its four neighbours; (3).
N_f, $\nu = 8N_f$, γ_μ, γ_4	The number of four-component flavours, the dimension of the spin–internal–colour–flavour fibre at one chamber, and the spin matrices; sections 2 and 5.
D_U, $D[U]$, m_0	D_U includes the bare mass; $D[U] = D_U - m_0$ is massless. The physical mass is $\mathsf{m} = a_K^{-1} m_0$; (4) and section 5.
$a_p(U)$, $A(U)$, β	The plaquette energy $1 - \frac{1}{2}\operatorname{tr} U_p$, its sum over positive plaquettes, and its coefficient $\beta = L^K \mathsf{b}$; section 2.
$\mathfrak{B}$, $\Phi_{\alpha,K}$, S^T_{n,K,ν_K}	The reflection-real meson basis, the action and meson insertions, and their joint cumulants. The index 0 denotes the action density; (1). The centered moments σ are defined in (2).
L, M, a_K, λ_j, n_j	The blocking factor, the polymer-block side in level-j units, the spacing $a_K = L^{-K}$, the spacing $\lambda_j = L^{j-K}$, and the site count $n_j = \ell^3 \lambda_j^{-3}$ on T_ℓ; section 4.
t_j, x_j, b_j	The calibration $\mathsf{t}_j = L^{K-j}\mathsf{b}$, the running action coefficient, and its extracted increment: $x_{j+1} = L^{-1}x_j - b_{j+1}$; section 4.
$\mathfrak{m}_j$, s_j, μ_j	The rescaled bare mass $\mathfrak{m}_j = \lambda_j \mathsf{m}$, the accumulated correction, and its increment $s_{j+1} = Ls_j + \mu_{j+1}$, $s_0 = 0$; section 4 and Prop. 8.7.
$R(t)$, $\epsilon(t)$, $\delta(t)$, $\mathfrak{b}(t)$	The range and gauge-field thresholds. The threshold for the rescaled fluctuation is $\mathfrak{b}(t) = t^{1/2}\delta(t)$; section 4.
$\phi(t)$, $\widehat{\phi}(t)$, $\phi_\sharp(t)$, $\psi_\sharp(t)$	The functions controlling logarithmic losses. They are $t^{-1/2}(1+\log t)^{\mathfrak{p}_0}$, $(1+\log t)^{\mathfrak{p}_0}$, $t^{-1/2}(1+\log t)^{3\mathfrak{p}_0}$, and $t^{-1/2}(1+\log t)^{9\mathfrak{p}_0}$, respectively; section 4.
$\mathcal{M}$, $U_j(V)$	The gauge block average and the critical gauge orbit on its fibre $\mathcal{M}^j U = V$; section 4.
$(\mathbf{U}, \mathbf{J})$, $(U, A', \mathbf{J})$, $\mathcal{U}_j^\circ(X; b_0, b_1)$	A complex pair, a representative with $\mathbf{U} = e^{iL^{-j}A'}U$ and U real, and its analytic core. The core includes bounds on the required block towers; section 4.
$\mathcal{P}_j$, $\mathcal{D}_j$, $d_j(X)$, $X^\sharp$	The partition into M-blocks, its nonempty wall-connected polymers, their block-tree length, and the prescribed enlargement used in localization; section 4.
ι, Π_0, Π_1	The normalized embedding of a site spinor into chamber constants, the projection onto these constants, and its complement; section 5.
$Q(V)$, R_k, q_{j+1}	The normalized site average, its iterated chamber average $R_k = Q(\mathcal{M}^{k-1}U)\cdots Q(U)\iota^*$, and $q_1 = L^{-1/2}Q(U)\iota^*$, $q_{j+1} = L^{-1/2}Q(\mathcal{M}^j U)$ for $j \ge 1$; Lems. 5.2 and 5.4.

Notation	Meaning and location (continued)
$\theta = b_F$, θ_k, $\mathfrak{a}_k$	The block penalty and the coefficients obtained by composing its Gaussian kernels. Here $\theta_k = b_F(1-L^{-1})/(1-L^{-k})$ and $\mathfrak{a}_k = L^{-k}\theta_k$; Lem. 5.4.
$\mathbb{D}_j$, Γ_j, Ψ^{bg}	The quadratic form after j integrations, the form on the variables integrated at the next step, and the background substitution obtained by completing the square. The fluctuation covariance is Γ_j^{-1}; the definitions precede Lem. 5.5.
$\mathcal{K}_i$, W_i	The operator on the original chamber variables with all the level-i block penalties added, and the sum of those penalties; section 6.
L_i, s_i, $d_{h,i}$	The layer scale, its Lipschitz minorant, and the path distance with bond length $1/\max(\mathsf{s}_i(x), \mathsf{s}_i(x'))$; section 6.
$\mathcal{H}_{(n)}$, S_j^{sym}	The background map from the stored and current variables to level n, and the symmetrized localized covariance; Prop. 6.7 and Lem. 6.8 and (26).
$\|F\|_h$, h_F, Γ_F	The Grassmann fibre norm with a weight for each generator, the fixed fermion weight, and the covariance allowance used in Gaussian contraction; Lem. 5.1 and Prop. 6.7 and the definitions preceding Lem. 7.3.
$\|F\|_{k;a}$, $h^{(k)}(n)$	The level-k family norm at pair radius a, with weight $e^{\kappa d_k(X)}$ in the polymer sum, and the leg weight $C_H L^{-(k-n)} h_F$; the definitions preceding Lem. 7.3.
$\|\mathbb{V}\|_{\star,r}$, $\|V\|_r$	The origin norm allowing quadratic fluctuation growth, and the norm that sums local potentials over polymers containing one specified cube; (29) and Lem. 7.5.
h, $\mathcal{S}(h)$, Ω_j, Λ_j, Z_j	A history, its discrete label, its nested regions, its integration regions, and $Z_j = \Lambda_j^c$. The label is separate from the free coordinates at which that history is evaluated; section 9.
$\mathcal{L}_j$, $\mathfrak{P}_h$, φ_j	The layers $\Omega_j \setminus \Omega_{j+1}$, with last layer Ω_k; their fermion places; and the fixed history cutoffs. The fermions on layers $j < k$ are stored; section 9.
$s = (w, J)$, n, $\mathfrak{c}_k(s)$	The physical action and matrix sources, the bare matrix $\mathsf{n} = -a_K J$, and the step coordinates $(w, \mathsf{n}, (\sigma_i)_{i\le k})$; section 10 and Lem. 10.4.
$\|w\|_{\sharp,j}$, $\|w\|_\sharp$, π_j	The norm $\max(\|w\|_\infty, 40M\lambda_j\pi_j \operatorname{Lip} w)$, its final-level value, and $\pi_j = (1+(K-j)/n_\pi)^{\mathfrak{p}_0}$; section 10.
$\mathscr{O}_j$, $\mathscr{D}_j$, $\mathscr{D}_j(a)$	The physical source domain, the domain of independent step coordinates, and the latter domain at pair radius a. A doubled domain doubles each radius; section 10.
σ_k, ν_{k+1}, $\mathcal{N}_k$, $\mathcal{Z}_k$	The source correction and its extracted increment, $\mathcal{N}_k = -\lambda_k J + \sigma_k$, and the response $\mathcal{Z}_k = 1 - L^{K-k}\partial_J \sigma_k|_0$; the definitions preceding Lem. 10.4.
$E_k(s)$, $Z_k^{\mathrm{sub}}[s]$	The extracted vacuum energy and the remaining partition function, with $Z[s] = e^{E_k(s)} Z_k^{\mathrm{sub}}[s]$; Prop. 10.14.

Conventions. The barred and unbarred fermions are independent Grassmann generators. We normalize Berezin integration by $\operatorname{Ber} e^{-\bar\Psi A \Psi} = \det A$. At complex gauge fields, q^+ denotes the holomorphic continuation of the adjoint at real fields. The barred background substitution is the row twin given before Lem. 5.5; it is not obtained by taking the adjoint of the unbarred substitution.

Operator formulas use lattice coordinates unless a physical scale is displayed. The physical torus has side ℓ. Gauge links are periodic, and spinors are antiperiodic in each coordinate. In the volume-first construction, one first takes torus limits at fixed K and then takes a subsequence as $K \to \infty$; see section 12.

Terms used in the history expansion. *Made and formed.* A made boundary entry is a term generated at an earlier step and retained near a boundary. A formed factor is one of the retained factors of the calibration construction. Their scalar contributions are listed separately in Def. 9.5; the corresponding historical families are those of [3, Prop. 4.126].

Tie. This is agreement at real triples representing the same pair, with the same stored data. Its preservation under substitution is proved in [3, Prop. 4.118]. For a term with external fermion legs, the coefficient and the legs must be transformed together, as in Lem. 9.7.

Tier. This is the prescribed range of a stored external field. Its bound depends on whether a bond lies in the collar, on the rim, or elsewhere; see [3, §4.14.7]. The superscript rad on a historical family norm restricts the stored coordinates to these ranges, as specified in [3, §4.15.15].

Guard. For a fine region $Z_\circ$, $\mathrm{Gd}(Z_\circ)$ is the union of coarse cubes in $Z_\circ^\sharp$ meeting the integration region. Localization parameters are fixed at one on this set and vary outside it; see [3, §4.14.3].

Pinned substitution and levelling function. A pinned substitution applies the shift and then the prescribed local gauge transformation u. This transformation removes the endpoint factors of the response and preserves the stored transports at the specified pinnable tops. It is the levelling function of [3, §4.15.6]; its use in an earlier term is described in [3, §4.15.9]. With fermion legs, the inverse action of the same u is included in their reading; see Lem. 9.7.

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[2] P. Etingof, *Lie Groups and Lie Algebras*, MIT lecture notes, 18.755, Spring 2024, Theorem 37.5.

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